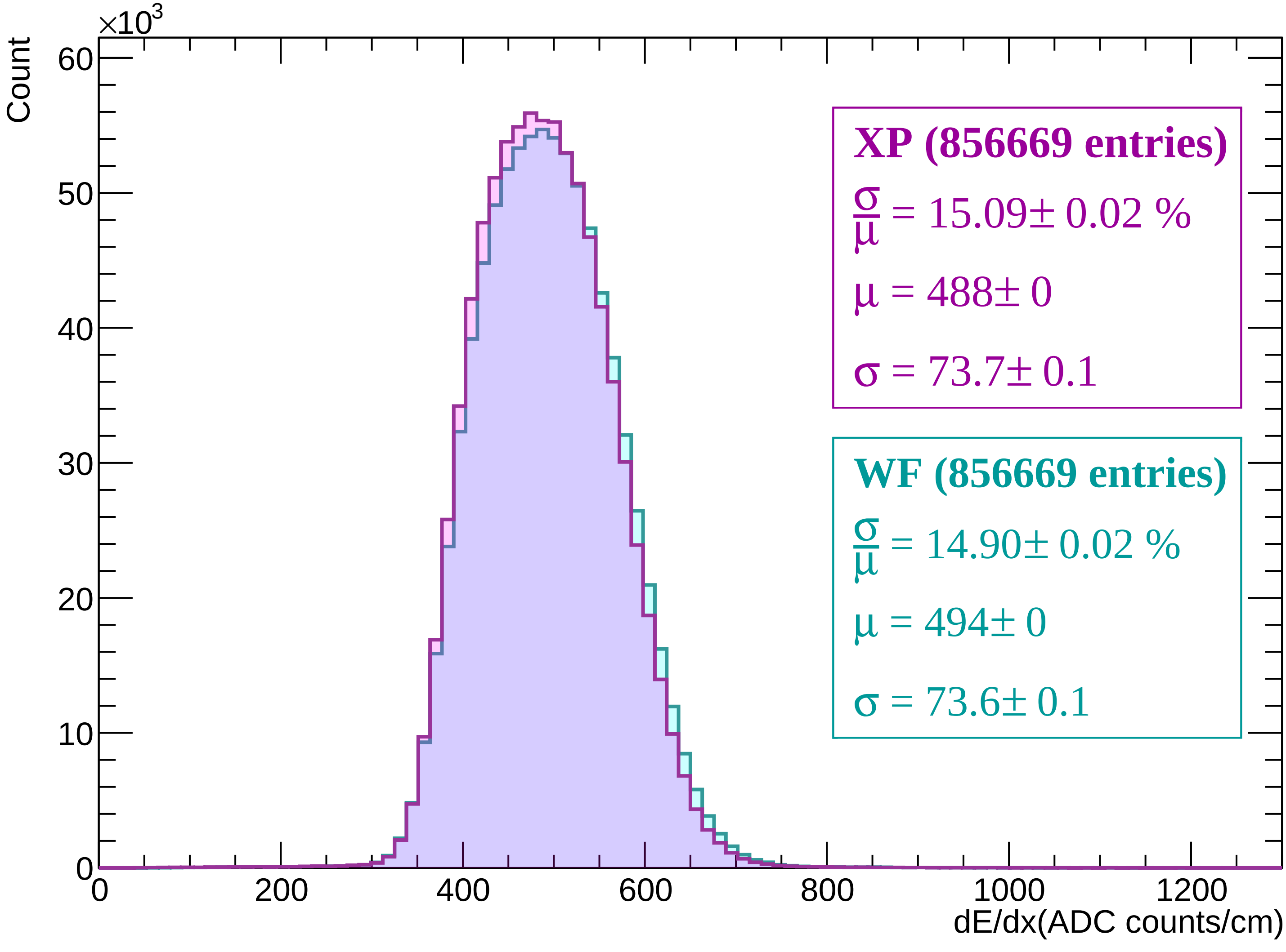
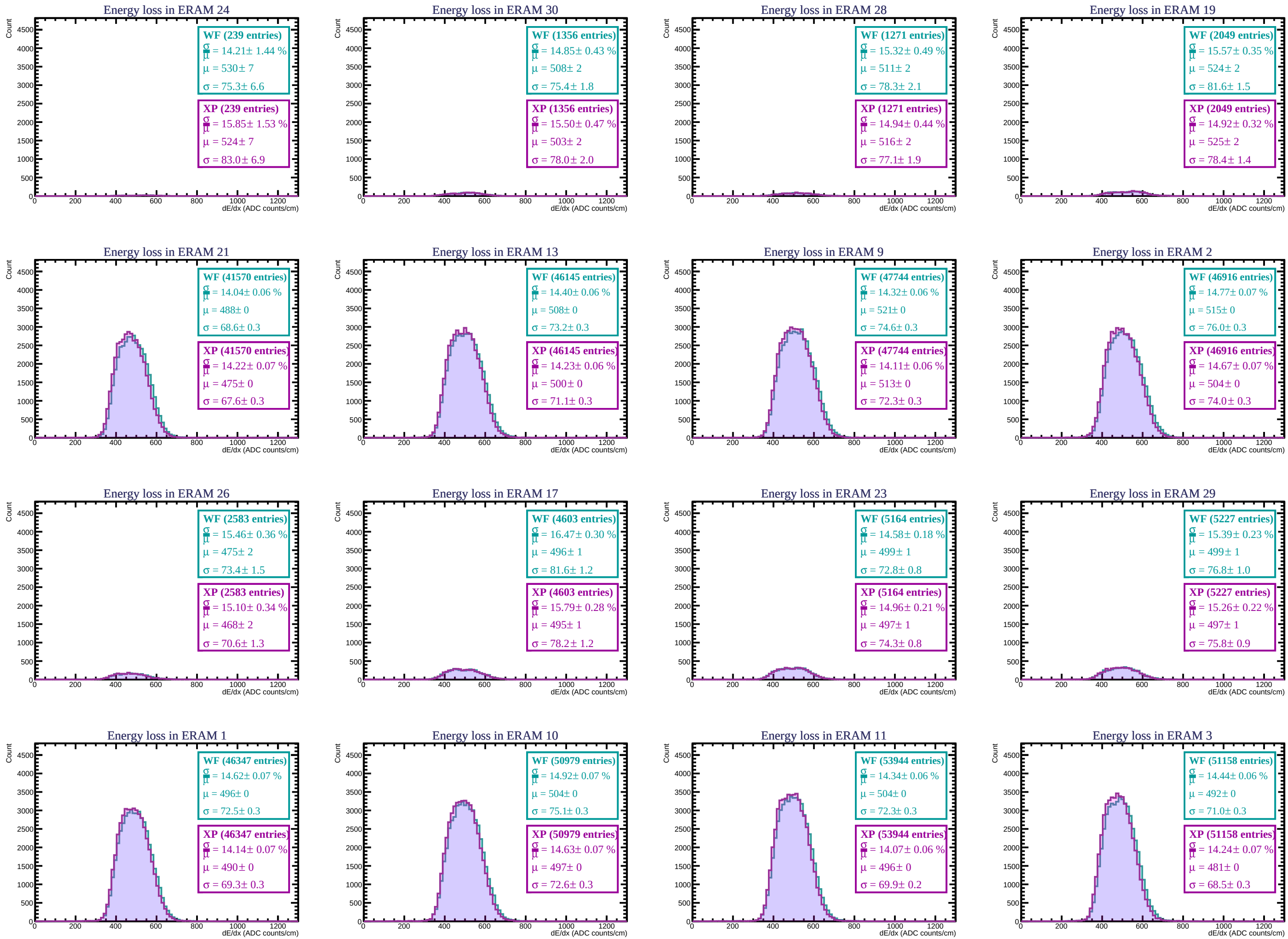
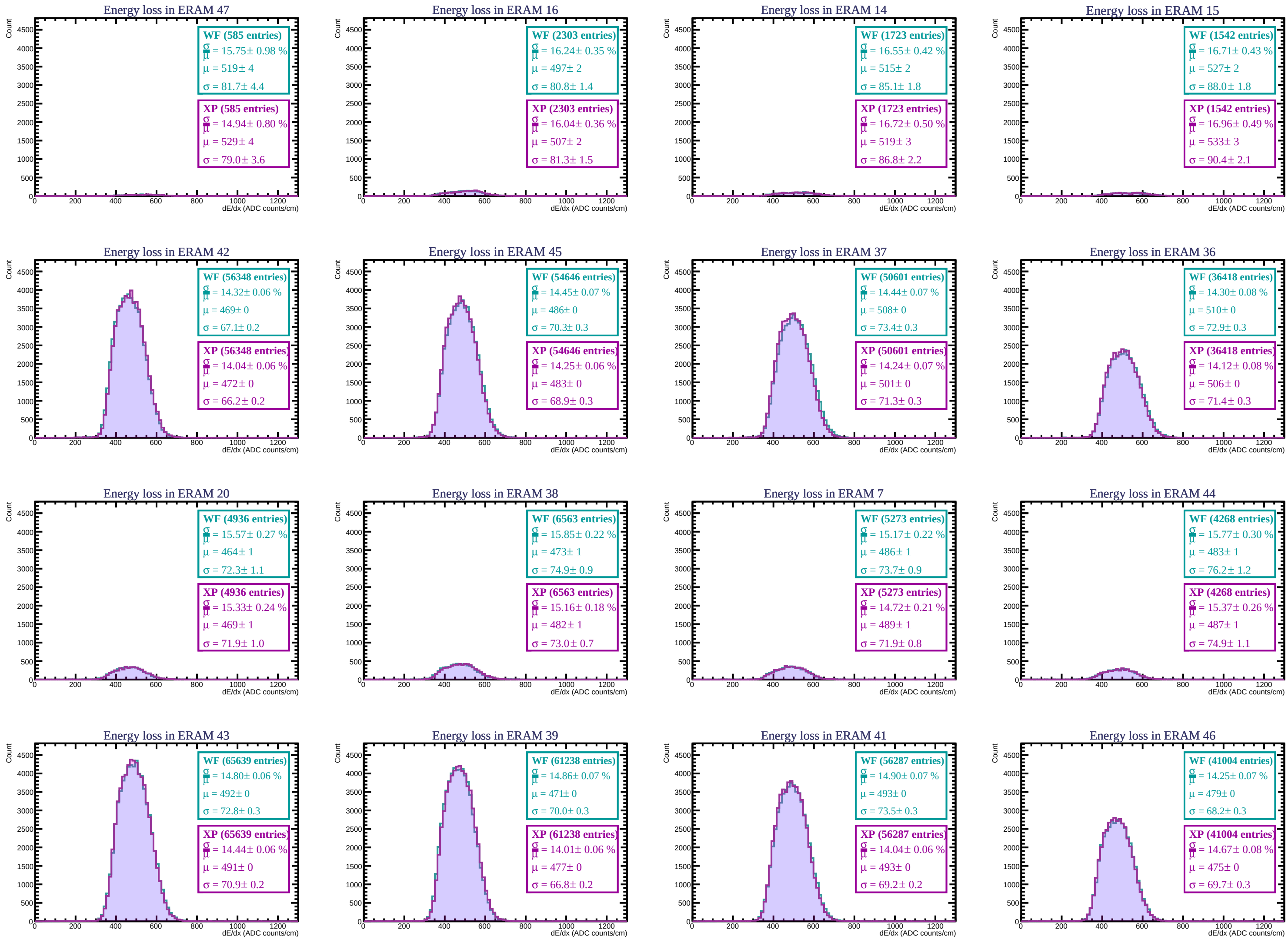
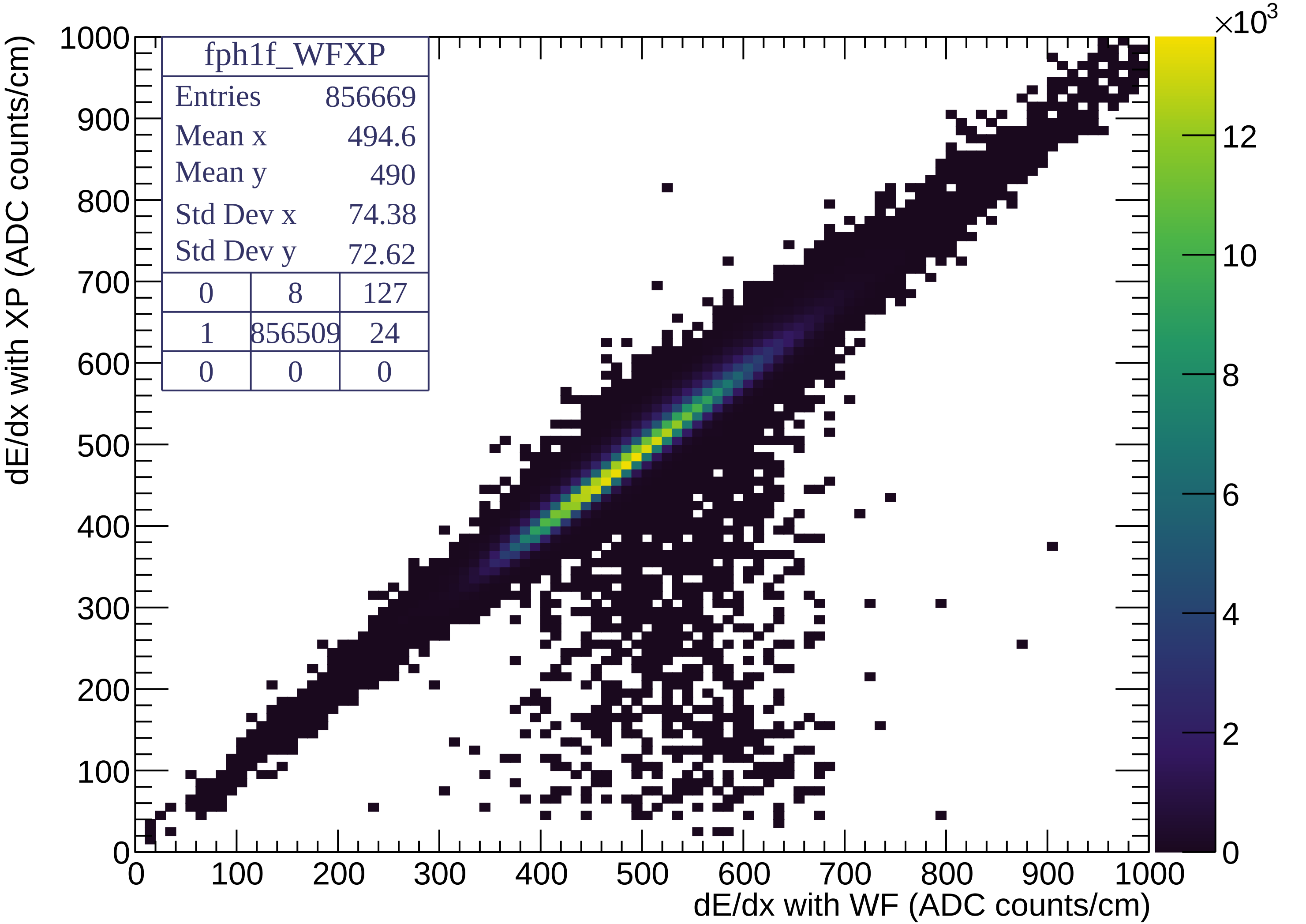


Count
×10^3
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (856669 entries)
σ/μ = 15.09± 0.02 %
μ = 488± 0
σ = 73.7± 0.1
WF (856669 entries)
σ/μ = 14.90± 0.02 %
μ = 494± 0
σ = 73.6± 0.1

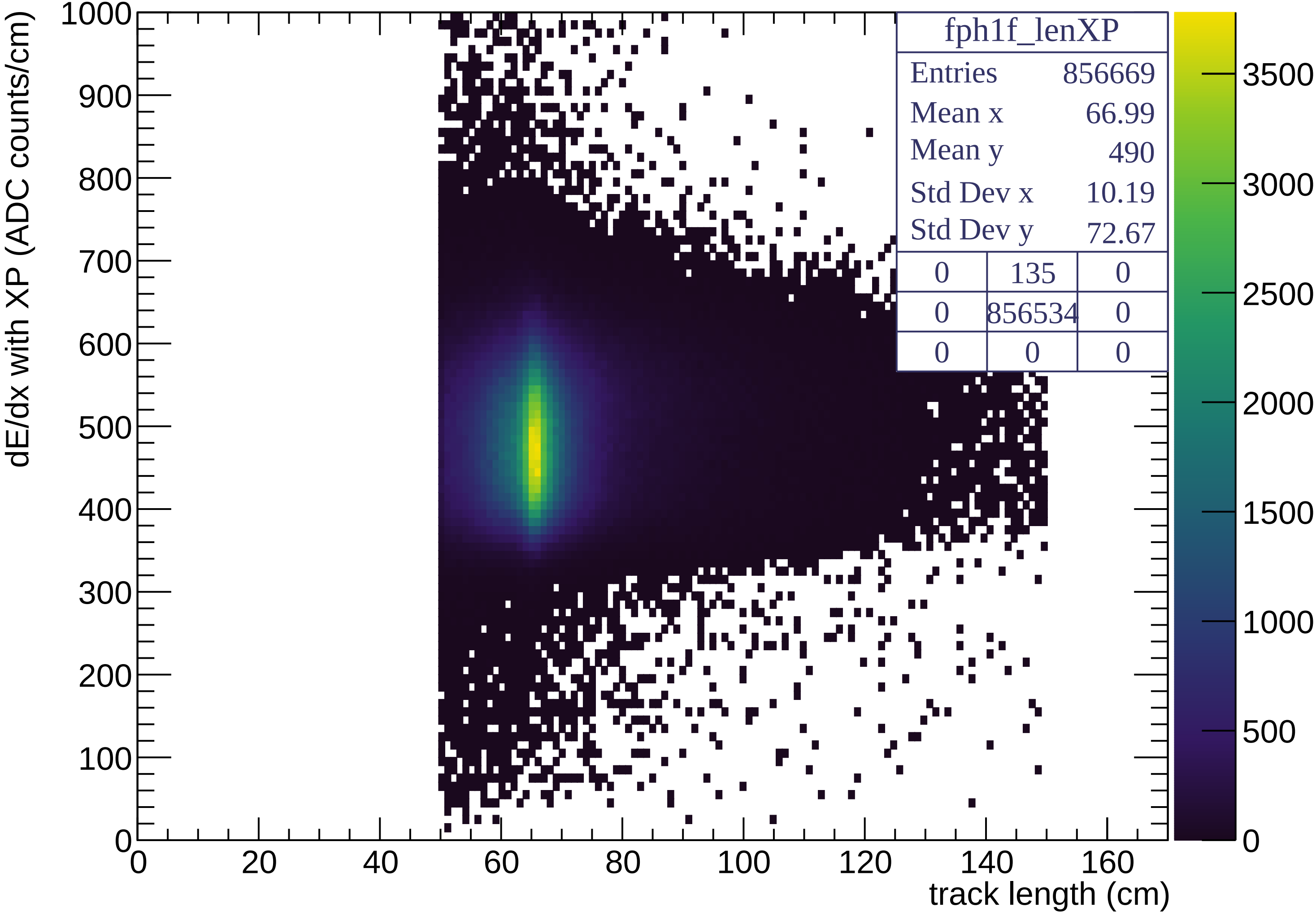

Energy loss in ERAM 24
WF (239 entries)
σ/μ = 14.21± 1.44 %
μ = 530± 7
σ = 75.3± 6.6
XP (239 entries)
σ/μ = 15.85± 1.53 %
μ = 524± 7
σ = 83.0± 6.9
Energy loss in ERAM 30
WF (1356 entries)
σ/μ = 14.85± 0.43 %
μ = 508± 2
σ = 75.4± 1.8
XP (1356 entries)
σ/μ = 15.50± 0.47 %
μ = 503± 2
σ = 78.0± 2.0
Energy loss in ERAM 28
WF (1271 entries)
σ/μ = 15.32± 0.49 %
μ = 511± 2
σ = 78.3± 2.1
XP (1271 entries)
σ/μ = 14.94± 0.44 %
μ = 516± 2
σ = 77.1± 1.9
Energy loss in ERAM 19
WF (2049 entries)
σ/μ = 15.57± 0.35 %
μ = 524± 2
σ = 81.6± 1.5
XP (2049 entries)
σ/μ = 14.92± 0.32 %
μ = 525± 2
σ = 78.4± 1.4
Energy loss in ERAM 21
WF (41570 entries)
σ/μ = 14.04± 0.06 %
μ = 488± 0
σ = 68.6± 0.3
XP (41570 entries)
σ/μ = 14.22± 0.07 %
μ = 475± 0
σ = 67.6± 0.3
Energy loss in ERAM 13
WF (46145 entries)
σ/μ = 14.40± 0.06 %
μ = 508± 0
σ = 73.2± 0.3
XP (46145 entries)
σ/μ = 14.23± 0.06 %
μ = 500± 0
σ = 71.1± 0.3
Energy loss in ERAM 9
WF (47744 entries)
σ/μ = 14.32± 0.06 %
μ = 521± 0
σ = 74.6± 0.3
XP (47744 entries)
σ/μ = 14.11± 0.06 %
μ = 513± 0
σ = 72.3± 0.3
Energy loss in ERAM 2
WF (46916 entries)
σ/μ = 14.77± 0.07 %
μ = 515± 0
σ = 76.0± 0.3
XP (46916 entries)
σ/μ = 14.67± 0.07 %
μ = 504± 0
σ = 74.0± 0.3
Energy loss in ERAM 26
WF (2583 entries)
σ/μ = 15.46± 0.36 %
μ = 475± 2
σ = 73.4± 1.5
XP (2583 entries)
σ/μ = 15.10± 0.34 %
μ = 468± 2
σ = 70.6± 1.3
Energy loss in ERAM 17
WF (4603 entries)
σ/μ = 16.47± 0.30 %
μ = 496± 1
σ = 81.6± 1.2
XP (4603 entries)
σ/μ = 15.79± 0.28 %
μ = 495± 1
σ = 78.2± 1.2
Energy loss in ERAM 23
WF (5164 entries)
σ/μ = 14.58± 0.18 %
μ = 499± 1
σ = 72.8± 0.8
XP (5164 entries)
σ/μ = 14.96± 0.21 %
μ = 497± 1
σ = 74.3± 0.8
Energy loss in ERAM 29
WF (5227 entries)
σ/μ = 15.39± 0.23 %
μ = 499± 1
σ = 76.8± 1.0
XP (5227 entries)
σ/μ = 15.26± 0.22 %
μ = 497± 1
σ = 75.8± 0.9
Energy loss in ERAM 1
WF (46347 entries)
σ/μ = 14.62± 0.07 %
μ = 496± 0
σ = 72.5± 0.3
XP (46347 entries)
σ/μ = 14.14± 0.07 %
μ = 490± 0
σ = 69.3± 0.3
Energy loss in ERAM 10
WF (50979 entries)
σ/μ = 14.92± 0.07 %
μ = 504± 0
σ = 75.1± 0.3
XP (50979 entries)
σ/μ = 14.63± 0.07 %
μ = 497± 0
σ = 72.6± 0.3
Energy loss in ERAM 11
WF (53944 entries)
σ/μ = 14.34± 0.06 %
μ = 504± 0
σ = 72.3± 0.3
XP (53944 entries)
σ/μ = 14.07± 0.06 %
μ = 496± 0
σ = 69.9± 0.2
Energy loss in ERAM 3
WF (51158 entries)
σ/μ = 14.44± 0.06 %
μ = 492± 0
σ = 71.0± 0.3
XP (51158 entries)
σ/μ = 14.24± 0.07 %
μ = 481± 0
σ = 68.5± 0.3
Count
dE/dx (ADC counts/cm)



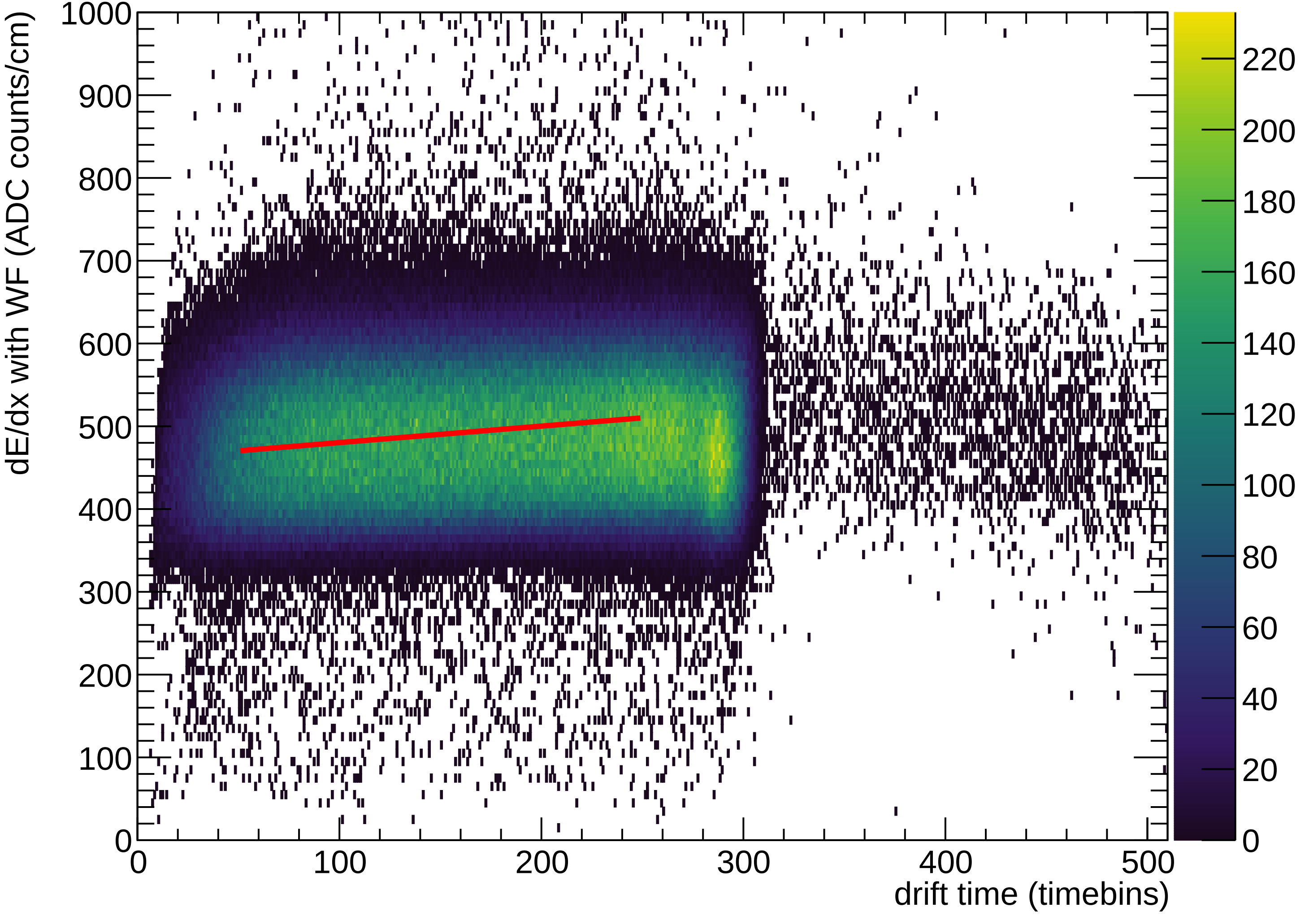
Energy loss in ERAM 47
WF (585 entries)
σ/μ = 15.75± 0.98 %
μ = 519± 4
σ = 81.7± 4.4
XP (585 entries)
σ/μ = 14.94± 0.80 %
μ = 529± 4
σ = 79.0± 3.6
Energy loss in ERAM 16
WF (2303 entries)
σ/μ = 16.24± 0.35 %
μ = 497± 2
σ = 80.8± 1.4
XP (2303 entries)
σ/μ = 16.04± 0.36 %
μ = 507± 2
σ = 81.3± 1.5
Energy loss in ERAM 14
WF (1723 entries)
σ/μ = 16.55± 0.42 %
μ = 515± 2
σ = 85.1± 1.8
XP (1723 entries)
σ/μ = 16.72± 0.50 %
μ = 519± 3
σ = 86.8± 2.2
Energy loss in ERAM 15
WF (1542 entries)
σ/μ = 16.71± 0.43 %
μ = 527± 2
σ = 88.0± 1.8
XP (1542 entries)
σ/μ = 16.96± 0.49 %
μ = 533± 3
σ = 90.4± 2.1
Energy loss in ERAM 42
WF (56348 entries)
σ/μ = 14.32± 0.06 %
μ = 469± 0
σ = 67.1± 0.2
XP (56348 entries)
σ/μ = 14.04± 0.06 %
μ = 472± 0
σ = 66.2± 0.2
Energy loss in ERAM 45
WF (54646 entries)
σ/μ = 14.45± 0.07 %
μ = 486± 0
σ = 70.3± 0.3
XP (54646 entries)
σ/μ = 14.25± 0.06 %
μ = 483± 0
σ = 68.9± 0.3
Energy loss in ERAM 37
WF (50601 entries)
σ/μ = 14.44± 0.07 %
μ = 508± 0
σ = 73.4± 0.3
XP (50601 entries)
σ/μ = 14.24± 0.07 %
μ = 501± 0
σ = 71.3± 0.3
Energy loss in ERAM 36
WF (36418 entries)
σ/μ = 14.30± 0.08 %
μ = 510± 0
σ = 72.9± 0.3
XP (36418 entries)
σ/μ = 14.12± 0.08 %
μ = 506± 0
σ = 71.4± 0.3
Energy loss in ERAM 20
WF (4936 entries)
σ/μ = 15.57± 0.27 %
μ = 464± 1
σ = 72.3± 1.1
XP (4936 entries)
σ/μ = 15.33± 0.24 %
μ = 469± 1
σ = 71.9± 1.0
Energy loss in ERAM 38
WF (6563 entries)
σ/μ = 15.85± 0.22 %
μ = 473± 1
σ = 74.9± 0.9
XP (6563 entries)
σ/μ = 15.16± 0.18 %
μ = 482± 1
σ = 73.0± 0.7
Energy loss in ERAM 7
WF (5273 entries)
σ/μ = 15.17± 0.22 %
μ = 486± 1
σ = 73.7± 0.9
XP (5273 entries)
σ/μ = 14.72± 0.21 %
μ = 489± 1
σ = 71.9± 0.8
Energy loss in ERAM 44
WF (4268 entries)
σ/μ = 15.77± 0.30 %
μ = 483± 1
σ = 76.2± 1.2
XP (4268 entries)
σ/μ = 15.37± 0.26 %
μ = 487± 1
σ = 74.9± 1.1
Energy loss in ERAM 43
WF (65639 entries)
σ/μ = 14.80± 0.06 %
μ = 492± 0
σ = 72.8± 0.3
XP (65639 entries)
σ/μ = 14.44± 0.06 %
μ = 491± 0
σ = 70.9± 0.2
Energy loss in ERAM 39
WF (61238 entries)
σ/μ = 14.86± 0.07 %
μ = 471± 0
σ = 70.0± 0.3
XP (61238 entries)
σ/μ = 14.01± 0.06 %
μ = 477± 0
σ = 66.8± 0.2
Energy loss in ERAM 41
WF (56287 entries)
σ/μ = 14.90± 0.07 %
μ = 493± 0
σ = 73.5± 0.3
XP (56287 entries)
σ/μ = 14.04± 0.06 %
μ = 493± 0
σ = 69.2± 0.2
Energy loss in ERAM 46
WF (41004 entries)
σ/μ = 14.25± 0.07 %
μ = 479± 0
σ = 68.2± 0.3
XP (41004 entries)
σ/μ = 14.67± 0.08 %
μ = 475± 0
σ = 69.7± 0.3
Count
dE/dx (ADC counts/cm)

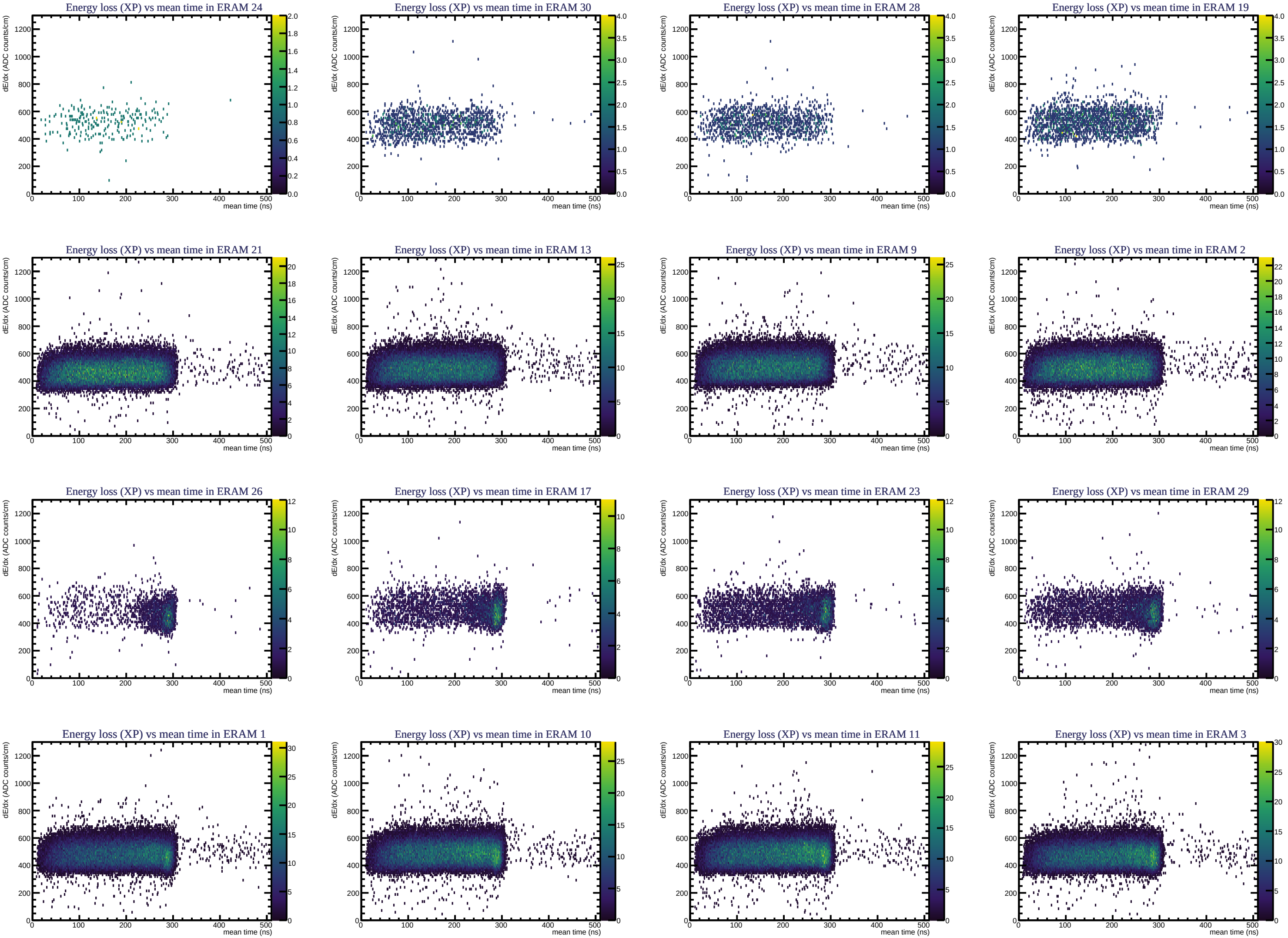


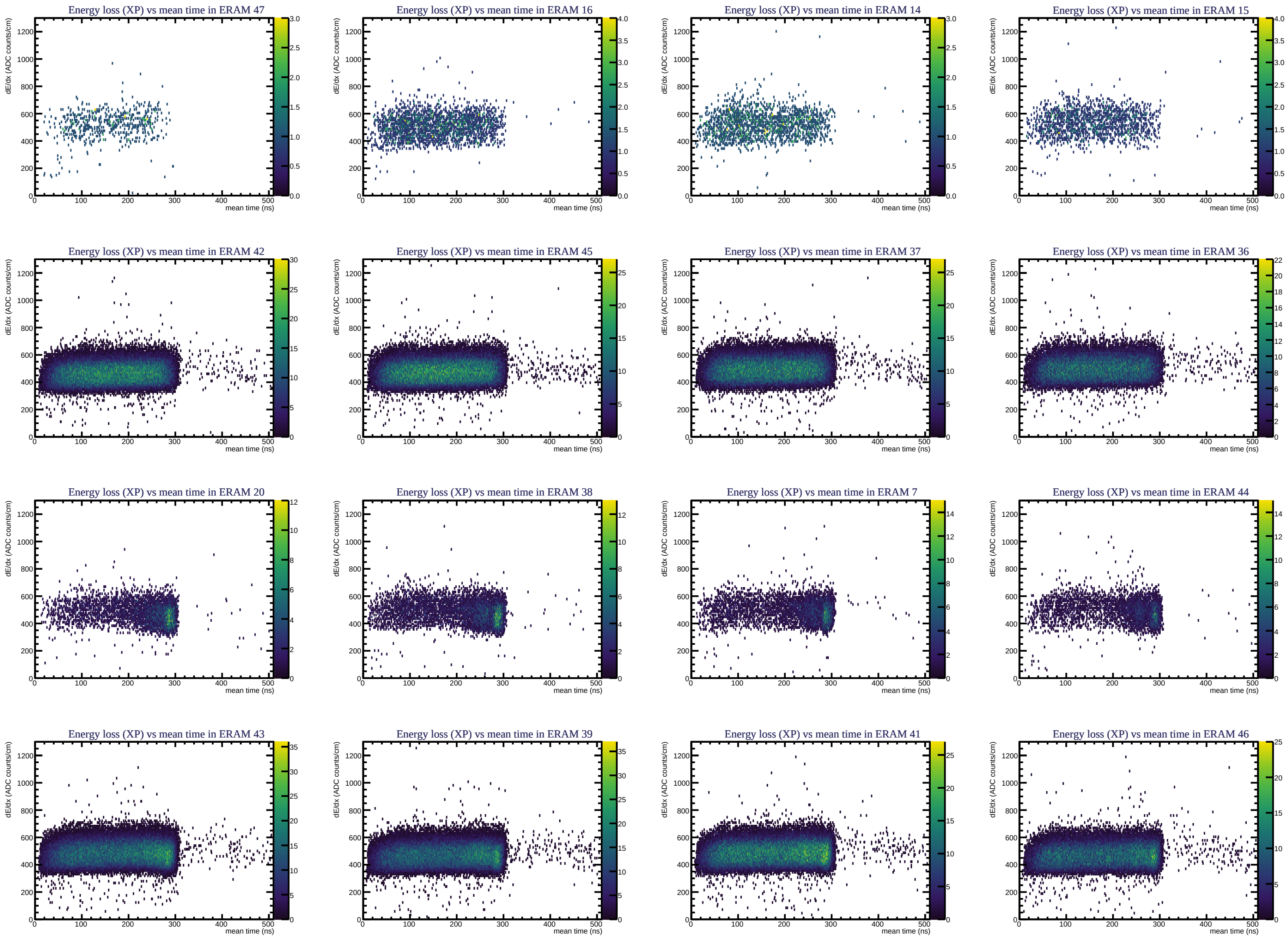
fph1f_WFXP
Entries 856669
Mean x 494.6
Mean y 490
Std Dev x 74.38
Std Dev y 72.62
0 8 127
1 856509 24
0 0 0
dE/dx with WF (ADC counts/cm)
dE/dx with XP (ADC counts/cm)
×10^3



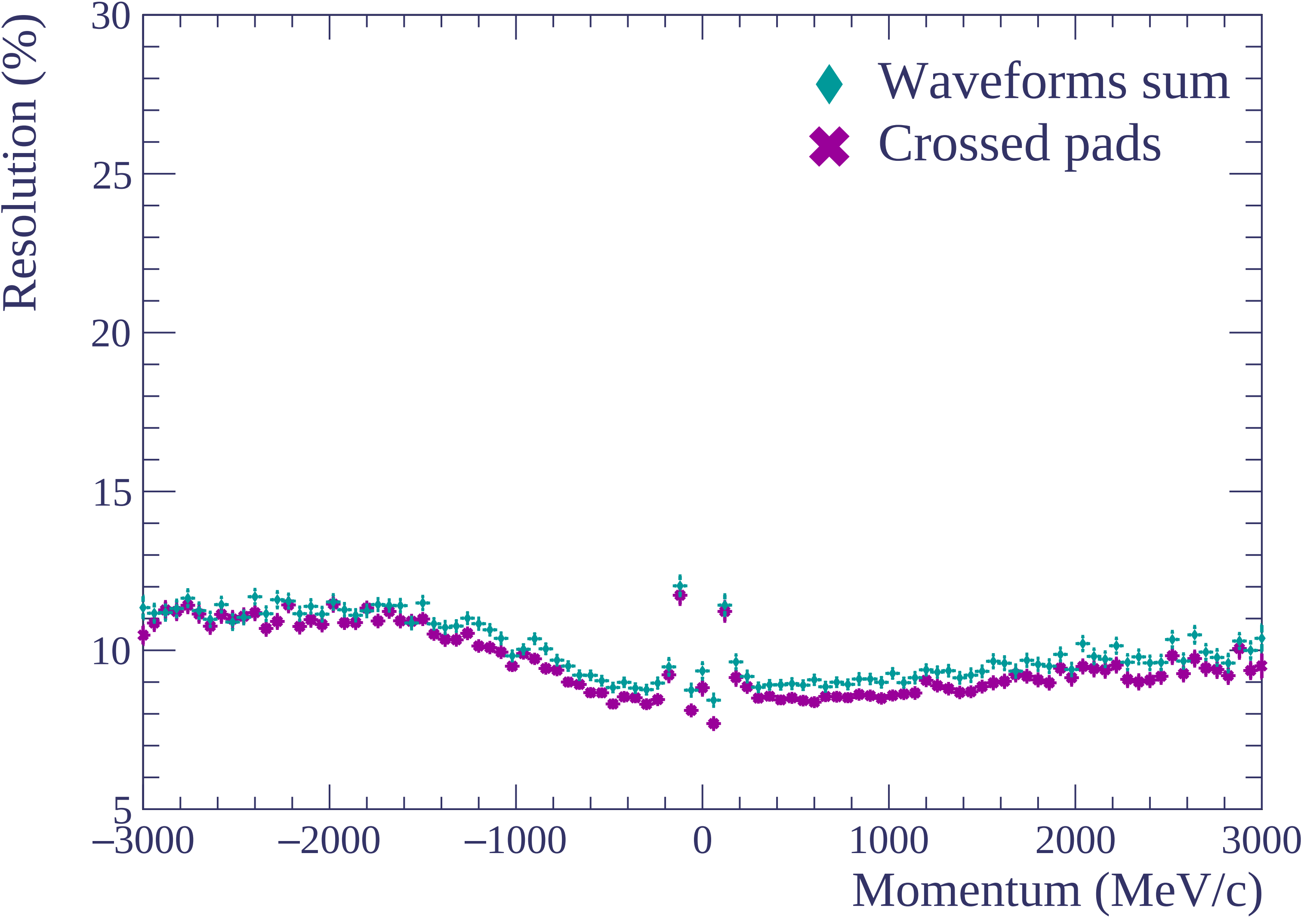
fph1f_lenXP
Entries 856669
Mean x 66.99
Mean y 490
Std Dev x 10.19
Std Dev y 72.67
0 135 0
0 856534 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
100
200
300
400
500
600
700
800
900
1000
500
1000
1500
2000
2500
3000
3500

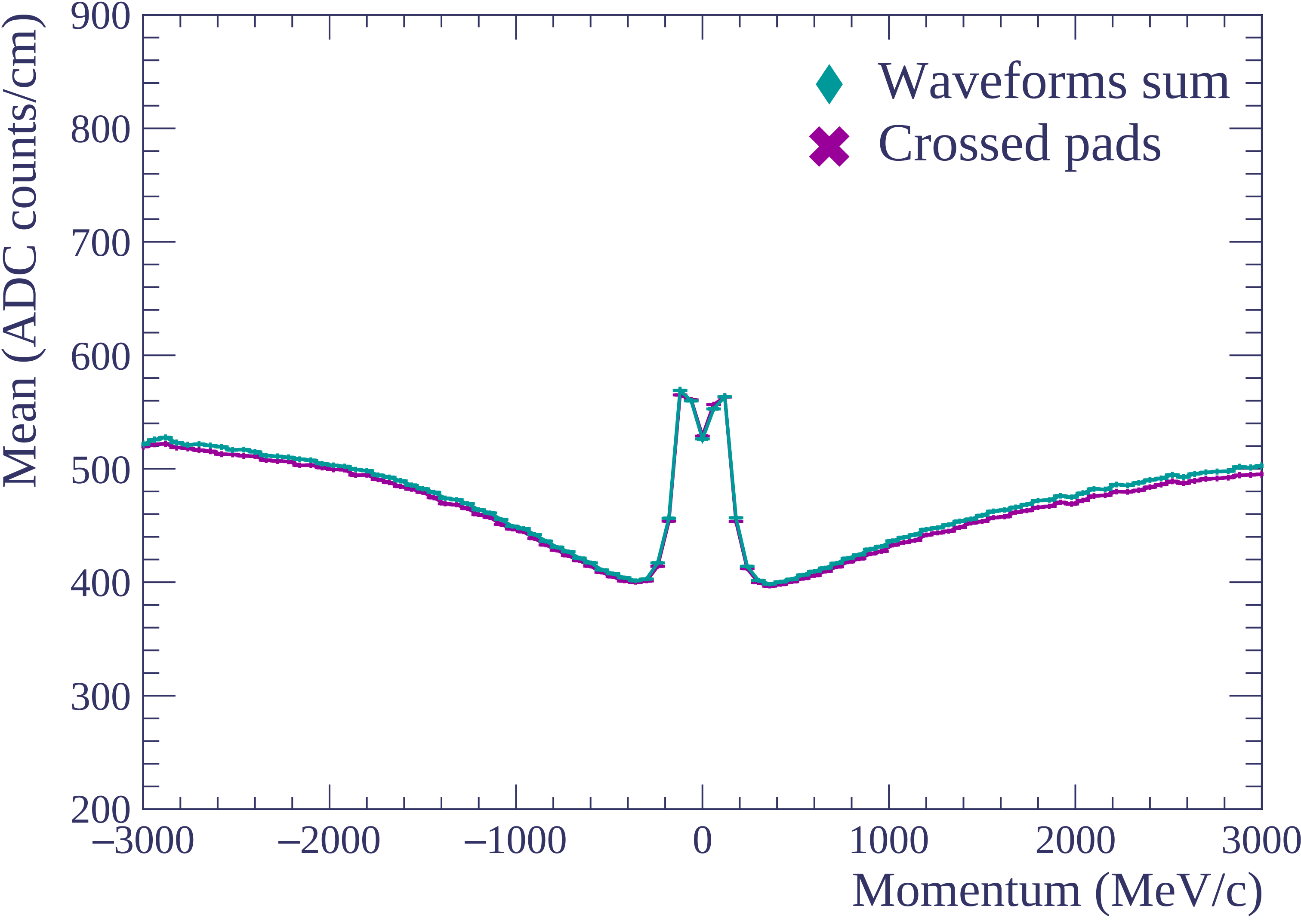

dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120
140
160
180
200
220


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

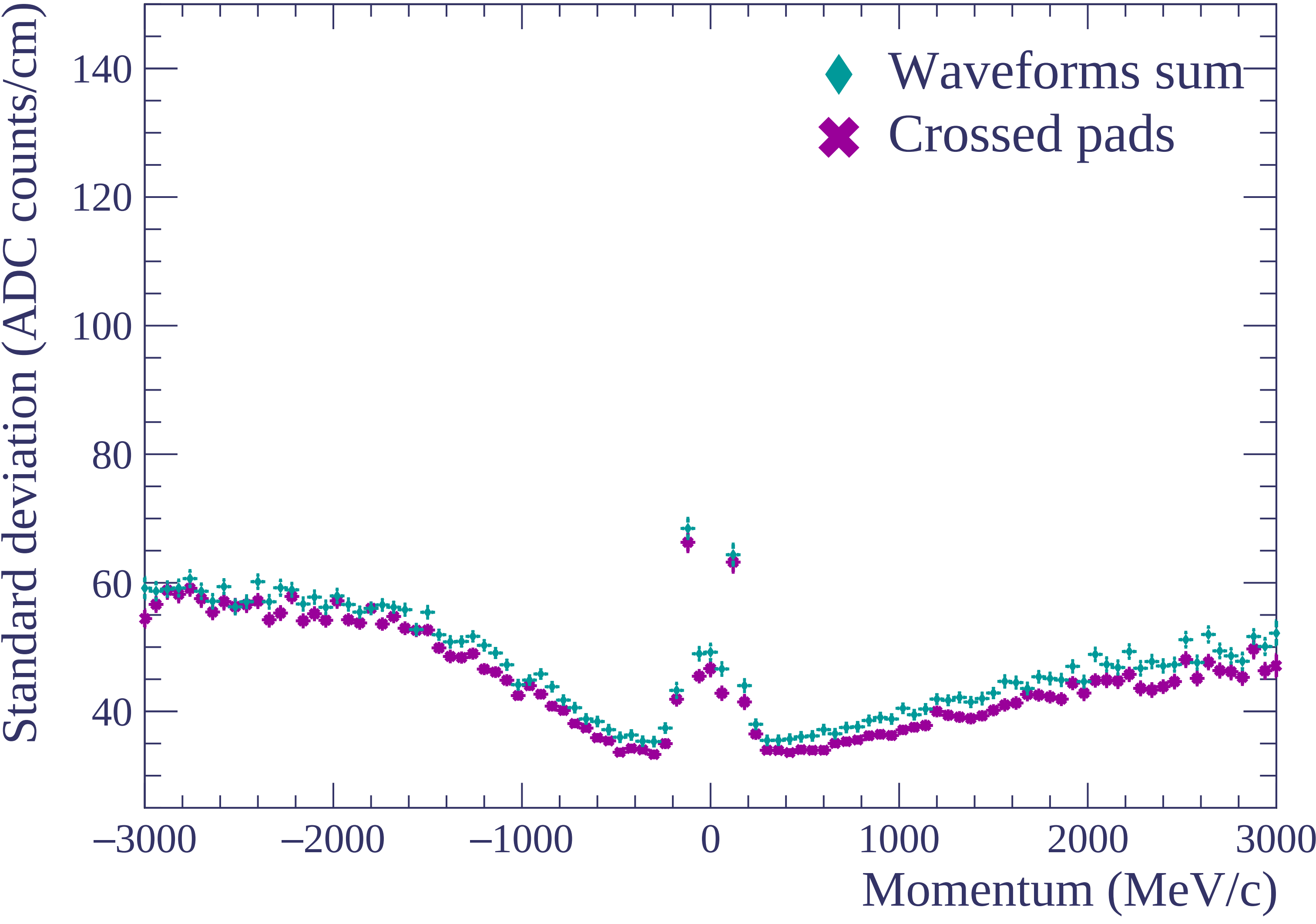


Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

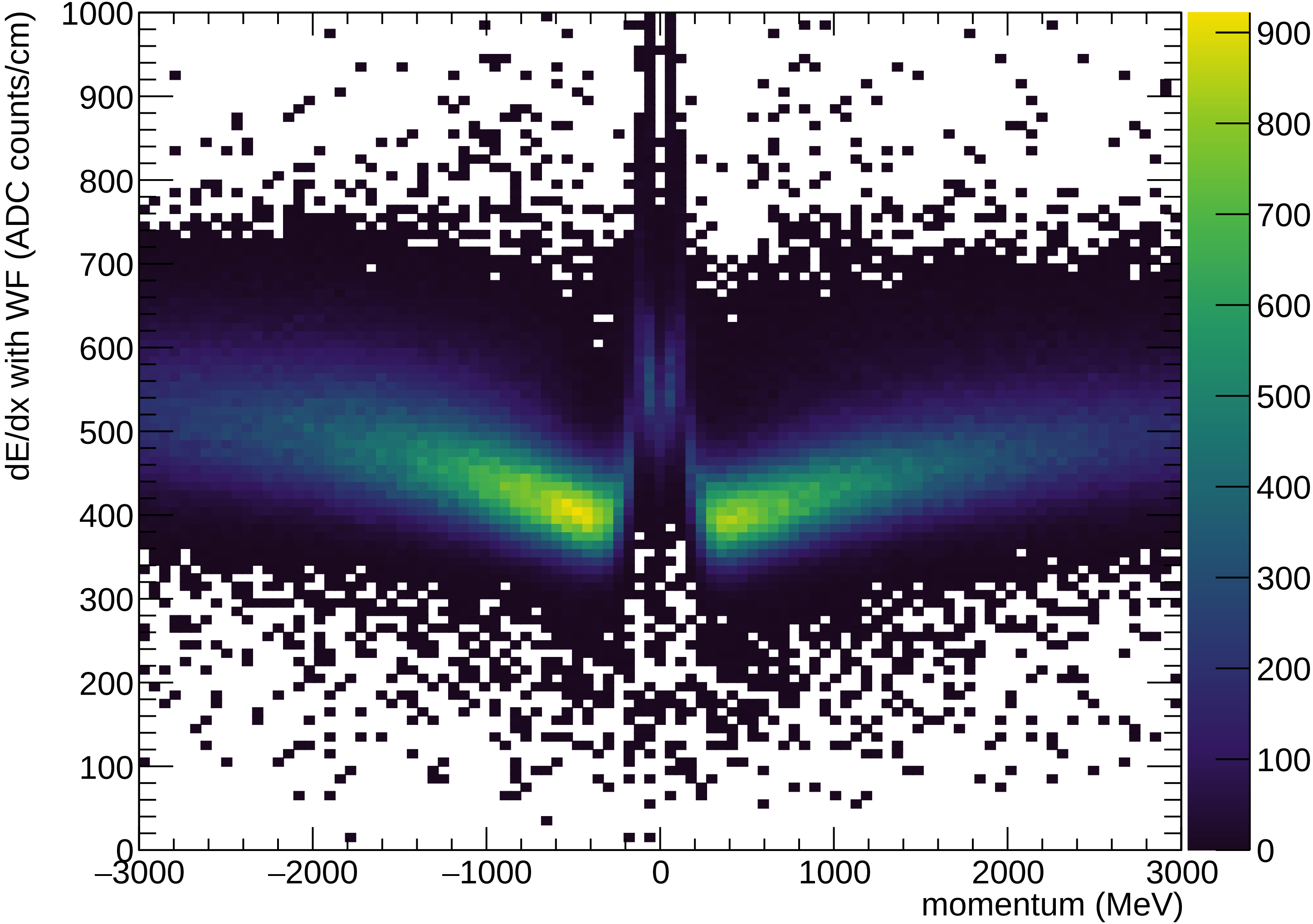

Resolution (%)
30
25
20
15
10
5
Waveforms sum
Crossed pads
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)

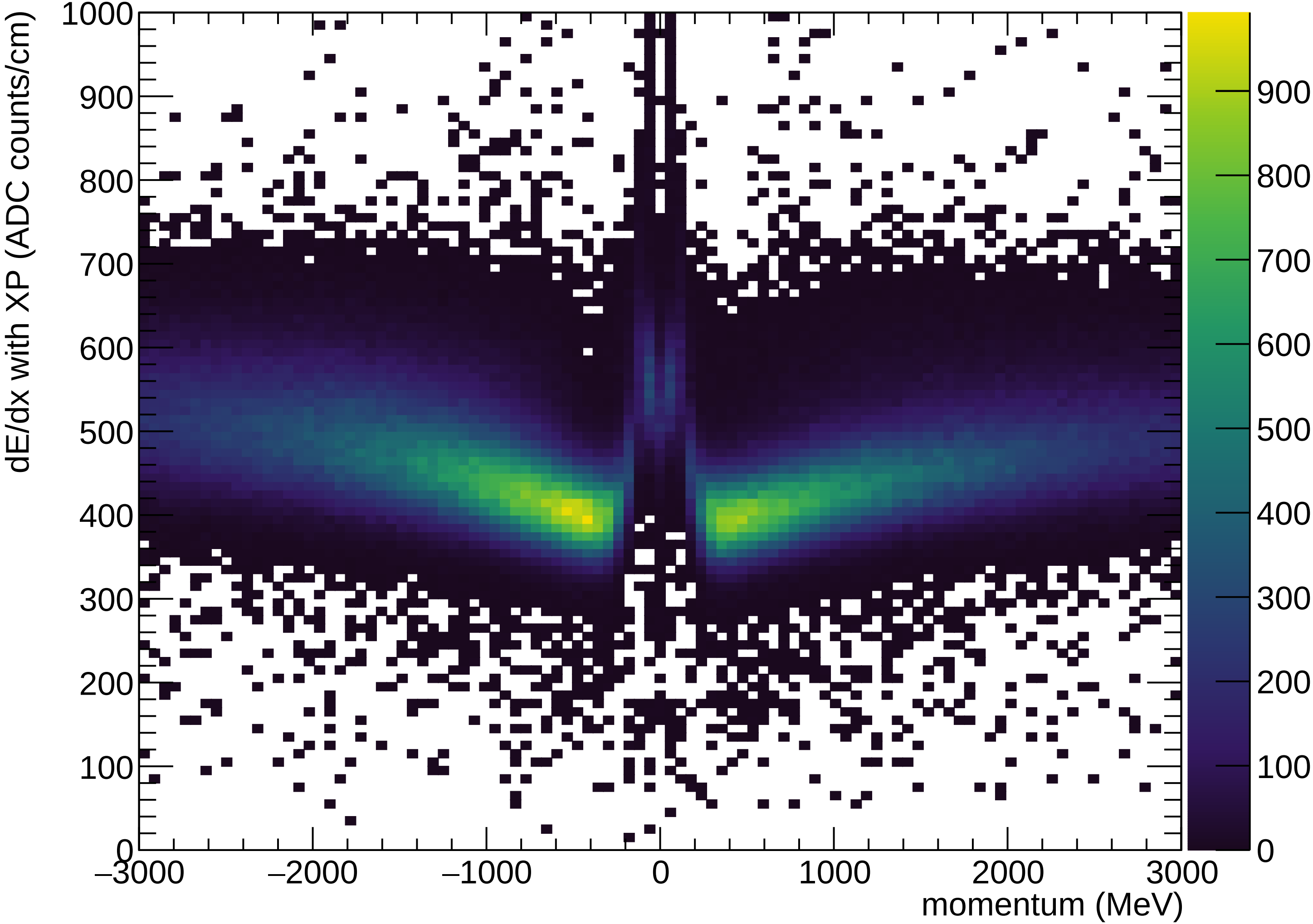


Mean (ADC counts/cm)
900
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

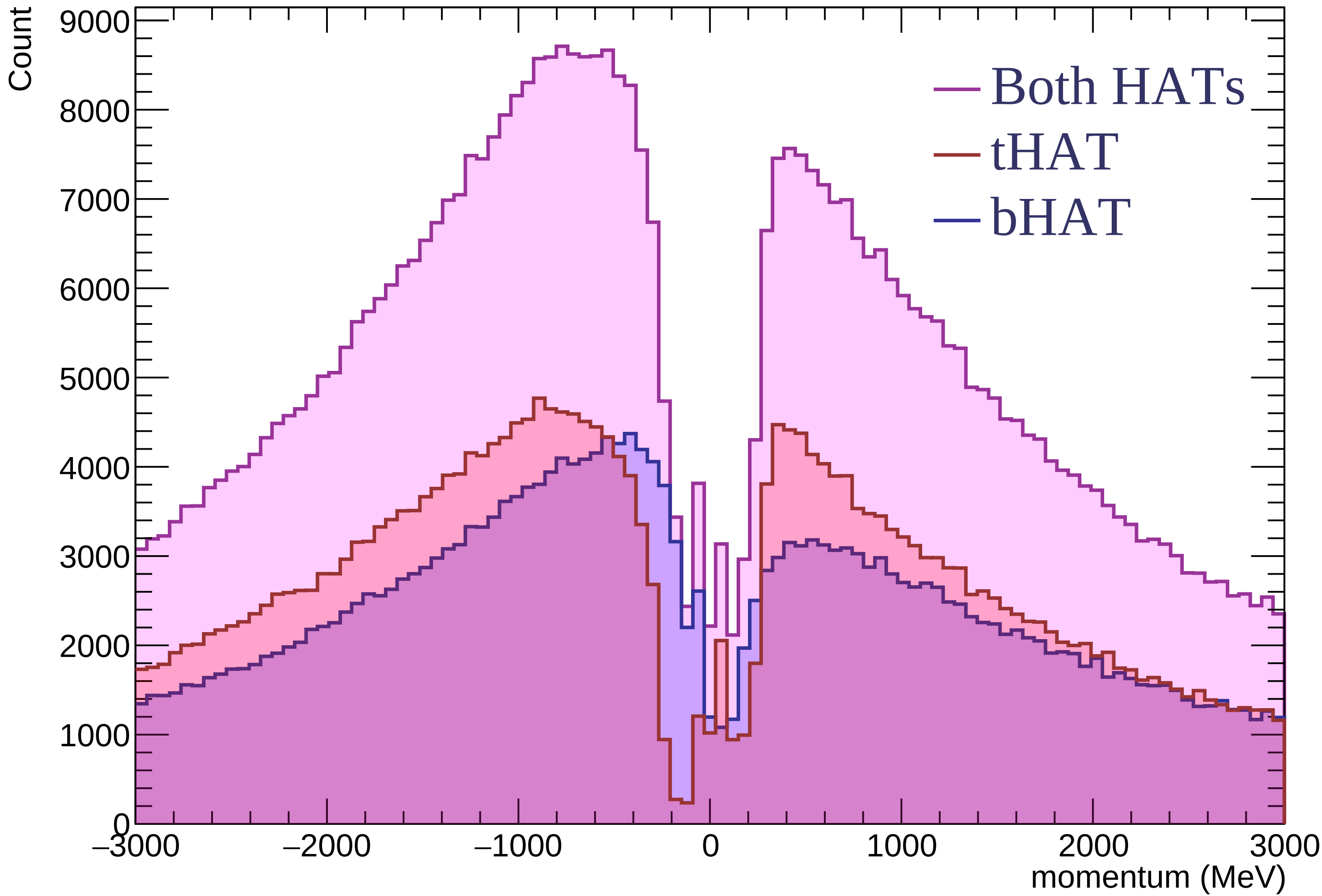


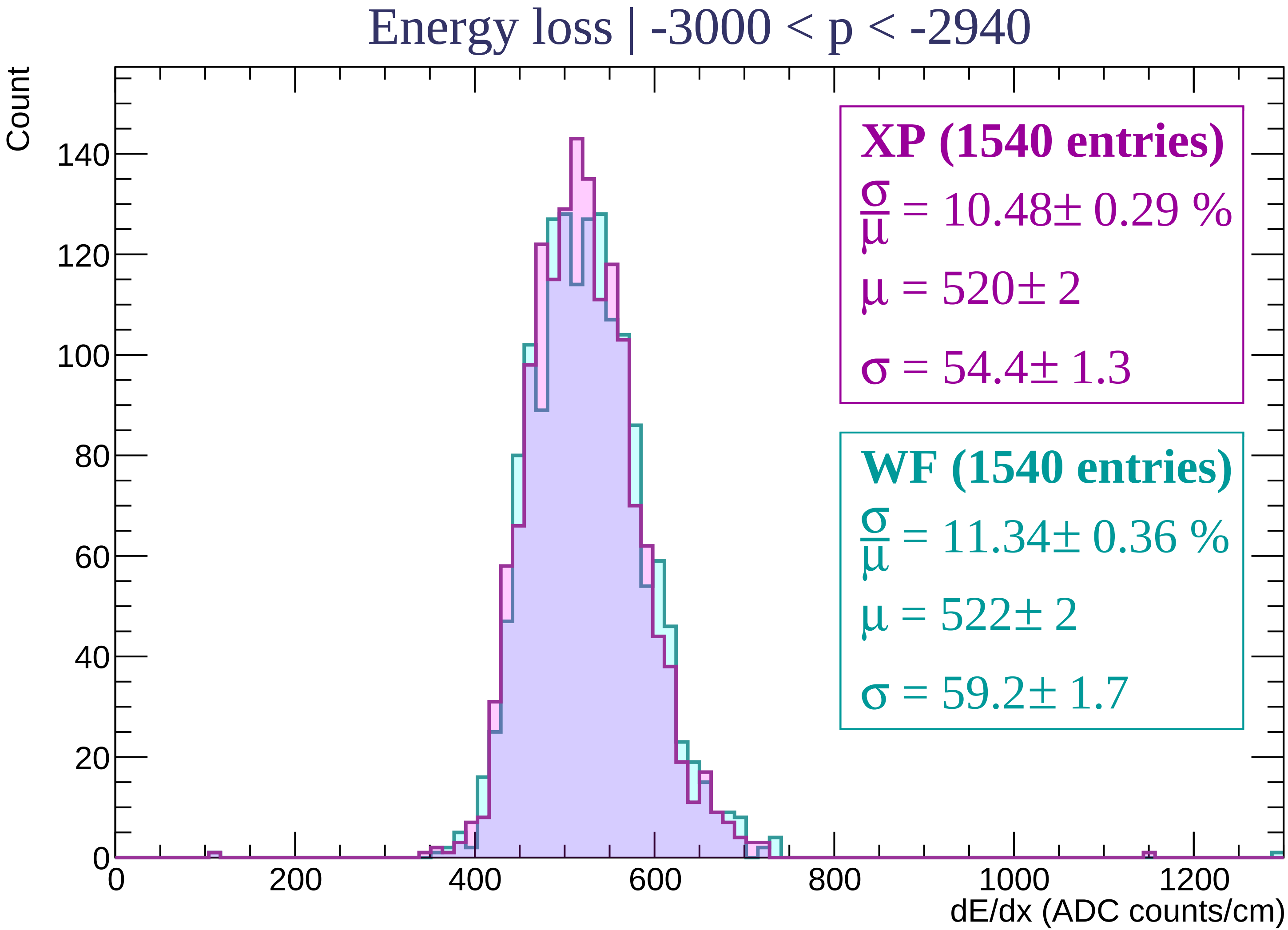
Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
140
120
100
80
60
40
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)


dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
900
800
700
600
500
400
300
200
100
0

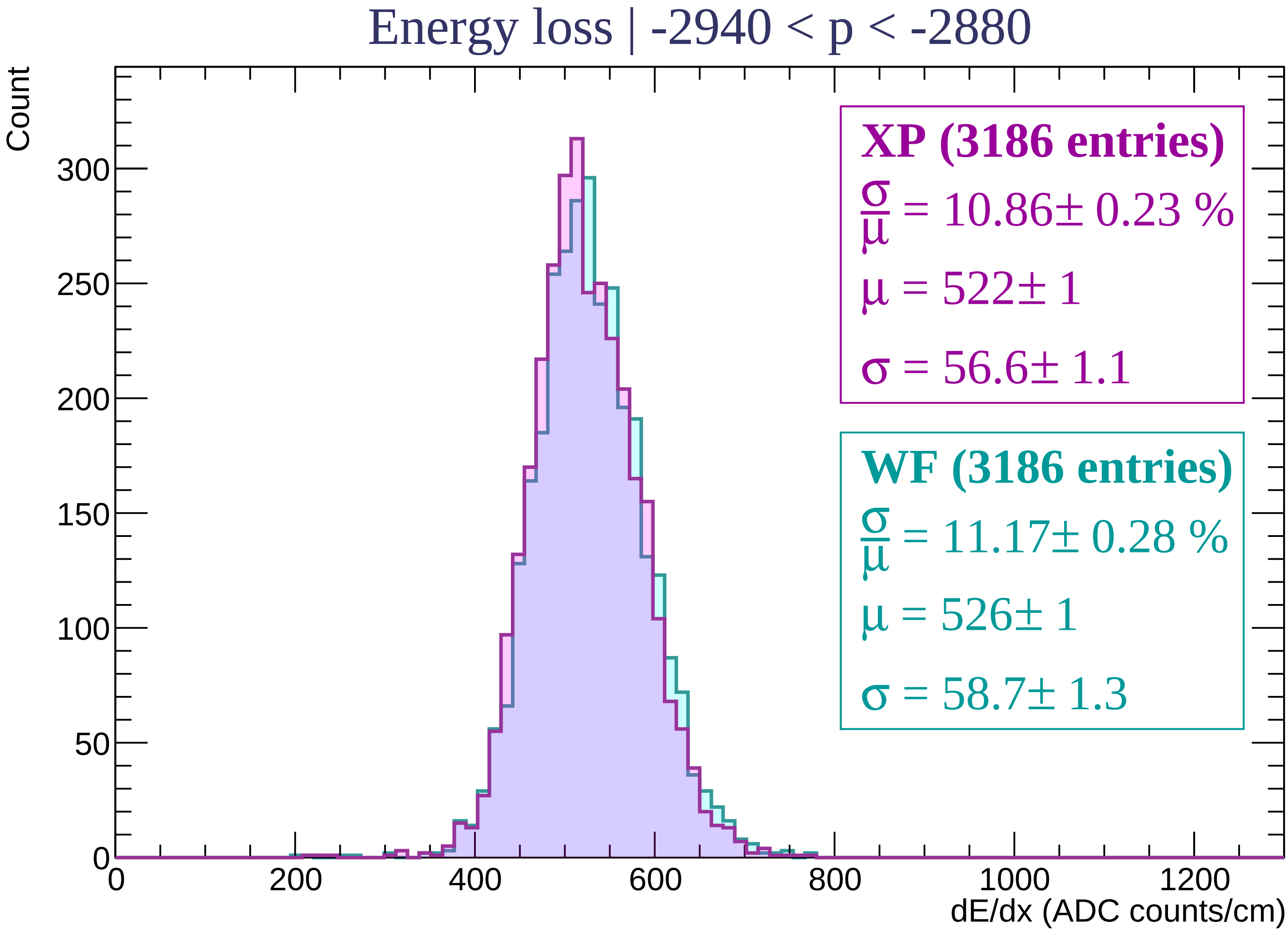


dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
–3000
–2000
–1000
0
1000
2000
3000
momentum (MeV)
900
800
700
600
500
400
300
200
100
0

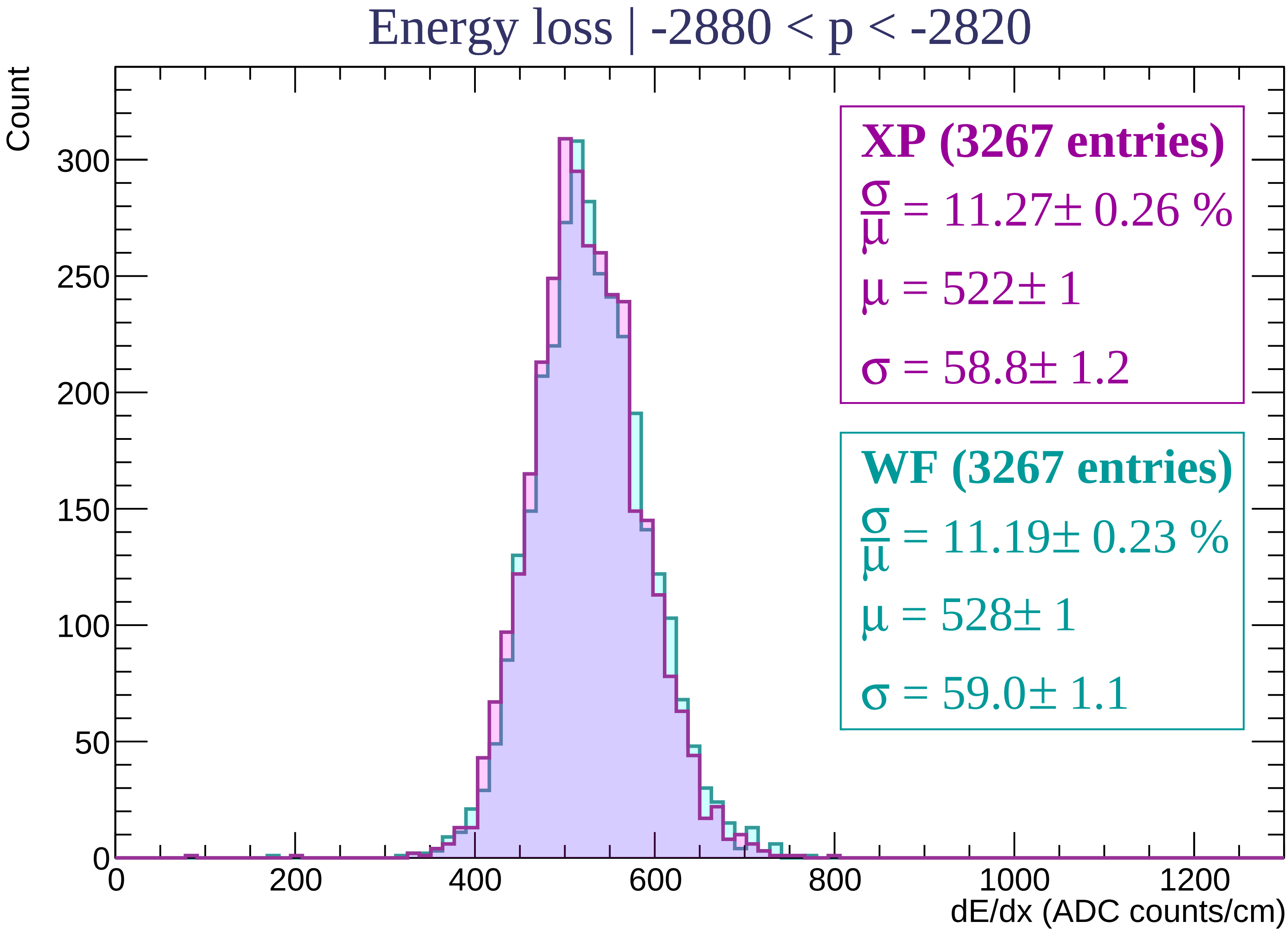

Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

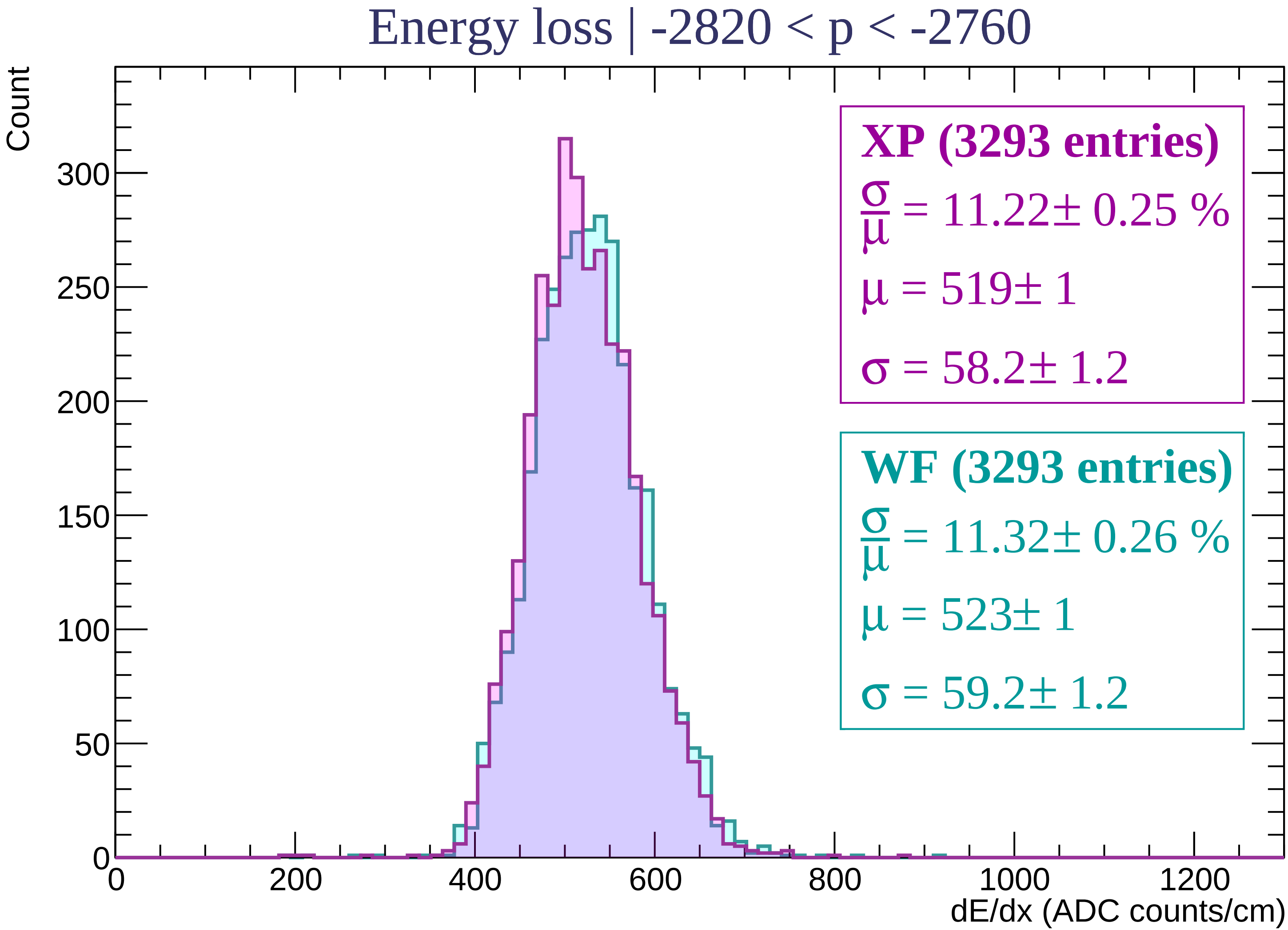


Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1540 entries)
σ/μ = 10.48± 0.29 %
μ = 520± 2
σ = 54.4± 1.3
WF (1540 entries)
σ/μ = 11.34± 0.36 %
μ = 522± 2
σ = 59.2± 1.7

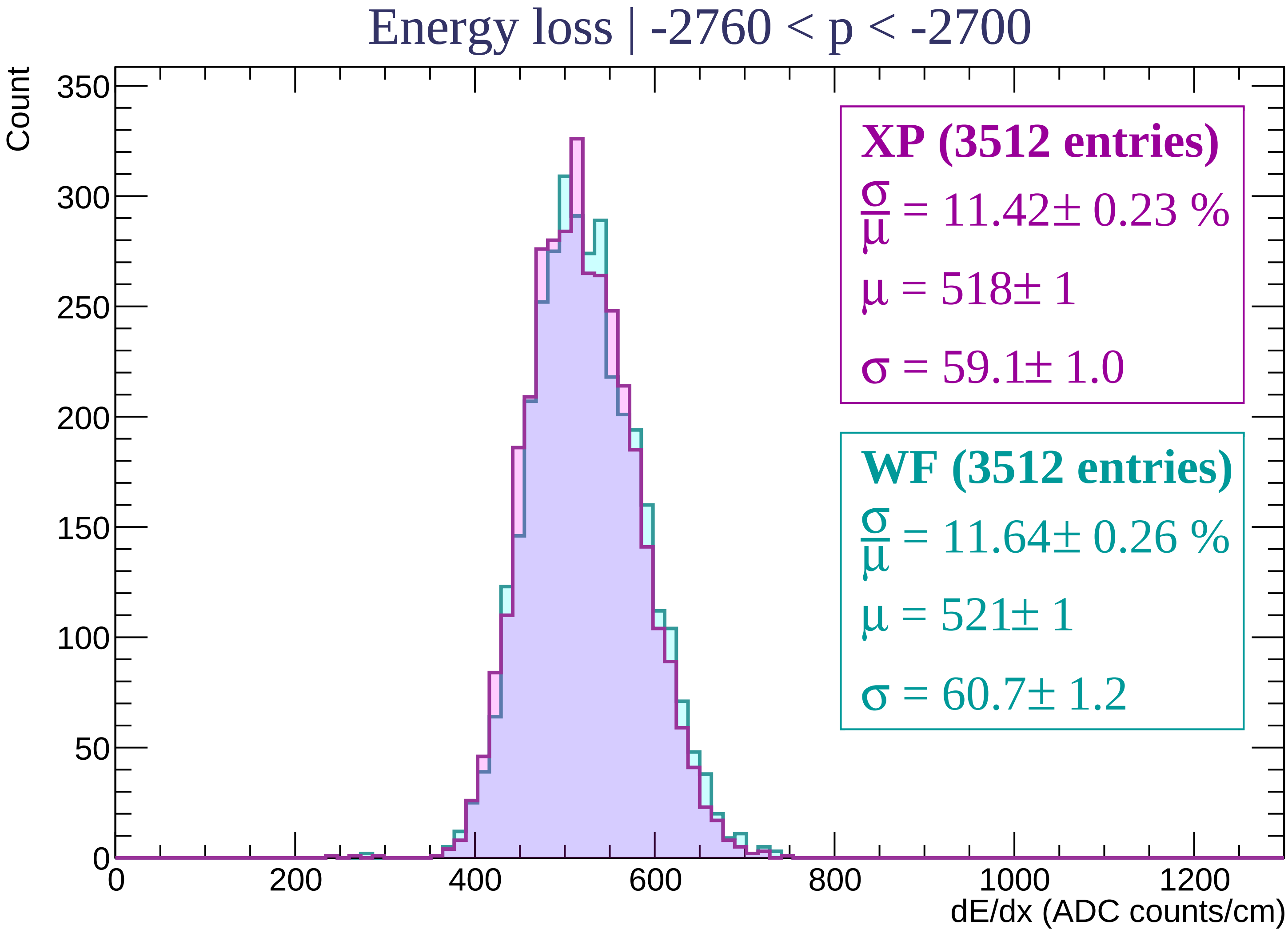


Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3186 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.23 %
μ = 522± 1
σ = 56.6± 1.1
WF (3186 entries)
$\frac{\sigma}{\mu}$ = 11.17± 0.28 %
μ = 526± 1
σ = 58.7± 1.3

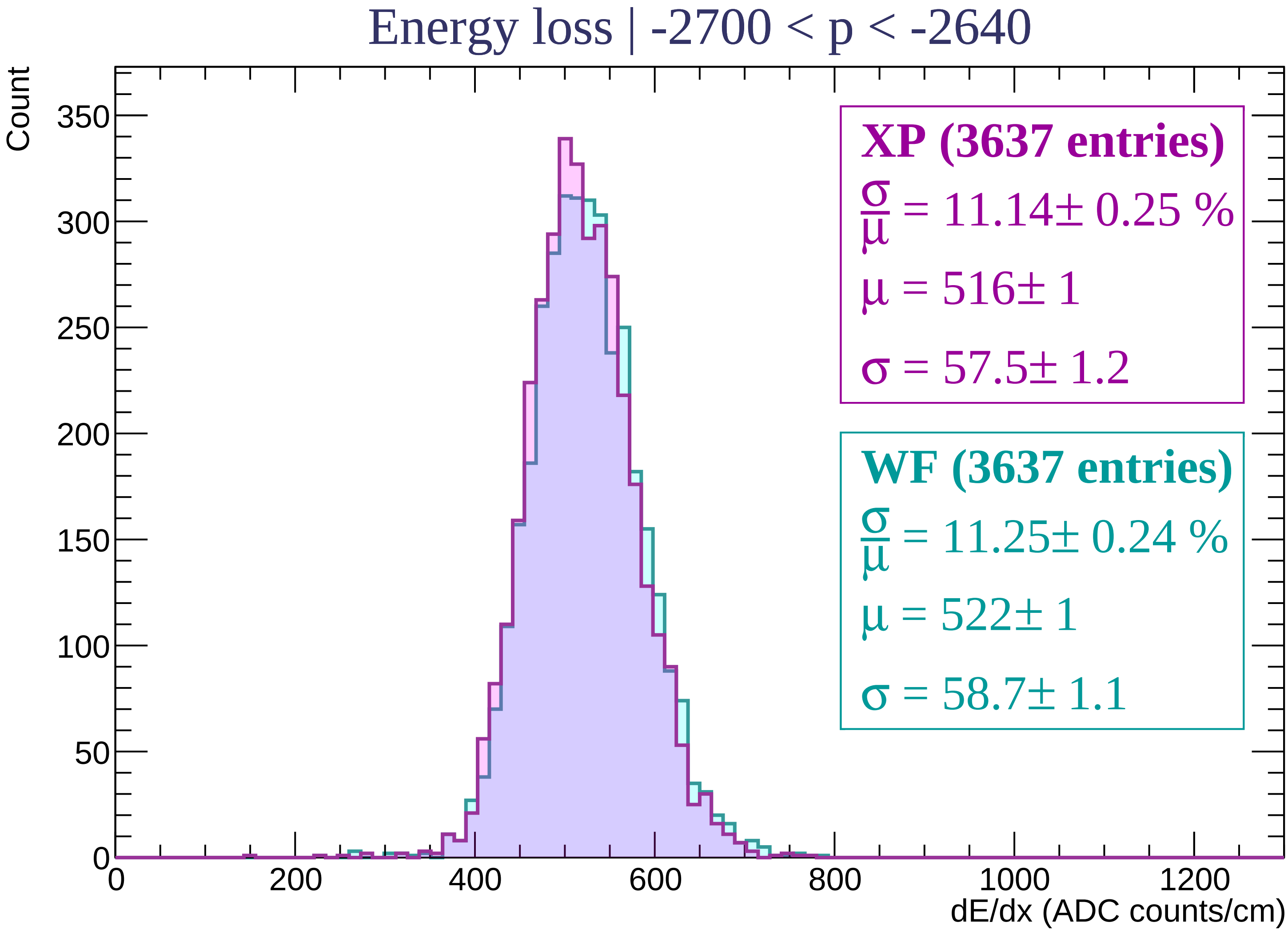

Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3267 entries)
σ/μ = 11.27± 0.26 %
μ = 522± 1
σ = 58.8± 1.2
WF (3267 entries)
σ/μ = 11.19± 0.23 %
μ = 528± 1
σ = 59.0± 1.1



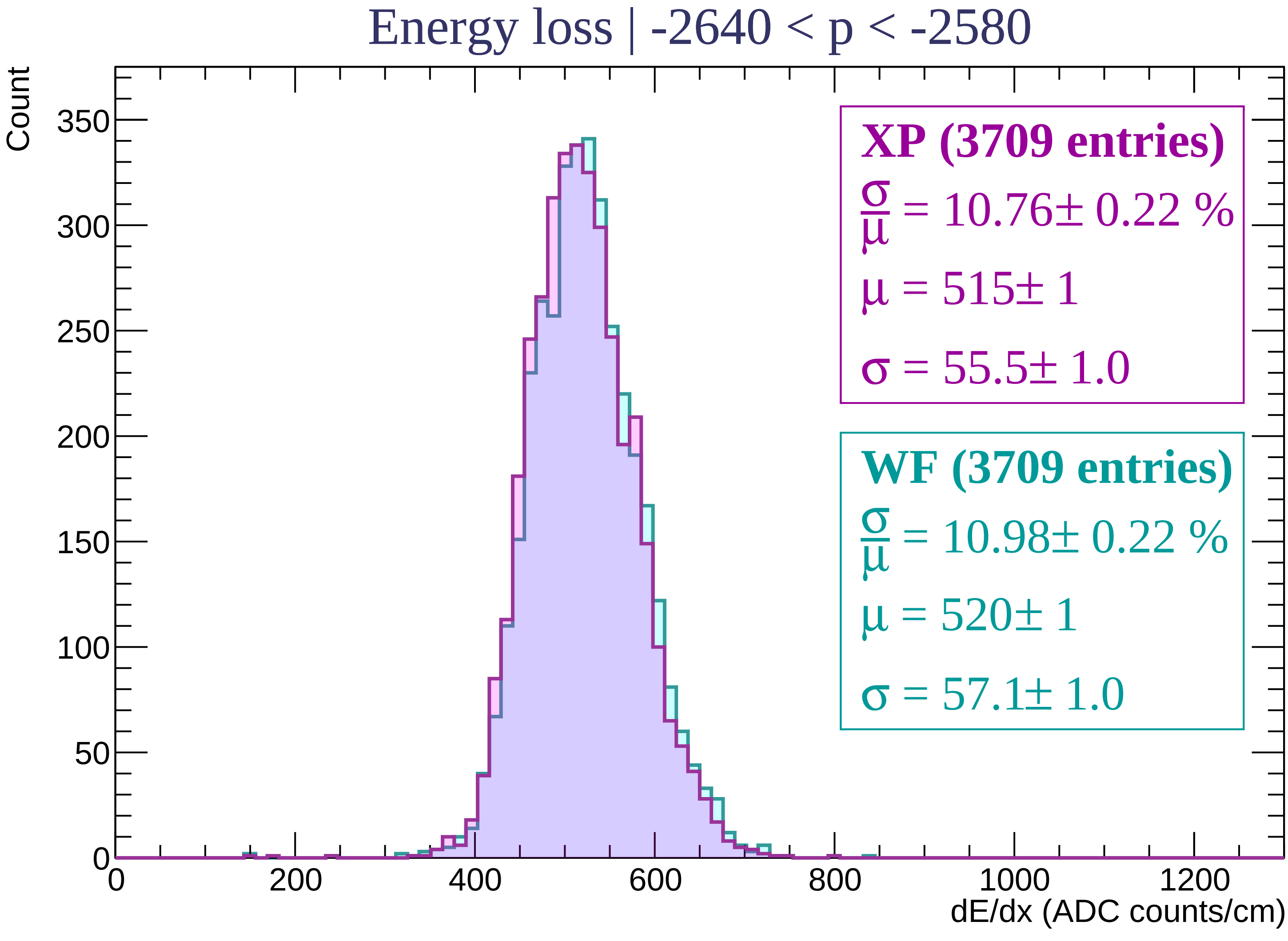
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
XP (3293 entries)
σ/μ = 11.22± 0.25 %
μ = 519± 1
σ = 58.2± 1.2
WF (3293 entries)
σ/μ = 11.32± 0.26 %
μ = 523± 1
σ = 59.2± 1.2

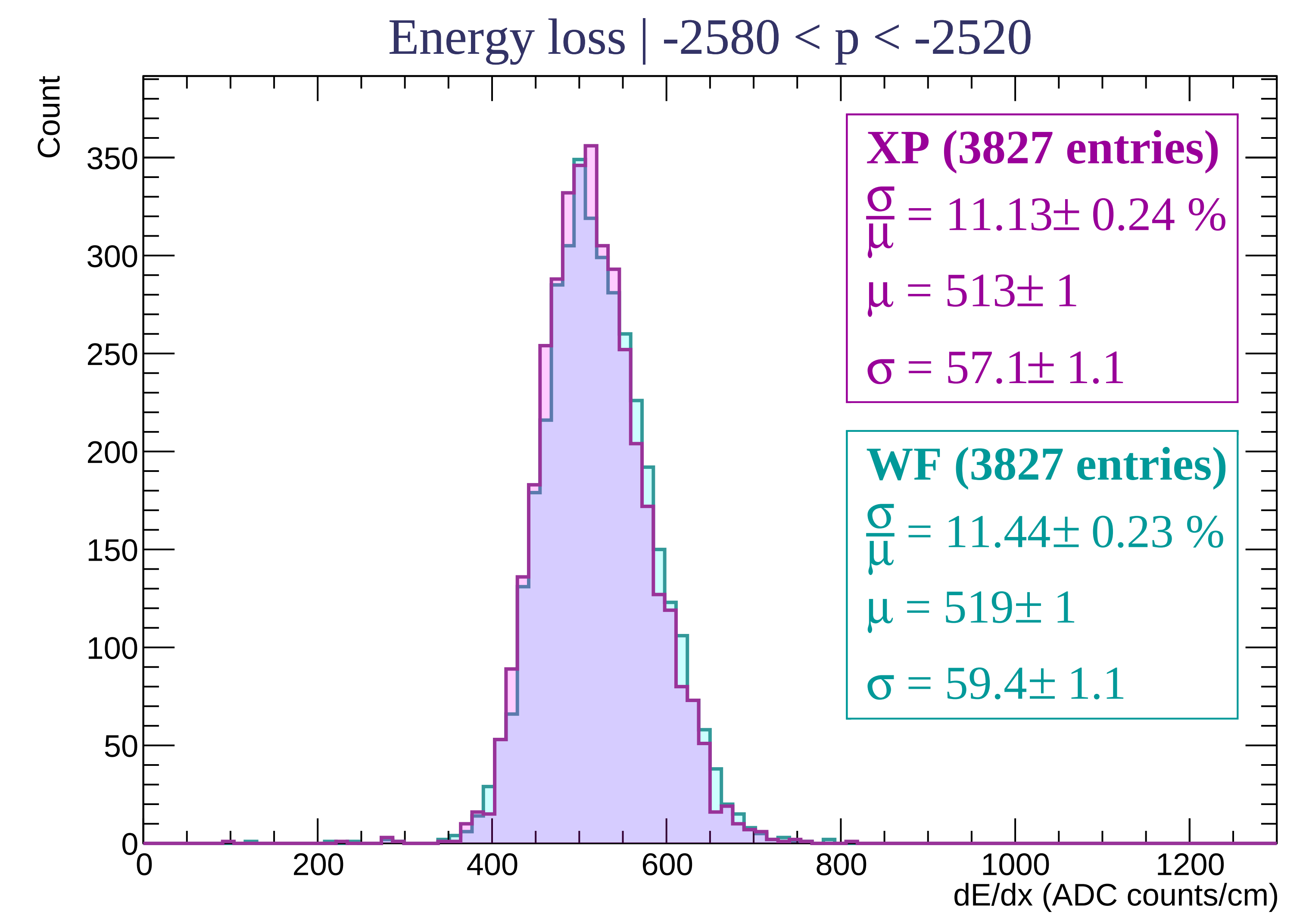


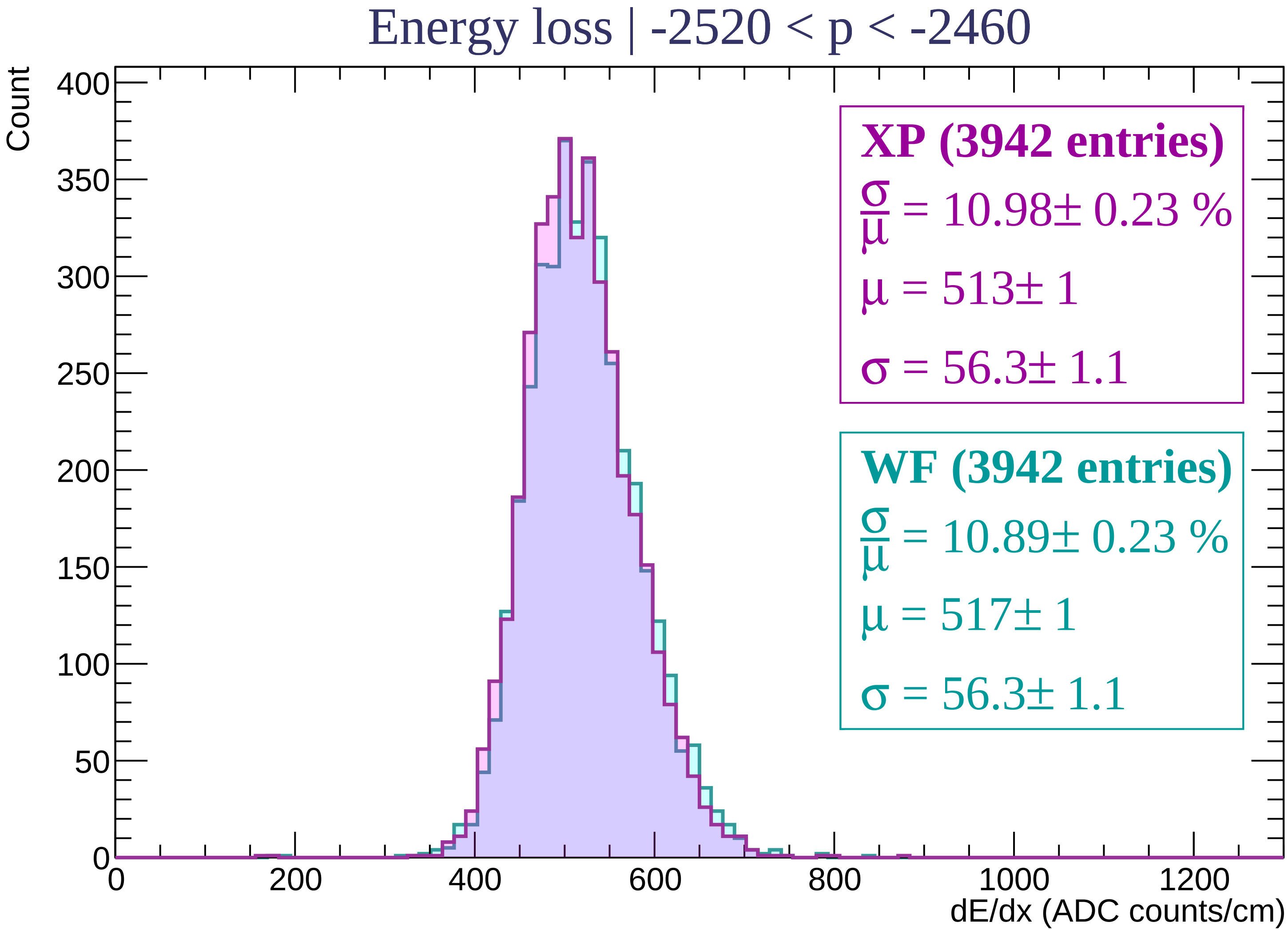
Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3512 entries)
σ/μ = 11.42± 0.23 %
μ = 518± 1
σ = 59.1± 1.0
WF (3512 entries)
σ/μ = 11.64± 0.26 %
μ = 521± 1
σ = 60.7± 1.2



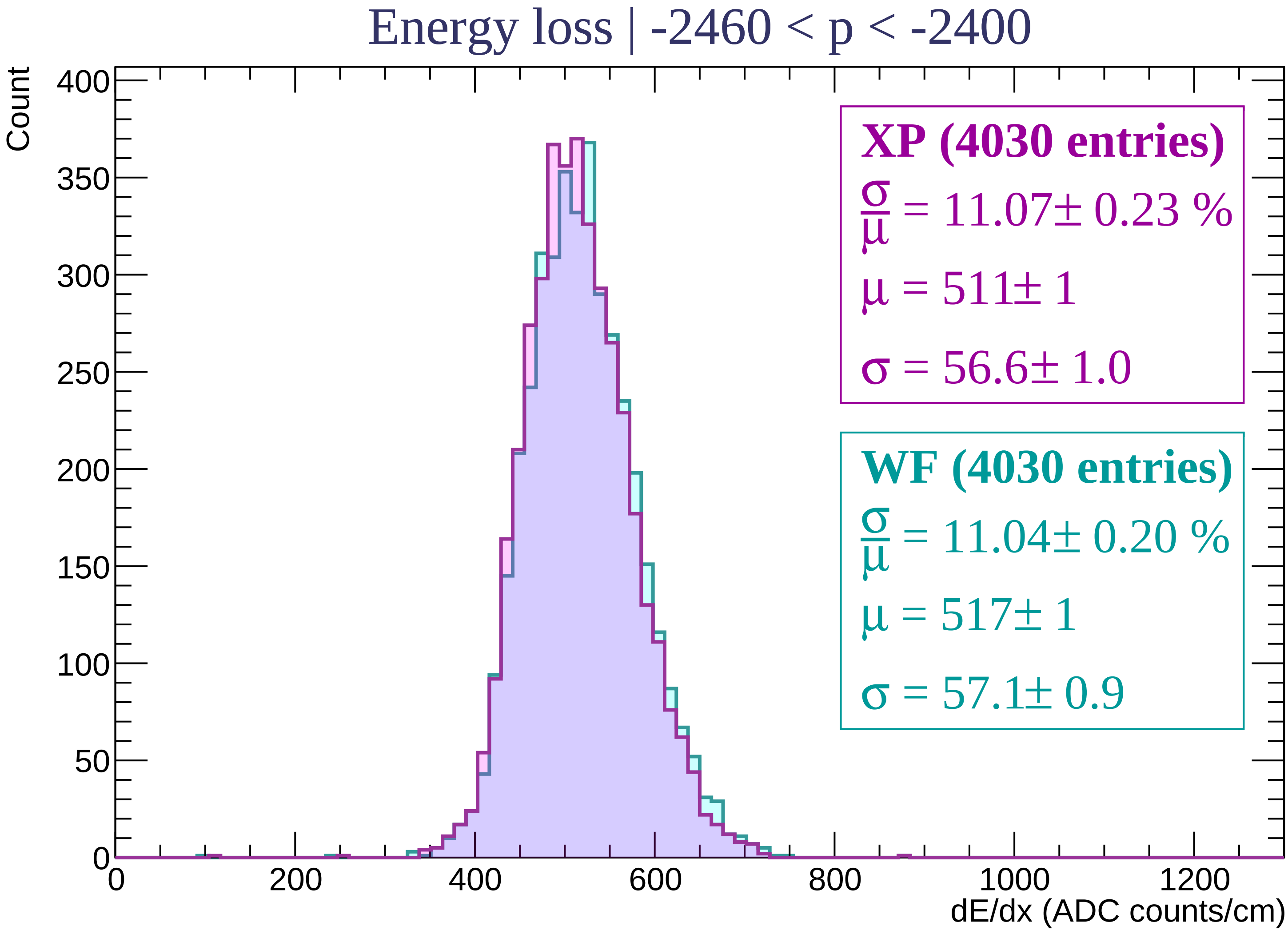
Energy loss | -2700 < p < -2640
Count
XP (3637 entries)
σ/μ = 11.14± 0.25 %
μ = 516± 1
σ = 57.5± 1.2
WF (3637 entries)
σ/μ = 11.25± 0.24 %
μ = 522± 1
σ = 58.7± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

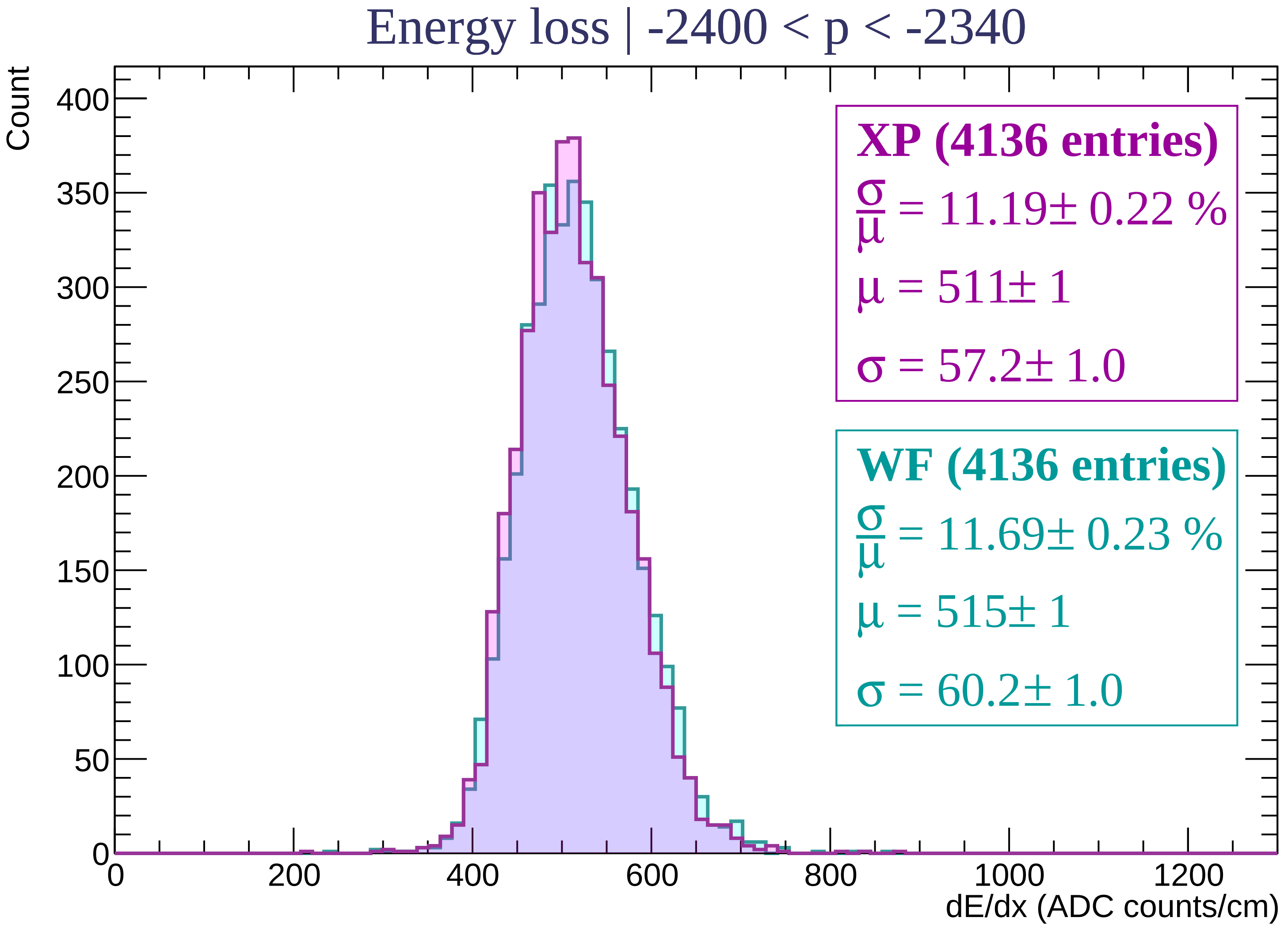

Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3709 entries)
σ/μ = 10.76± 0.22 %
μ = 515± 1
σ = 55.5± 1.0
WF (3709 entries)
σ/μ = 10.98± 0.22 %
μ = 520± 1
σ = 57.1± 1.0


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP (3827 entries)
σ/μ = 11.13± 0.24 %
μ = 513± 1
σ = 57.1± 1.1
WF (3827 entries)
σ/μ = 11.44± 0.23 %
μ = 519± 1
σ = 59.4± 1.1

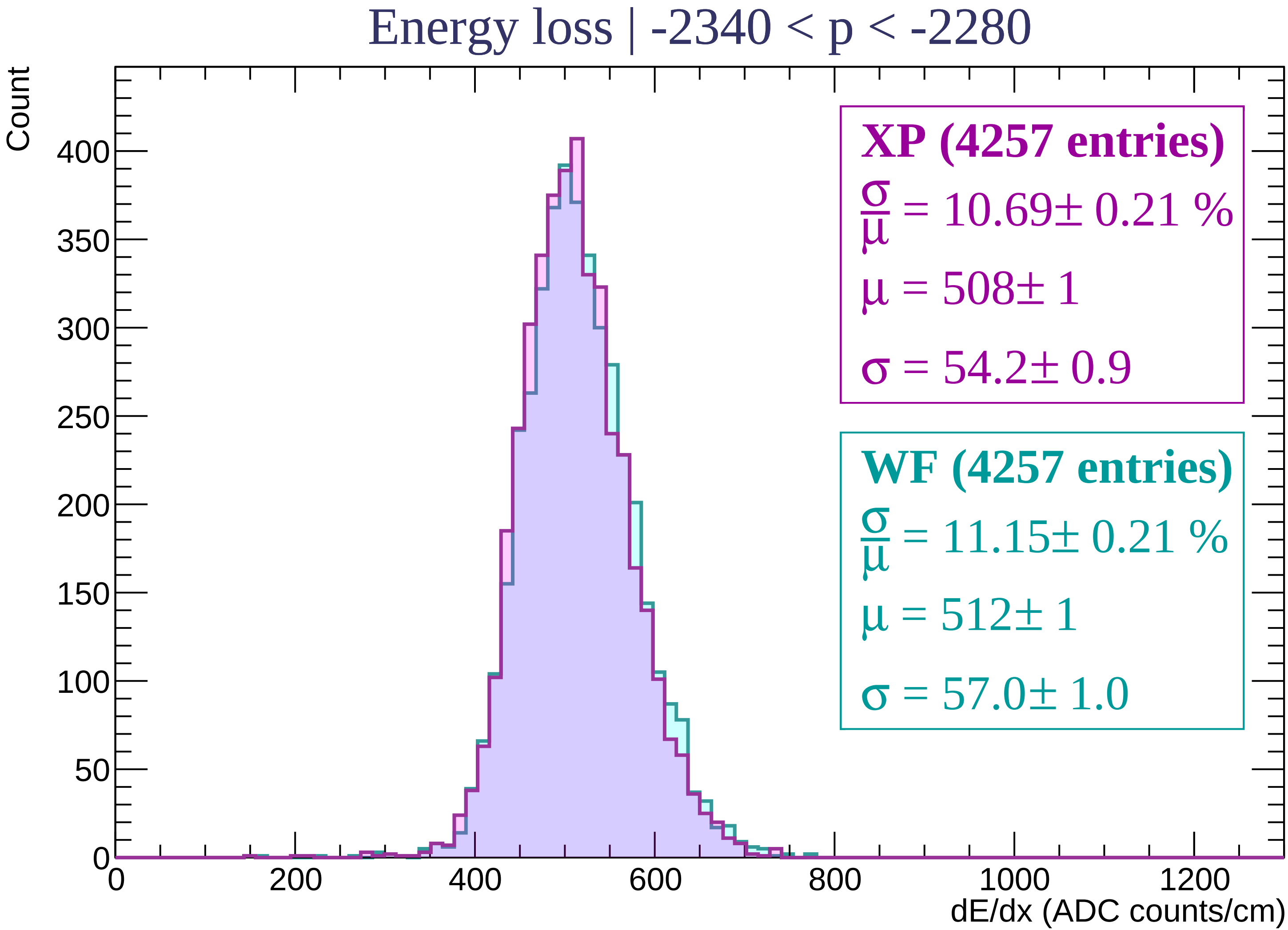


Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3942 entries)
$\frac{\sigma}{\mu}$ = 10.98± 0.23 %
μ = 513± 1
σ = 56.3± 1.1
WF (3942 entries)
$\frac{\sigma}{\mu}$ = 10.89± 0.23 %
μ = 517± 1
σ = 56.3± 1.1

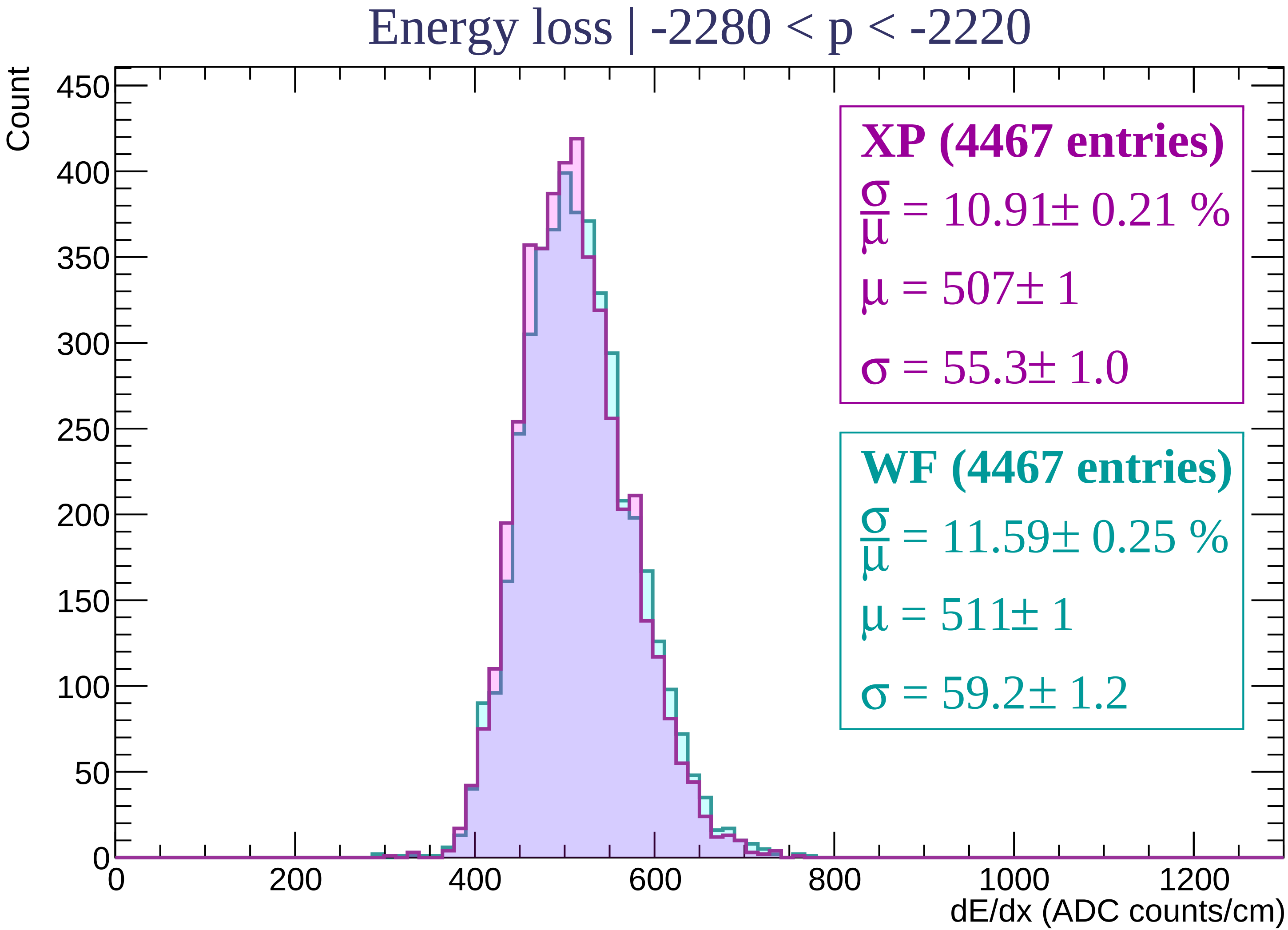

Energy loss | -2460 < p < -2400
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4030 entries)
σ/μ = 11.07± 0.23 %
μ = 511± 1
σ = 56.6± 1.0
WF (4030 entries)
σ/μ = 11.04± 0.20 %
μ = 517± 1
σ = 57.1± 0.9

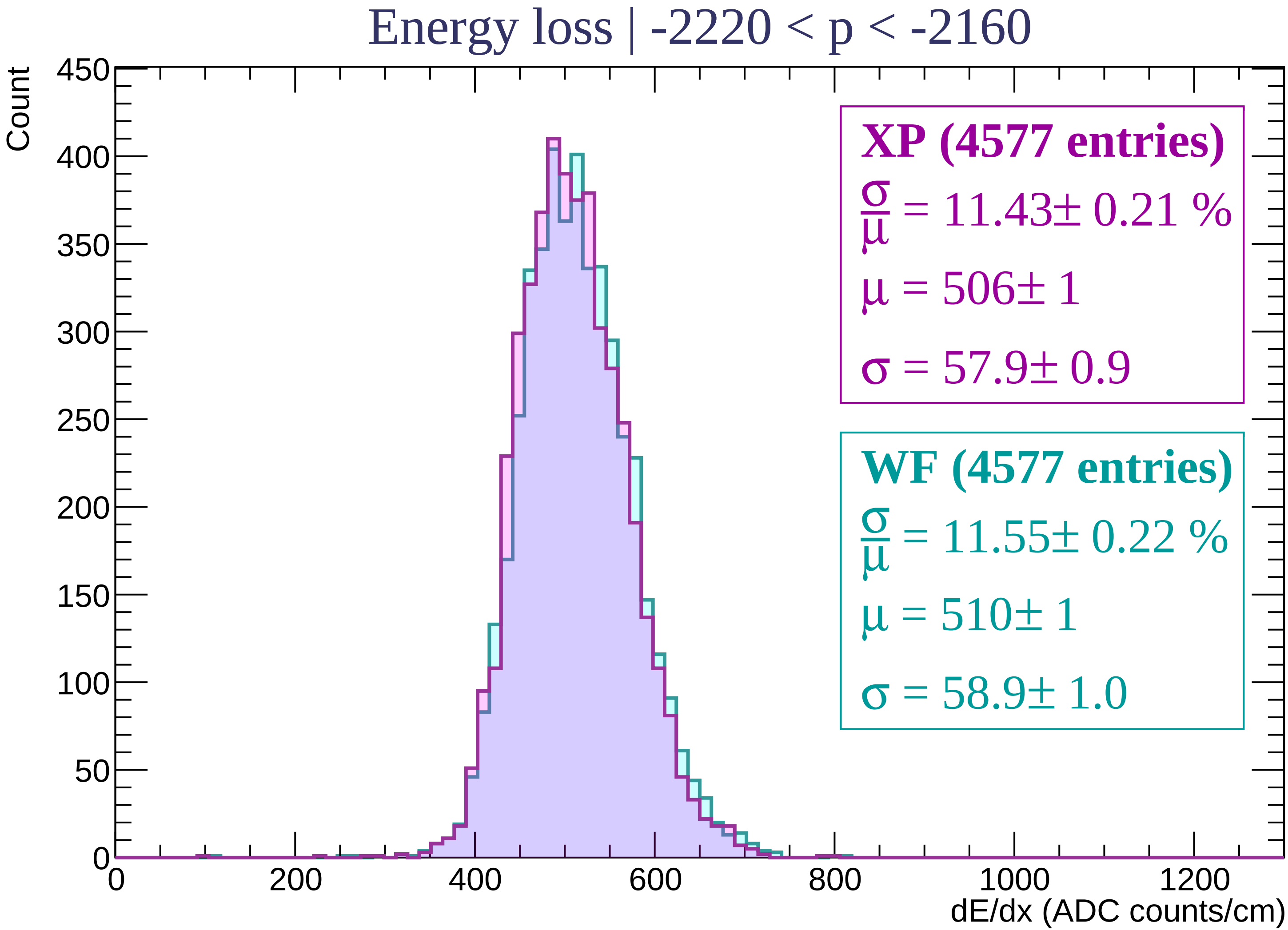


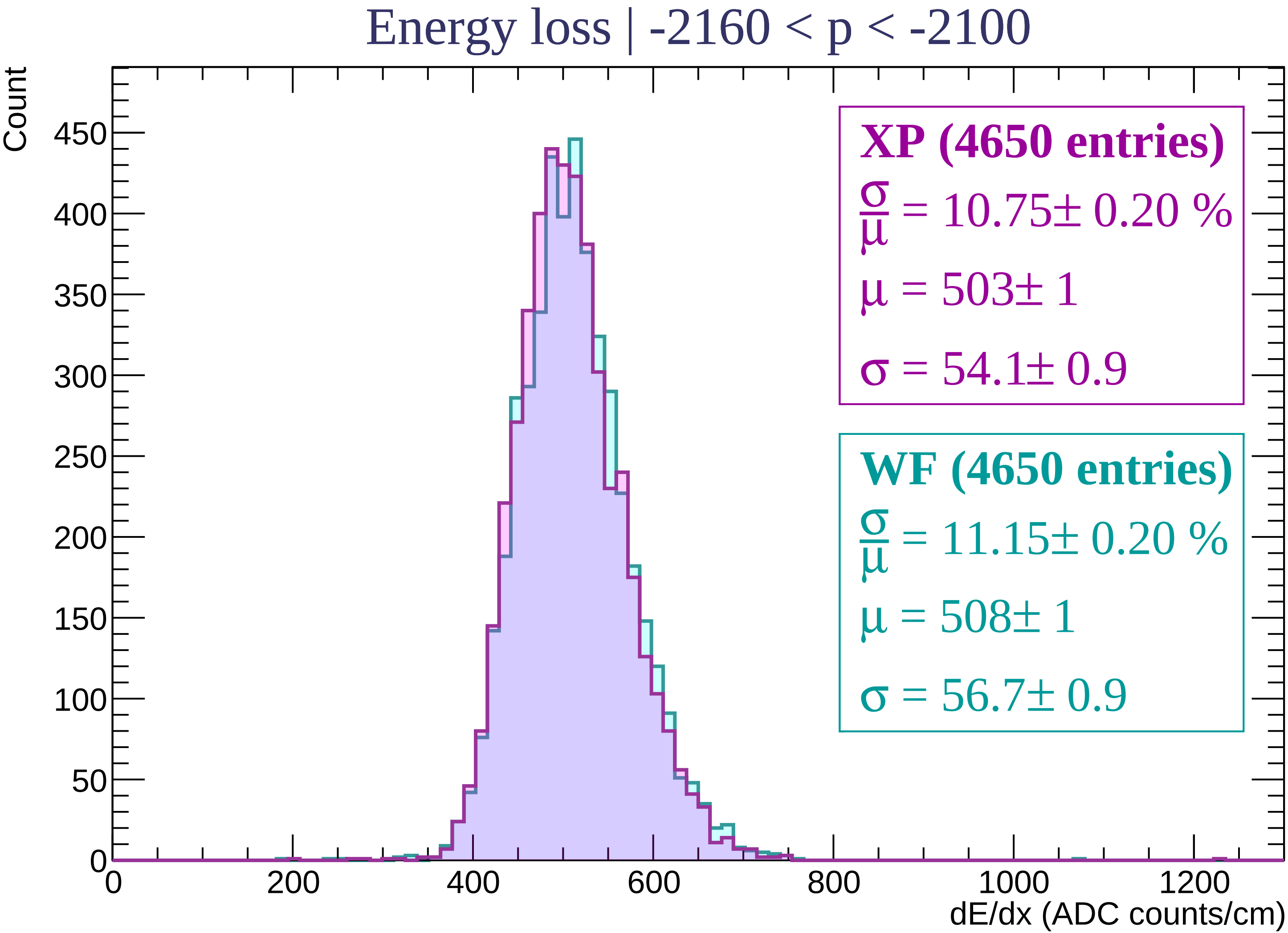
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP (4136 entries)
σ/μ = 11.19± 0.22 %
μ = 511± 1
σ = 57.2± 1.0
WF (4136 entries)
σ/μ = 11.69± 0.23 %
μ = 515± 1
σ = 60.2± 1.0



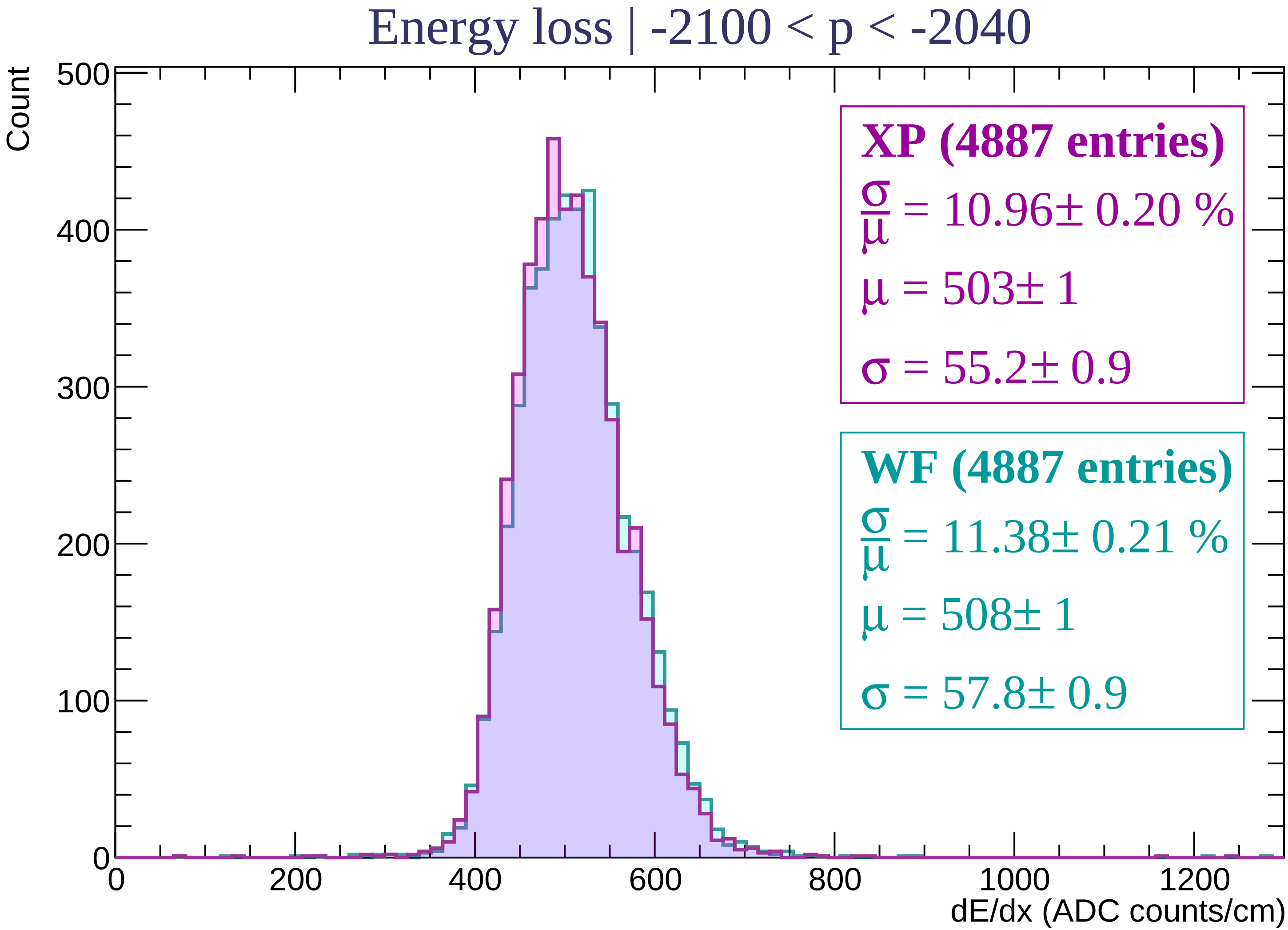
Energy loss | -2340 < p < -2280
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4257 entries)
σ/μ = 10.69± 0.21 %
μ = 508± 1
σ = 54.2± 0.9
WF (4257 entries)
σ/μ = 11.15± 0.21 %
μ = 512± 1
σ = 57.0± 1.0


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP (4467 entries)
σ/μ = 10.91± 0.21 %
μ = 507± 1
σ = 55.3± 1.0
WF (4467 entries)
σ/μ = 11.59± 0.25 %
μ = 511± 1
σ = 59.2± 1.2

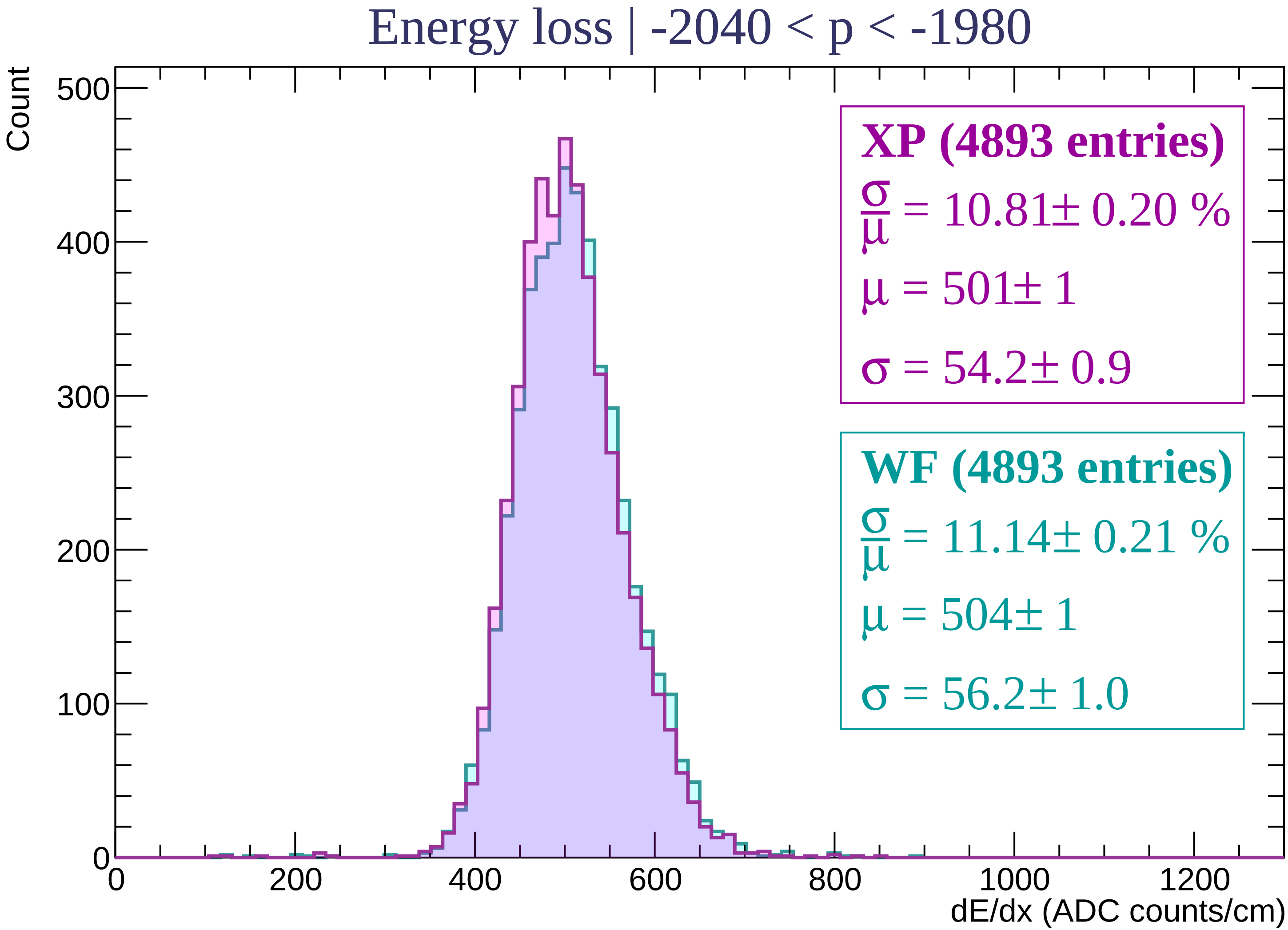

Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
XP (4577 entries)
$\frac{\sigma}{\mu}$ = 11.43± 0.21 %
μ = 506± 1
σ = 57.9± 0.9
WF (4577 entries)
$\frac{\sigma}{\mu}$ = 11.55± 0.22 %
μ = 510± 1
σ = 58.9± 1.0

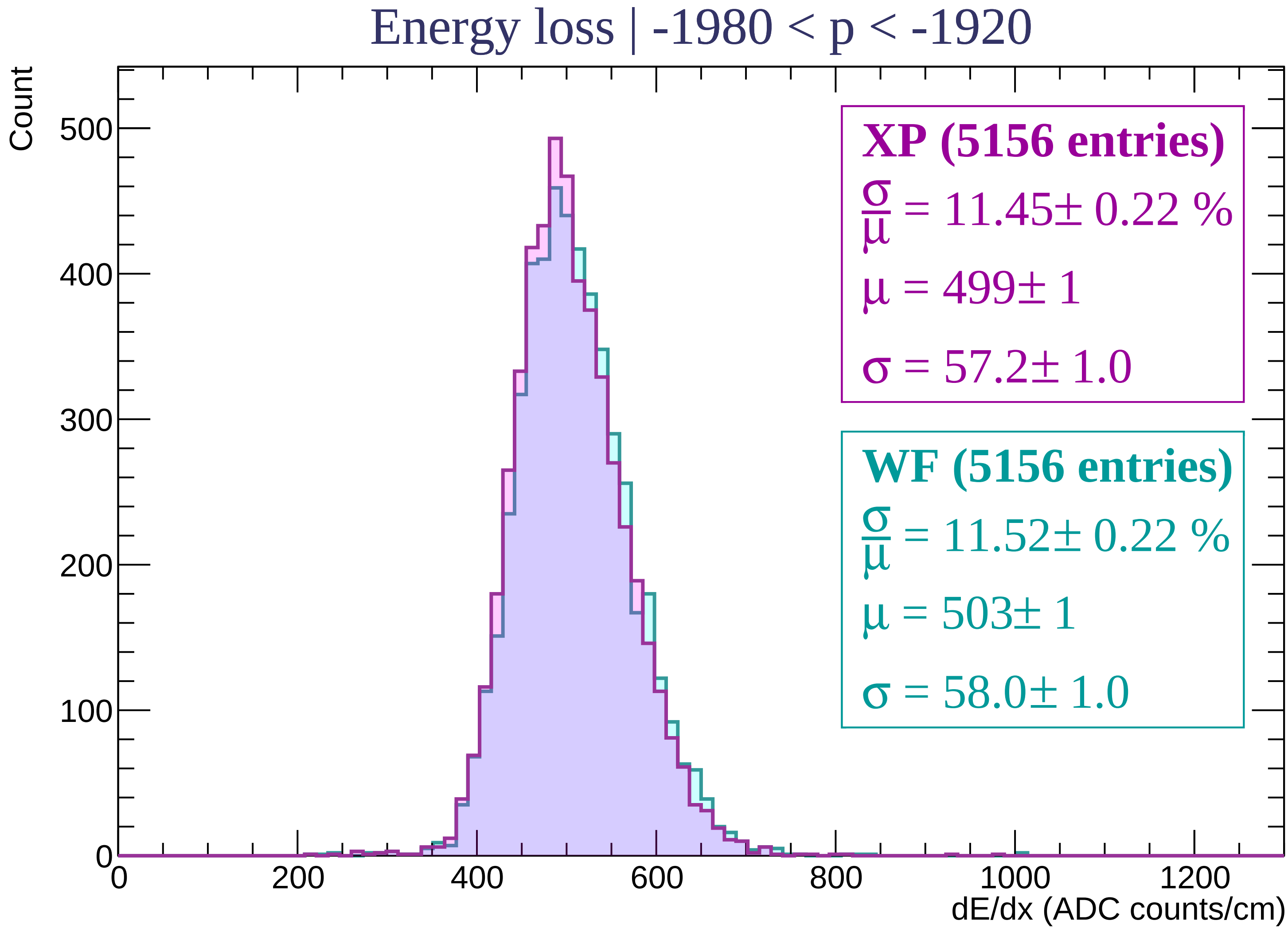


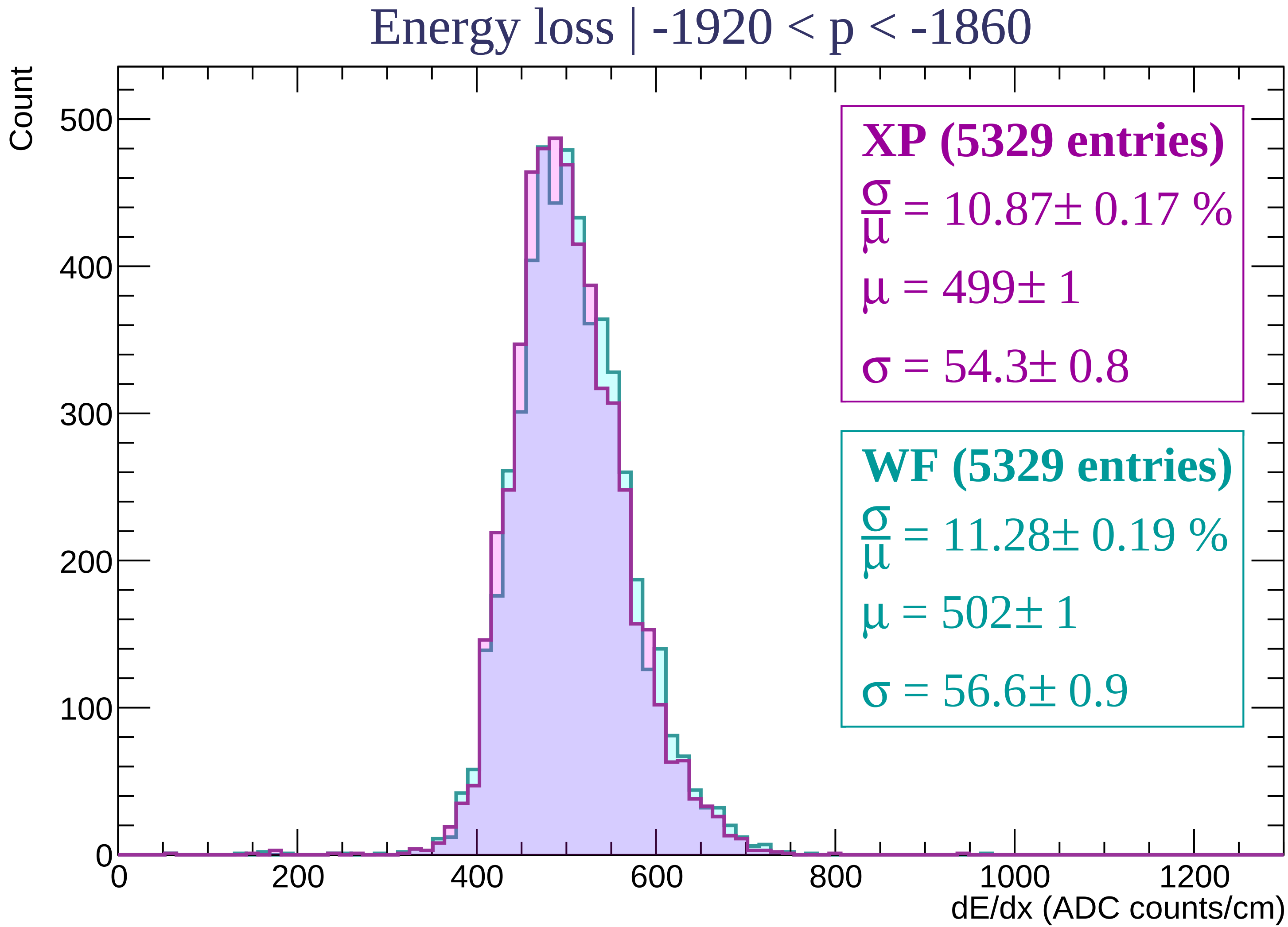
Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4650 entries)
$\frac{\sigma}{\mu}$ = 10.75± 0.20 %
μ = 503± 1
σ = 54.1± 0.9
WF (4650 entries)
$\frac{\sigma}{\mu}$ = 11.15± 0.20 %
μ = 508± 1
σ = 56.7± 0.9



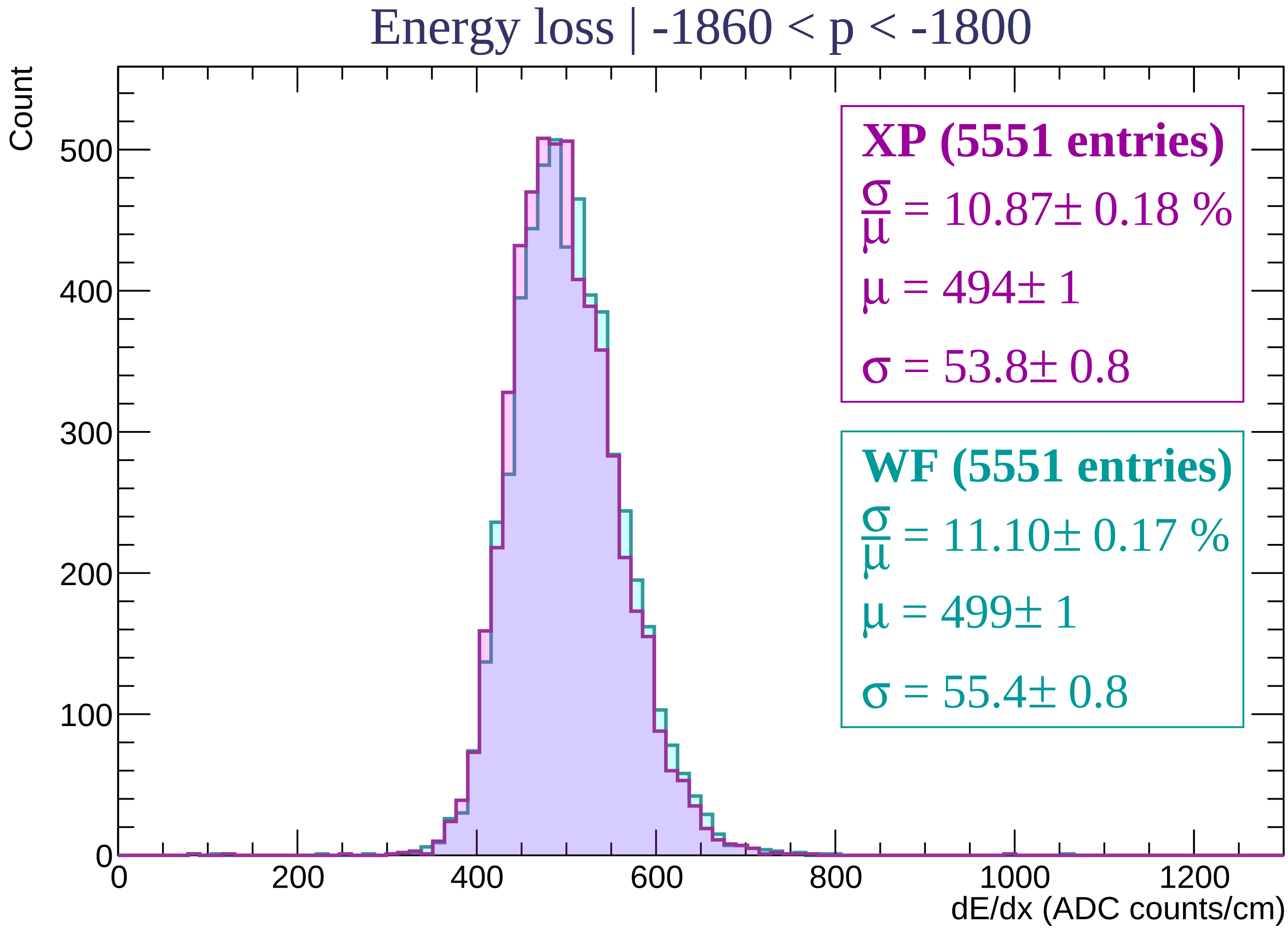
Energy loss | -2100 < p < -2040
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4887 entries)
σ/μ = 10.96± 0.20 %
μ = 503± 1
σ = 55.2± 0.9
WF (4887 entries)
σ/μ = 11.38± 0.21 %
μ = 508± 1
σ = 57.8± 0.9


Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4893 entries)
σ/μ = 10.81± 0.20 %
μ = 501± 1
σ = 54.2± 0.9
WF (4893 entries)
σ/μ = 11.14± 0.21 %
μ = 504± 1
σ = 56.2± 1.0

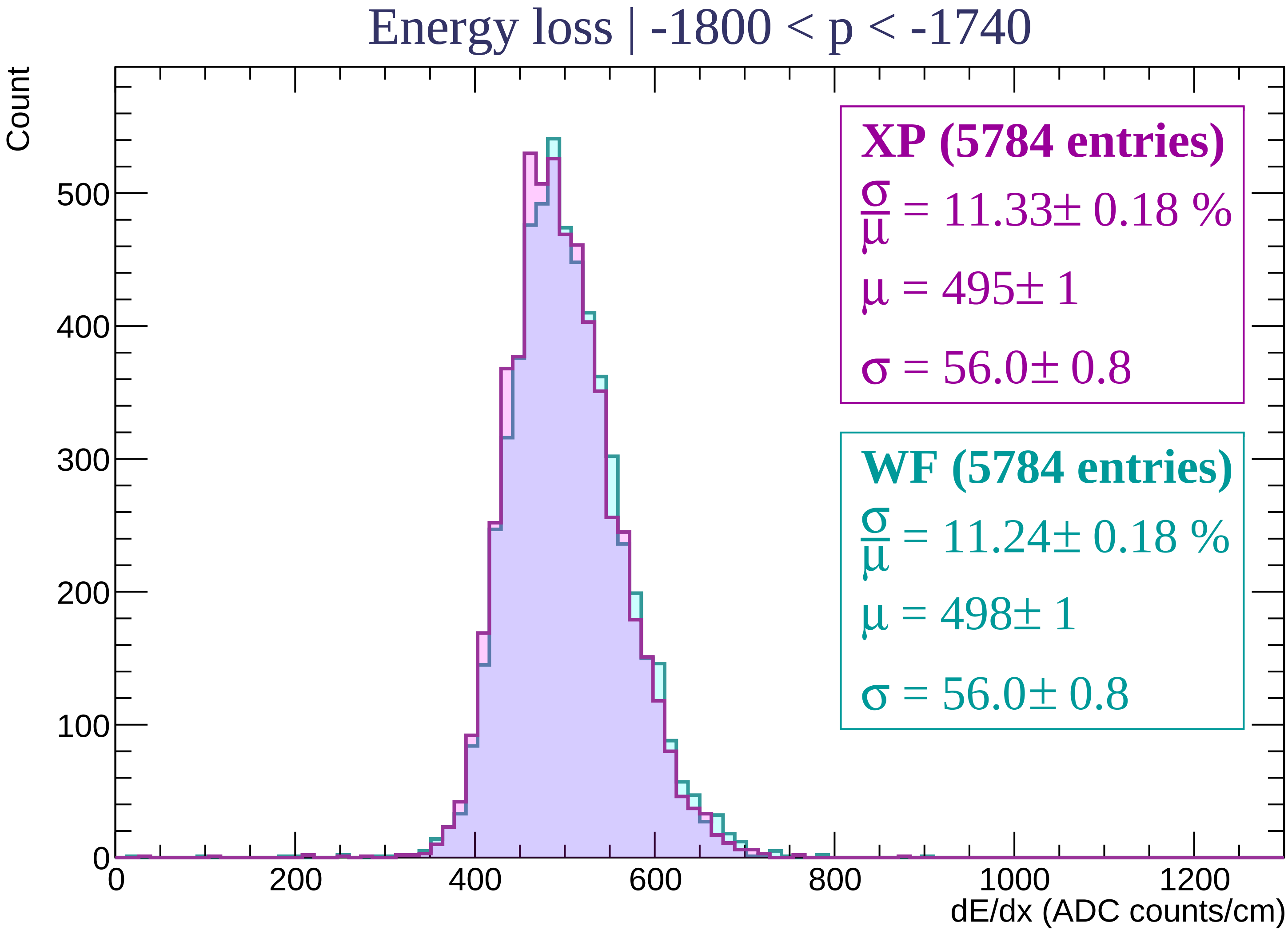

Energy loss | -1980 < p < -1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (5156 entries)
$\frac{\sigma}{\mu}$ = 11.45± 0.22 %
μ = 499± 1
σ = 57.2± 1.0
WF (5156 entries)
$\frac{\sigma}{\mu}$ = 11.52± 0.22 %
μ = 503± 1
σ = 58.0± 1.0



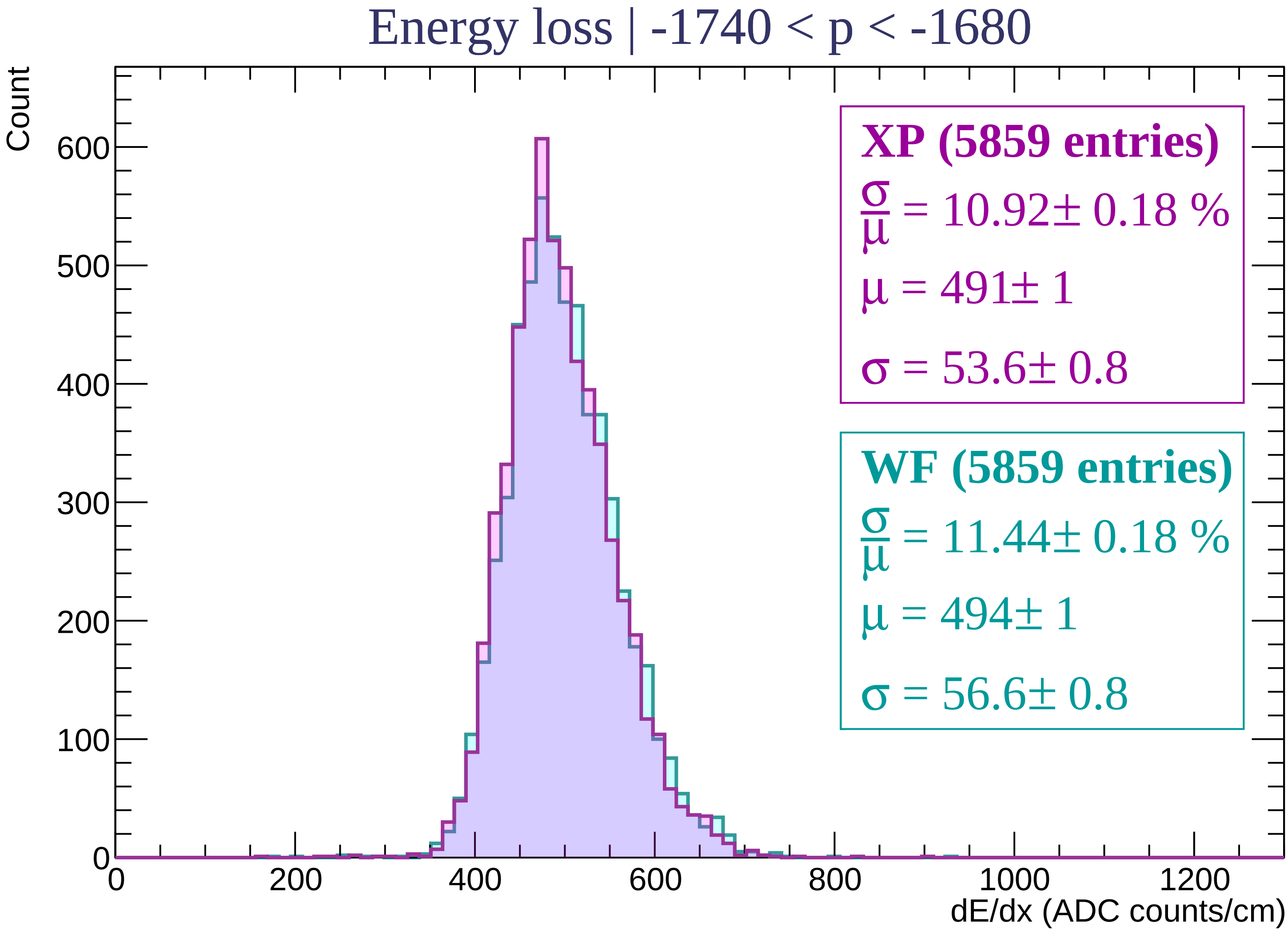
Energy loss | -1920 < p < -1860
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5329 entries)
$\frac{\sigma}{\mu}$ = 10.87± 0.17 %
μ = 499± 1
σ = 54.3± 0.8
WF (5329 entries)
$\frac{\sigma}{\mu}$ = 11.28± 0.19 %
μ = 502± 1
σ = 56.6± 0.9

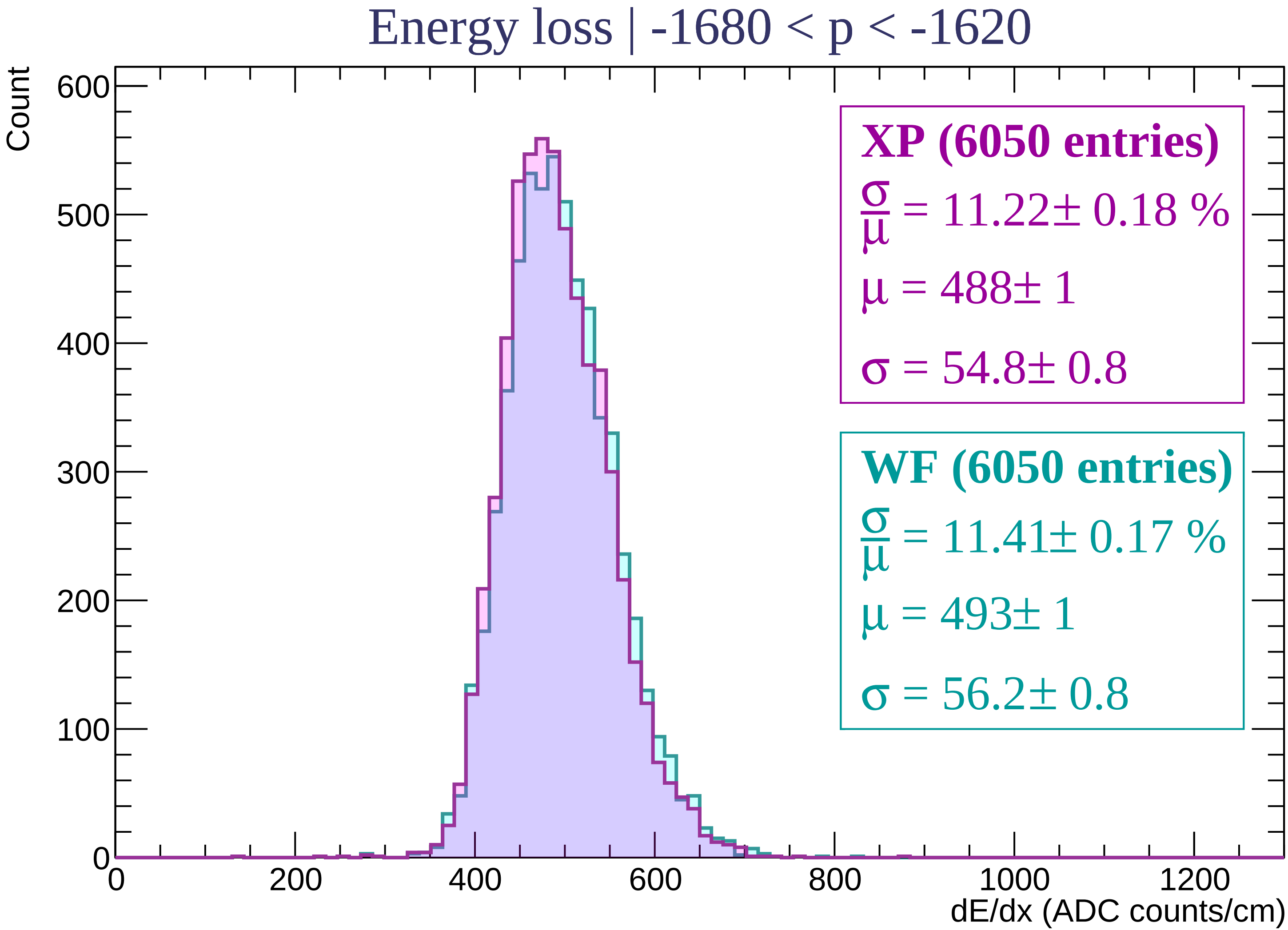


Energy loss | -1860 < p < -1800
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5551 entries)
$\frac{\sigma}{\mu}$ = 10.87± 0.18 %
μ = 494± 1
σ = 53.8± 0.8
WF (5551 entries)
$\frac{\sigma}{\mu}$ = 11.10± 0.17 %
μ = 499± 1
σ = 55.4± 0.8

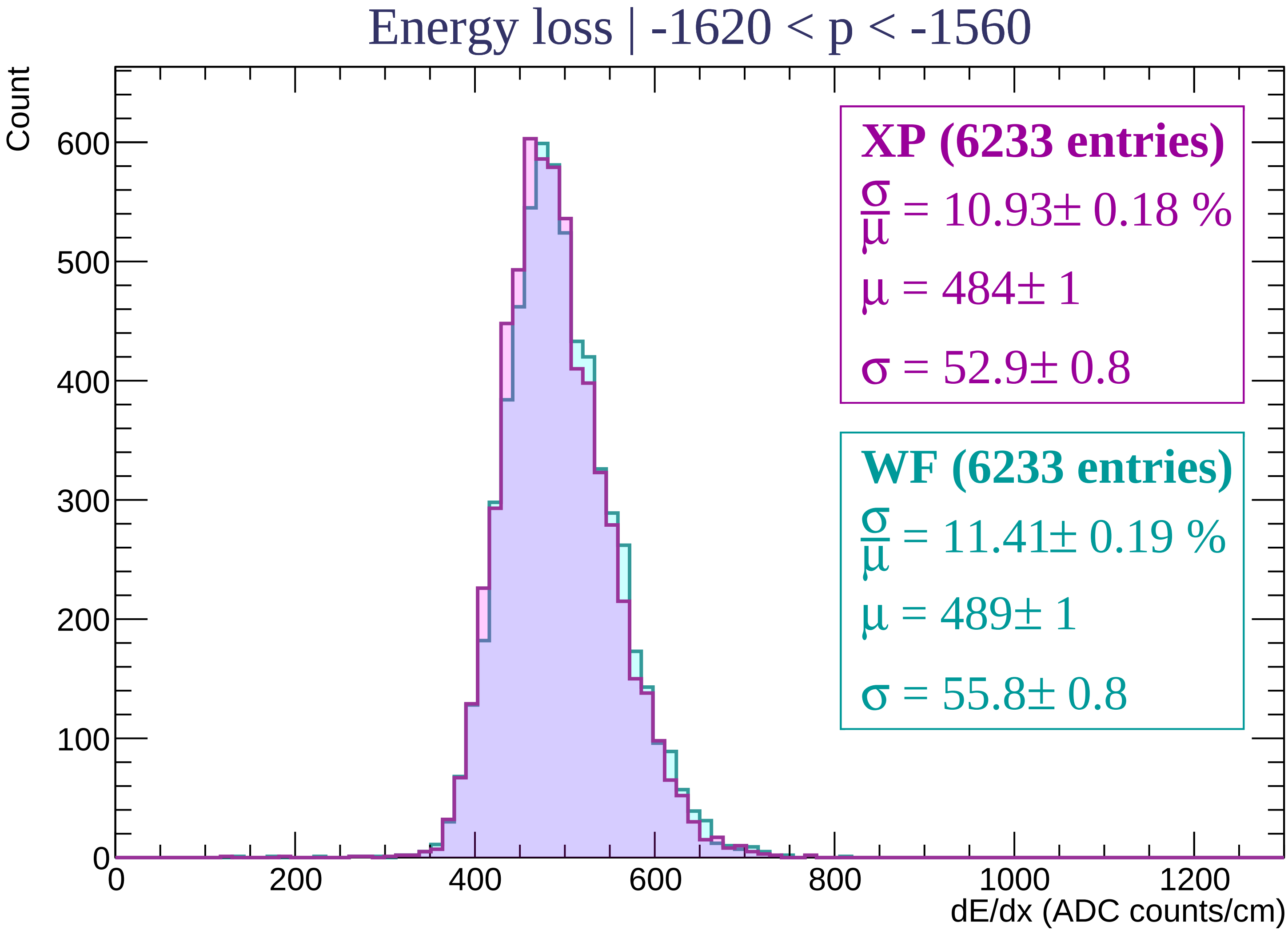


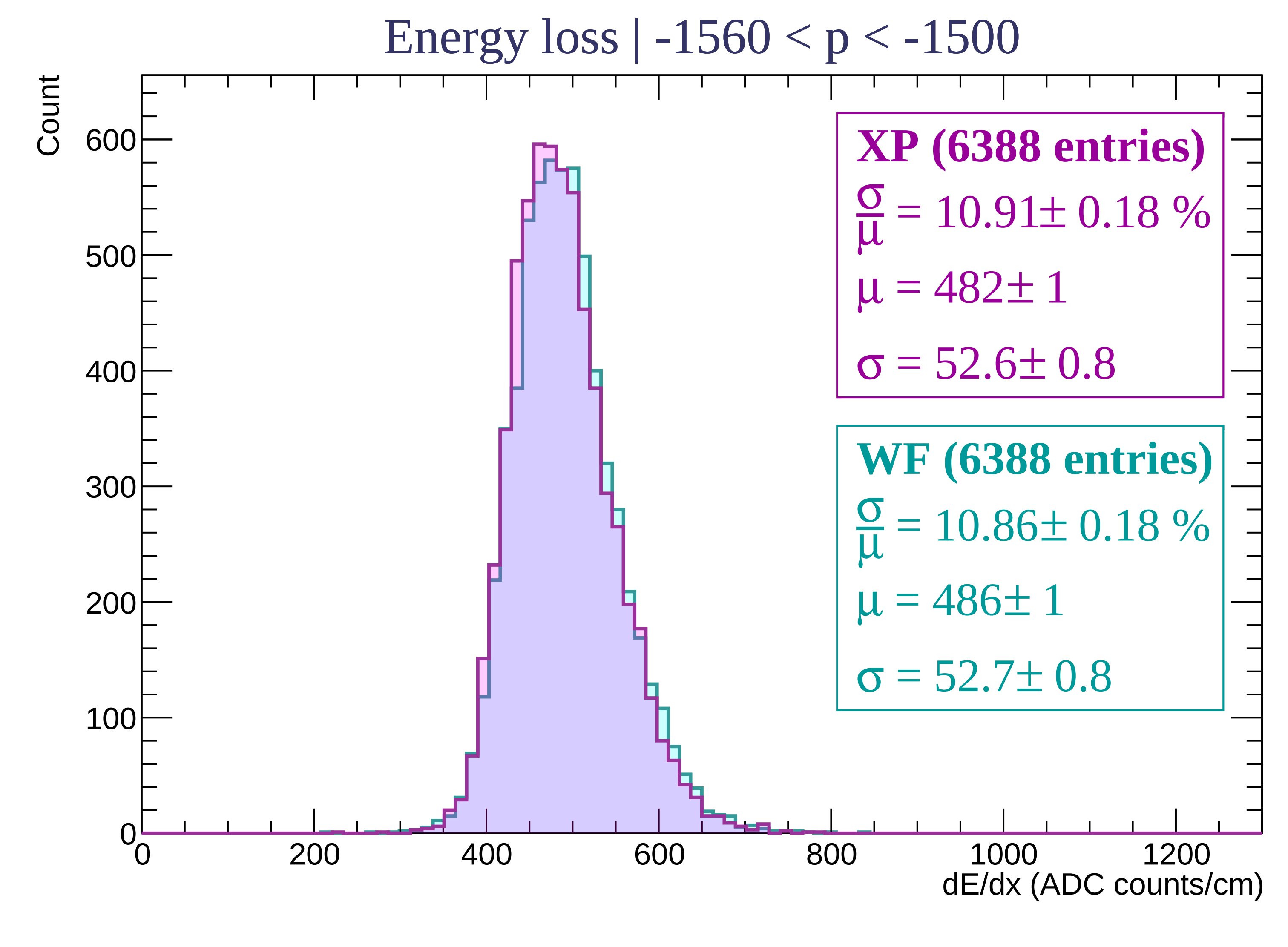
Energy loss | -1800 < p < -1740
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5784 entries)
σ/μ = 11.33± 0.18 %
μ = 495± 1
σ = 56.0± 0.8
WF (5784 entries)
σ/μ = 11.24± 0.18 %
μ = 498± 1
σ = 56.0± 0.8


Energy loss | -1740 < p < -1680
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5859 entries)
σ/μ = 10.92± 0.18 %
μ = 491± 1
σ = 53.6± 0.8
WF (5859 entries)
σ/μ = 11.44± 0.18 %
μ = 494± 1
σ = 56.6± 0.8

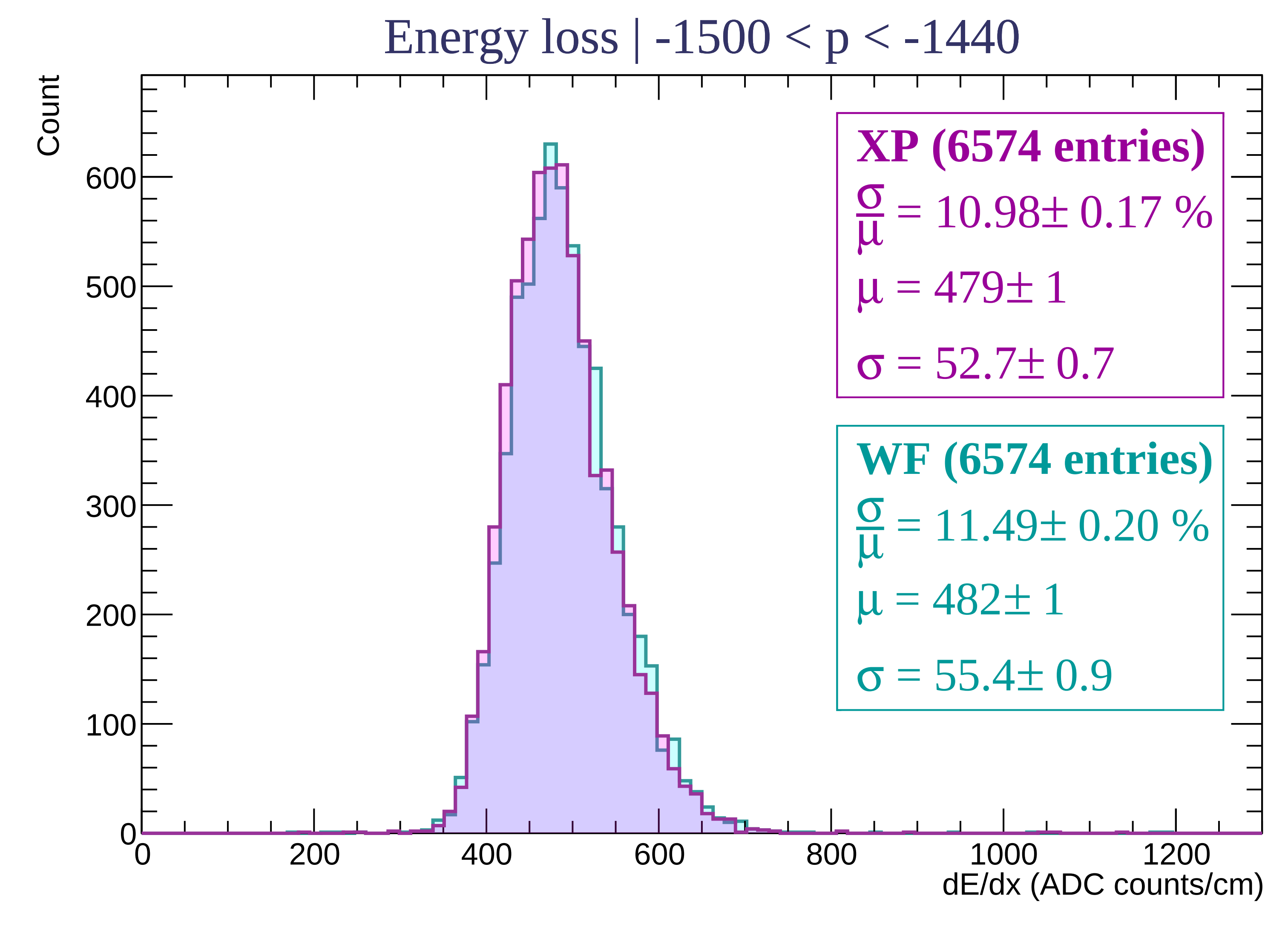


Energy loss | -1680 < p < -1620
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6050 entries)
$\frac{\sigma}{\mu}$ = 11.22± 0.18 %
μ = 488± 1
σ = 54.8± 0.8
WF (6050 entries)
$\frac{\sigma}{\mu}$ = 11.41± 0.17 %
μ = 493± 1
σ = 56.2± 0.8

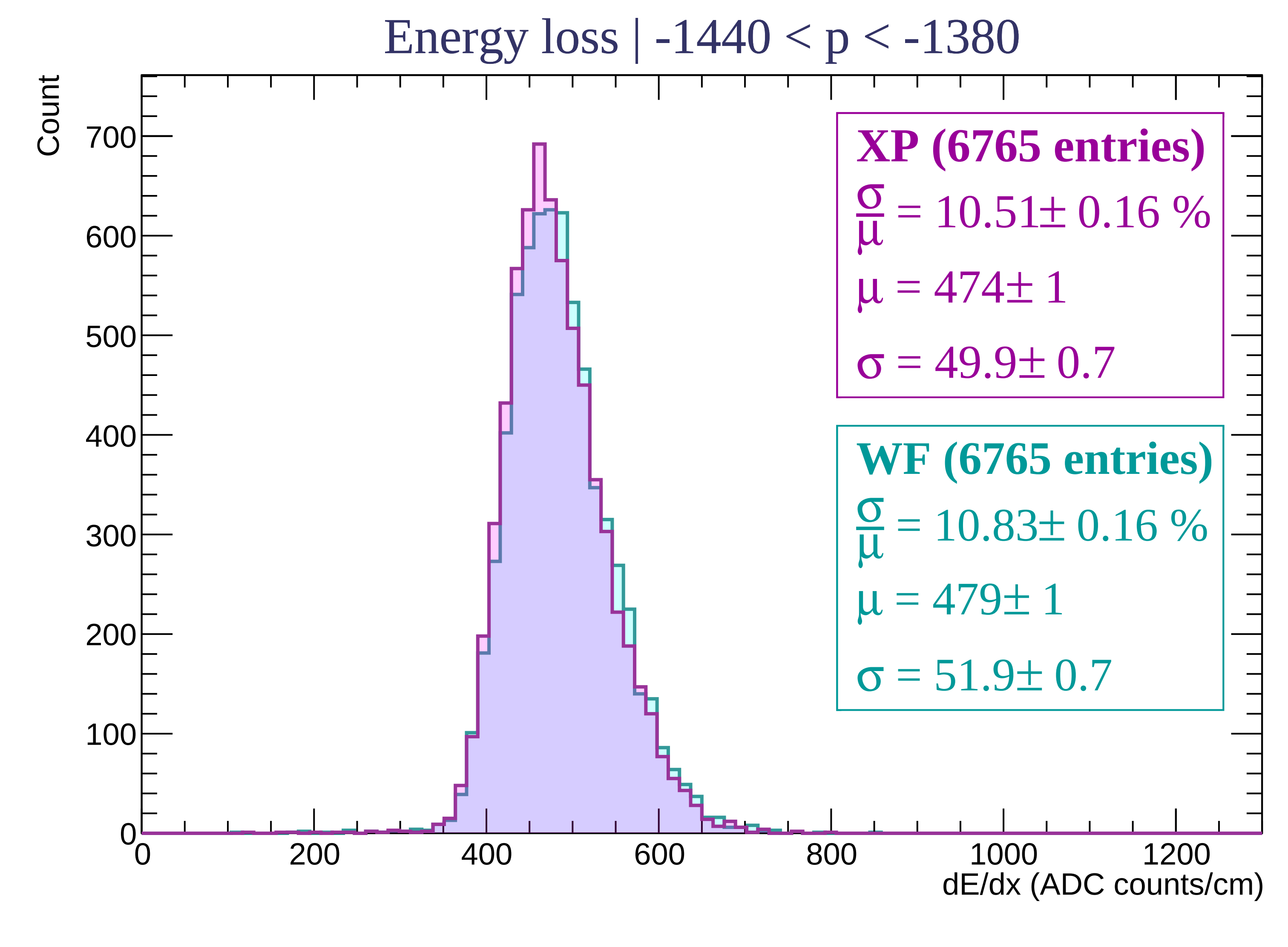

Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (6233 entries)
σ/μ = 10.93± 0.18 %
μ = 484± 1
σ = 52.9± 0.8
WF (6233 entries)
σ/μ = 11.41± 0.19 %
μ = 489± 1
σ = 55.8± 0.8



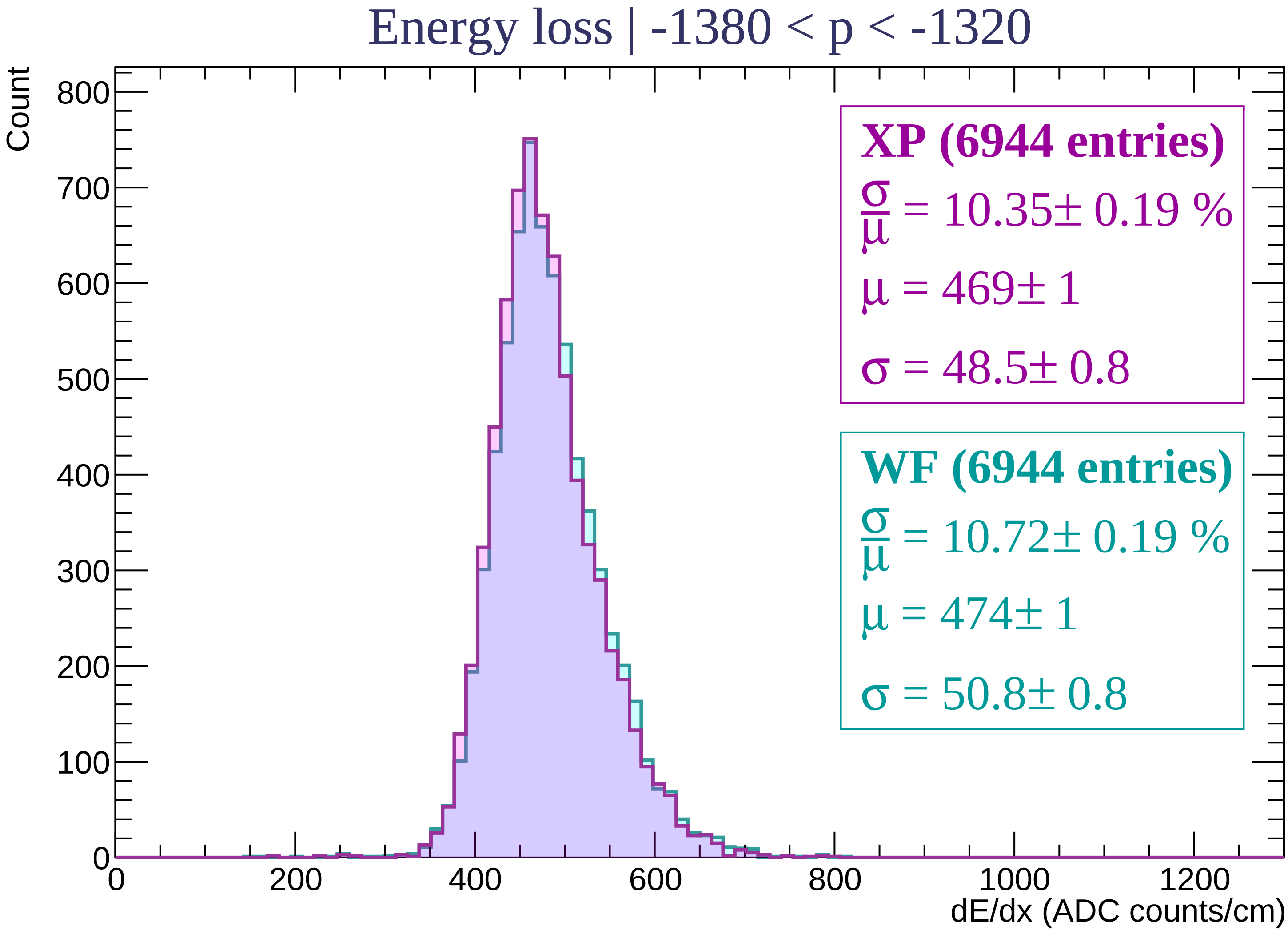
Energy loss | -1560 < p < -1500
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6388 entries)
$\frac{\sigma}{\mu}$ = 10.91± 0.18 %
μ = 482± 1
σ = 52.6± 0.8
WF (6388 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.18 %
μ = 486± 1
σ = 52.7± 0.8

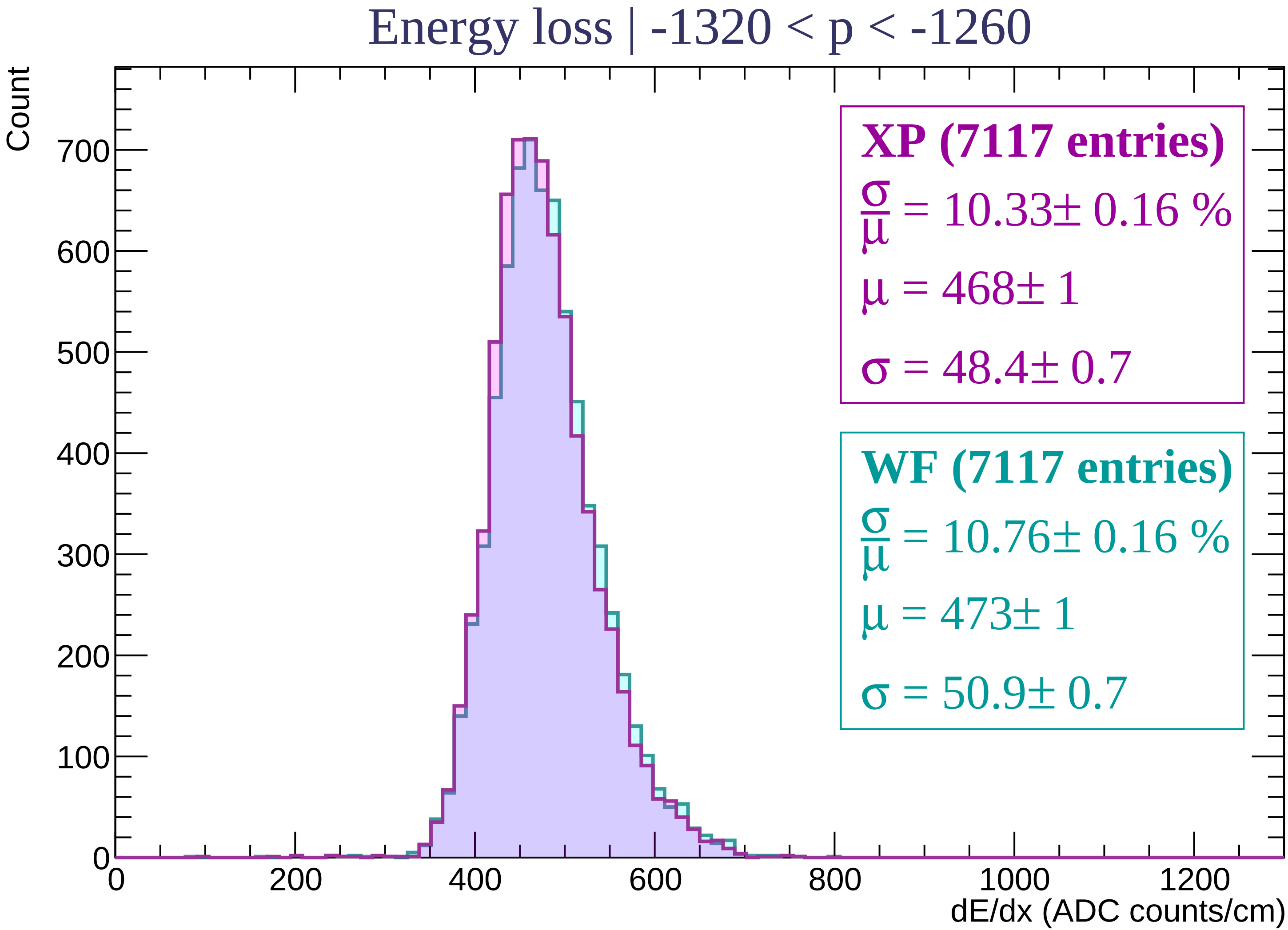


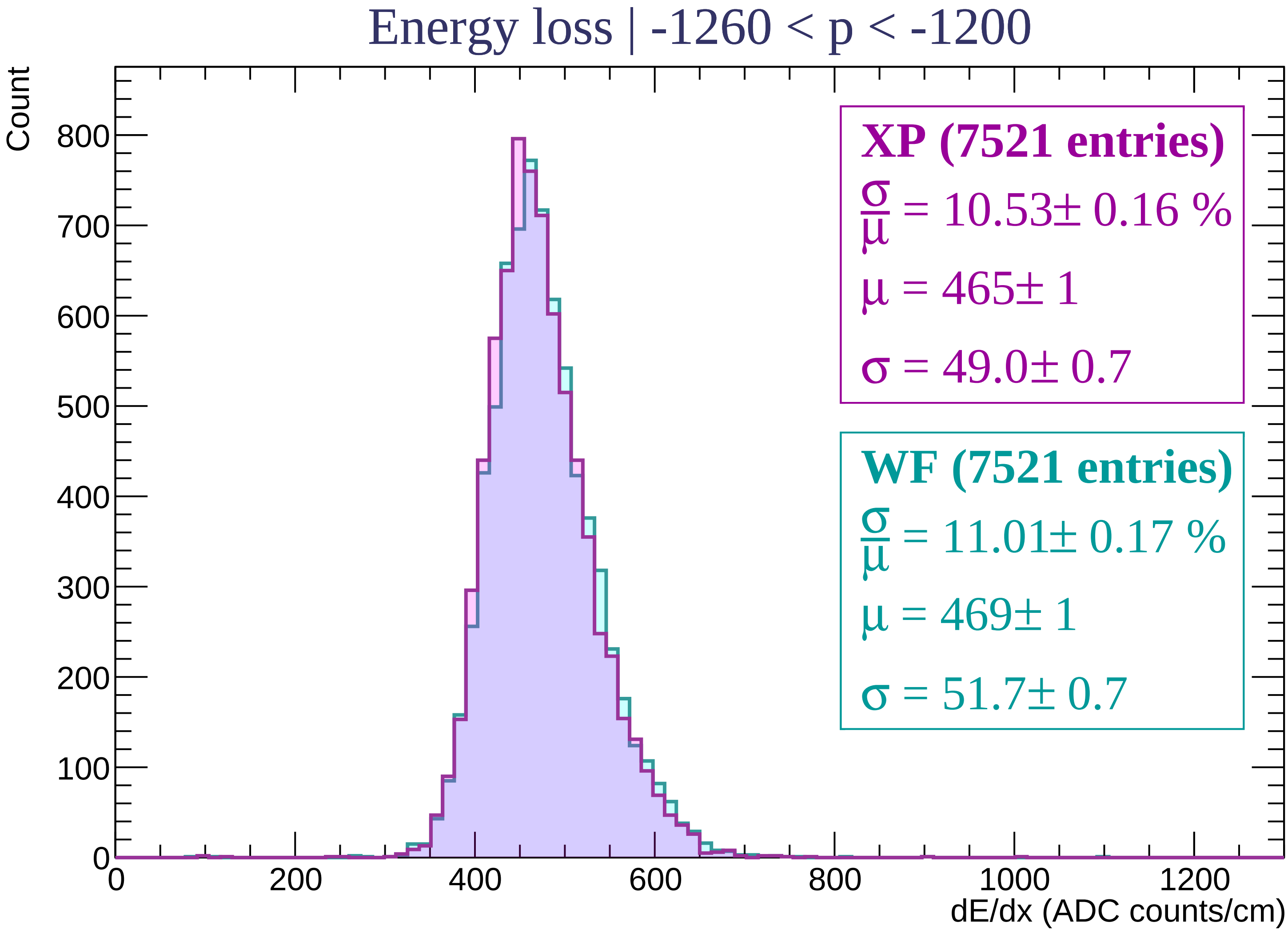
Energy loss | -1500 < p < -1440
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6574 entries)
σ/μ = 10.98± 0.17 %
μ = 479± 1
σ = 52.7± 0.7
WF (6574 entries)
σ/μ = 11.49± 0.20 %
μ = 482± 1
σ = 55.4± 0.9

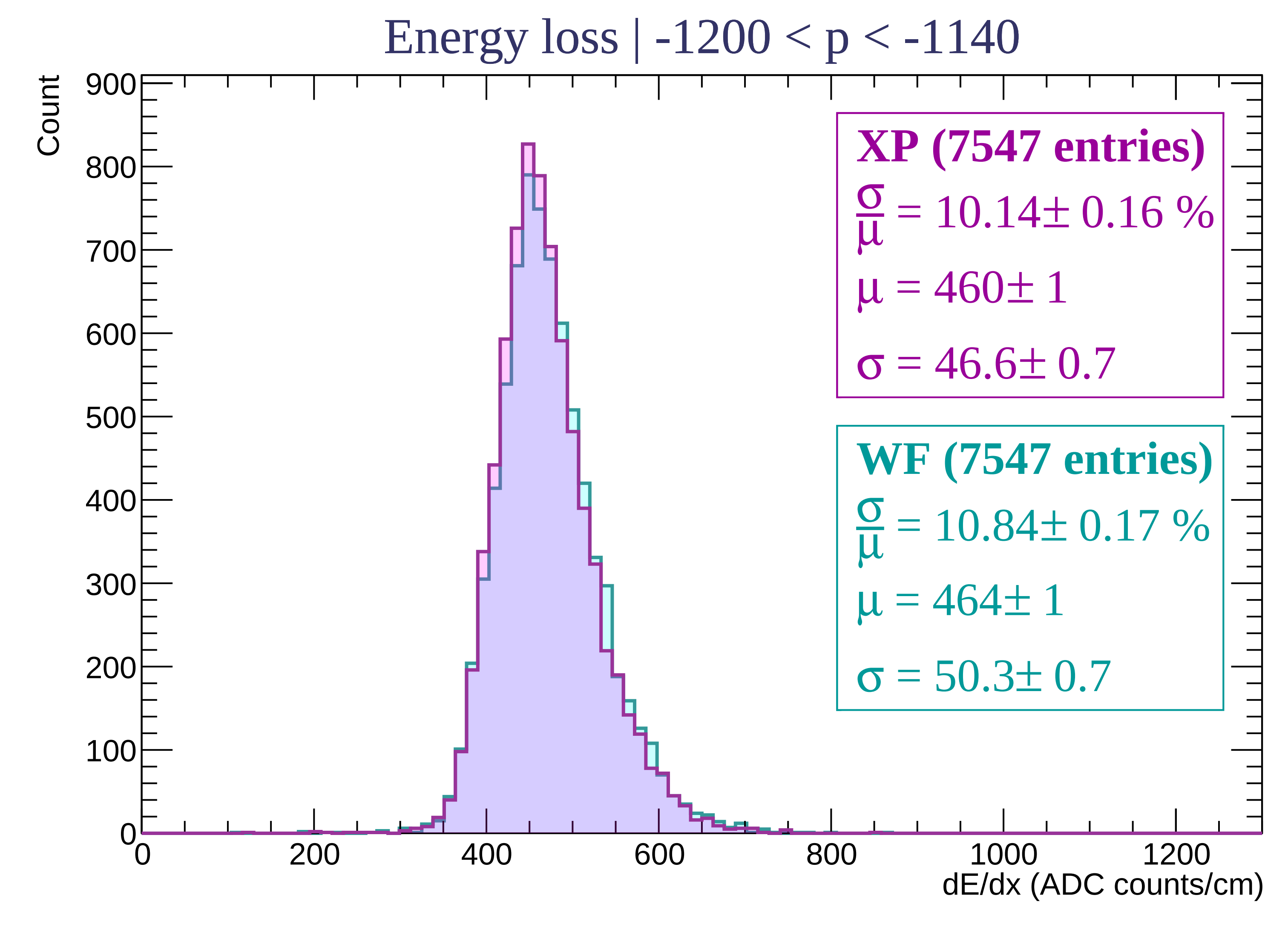


Energy loss | -1440 < p < -1380
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6765 entries)
σ/μ = 10.51± 0.16 %
μ = 474± 1
σ = 49.9± 0.7
WF (6765 entries)
σ/μ = 10.83± 0.16 %
μ = 479± 1
σ = 51.9± 0.7

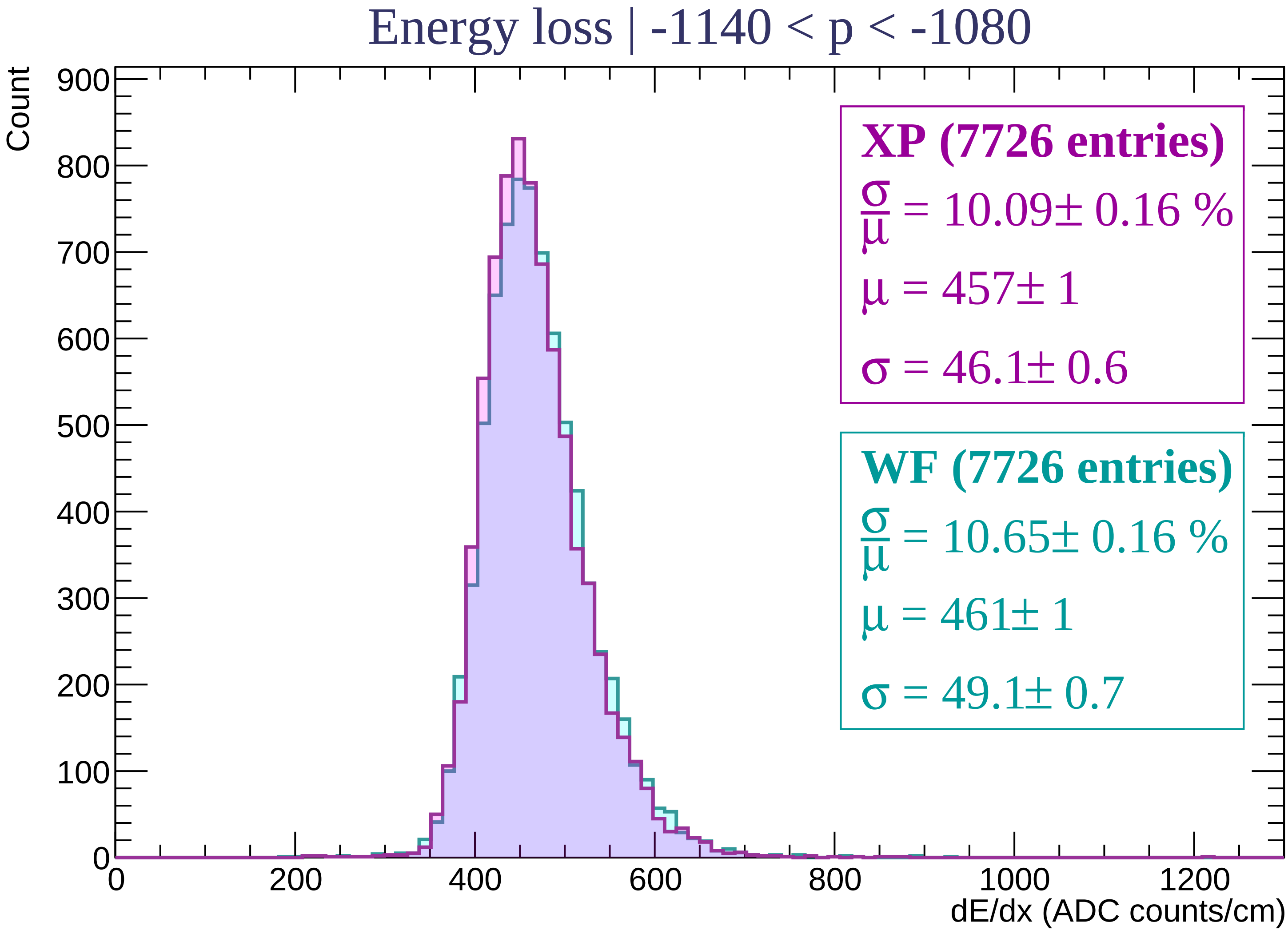

Energy loss | -1380 < p < -1320
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6944 entries)
$\frac{\sigma}{\mu}$ = 10.35± 0.19 %
μ = 469± 1
σ = 48.5± 0.8
WF (6944 entries)
$\frac{\sigma}{\mu}$ = 10.72± 0.19 %
μ = 474± 1
σ = 50.8± 0.8


Energy loss | -1320 < p < -1260
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7117 entries)
σ/μ = 10.33± 0.16 %
μ = 468± 1
σ = 48.4± 0.7
WF (7117 entries)
σ/μ = 10.76± 0.16 %
μ = 473± 1
σ = 50.9± 0.7

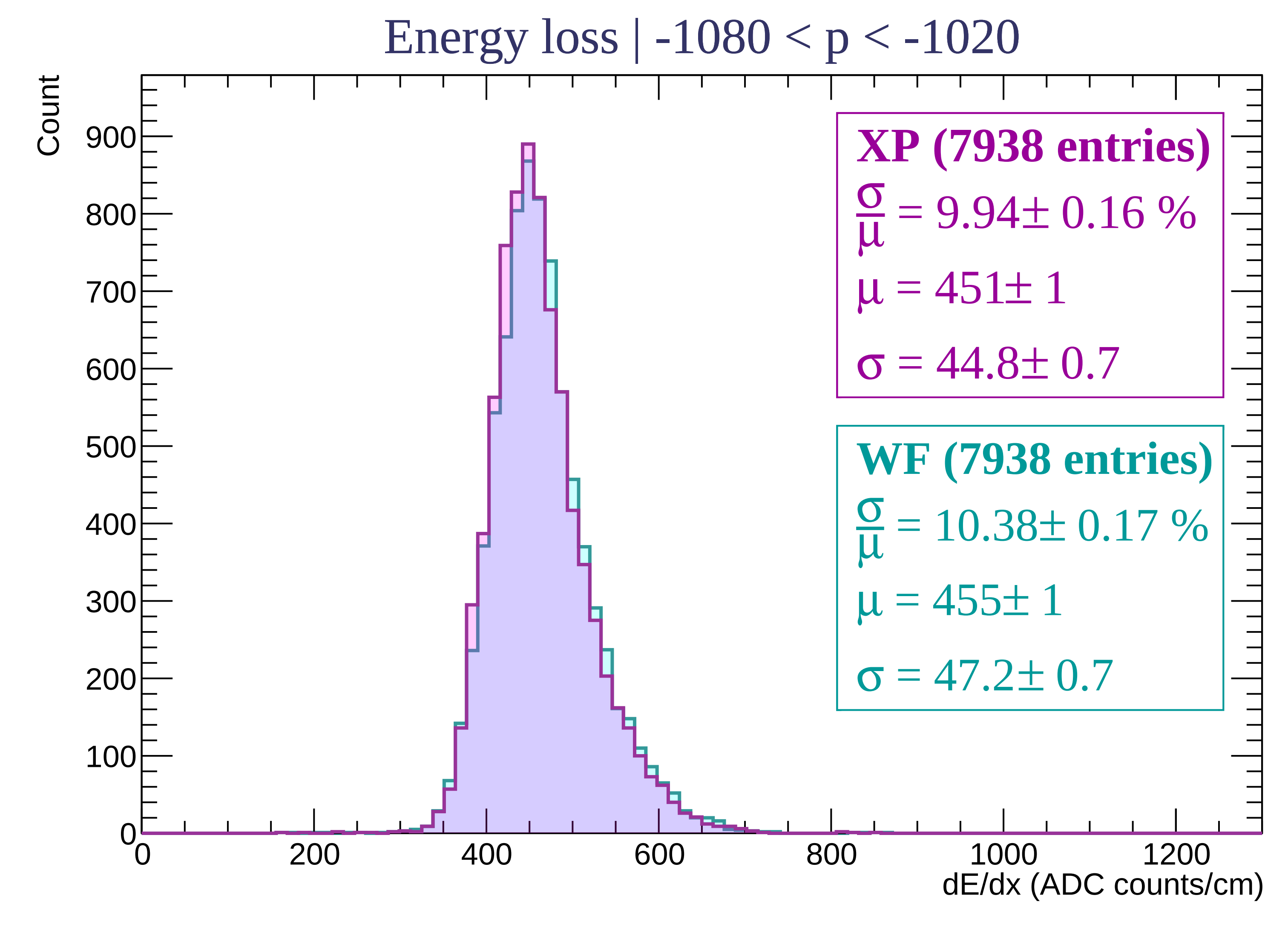

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (7521 entries)
$\frac{\sigma}{\mu}$ = 10.53± 0.16 %
μ = 465± 1
σ = 49.0± 0.7
WF (7521 entries)
$\frac{\sigma}{\mu}$ = 11.01± 0.17 %
μ = 469± 1
σ = 51.7± 0.7



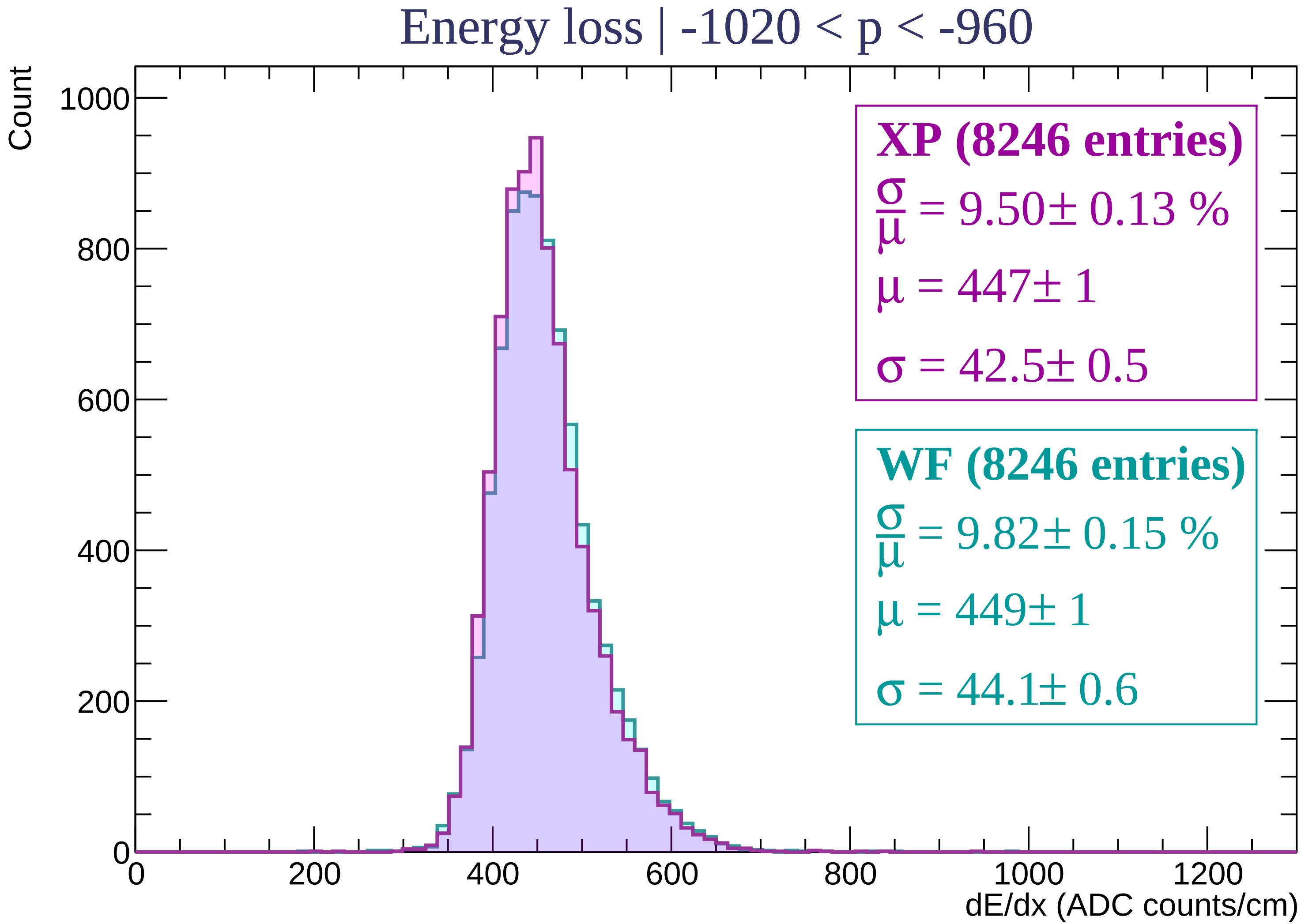
Energy loss | -1200 < p < -1140
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7547 entries)
$\frac{\sigma}{\mu}$ = 10.14± 0.16 %
μ = 460± 1
σ = 46.6± 0.7
WF (7547 entries)
$\frac{\sigma}{\mu}$ = 10.84± 0.17 %
μ = 464± 1
σ = 50.3± 0.7



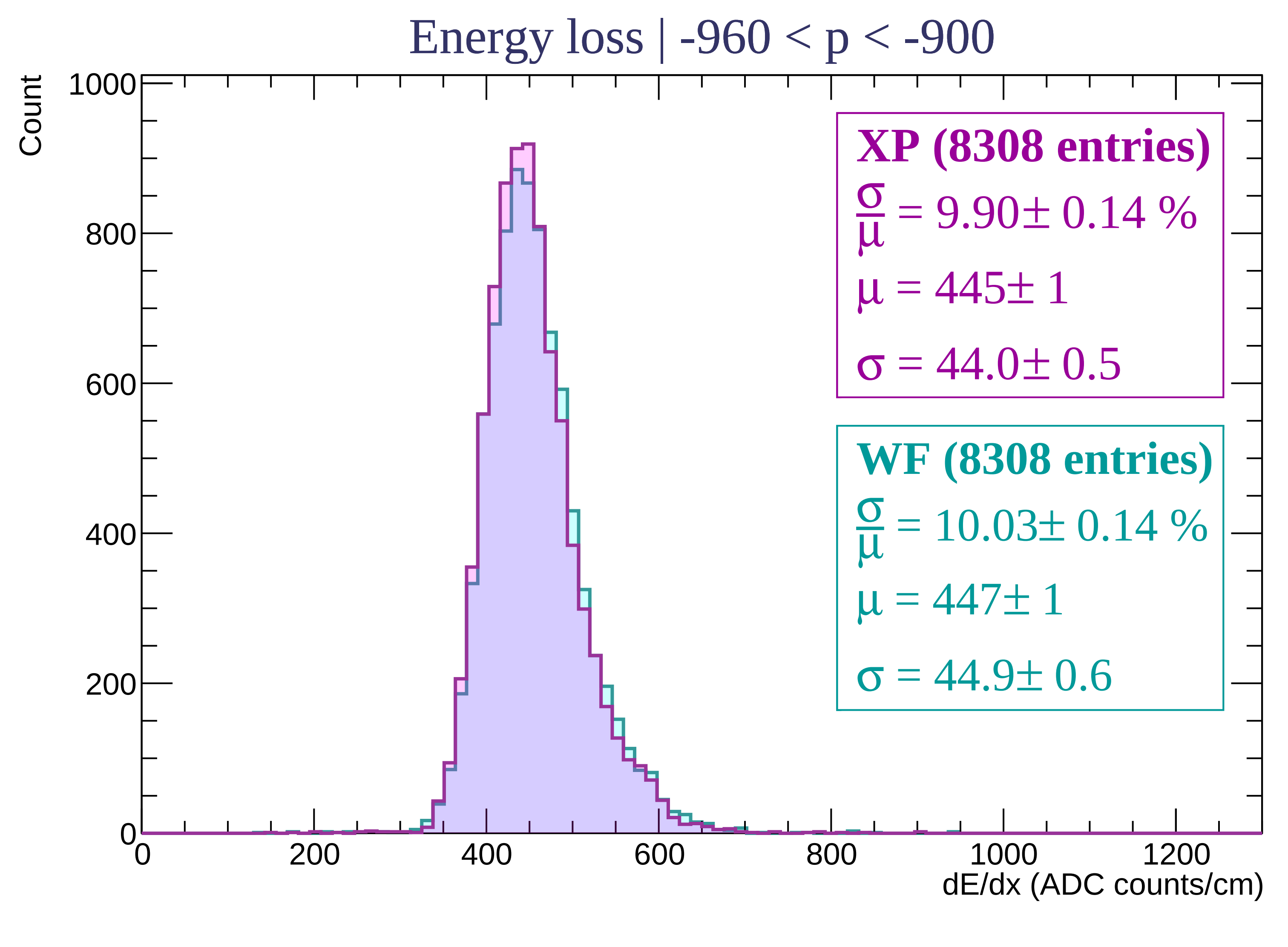
Energy loss | -1140 < p < -1080
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7726 entries)
σ/μ = 10.09± 0.16 %
μ = 457± 1
σ = 46.1± 0.6
WF (7726 entries)
σ/μ = 10.65± 0.16 %
μ = 461± 1
σ = 49.1± 0.7

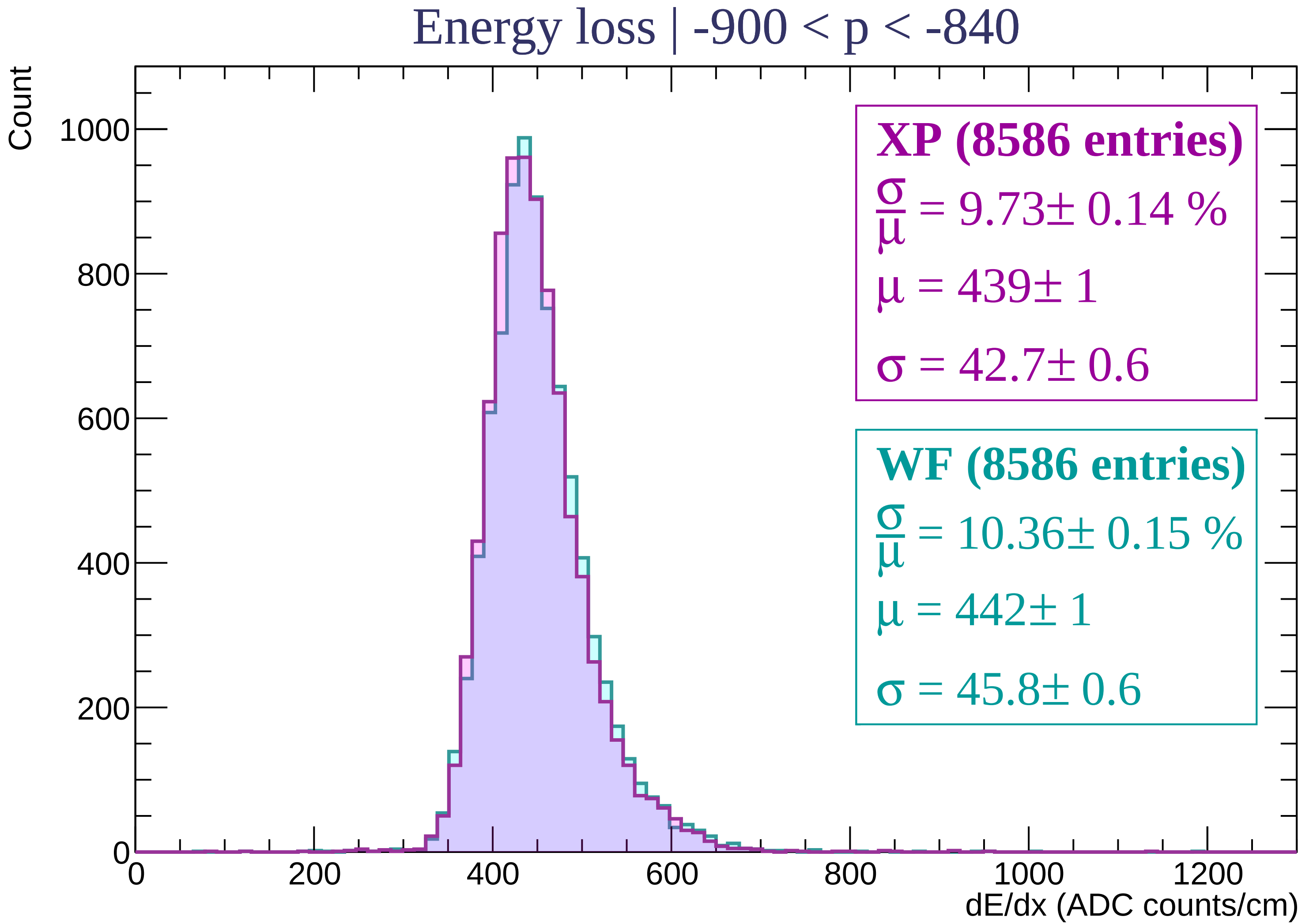


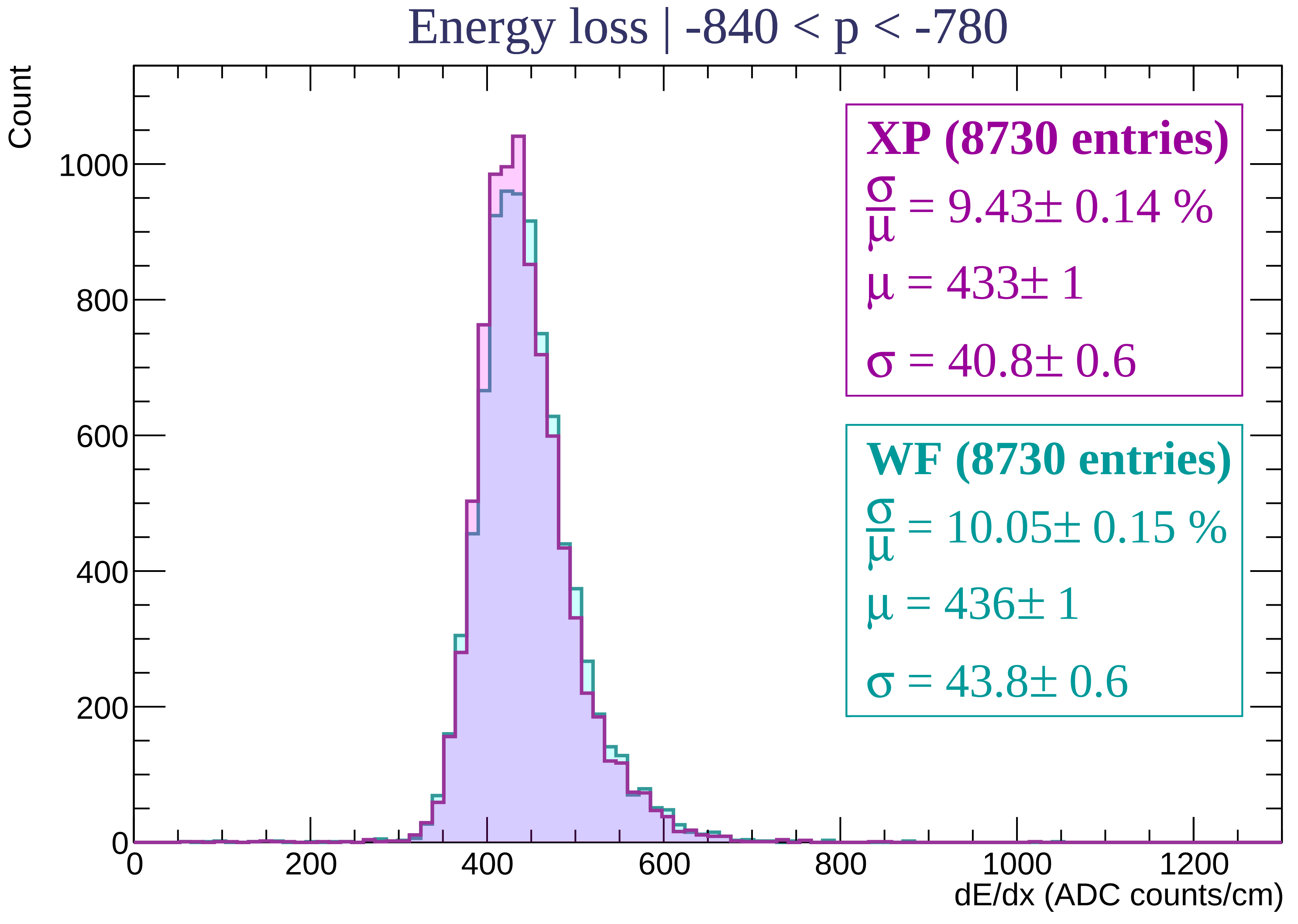
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
XP (7938 entries)
σ/μ = 9.94± 0.16 %
μ = 451± 1
σ = 44.8± 0.7
WF (7938 entries)
σ/μ = 10.38± 0.17 %
μ = 455± 1
σ = 47.2± 0.7
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200

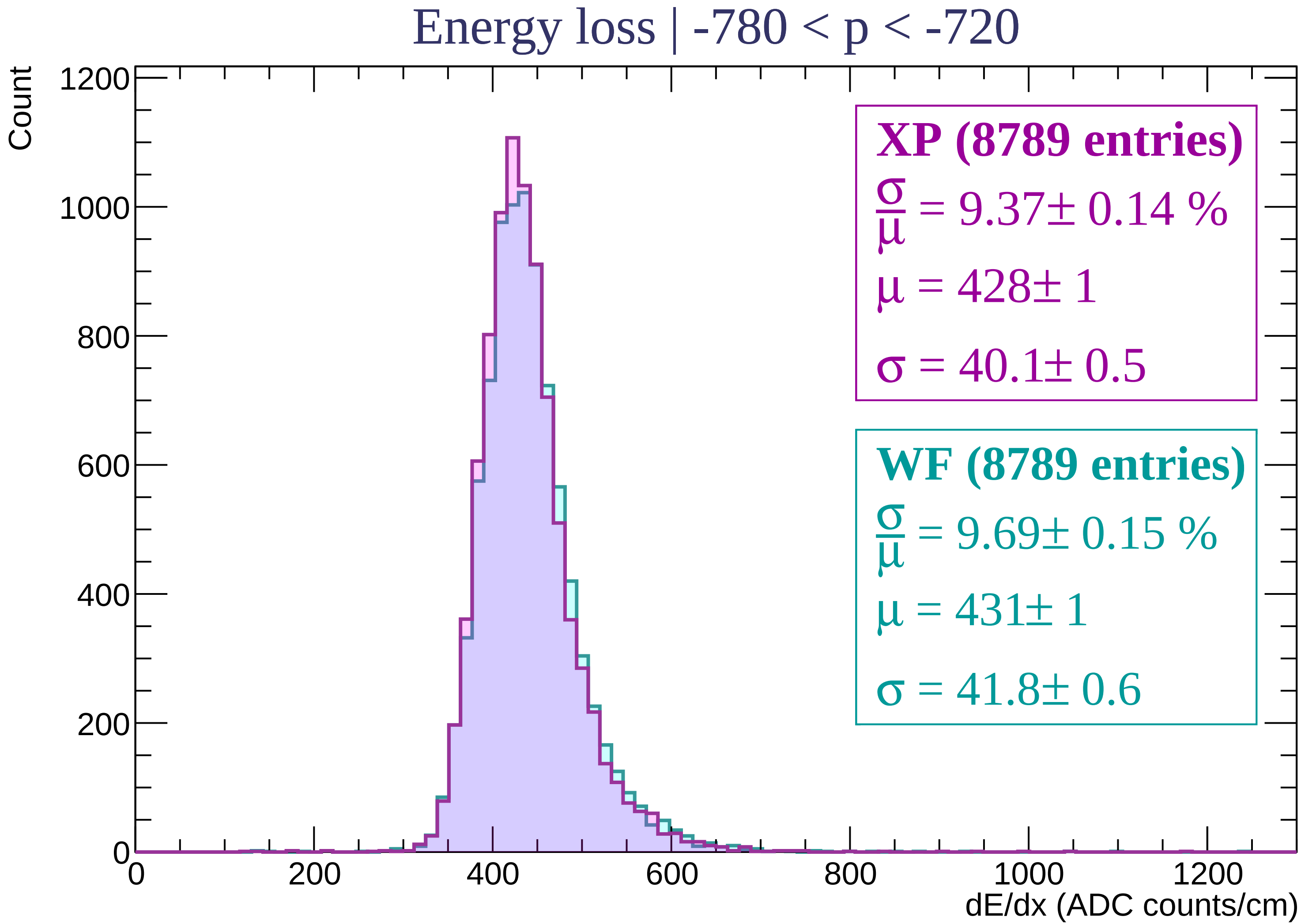


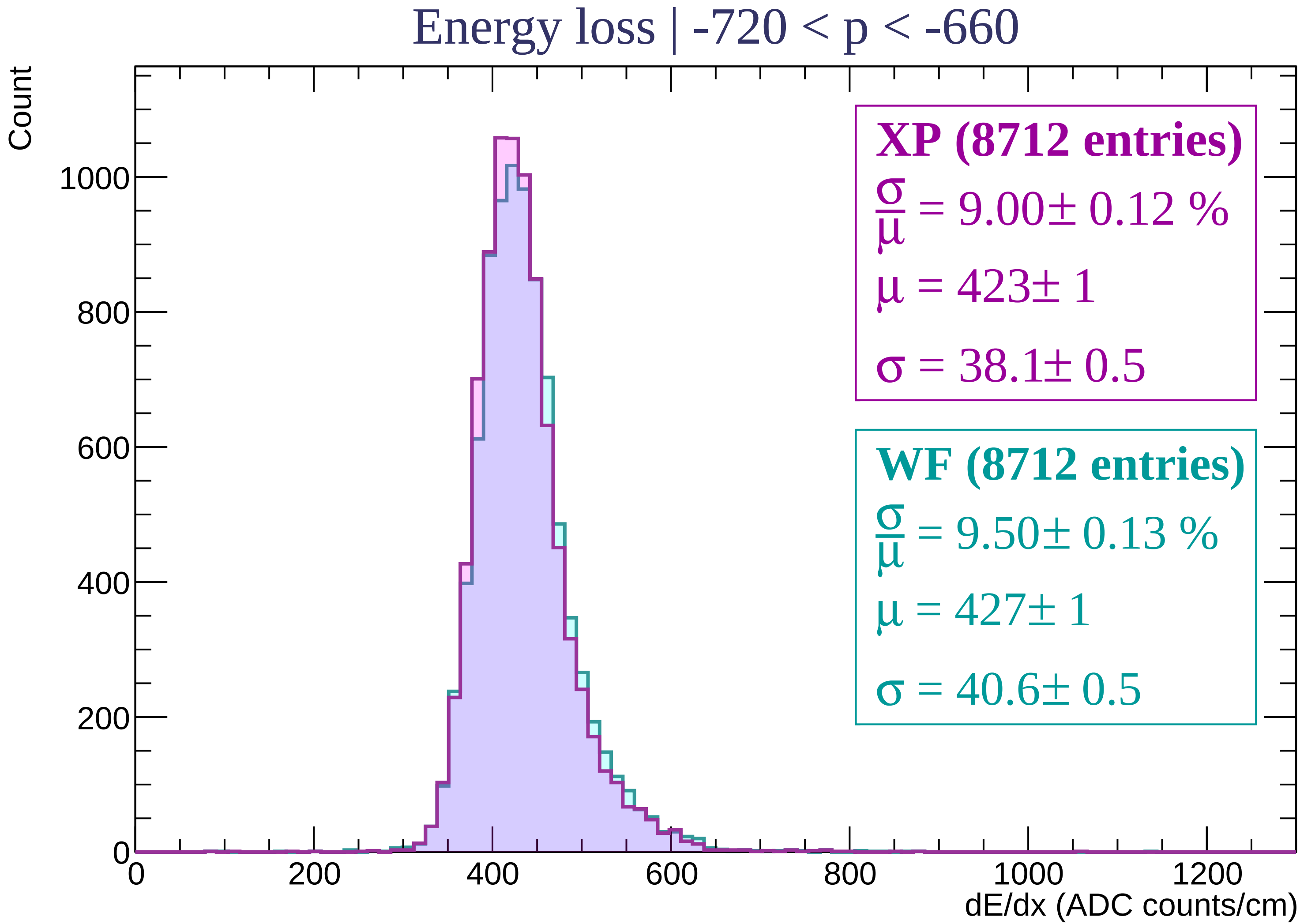
Energy loss | -1020 < p < -960
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8246 entries)
σ/μ = 9.50± 0.13 %
μ = 447± 1
σ = 42.5± 0.5
WF (8246 entries)
σ/μ = 9.82± 0.15 %
μ = 449± 1
σ = 44.1± 0.6


Energy loss | -960 < p < -900
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8308 entries)
$\frac{\sigma}{\mu}$ = 9.90± 0.14 %
μ = 445± 1
σ = 44.0± 0.5
WF (8308 entries)
$\frac{\sigma}{\mu}$ = 10.03± 0.14 %
μ = 447± 1
σ = 44.9± 0.6

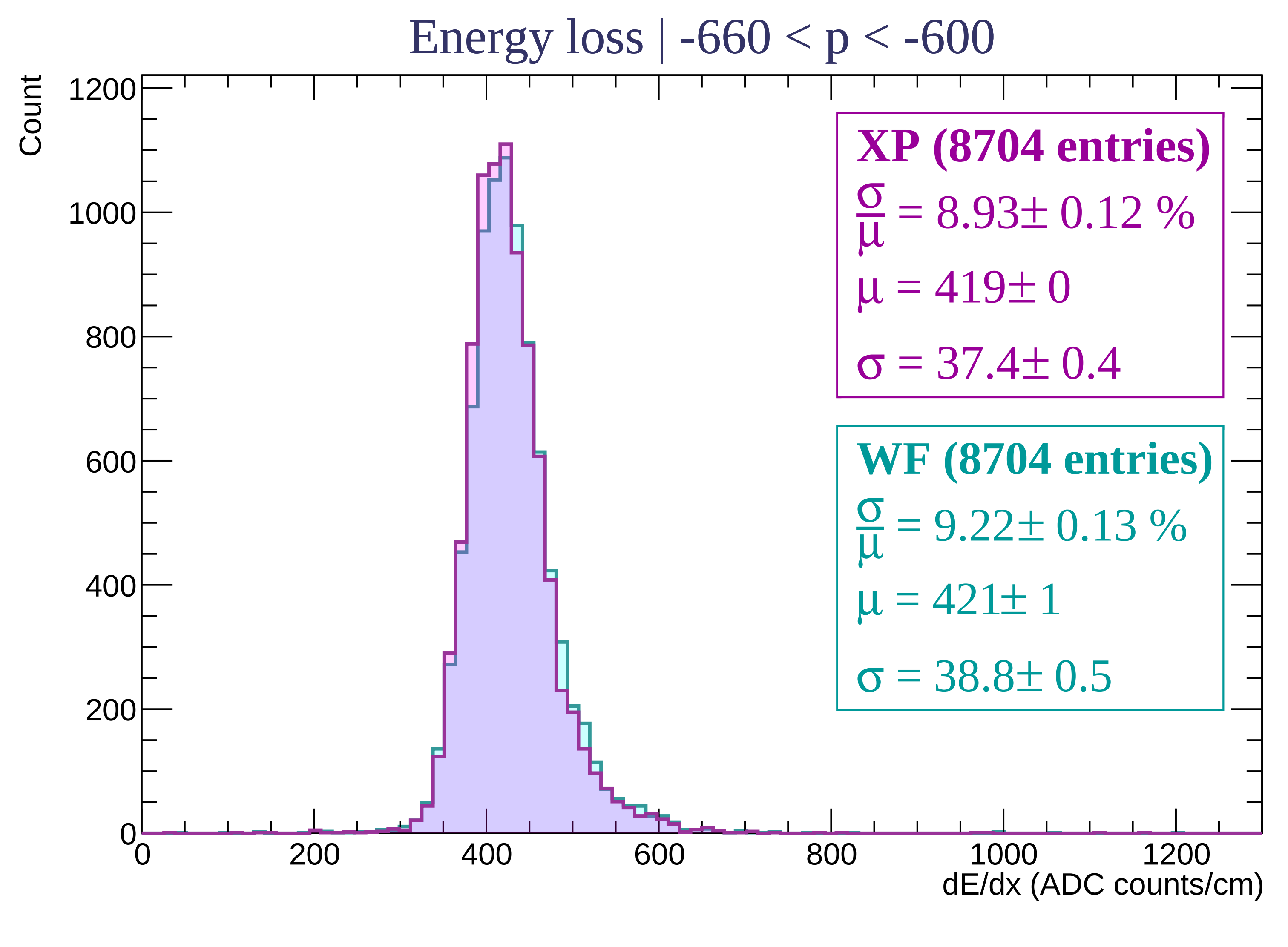

Energy loss | -900 < p < -840
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8586 entries)
σ/μ = 9.73± 0.14 %
μ = 439± 1
σ = 42.7± 0.6
WF (8586 entries)
σ/μ = 10.36± 0.15 %
μ = 442± 1
σ = 45.8± 0.6


Energy loss | -840 < p < -780
Count
0
200
400
600
800
1000
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8730 entries)
$\frac{\sigma}{\mu}$ = 9.43± 0.14 %
μ = 433± 1
σ = 40.8± 0.6
WF (8730 entries)
$\frac{\sigma}{\mu}$ = 10.05± 0.15 %
μ = 436± 1
σ = 43.8± 0.6

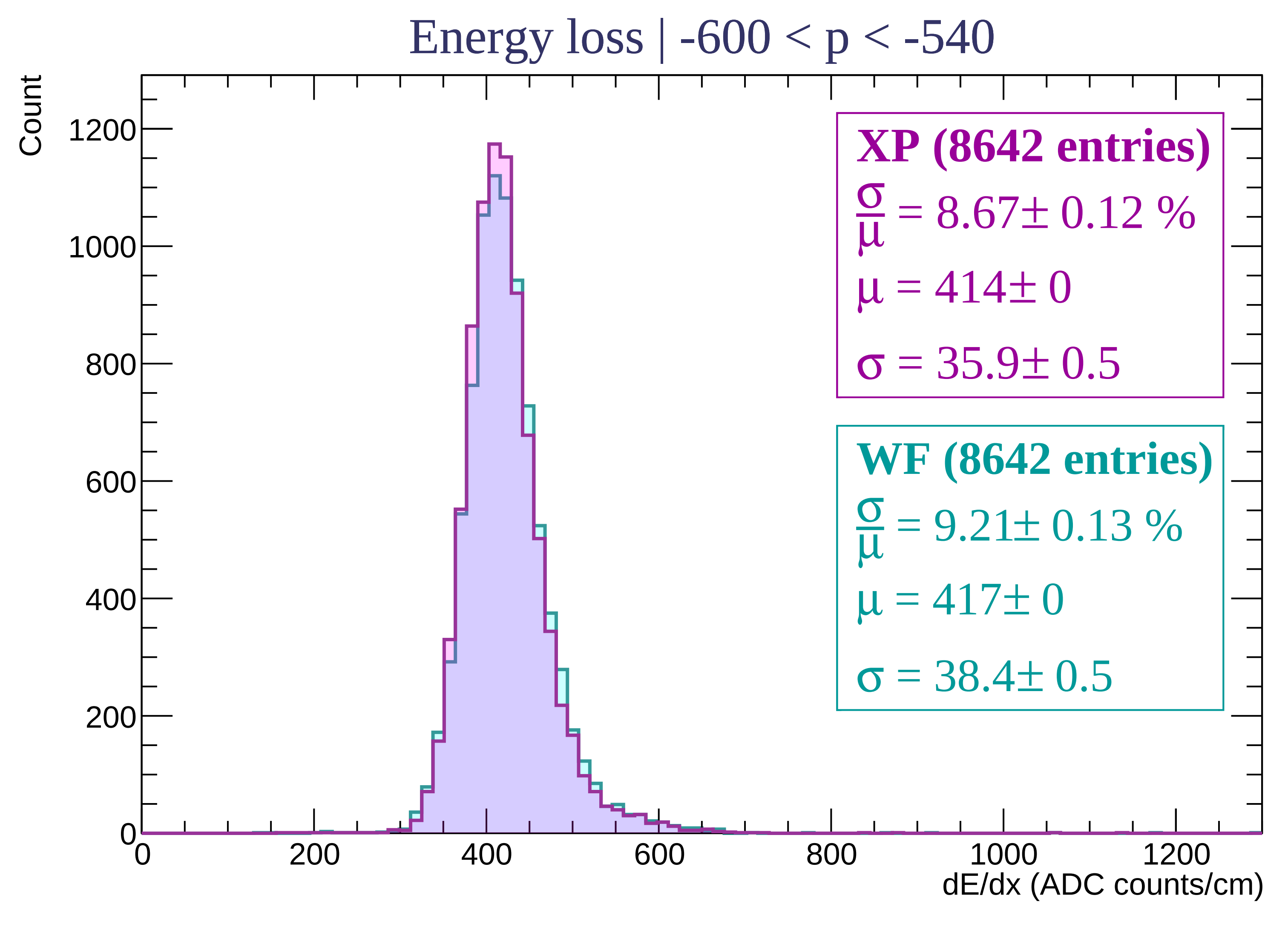

Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP (8789 entries)
σ/μ = 9.37± 0.14 %
μ = 428± 1
σ = 40.1± 0.5
WF (8789 entries)
σ/μ = 9.69± 0.15 %
μ = 431± 1
σ = 41.8± 0.6
0
200
400
600
800
1000
1200



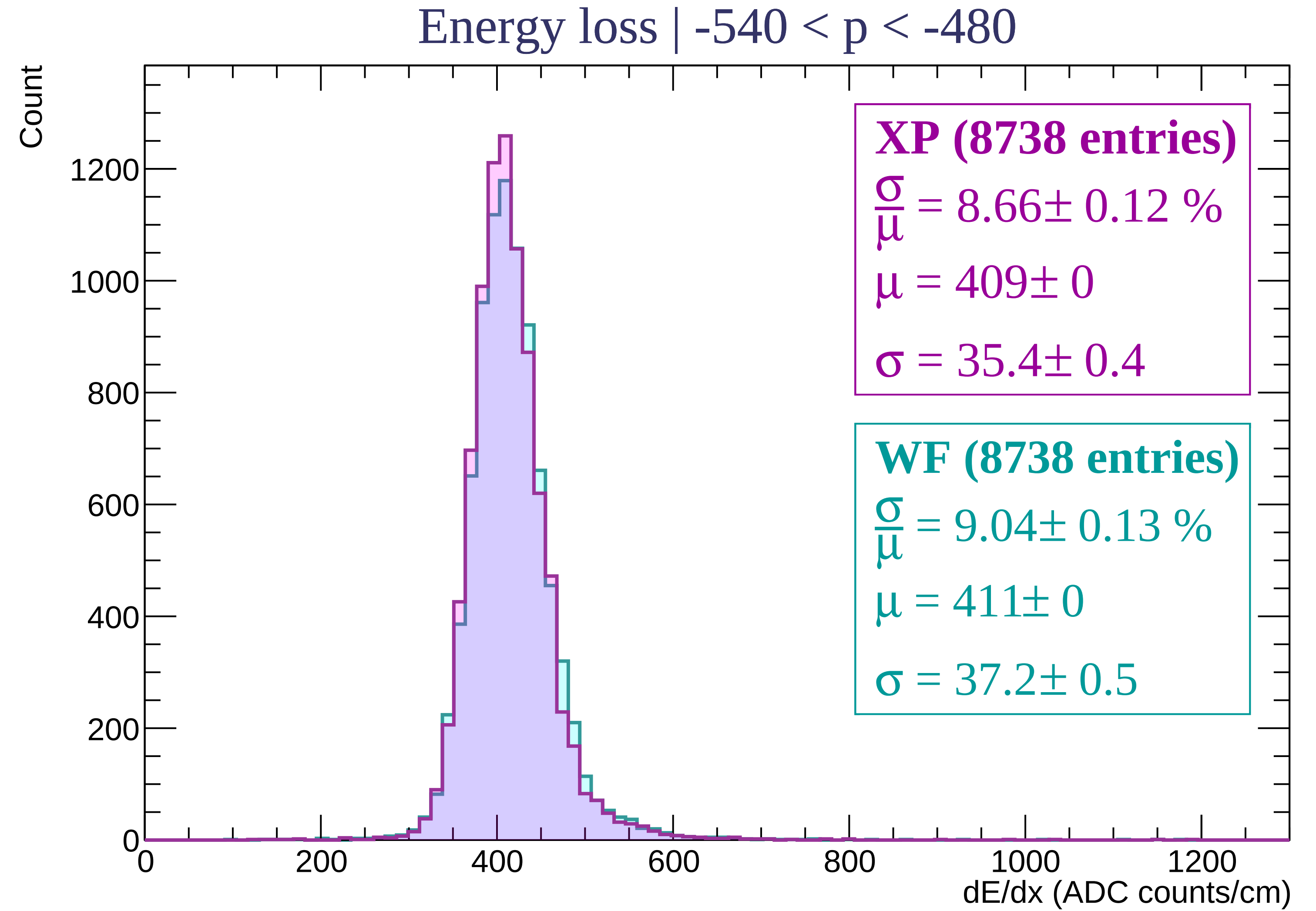
Energy loss | -720 < p < -660
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8712 entries)
σ/μ = 9.00± 0.12 %
μ = 423± 1
σ = 38.1± 0.5
WF (8712 entries)
σ/μ = 9.50± 0.13 %
μ = 427± 1
σ = 40.6± 0.5



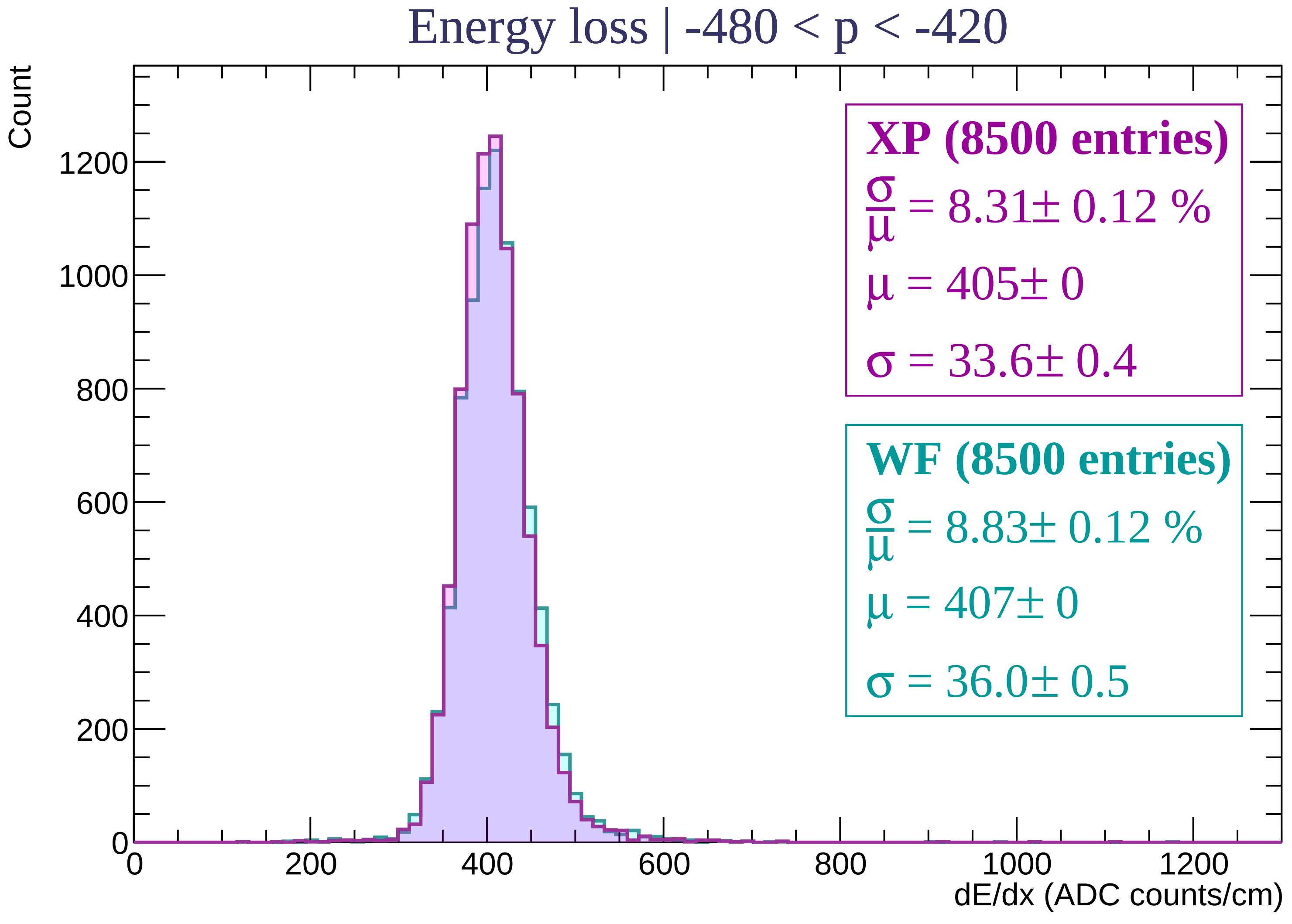
Energy loss | -660 < p < -600
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8704 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.12 %
μ = 419± 0
σ = 37.4± 0.4
WF (8704 entries)
$\frac{\sigma}{\mu}$ = 9.22± 0.13 %
μ = 421± 1
σ = 38.8± 0.5

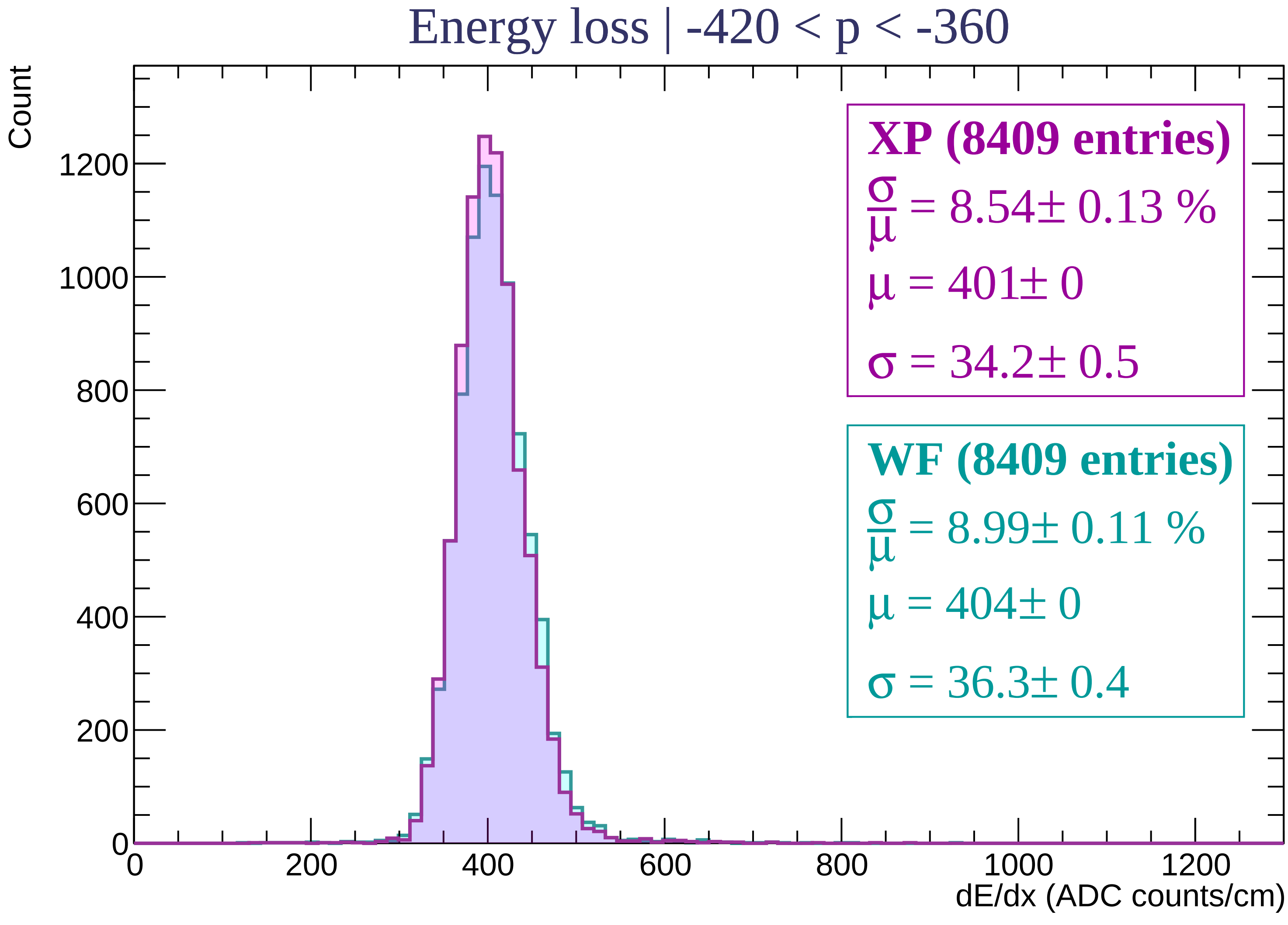


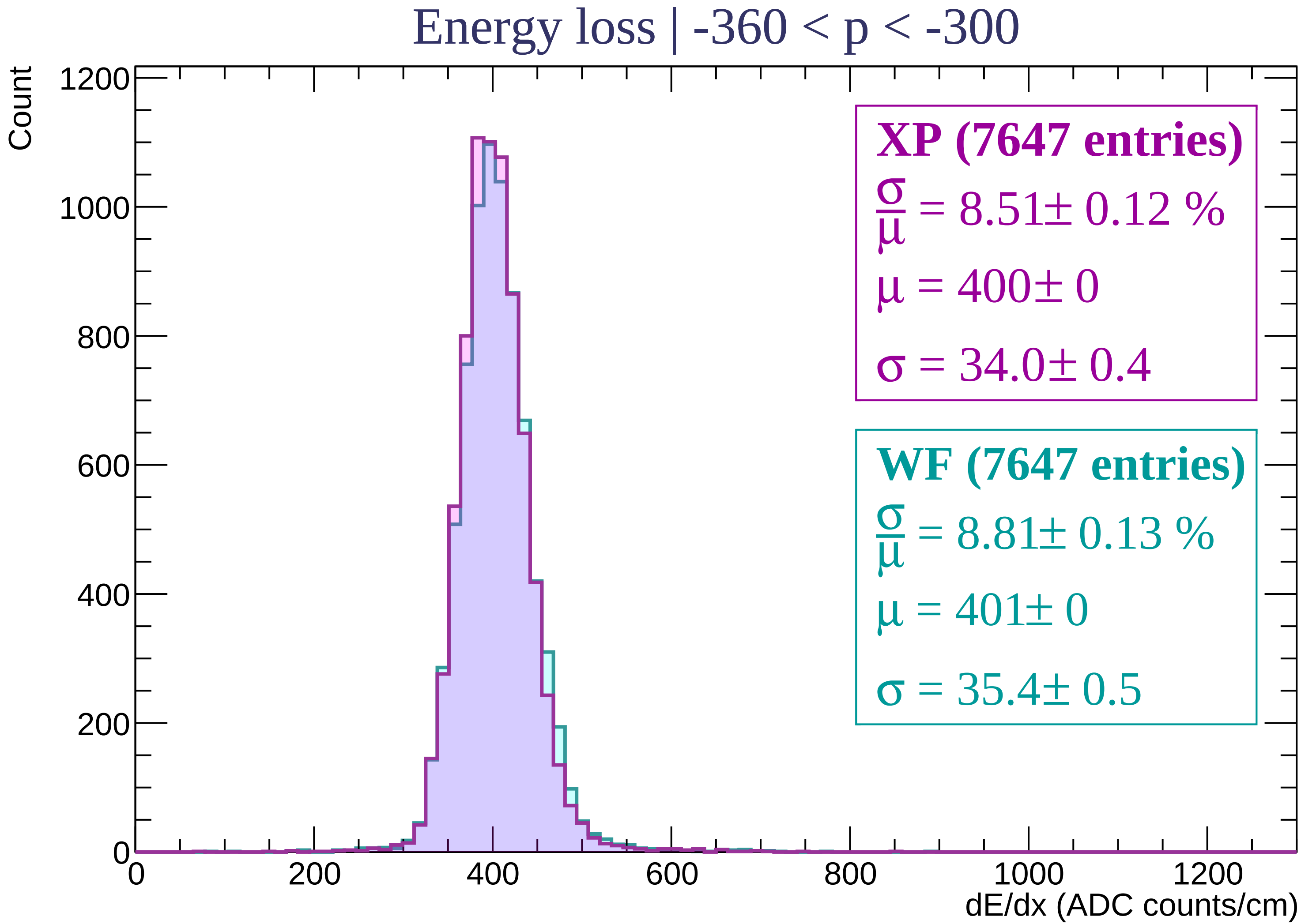
Energy loss | -600 < p < -540
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8642 entries)
σ/μ = 8.67± 0.12 %
μ = 414± 0
σ = 35.9± 0.5
WF (8642 entries)
σ/μ = 9.21± 0.13 %
μ = 417± 0
σ = 38.4± 0.5



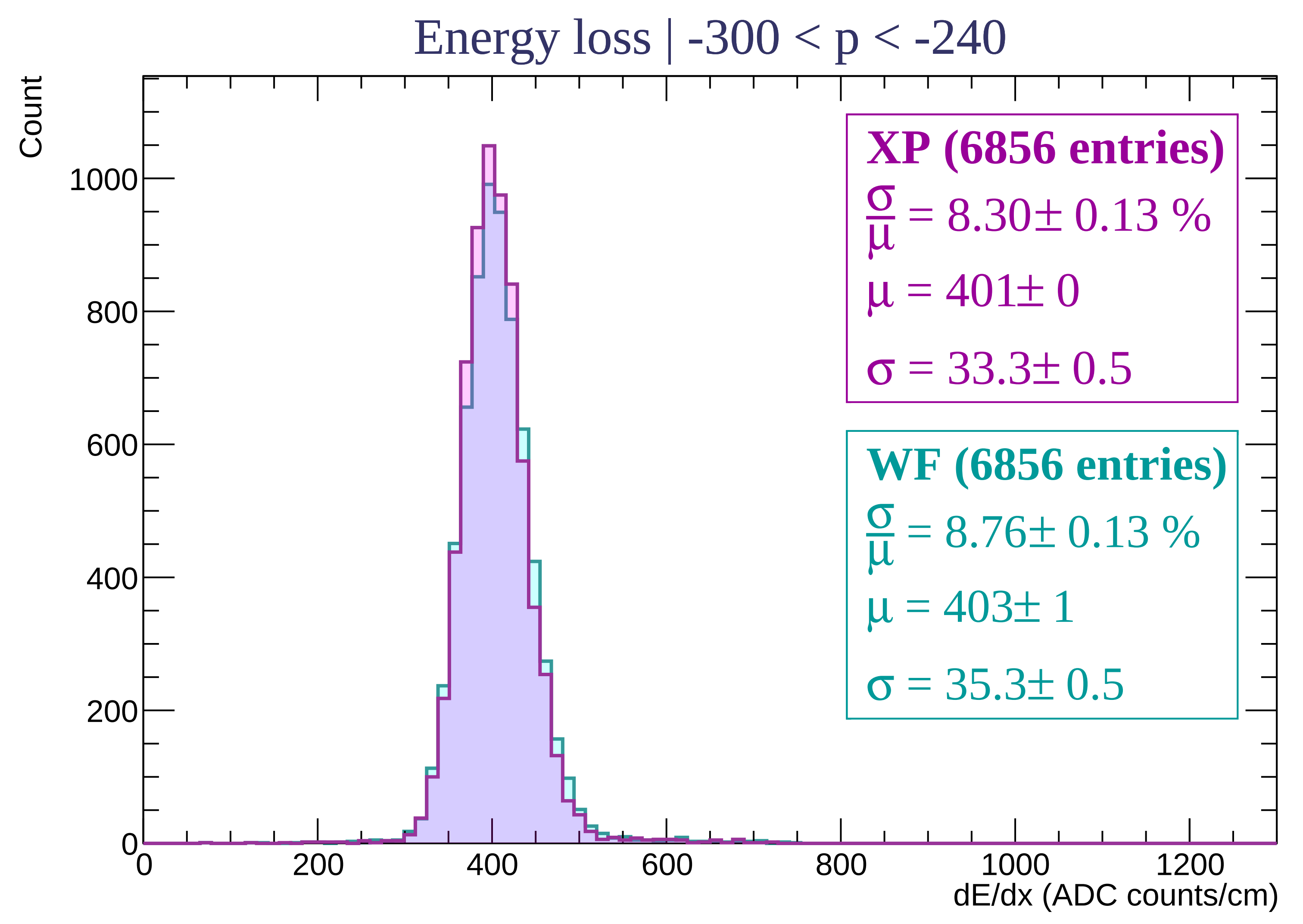
Energy loss | -540 < p < -480
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (8738 entries)
σ/μ = 8.66± 0.12 %
μ = 409± 0
σ = 35.4± 0.4
WF (8738 entries)
σ/μ = 9.04± 0.13 %
μ = 411± 0
σ = 37.2± 0.5

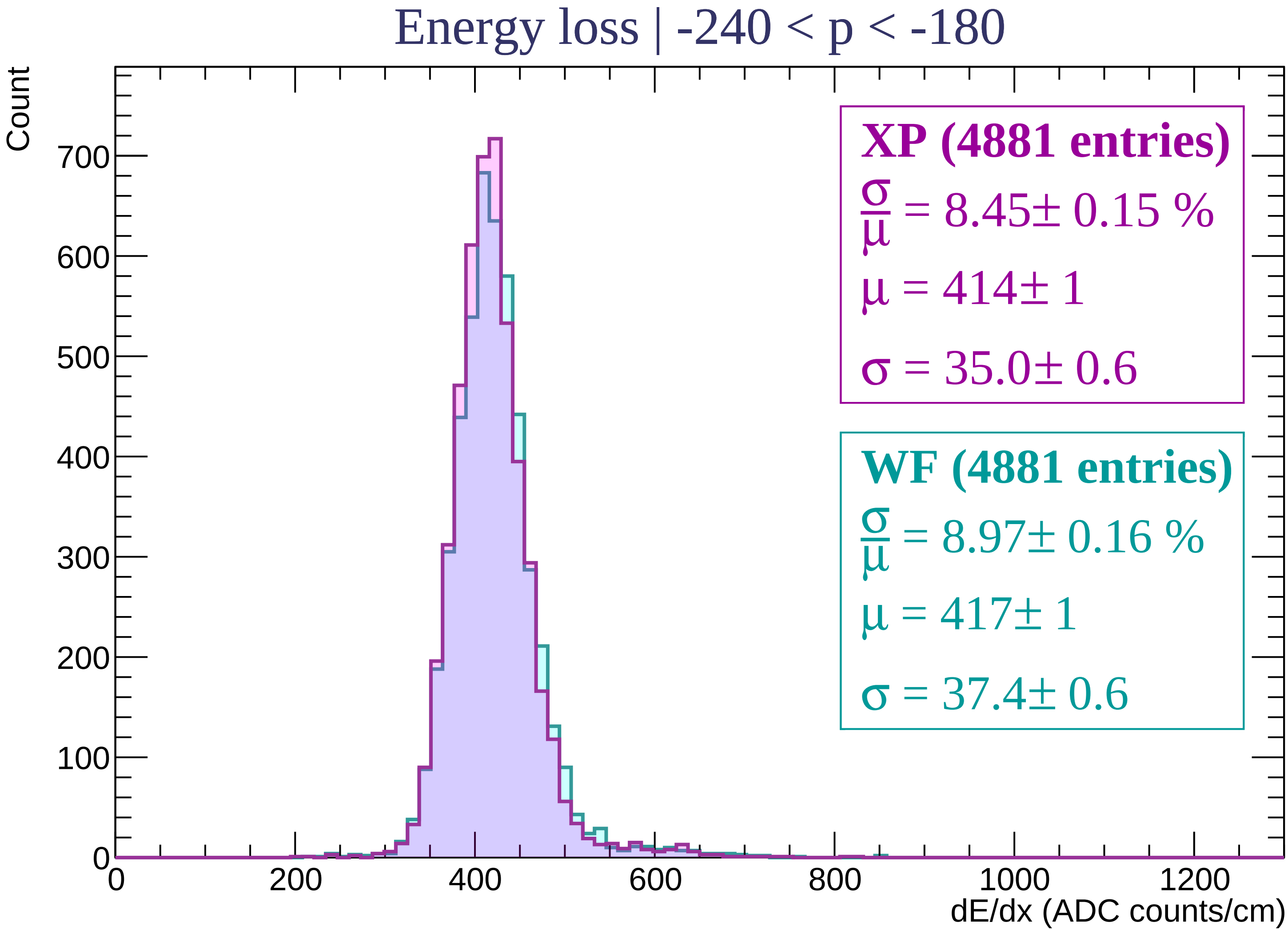

Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (8500 entries)
σ/μ = 8.31± 0.12 %
μ = 405± 0
σ = 33.6± 0.4
WF (8500 entries)
σ/μ = 8.83± 0.12 %
μ = 407± 0
σ = 36.0± 0.5

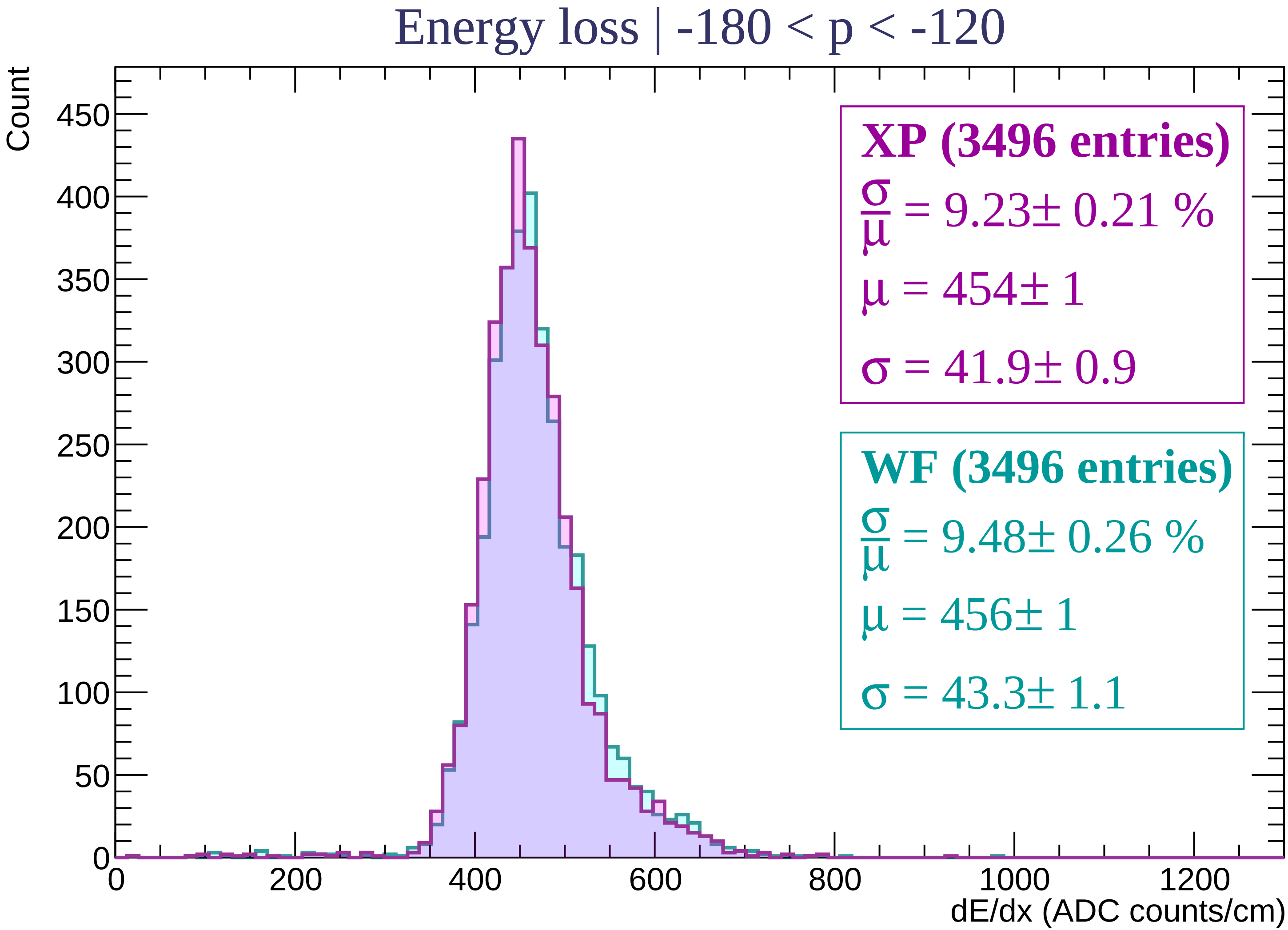

Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
XP (8409 entries)
$\frac{\sigma}{\mu}$ = 8.54± 0.13 %
μ = 401± 0
σ = 34.2± 0.5
WF (8409 entries)
$\frac{\sigma}{\mu}$ = 8.99± 0.11 %
μ = 404± 0
σ = 36.3± 0.4
0
200
400
600
800
1000
1200

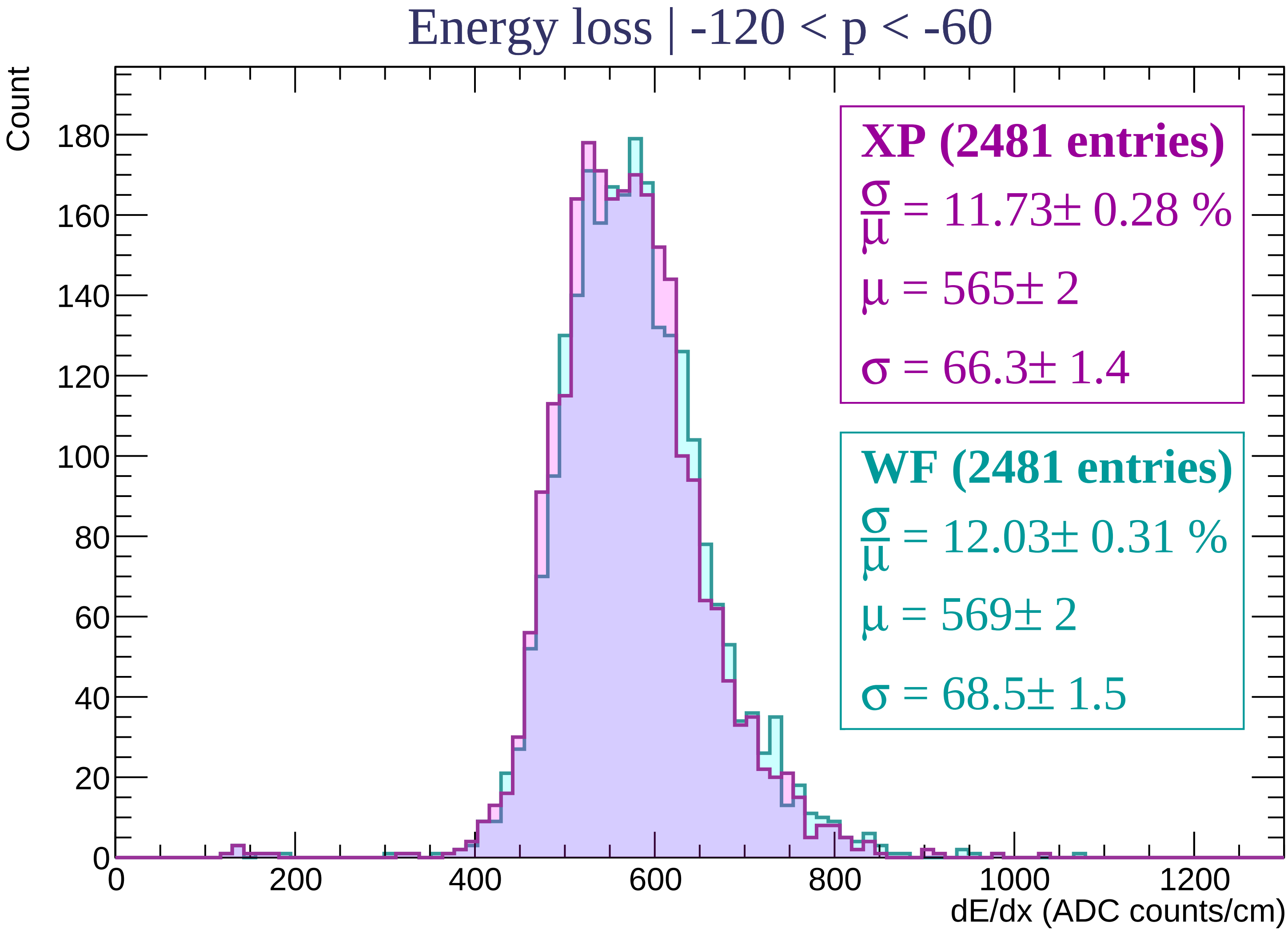


Energy loss | -360 < p < -300
Count
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7647 entries)
σ/μ = 8.51± 0.12 %
μ = 400± 0
σ = 34.0± 0.4
WF (7647 entries)
σ/μ = 8.81± 0.13 %
μ = 401± 0
σ = 35.4± 0.5

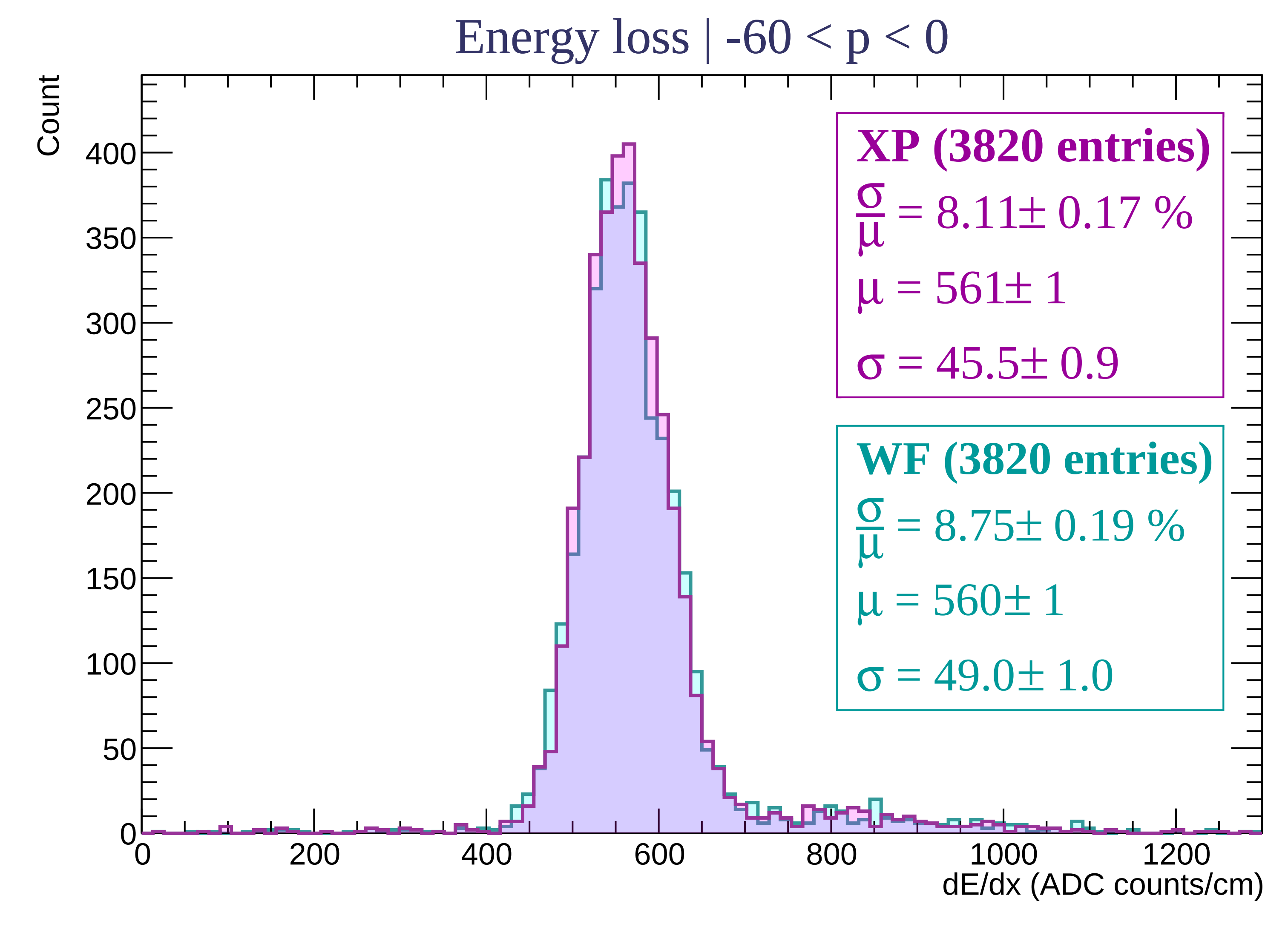

Energy loss | -300 < p < -240
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6856 entries)
σ/μ = 8.30± 0.13 %
μ = 401± 0
σ = 33.3± 0.5
WF (6856 entries)
σ/μ = 8.76± 0.13 %
μ = 403± 1
σ = 35.3± 0.5


Energy loss | -240 < p < -180
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4881 entries)
σ/μ = 8.45± 0.15 %
μ = 414± 1
σ = 35.0± 0.6
WF (4881 entries)
σ/μ = 8.97± 0.16 %
μ = 417± 1
σ = 37.4± 0.6

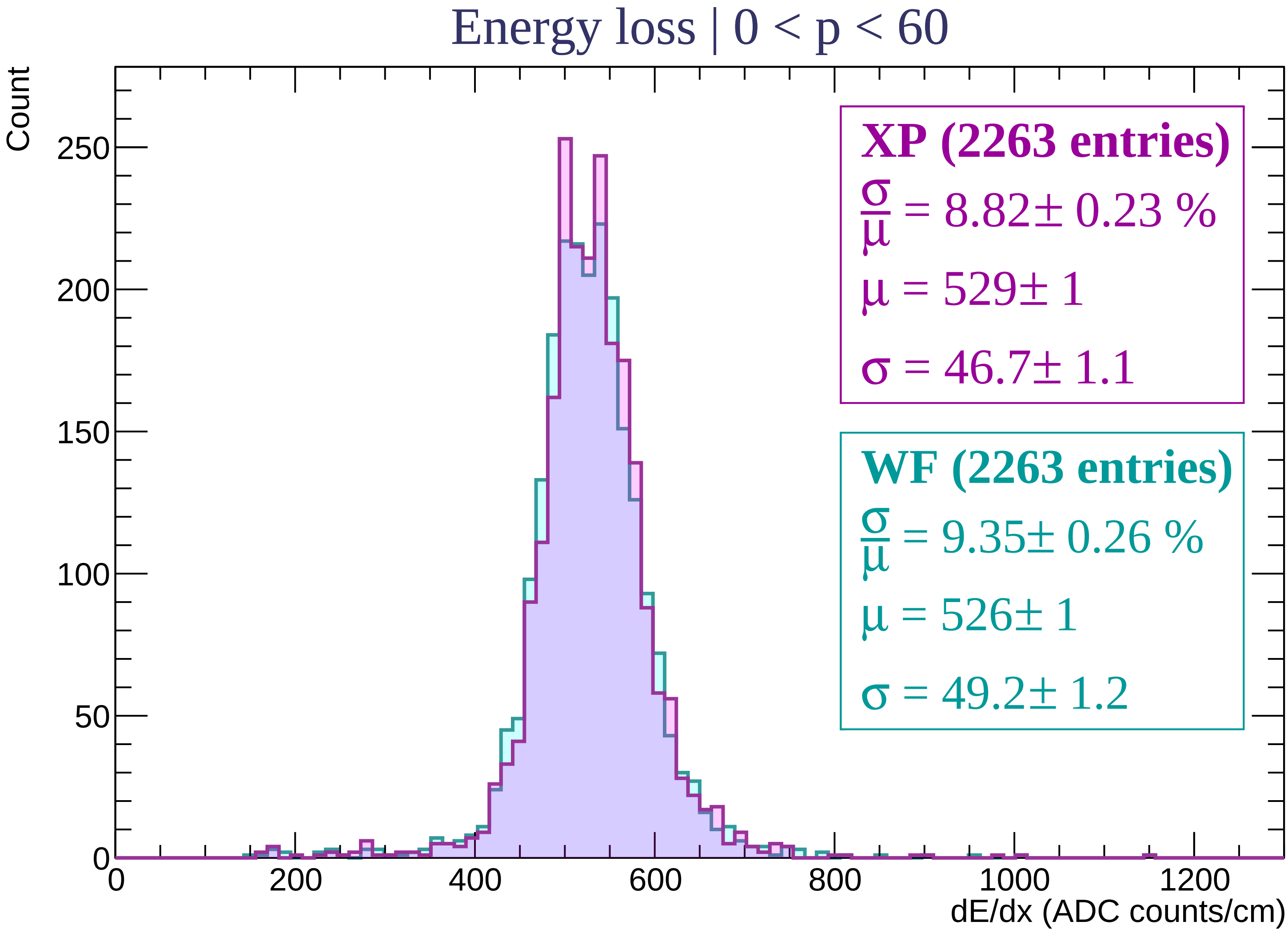

Energy loss | -180 < p < -120
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3496 entries)
σ/μ = 9.23± 0.21 %
μ = 454± 1
σ = 41.9± 0.9
WF (3496 entries)
σ/μ = 9.48± 0.26 %
μ = 456± 1
σ = 43.3± 1.1

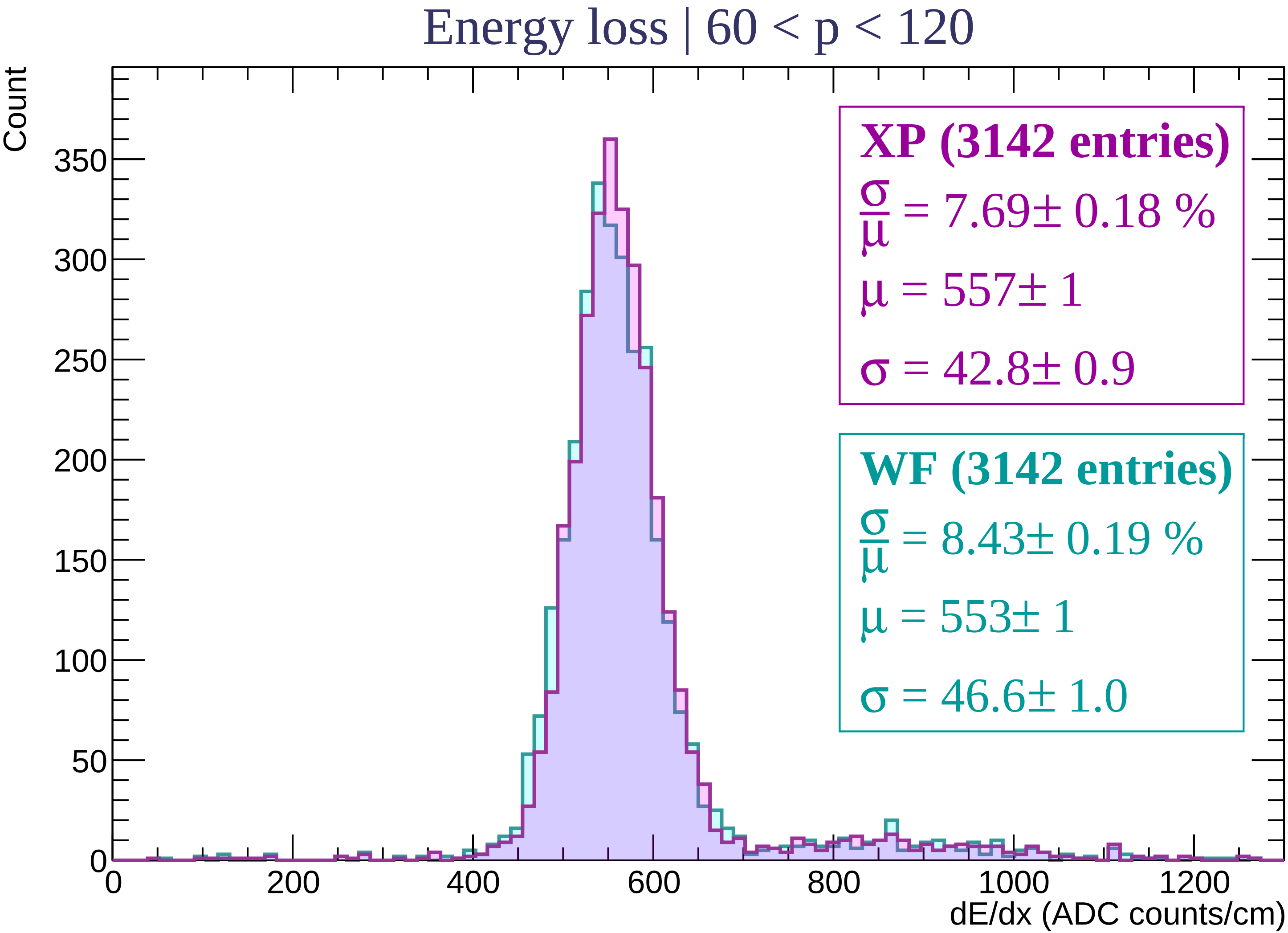


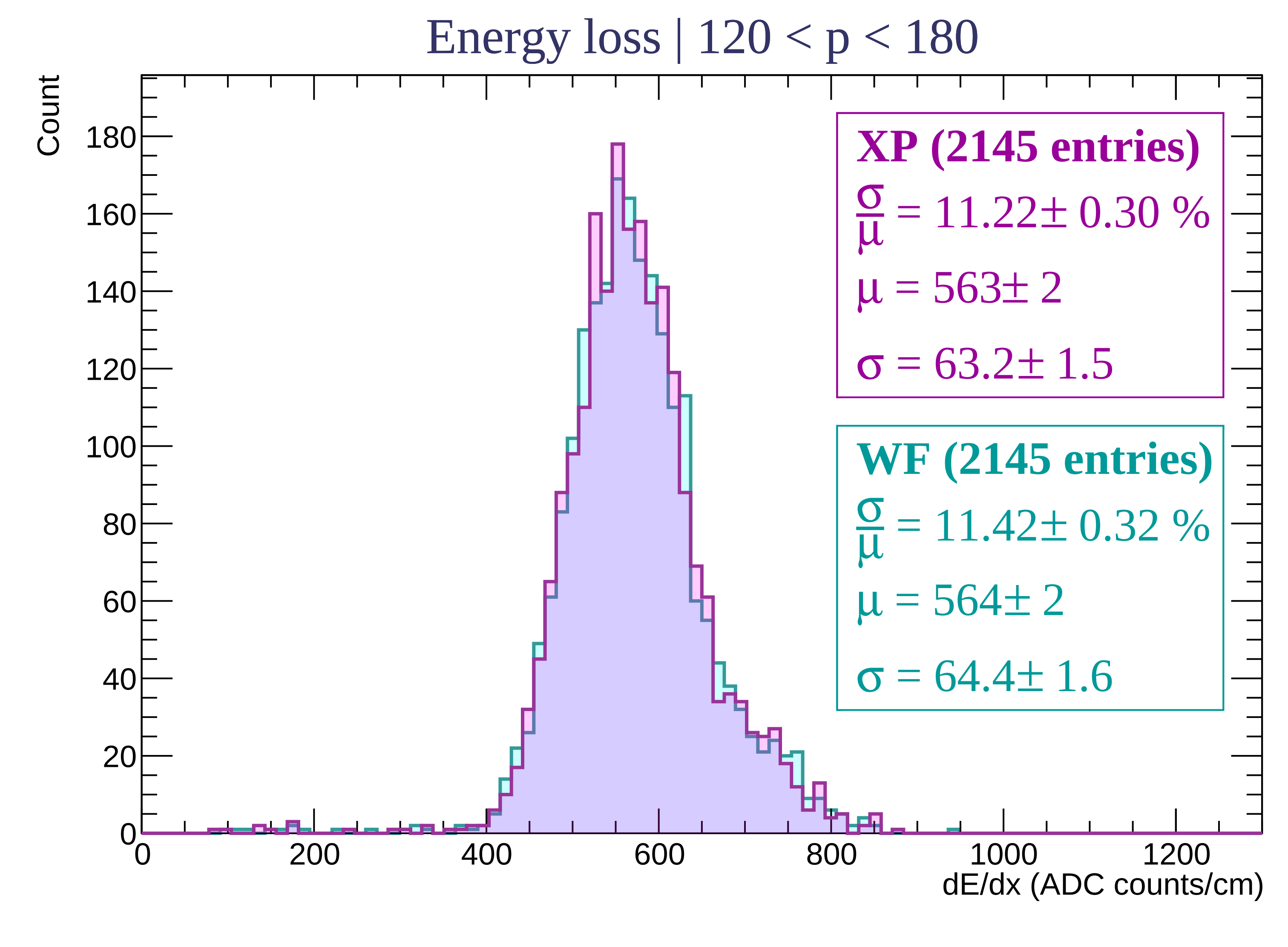
Energy loss | -120 < p < -60
Count
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2481 entries)
σ/μ = 11.73± 0.28 %
μ = 565± 2
σ = 66.3± 1.4
WF (2481 entries)
σ/μ = 12.03± 0.31 %
μ = 569± 2
σ = 68.5± 1.5



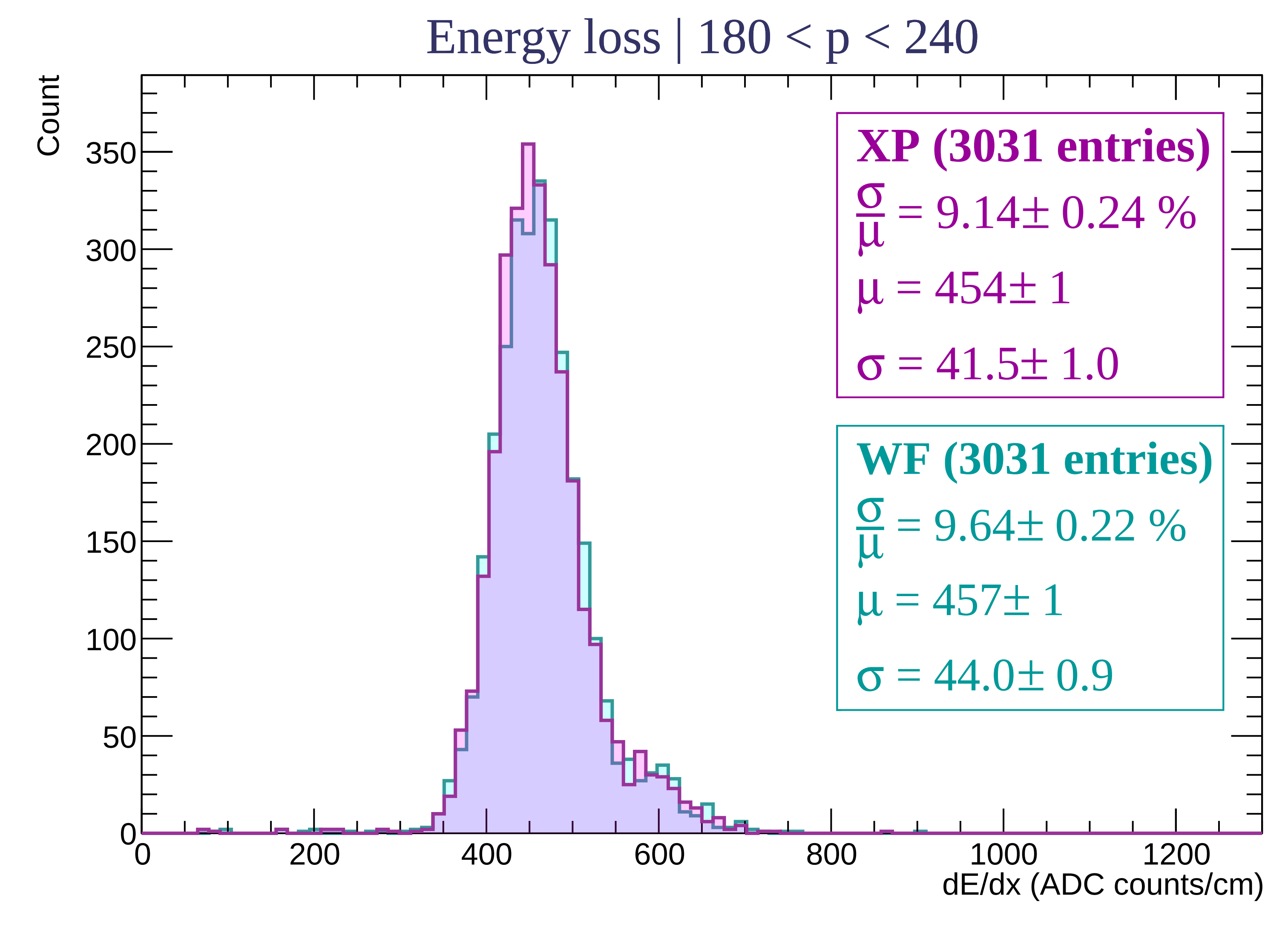
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3820 entries)
σ/μ = 8.11± 0.17 %
μ = 561± 1
σ = 45.5± 0.9
WF (3820 entries)
σ/μ = 8.75± 0.19 %
μ = 560± 1
σ = 49.0± 1.0


Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2263 entries)
$\frac{\sigma}{\mu}$ = 8.82± 0.23 %
μ = 529± 1
σ = 46.7± 1.1
WF (2263 entries)
$\frac{\sigma}{\mu}$ = 9.35± 0.26 %
μ = 526± 1
σ = 49.2± 1.2

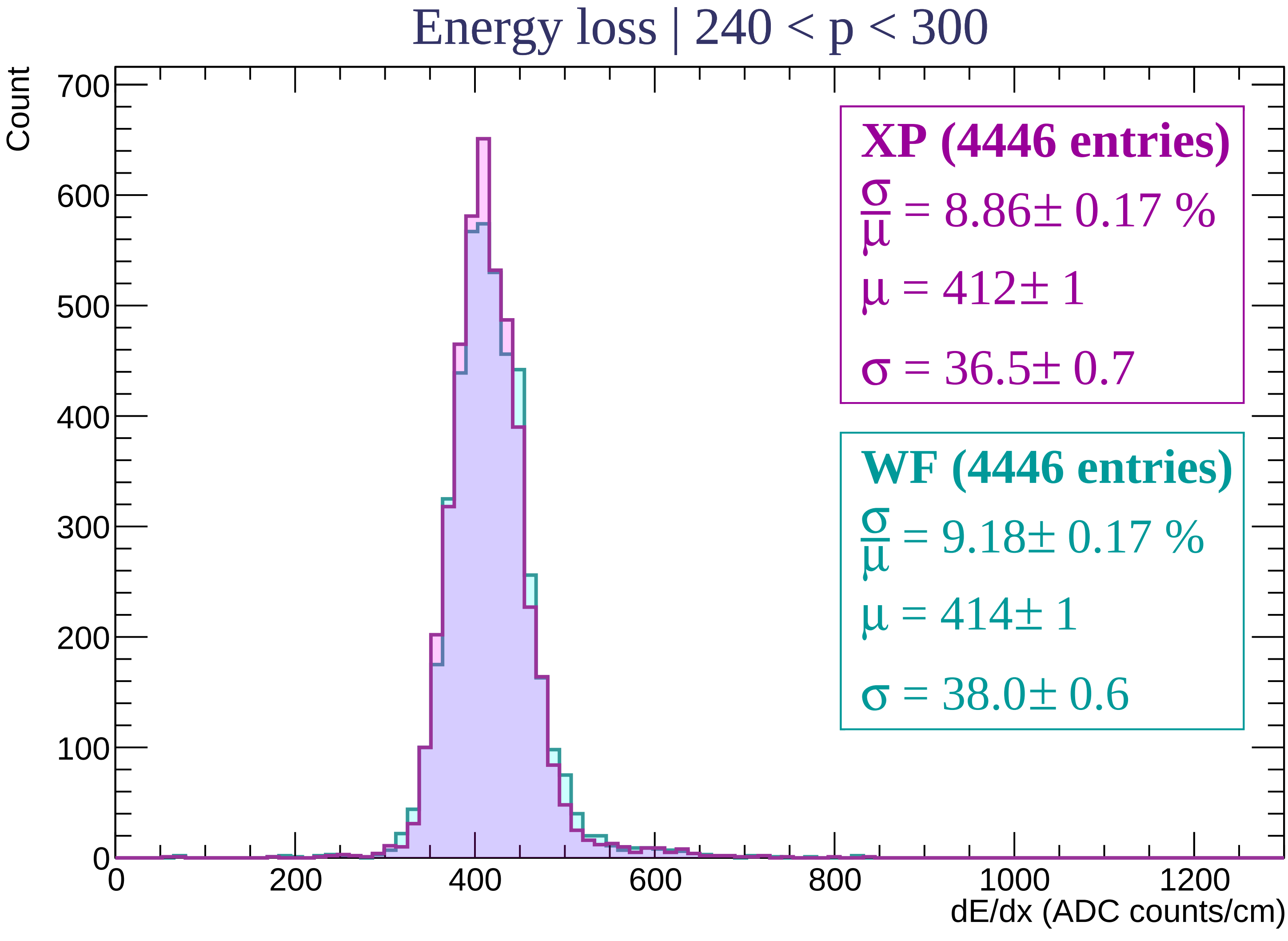

Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3142 entries)
$\frac{\sigma}{\mu} = 7.69 \pm 0.18$ %
$\mu = 557 \pm 1$
$\sigma = 42.8 \pm 0.9$
WF (3142 entries)
$\frac{\sigma}{\mu} = 8.43 \pm 0.19$ %
$\mu = 553 \pm 1$
$\sigma = 46.6 \pm 1.0$

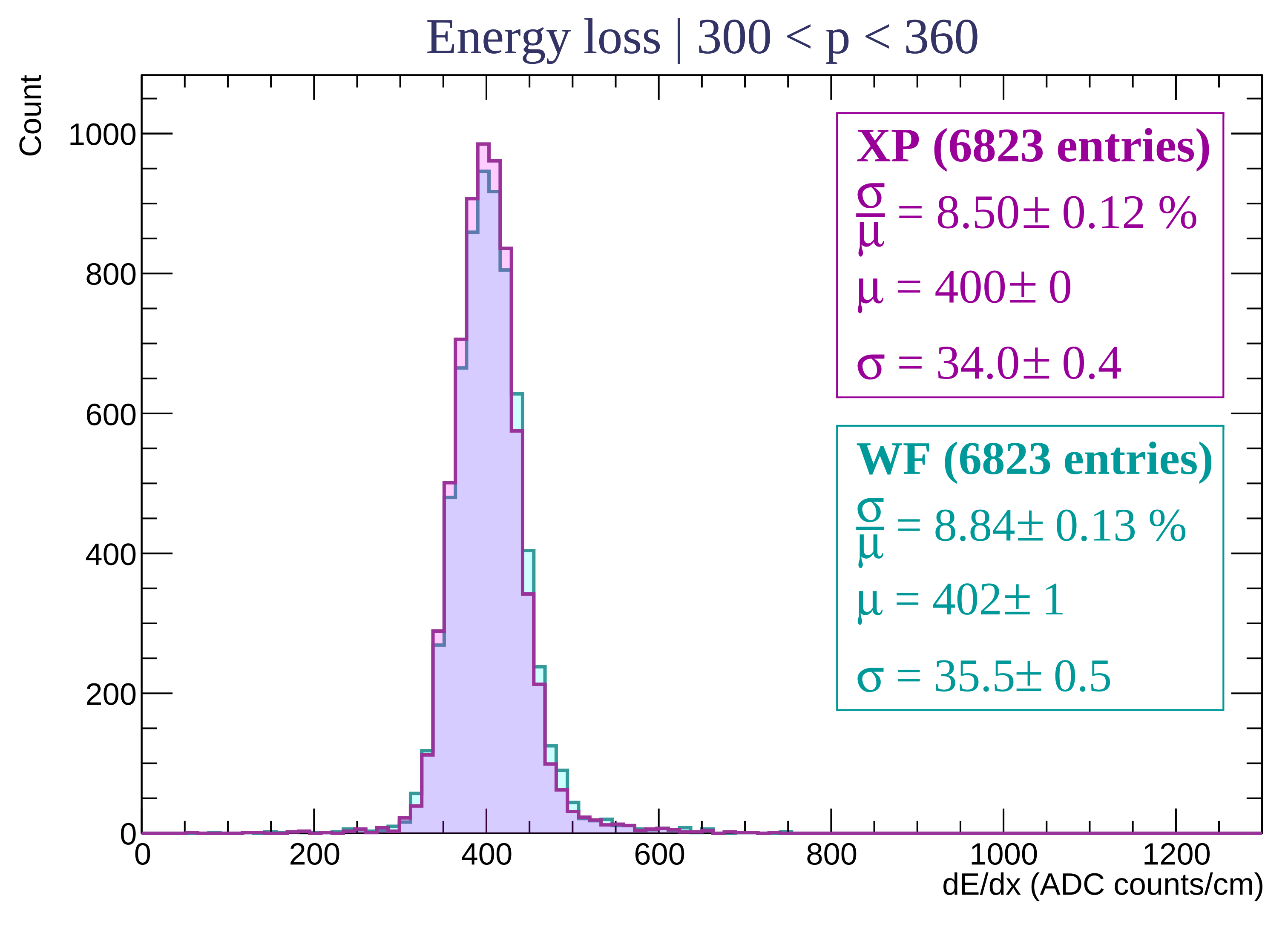


Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (2145 entries)
$\frac{\sigma}{\mu}$ = 11.22± 0.30 %
μ = 563± 2
σ = 63.2± 1.5
WF (2145 entries)
$\frac{\sigma}{\mu}$ = 11.42± 0.32 %
μ = 564± 2
σ = 64.4± 1.6

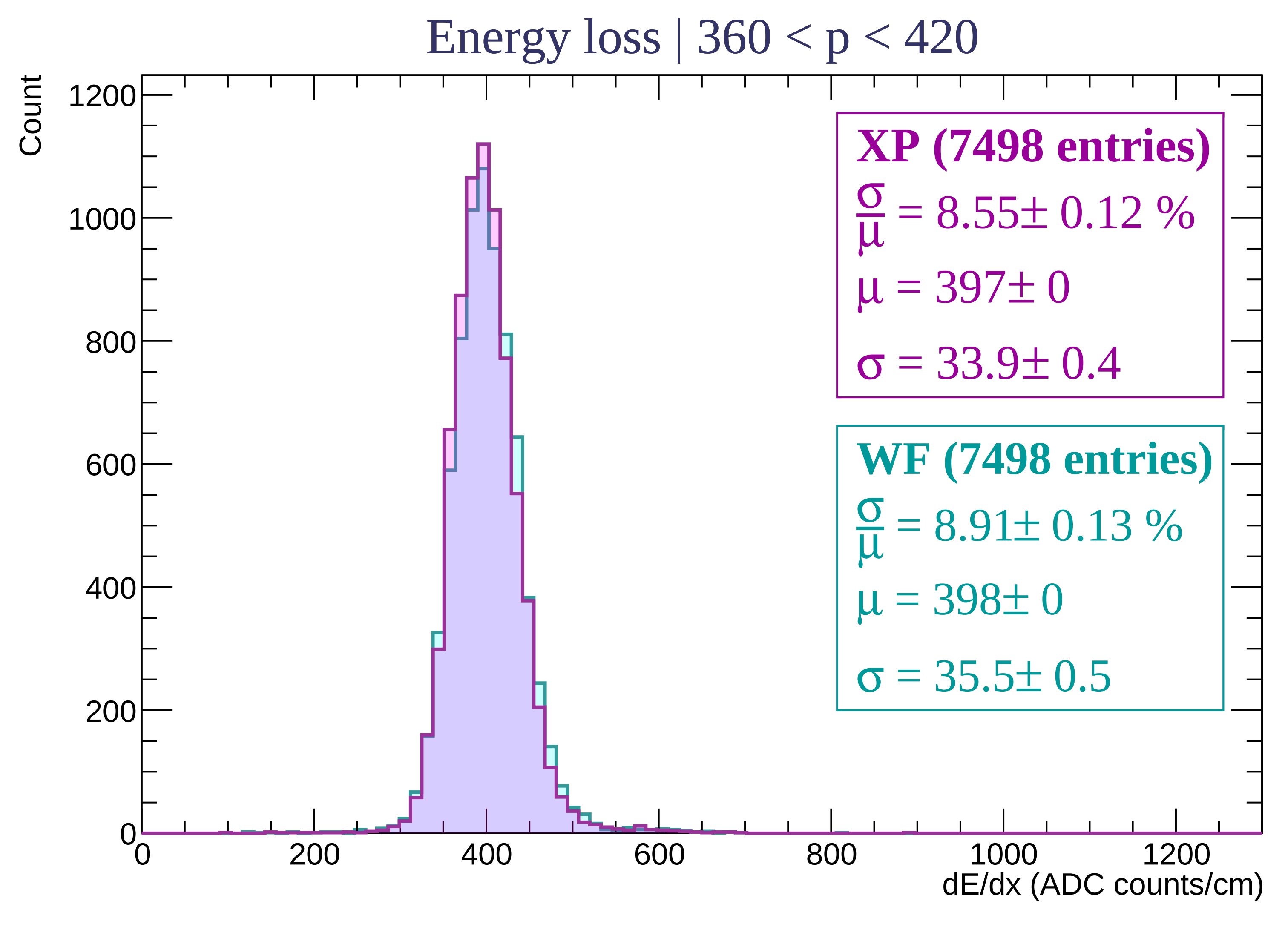


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3031 entries)
$\frac{\sigma}{\mu}$ = 9.14± 0.24 %
μ = 454± 1
σ = 41.5± 1.0
WF (3031 entries)
$\frac{\sigma}{\mu}$ = 9.64± 0.22 %
μ = 457± 1
σ = 44.0± 0.9

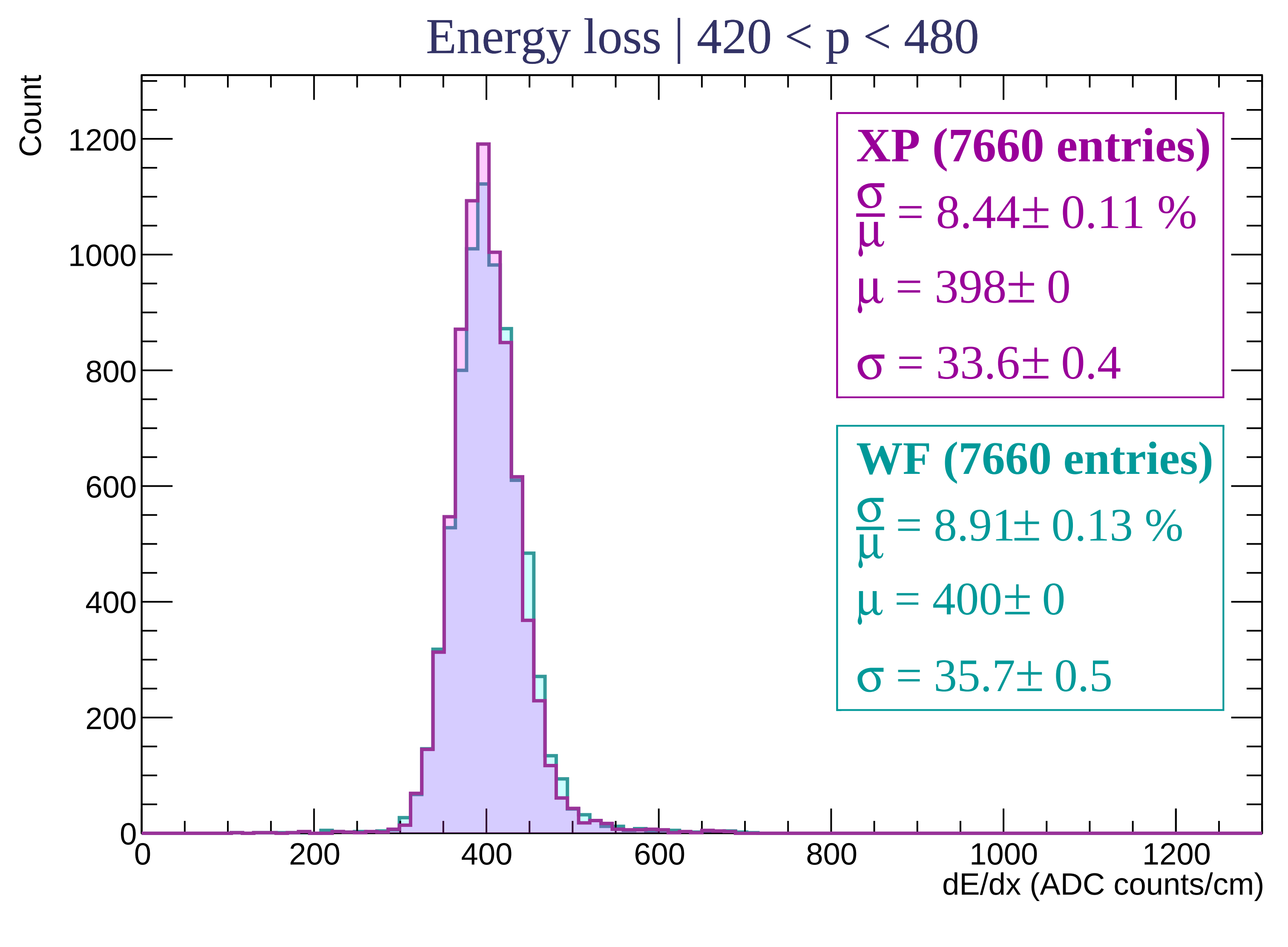

Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4446 entries)
$\frac{\sigma}{\mu}$ = 8.86± 0.17 %
μ = 412± 1
σ = 36.5± 0.7
WF (4446 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.17 %
μ = 414± 1
σ = 38.0± 0.6



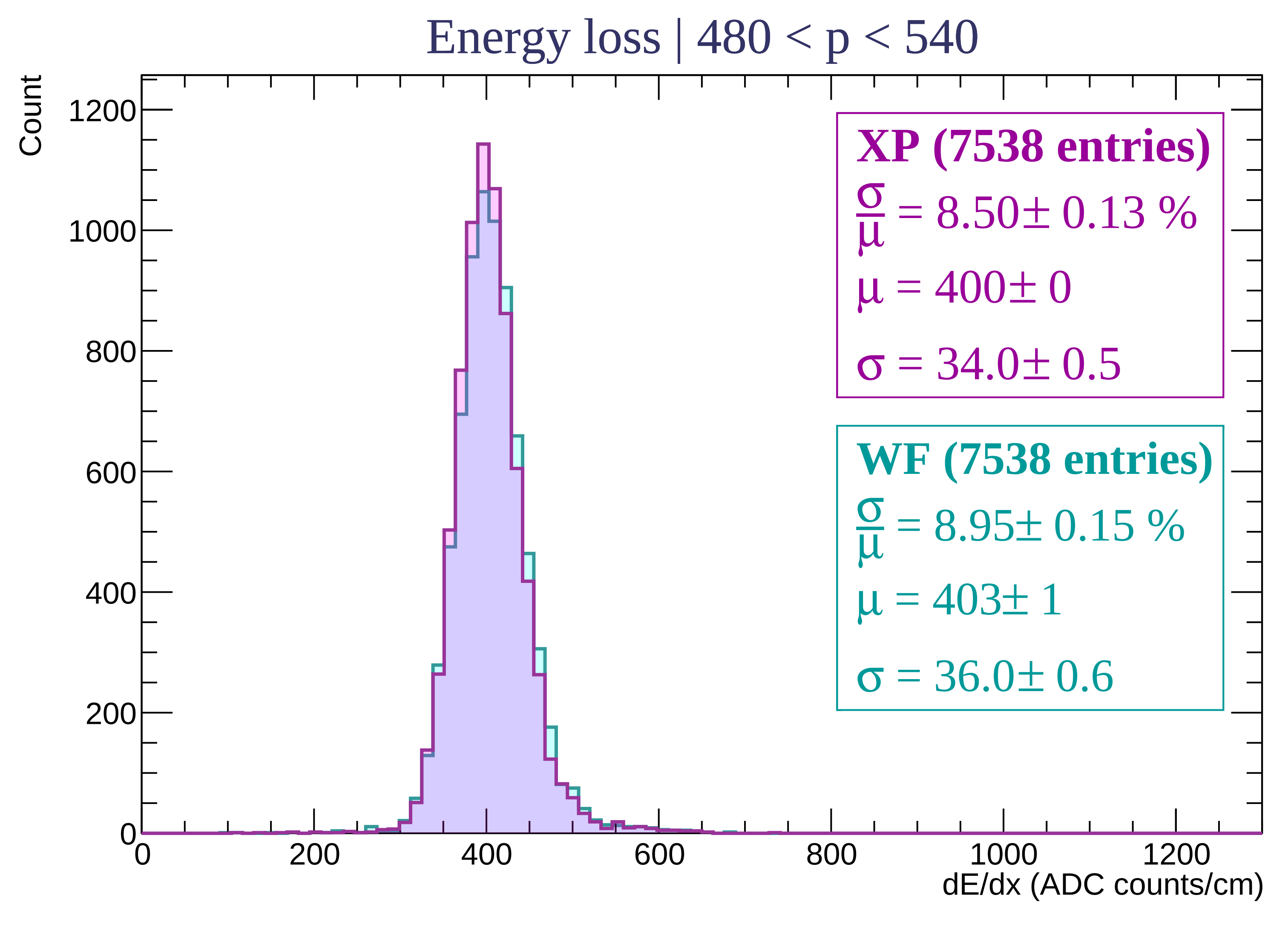
Energy loss | 300 < p < 360
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6823 entries)
$\frac{\sigma}{\mu}$ = 8.50± 0.12 %
μ = 400± 0
σ = 34.0± 0.4
WF (6823 entries)
$\frac{\sigma}{\mu}$ = 8.84± 0.13 %
μ = 402± 1
σ = 35.5± 0.5

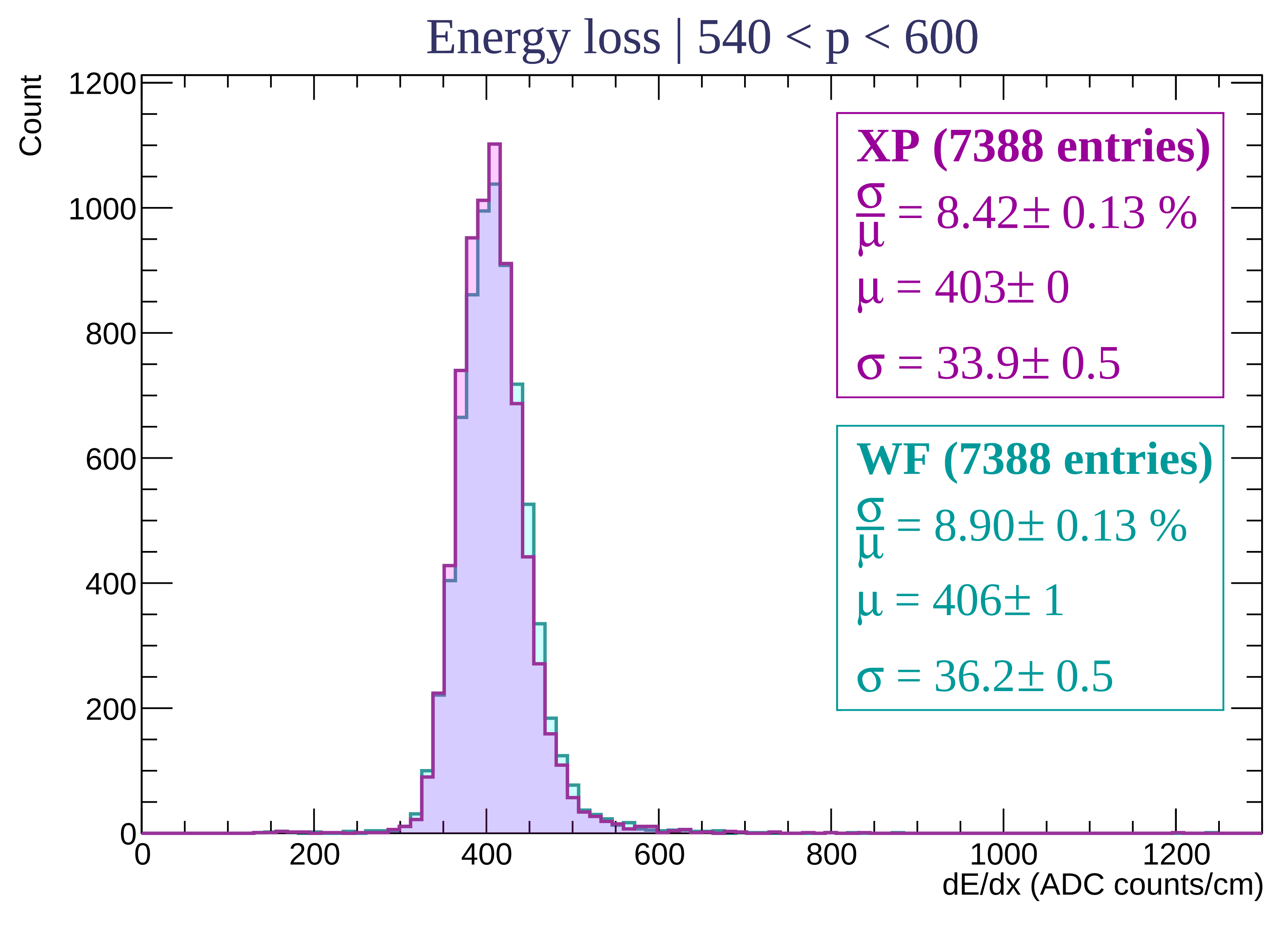


Energy loss | 360 < p < 420
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7498 entries)
σ/μ = 8.55± 0.12 %
μ = 397± 0
σ = 33.9± 0.4
WF (7498 entries)
σ/μ = 8.91± 0.13 %
μ = 398± 0
σ = 35.5± 0.5

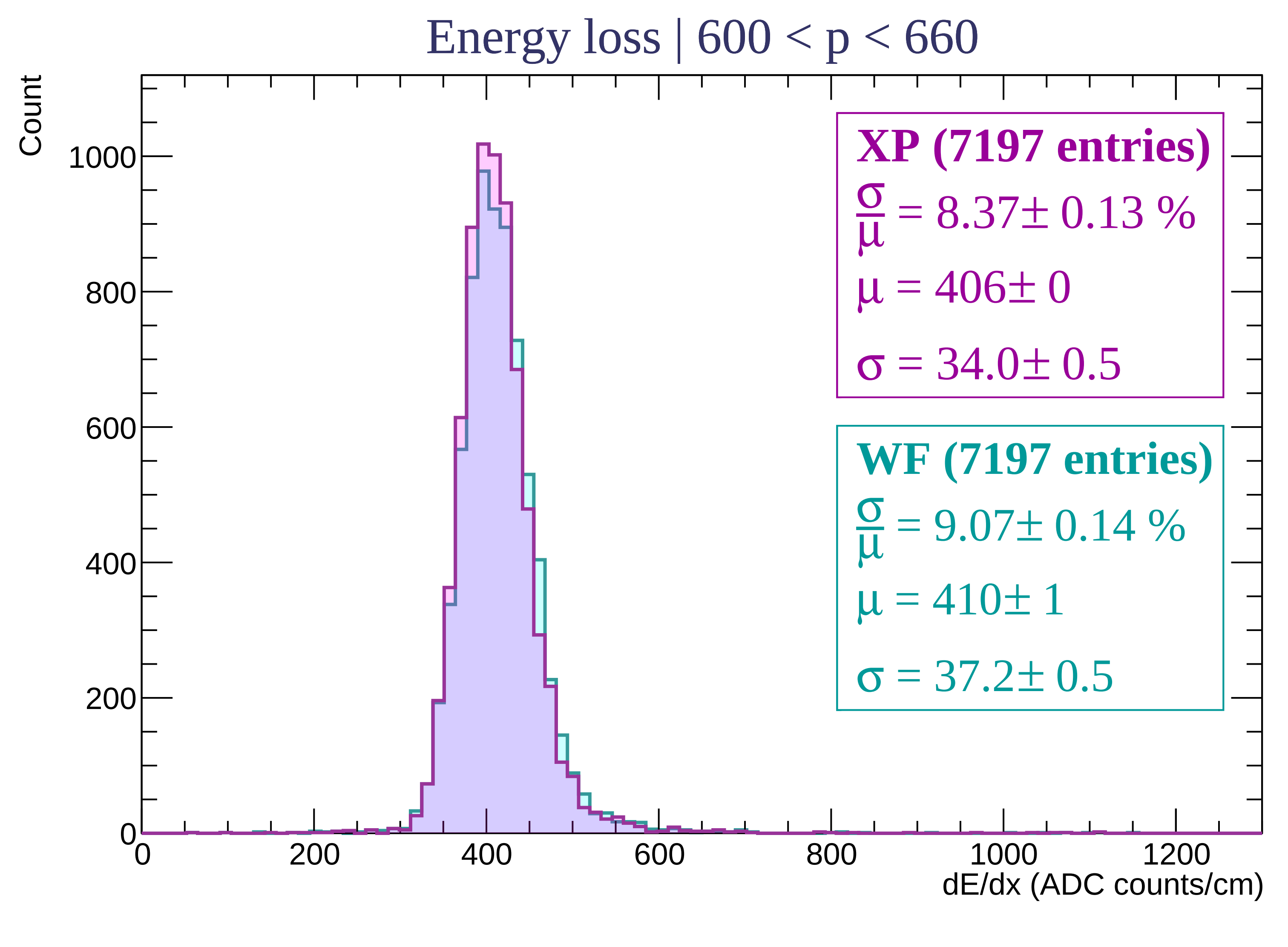


Energy loss | 420 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7660 entries)
σ/μ = 8.44± 0.11 %
μ = 398± 0
σ = 33.6± 0.4
WF (7660 entries)
σ/μ = 8.91± 0.13 %
μ = 400± 0
σ = 35.7± 0.5

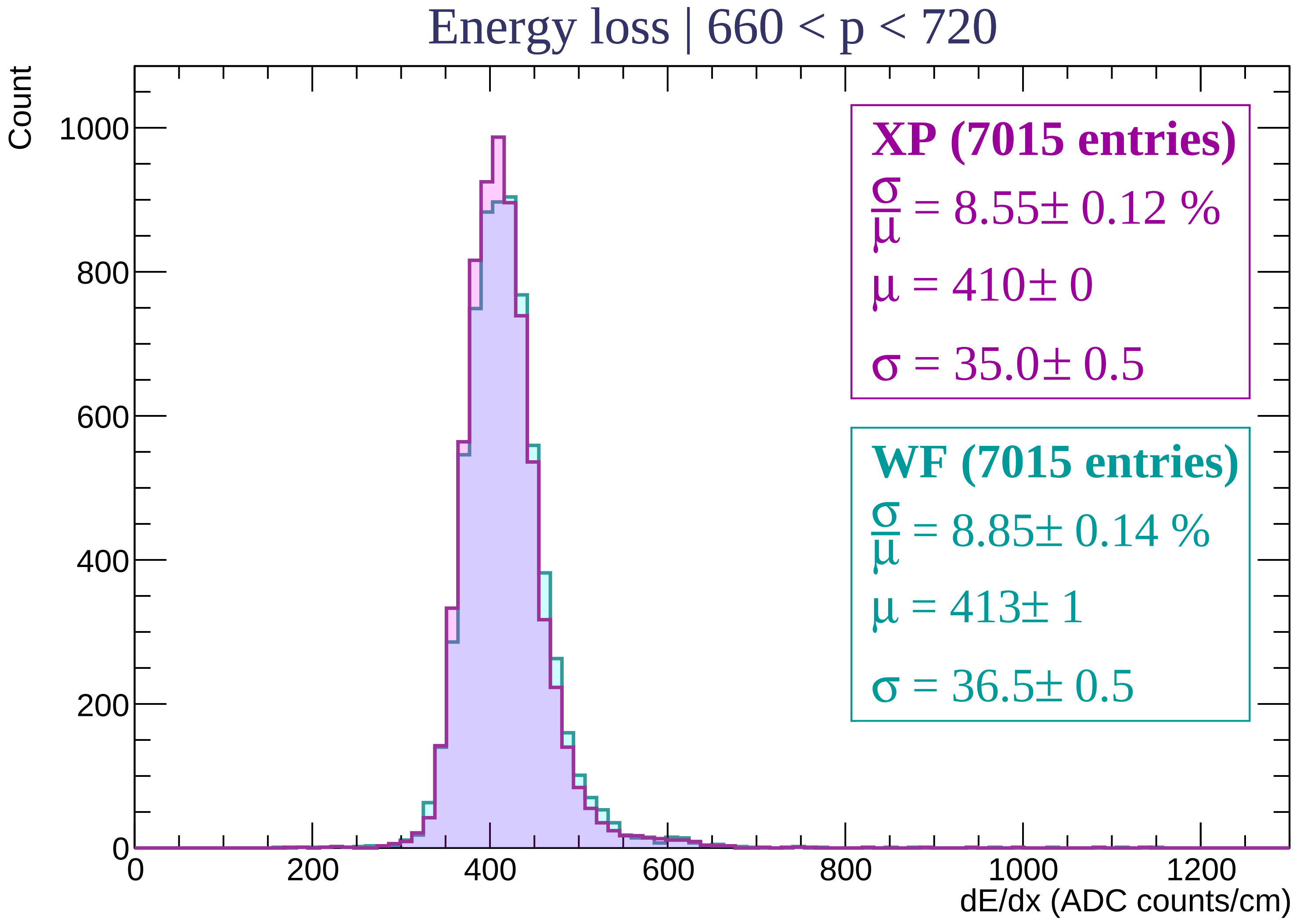

Energy loss | 480 < p < 540
Count
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7538 entries)
$\frac{\sigma}{\mu}$ = 8.50± 0.13 %
μ = 400± 0
σ = 34.0± 0.5
WF (7538 entries)
$\frac{\sigma}{\mu}$ = 8.95± 0.15 %
μ = 403± 1
σ = 36.0± 0.6

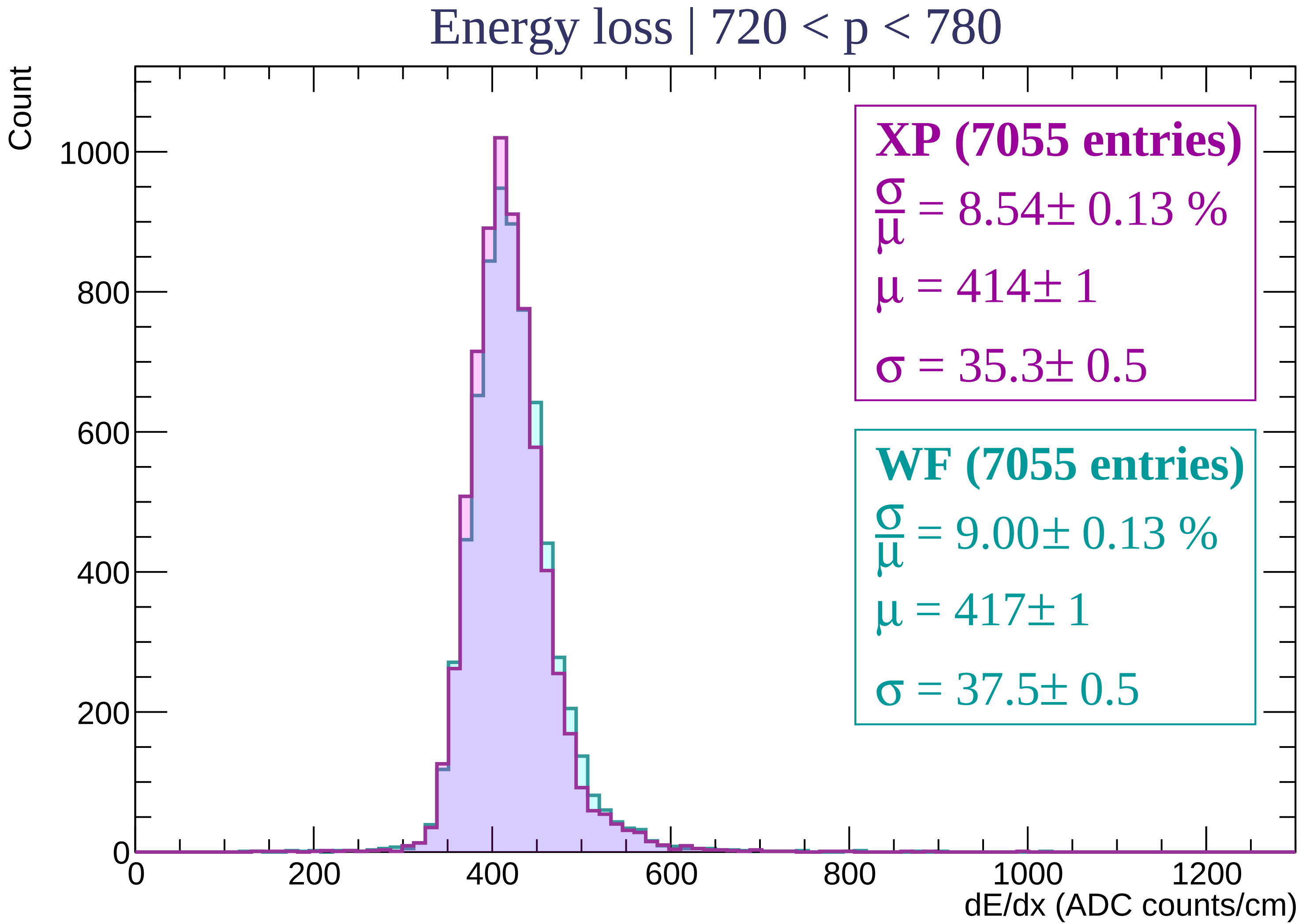


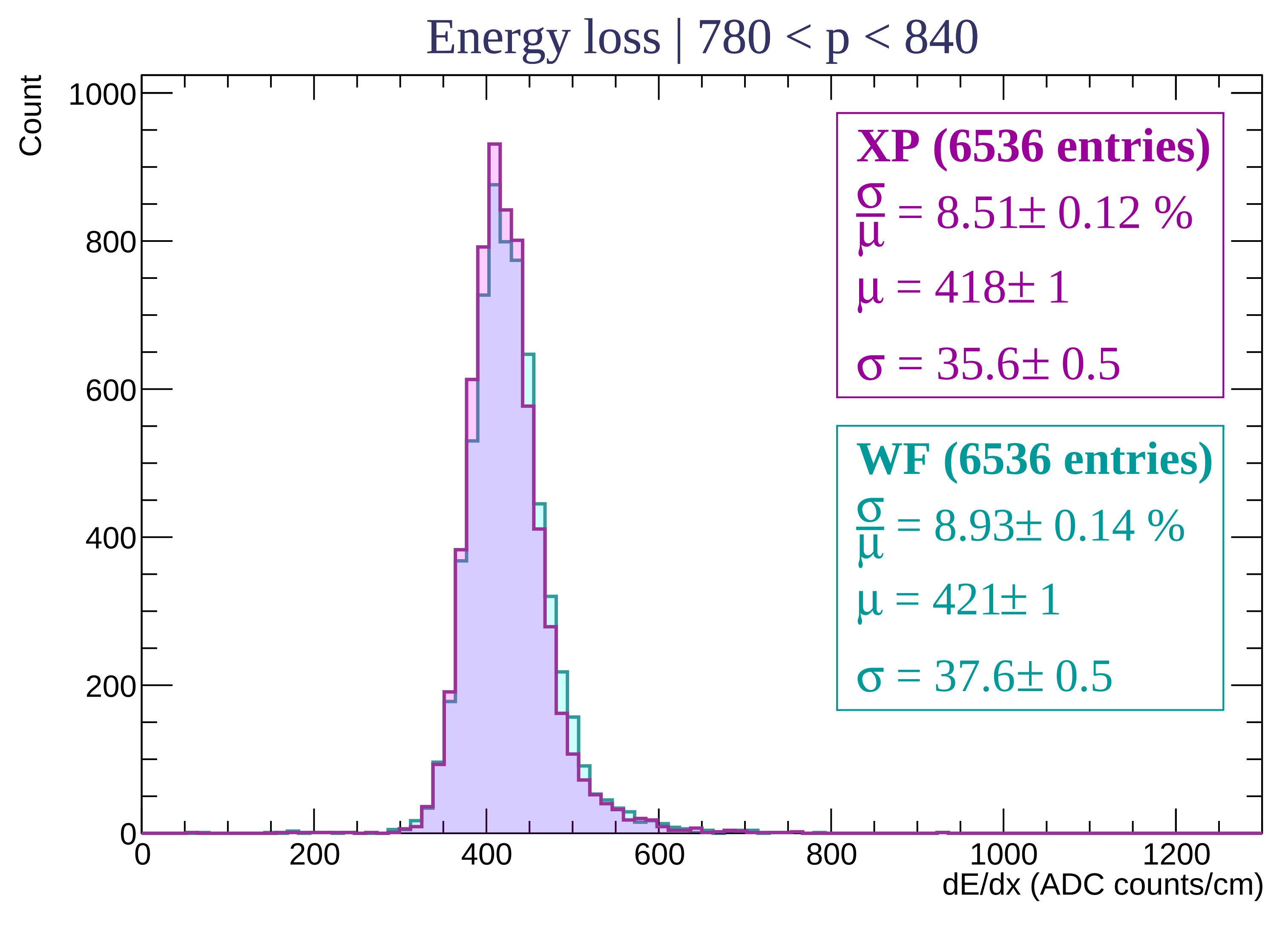
Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (7388 entries)
σ/μ = 8.42± 0.13 %
μ = 403± 0
σ = 33.9± 0.5
WF (7388 entries)
σ/μ = 8.90± 0.13 %
μ = 406± 1
σ = 36.2± 0.5



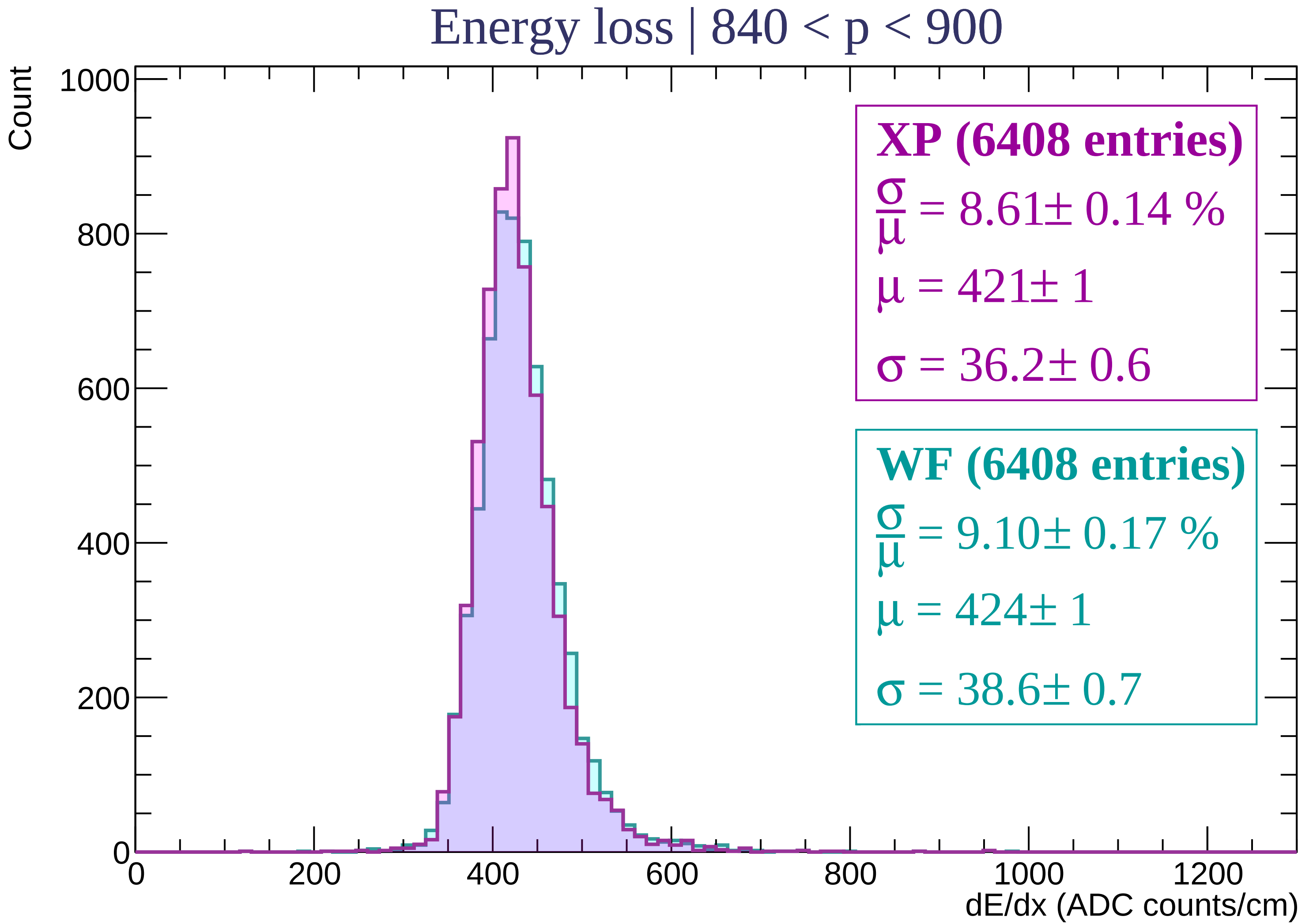
Energy loss | 600 < p < 660
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7197 entries)
$\frac{\sigma}{\mu}$ = 8.37± 0.13 %
μ = 406± 0
σ = 34.0± 0.5
WF (7197 entries)
$\frac{\sigma}{\mu}$ = 9.07± 0.14 %
μ = 410± 1
σ = 37.2± 0.5


Energy loss | 660 < p < 720
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7015 entries)
σ/μ = 8.55± 0.12 %
μ = 410± 0
σ = 35.0± 0.5
WF (7015 entries)
σ/μ = 8.85± 0.14 %
μ = 413± 1
σ = 36.5± 0.5

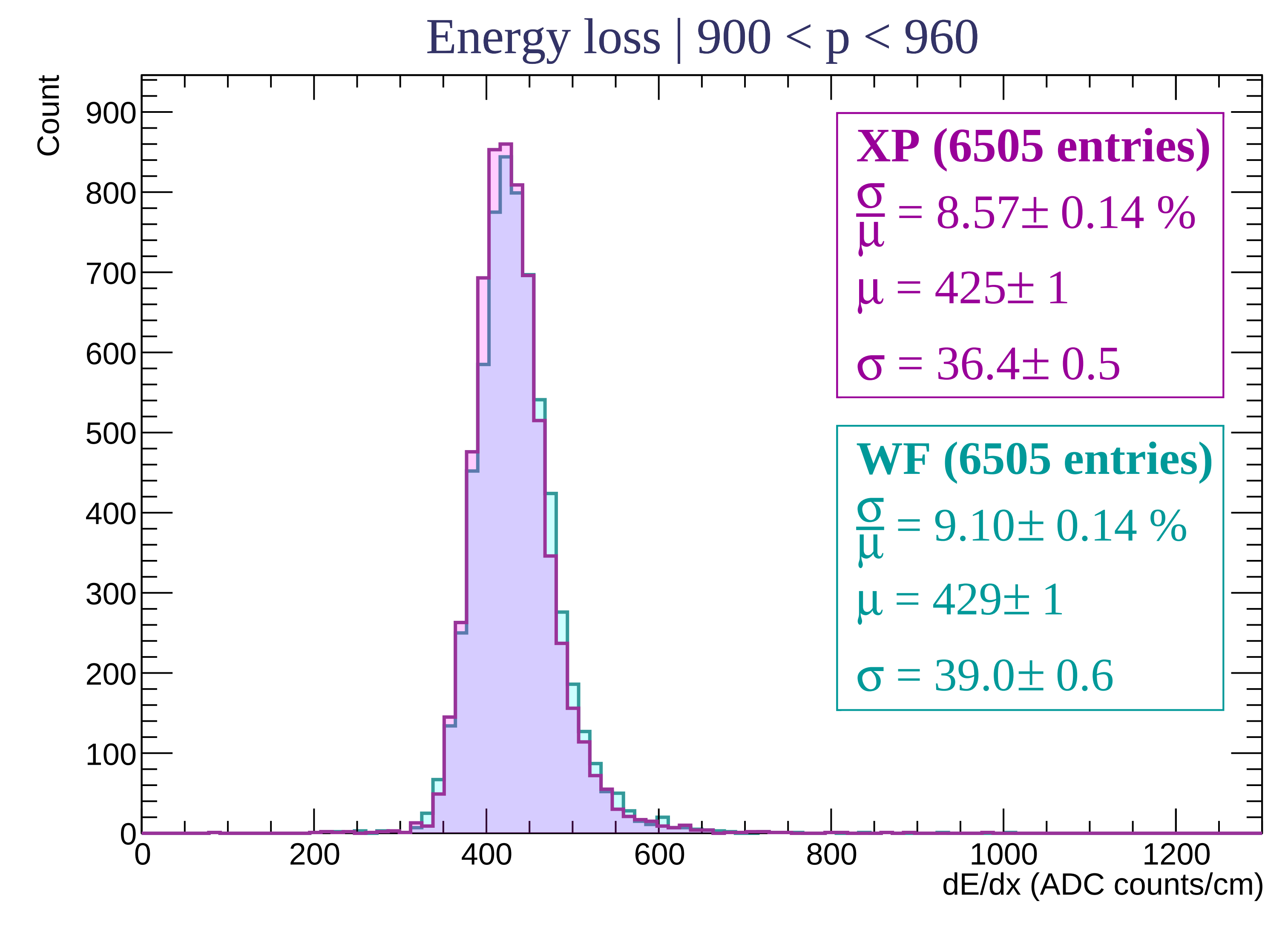

Energy loss | 720 < p < 780
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (7055 entries)
$\frac{\sigma}{\mu}$ = 8.54± 0.13 %
μ = 414± 1
σ = 35.3± 0.5
WF (7055 entries)
$\frac{\sigma}{\mu}$ = 9.00± 0.13 %
μ = 417± 1
σ = 37.5± 0.5

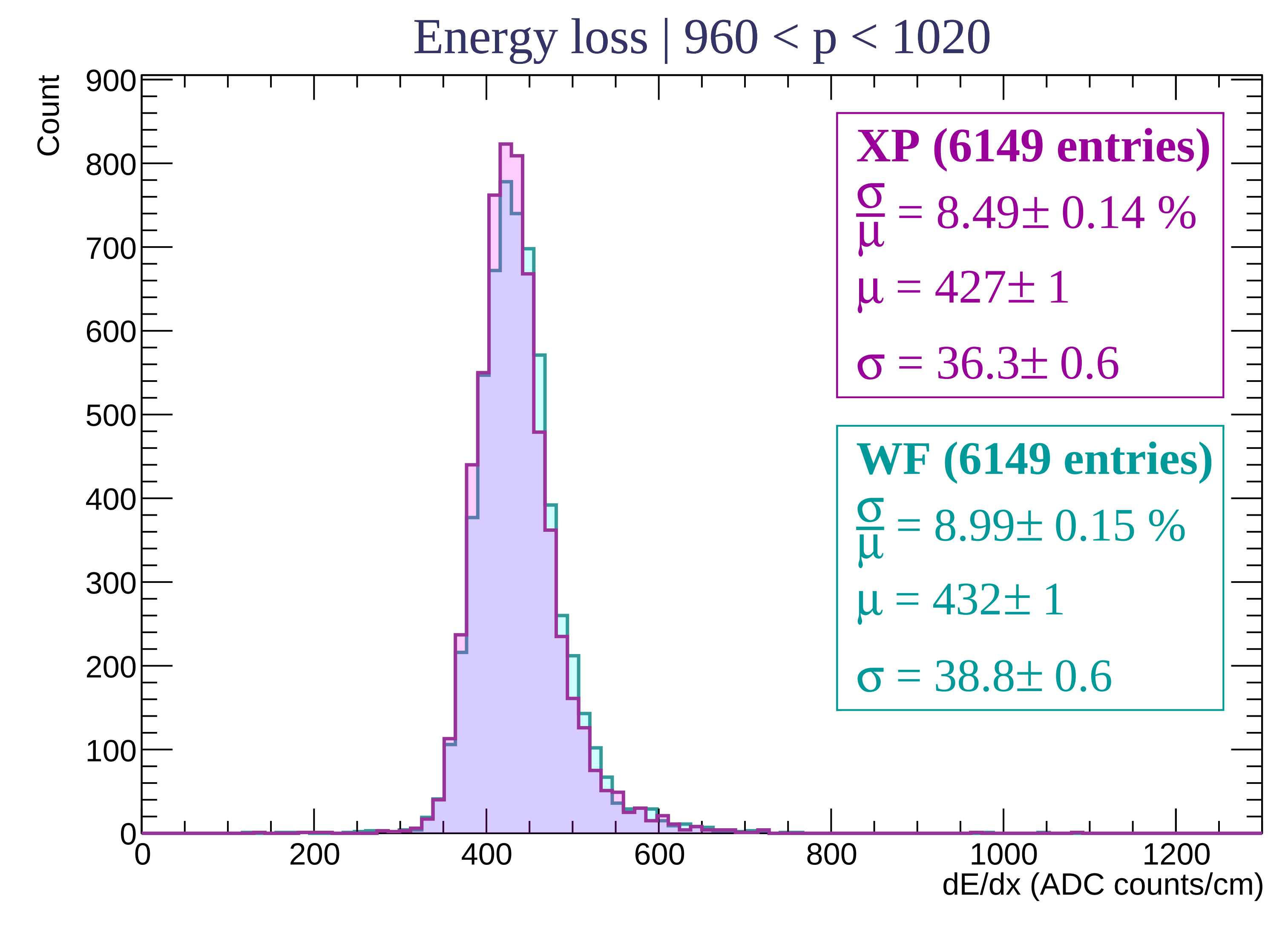


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6536 entries)
σ/μ = 8.51± 0.12 %
μ = 418± 1
σ = 35.6± 0.5
WF (6536 entries)
σ/μ = 8.93± 0.14 %
μ = 421± 1
σ = 37.6± 0.5

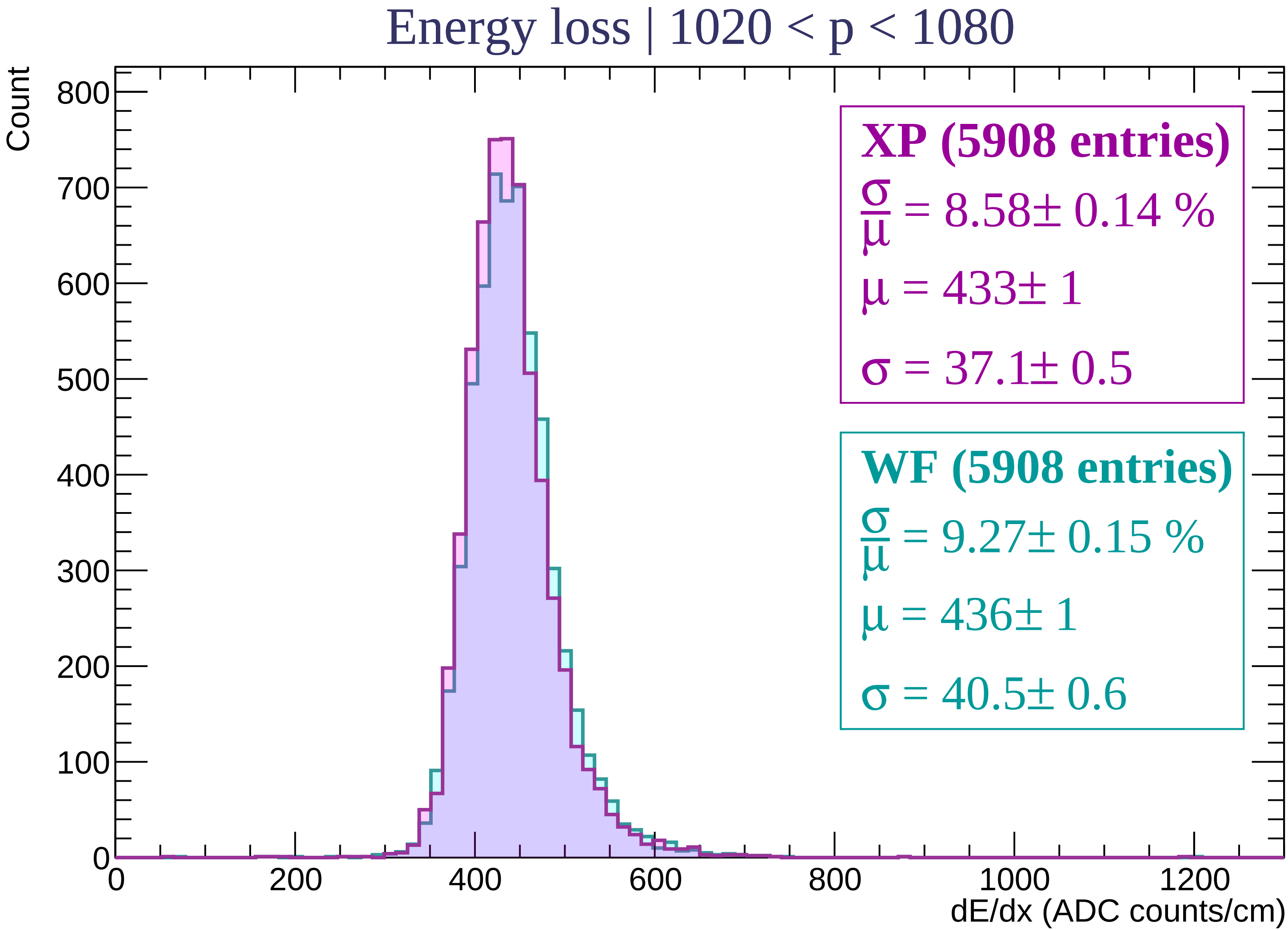


Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP (6408 entries)
$\frac{\sigma}{\mu}$ = 8.61± 0.14 %
μ = 421± 1
σ = 36.2± 0.6
WF (6408 entries)
$\frac{\sigma}{\mu}$ = 9.10± 0.17 %
μ = 424± 1
σ = 38.6± 0.7

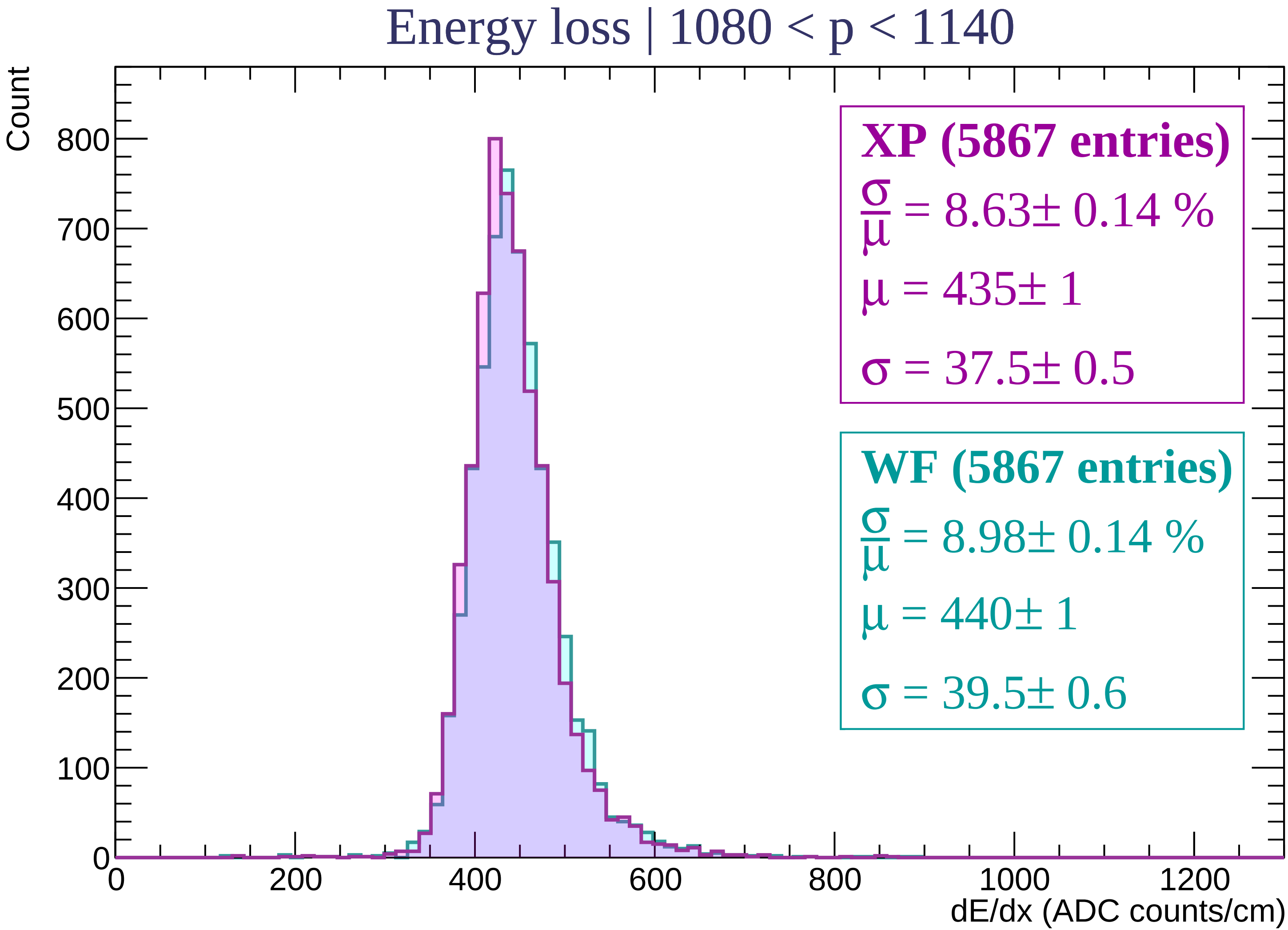

Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
XP (6505 entries)
σ/μ = 8.57± 0.14 %
μ = 425± 1
σ = 36.4± 0.5
WF (6505 entries)
σ/μ = 9.10± 0.14 %
μ = 429± 1
σ = 39.0± 0.6

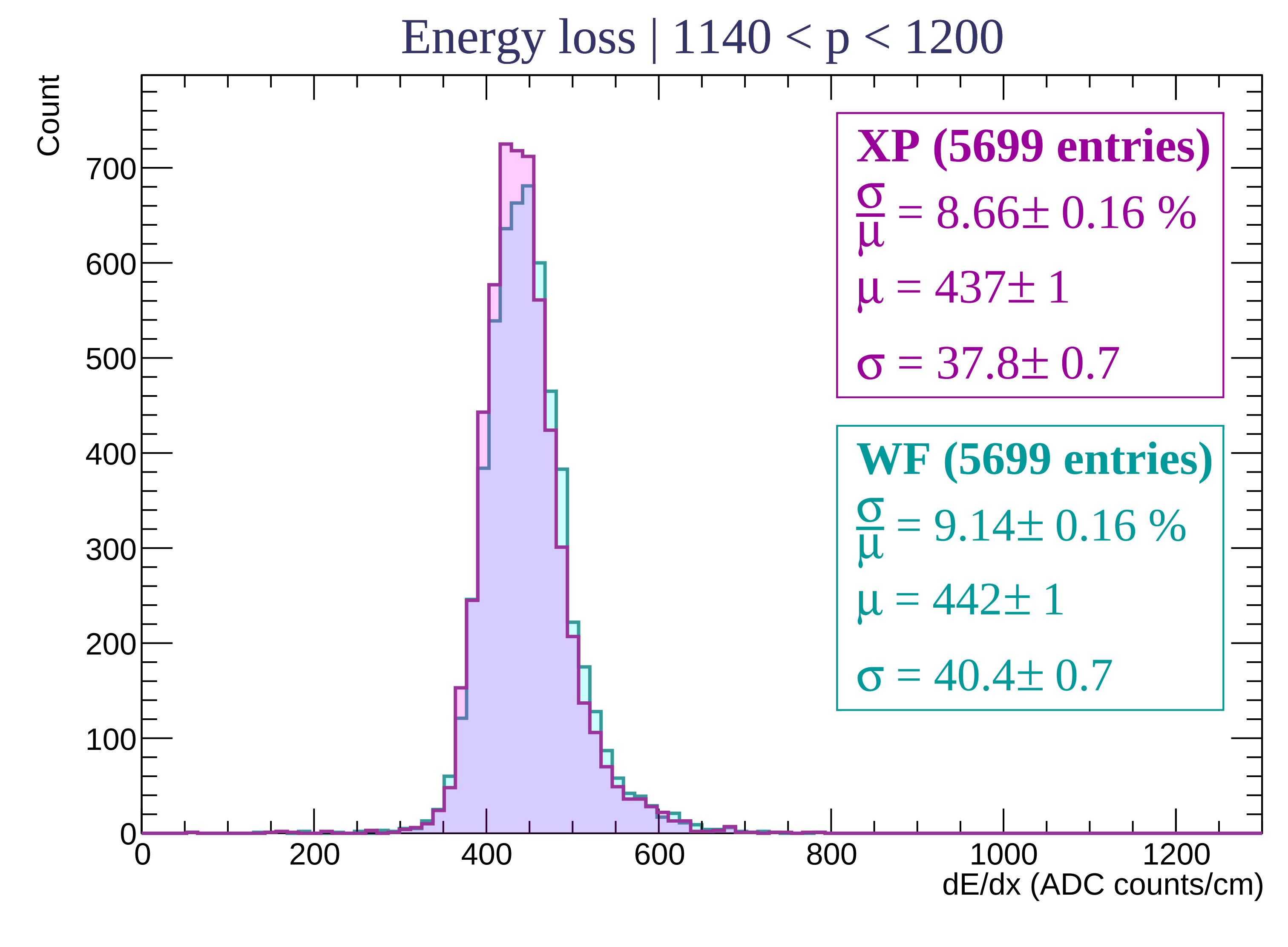


Energy loss | 960 < p < 1020
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (6149 entries)
σ/μ = 8.49± 0.14 %
μ = 427± 1
σ = 36.3± 0.6
WF (6149 entries)
σ/μ = 8.99± 0.15 %
μ = 432± 1
σ = 38.8± 0.6

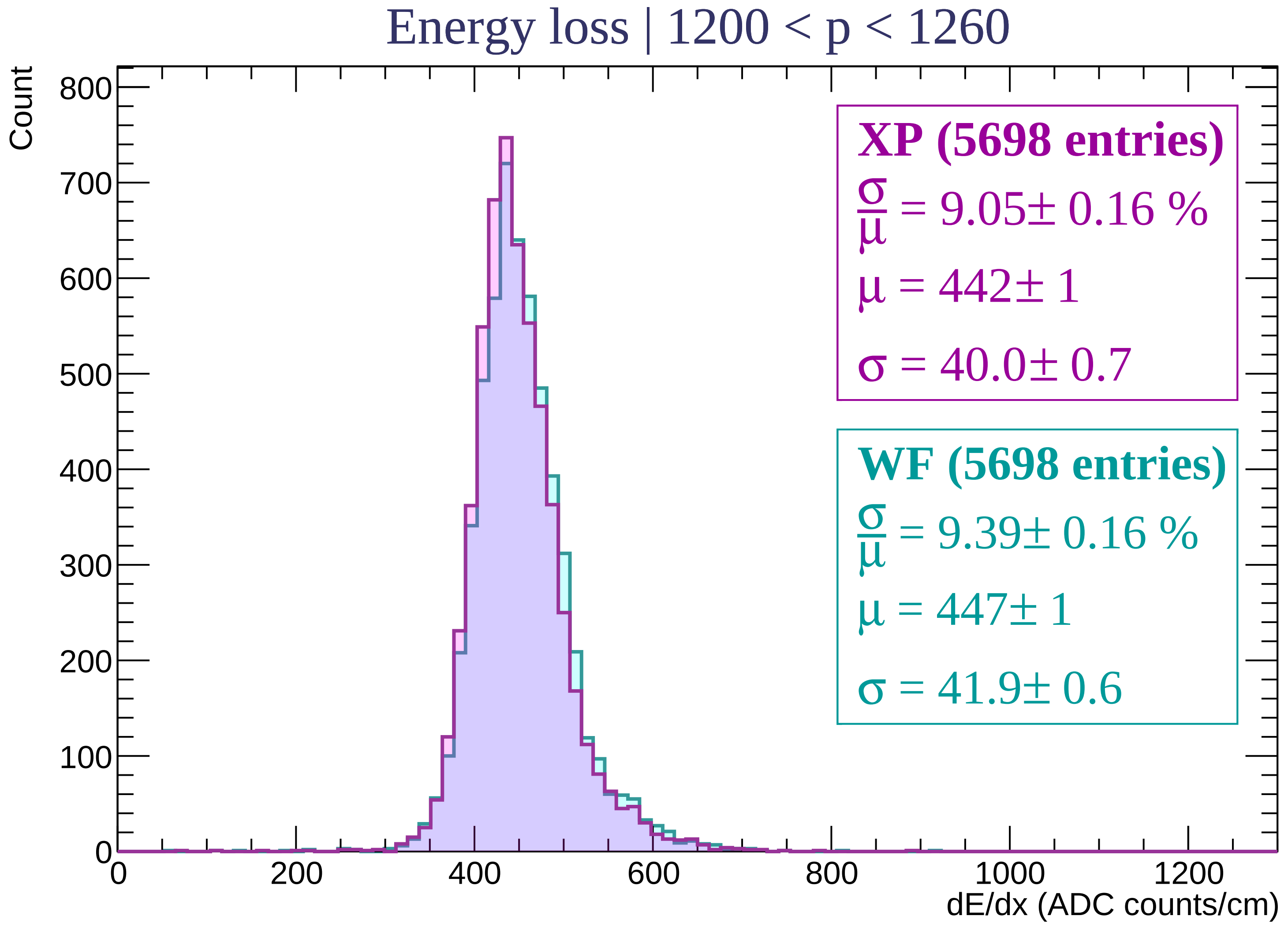


Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
XP (5908 entries)
$\frac{\sigma}{\mu}$ = 8.58± 0.14 %
μ = 433± 1
σ = 37.1± 0.5
WF (5908 entries)
$\frac{\sigma}{\mu}$ = 9.27± 0.15 %
μ = 436± 1
σ = 40.5± 0.6

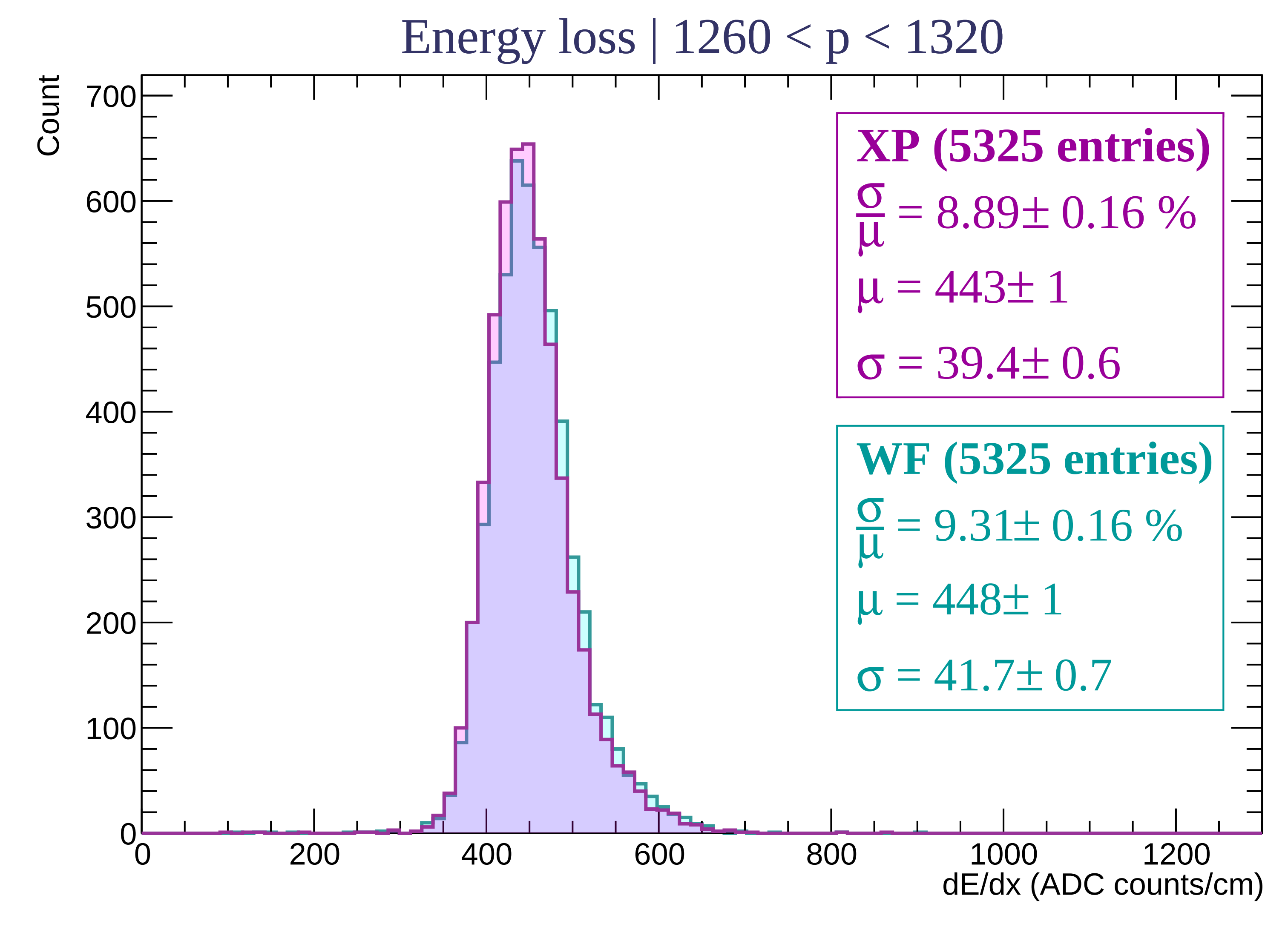

Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP (5867 entries)
σ/μ = 8.63± 0.14 %
μ = 435± 1
σ = 37.5± 0.5
WF (5867 entries)
σ/μ = 8.98± 0.14 %
μ = 440± 1
σ = 39.5± 0.6

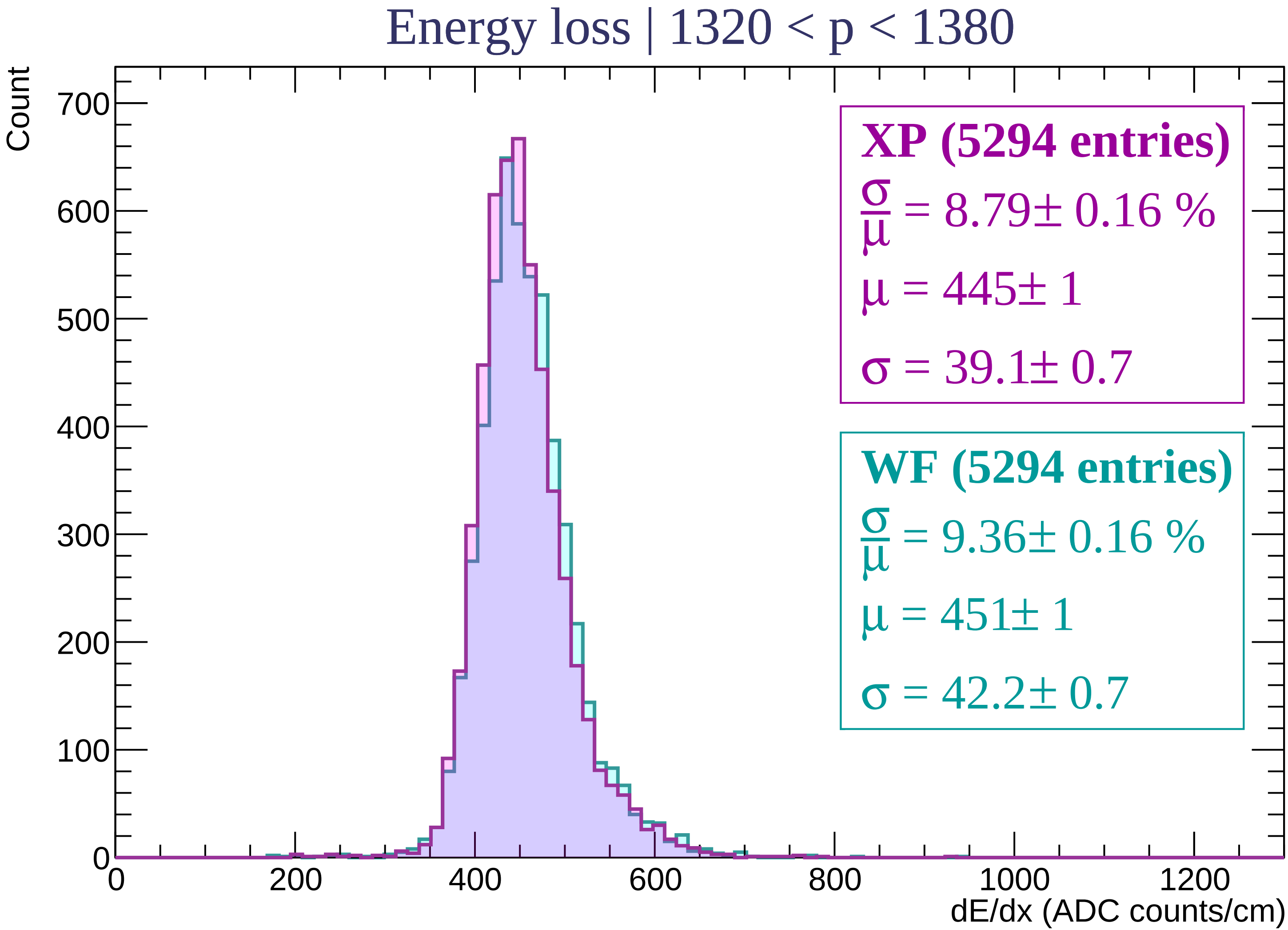


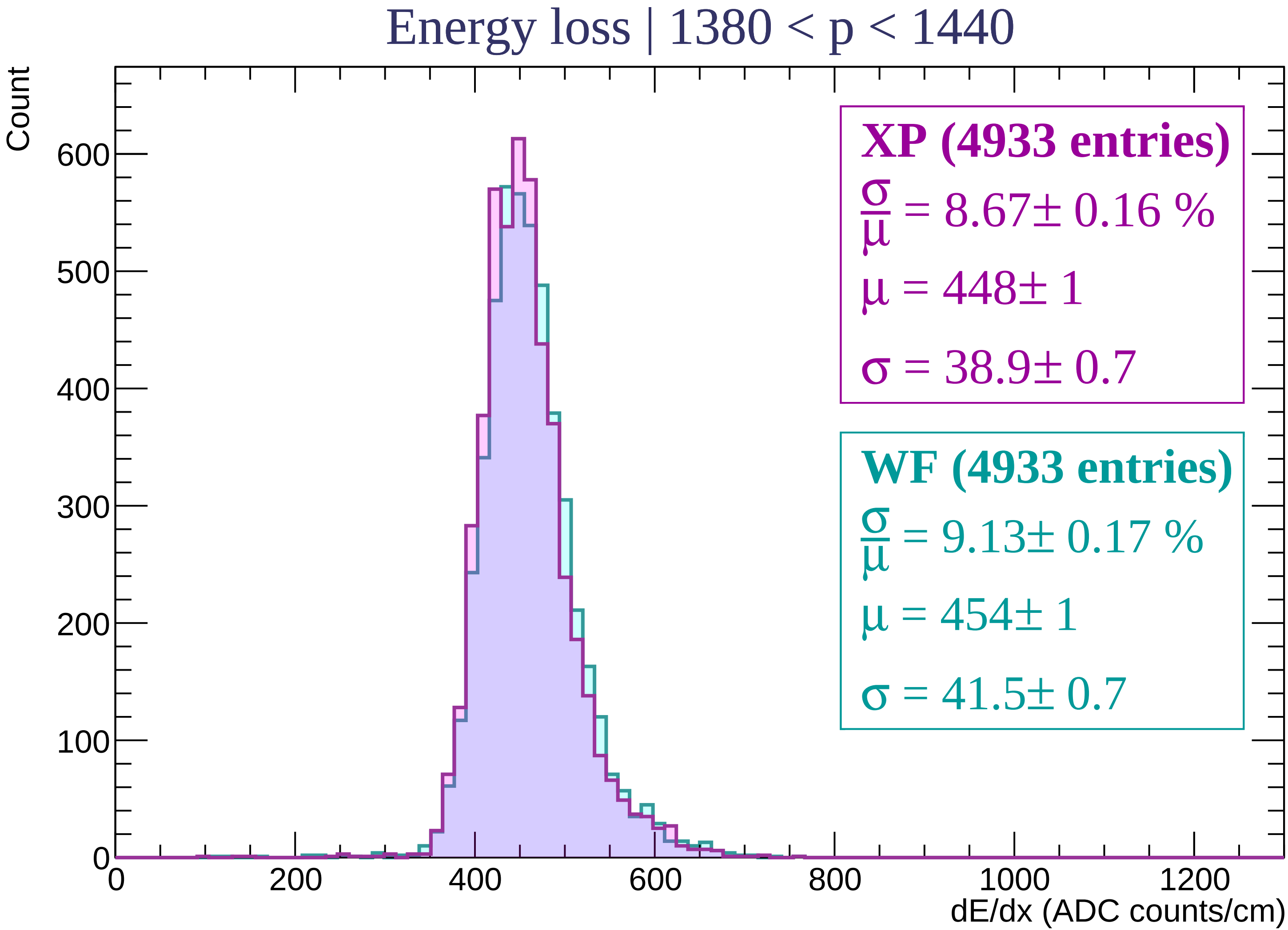
Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5699 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.16 %
μ = 437± 1
σ = 37.8± 0.7
WF (5699 entries)
$\frac{\sigma}{\mu}$ = 9.14± 0.16 %
μ = 442± 1
σ = 40.4± 0.7



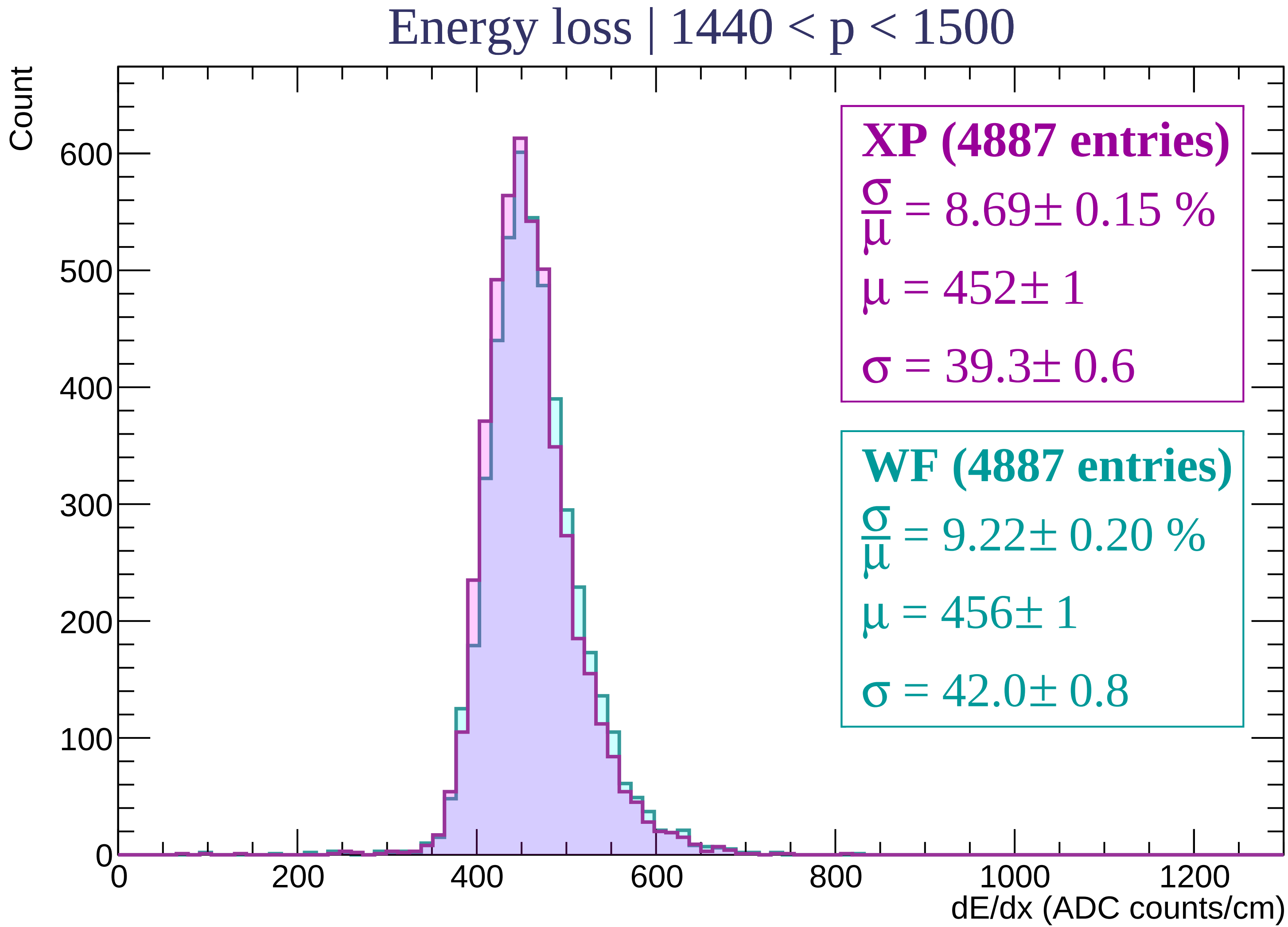
Energy loss | 1200 < p < 1260
Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5698 entries)
$\frac{\sigma}{\mu}$ = 9.05± 0.16 %
μ = 442± 1
σ = 40.0± 0.7
WF (5698 entries)
$\frac{\sigma}{\mu}$ = 9.39± 0.16 %
μ = 447± 1
σ = 41.9± 0.6


Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (5325 entries)
σ/μ = 8.89± 0.16 %
μ = 443± 1
σ = 39.4± 0.6
WF (5325 entries)
σ/μ = 9.31± 0.16 %
μ = 448± 1
σ = 41.7± 0.7

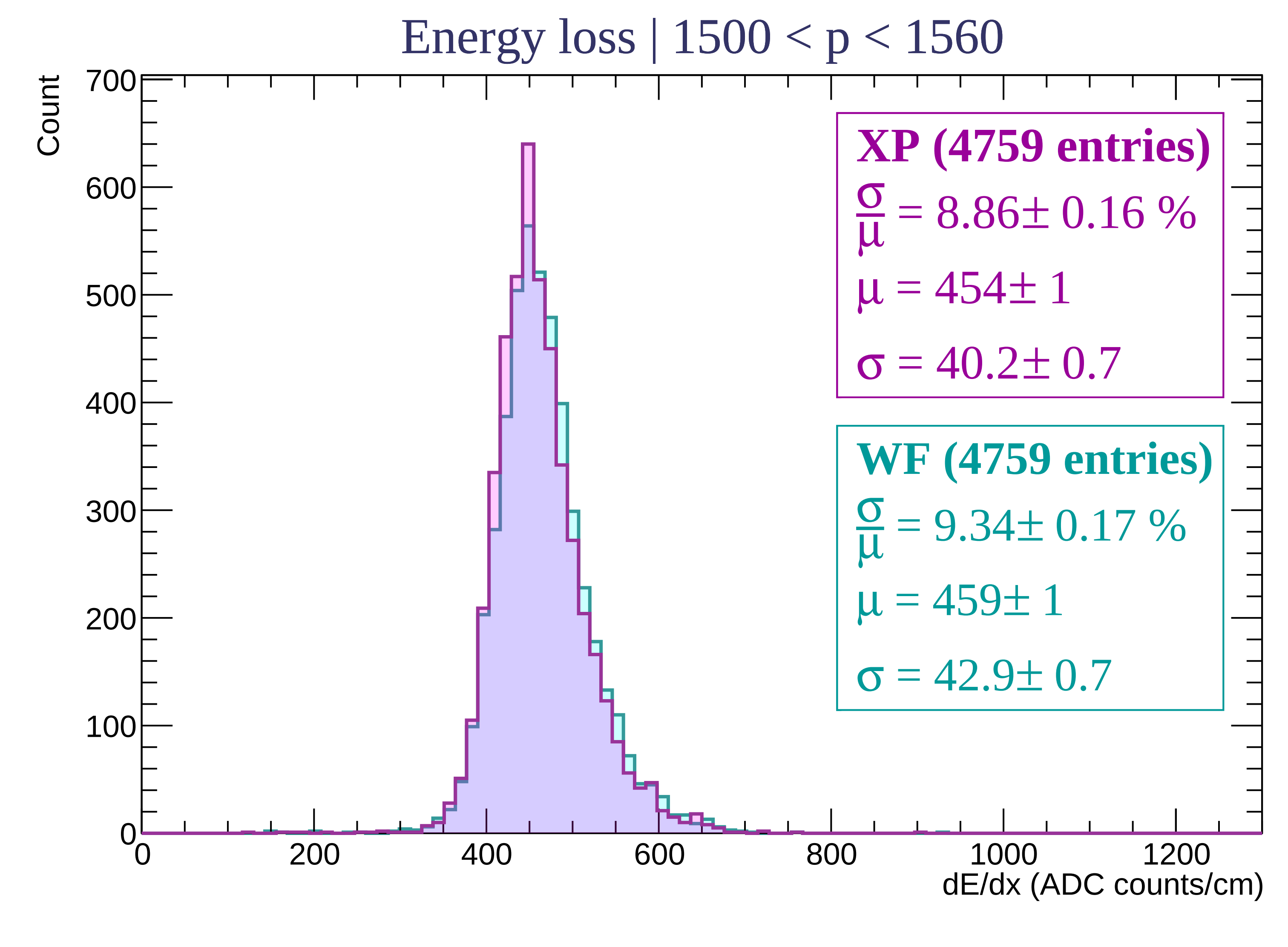

Energy loss | 1320 < p < 1380
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (5294 entries)
σ/μ = 8.79± 0.16 %
μ = 445± 1
σ = 39.1± 0.7
WF (5294 entries)
σ/μ = 9.36± 0.16 %
μ = 451± 1
σ = 42.2± 0.7



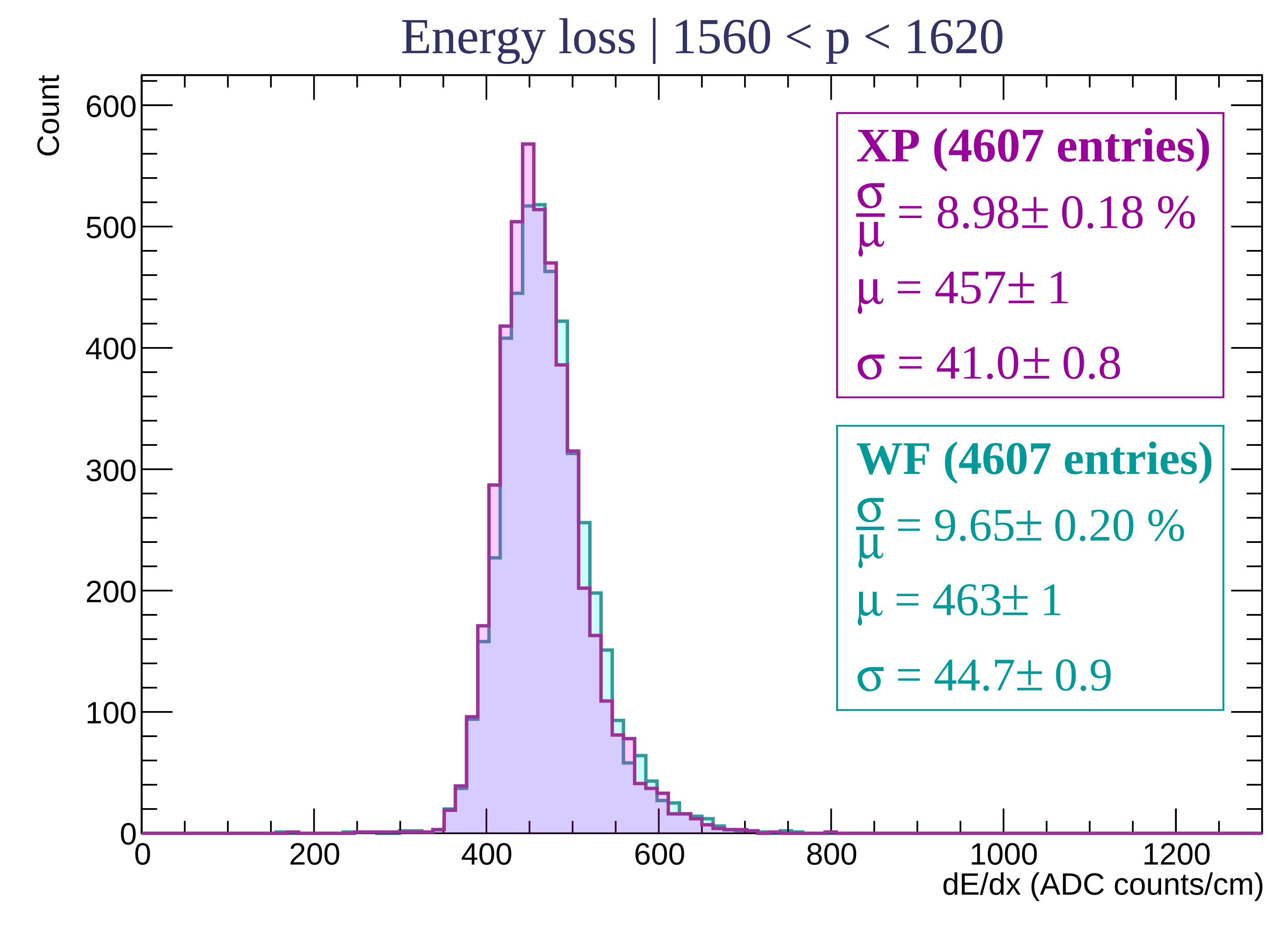
Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4933 entries)
σ/μ = 8.67± 0.16 %
μ = 448± 1
σ = 38.9± 0.7
WF (4933 entries)
σ/μ = 9.13± 0.17 %
μ = 454± 1
σ = 41.5± 0.7



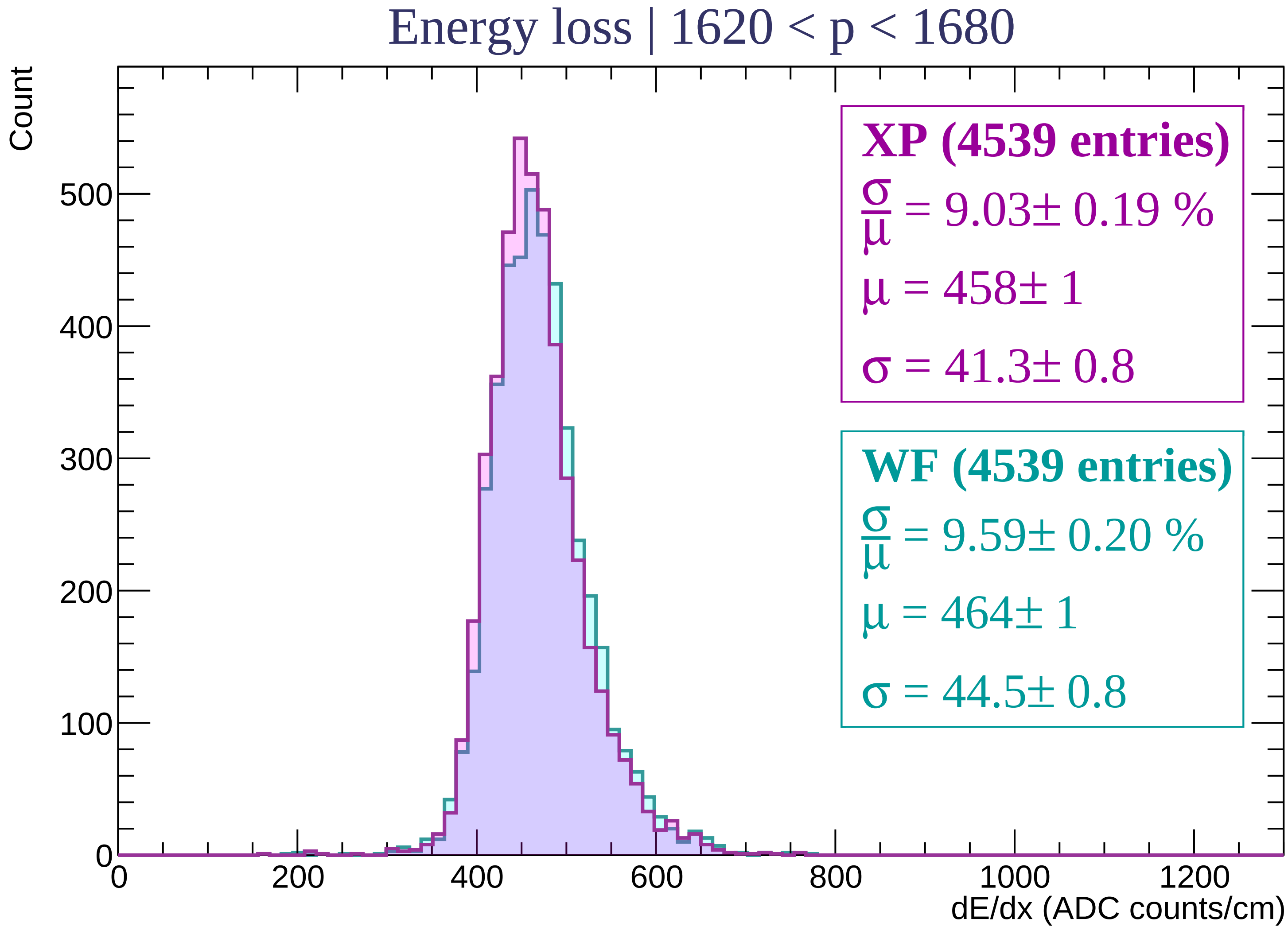
Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4887 entries)
$\frac{\sigma}{\mu}$ = 8.69± 0.15 %
μ = 452± 1
σ = 39.3± 0.6
WF (4887 entries)
$\frac{\sigma}{\mu}$ = 9.22± 0.20 %
μ = 456± 1
σ = 42.0± 0.8

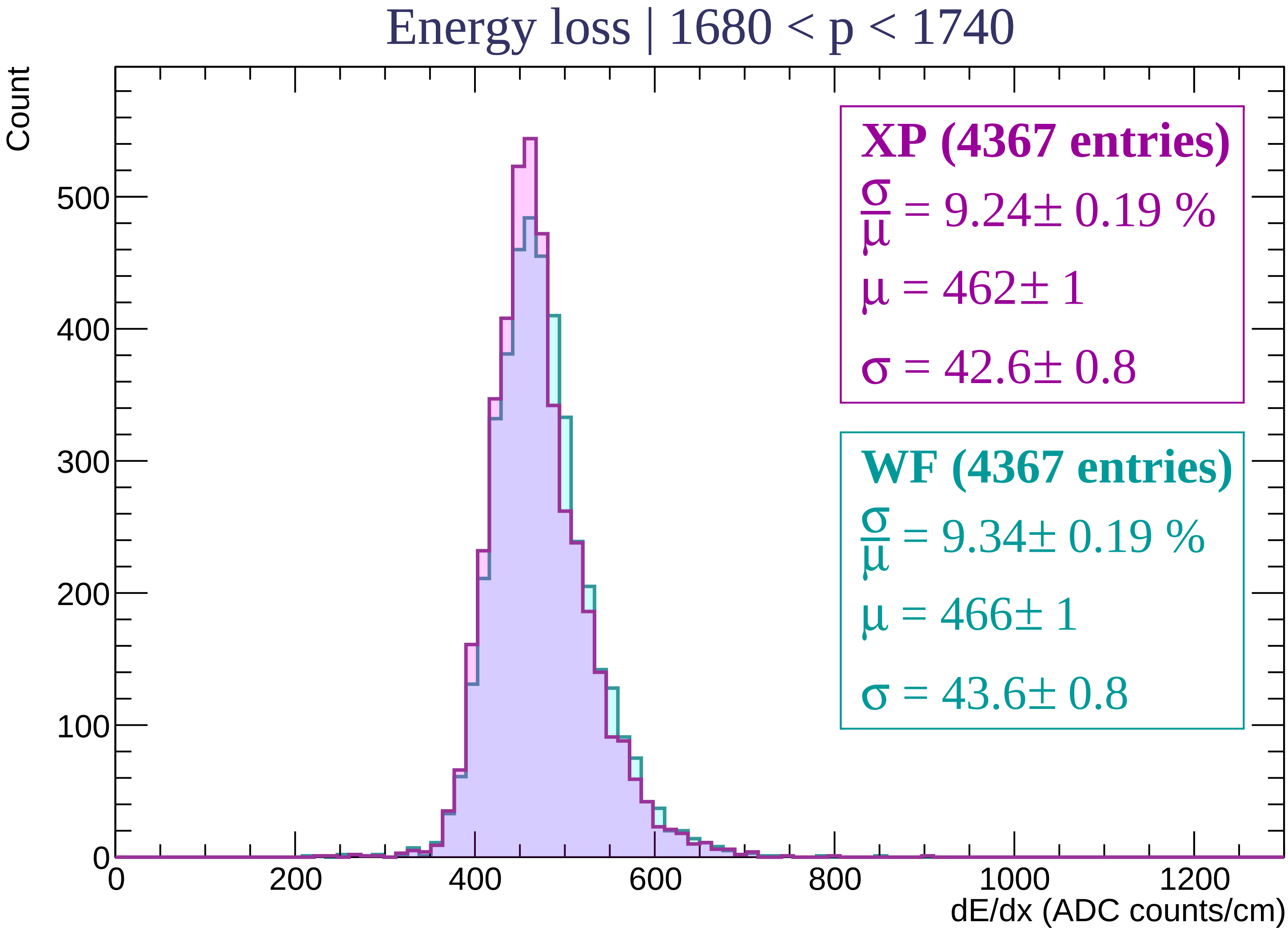


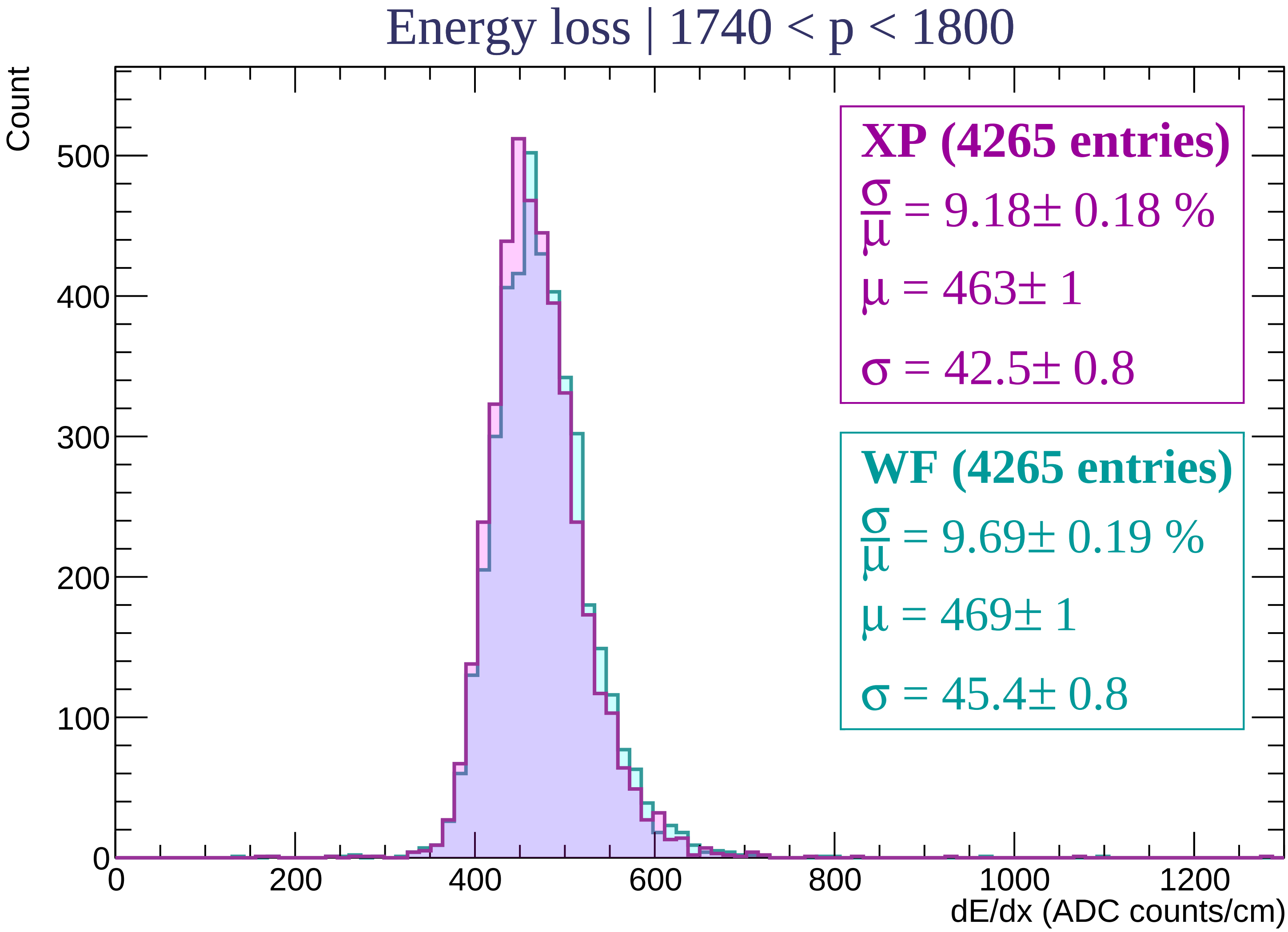
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4759 entries)
$\frac{\sigma}{\mu}$ = 8.86± 0.16 %
μ = 454± 1
σ = 40.2± 0.7
WF (4759 entries)
$\frac{\sigma}{\mu}$ = 9.34± 0.17 %
μ = 459± 1
σ = 42.9± 0.7

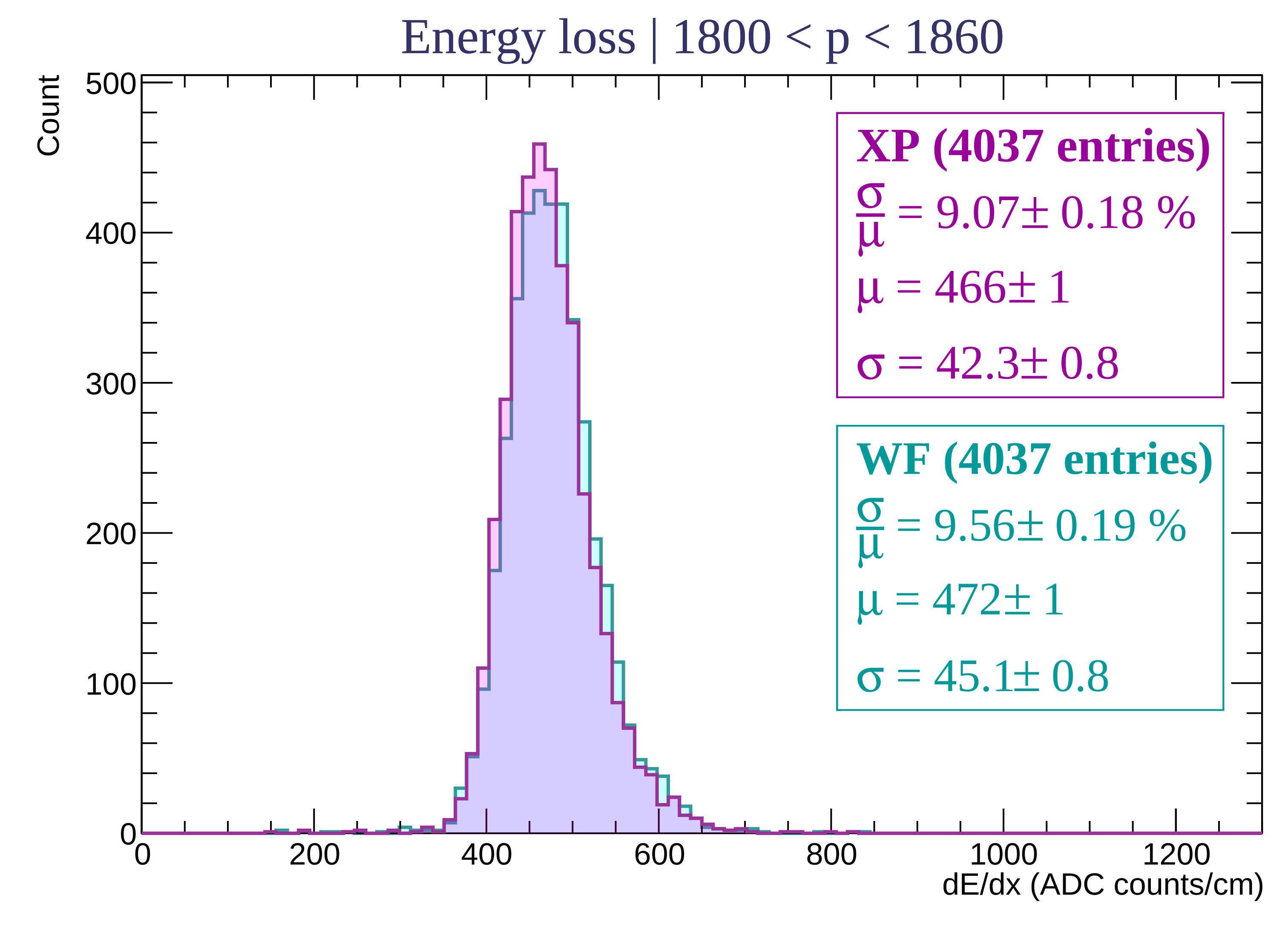


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4607 entries)
$\frac{\sigma}{\mu}$ = 8.98± 0.18 %
μ = 457± 1
σ = 41.0± 0.8
WF (4607 entries)
$\frac{\sigma}{\mu}$ = 9.65± 0.20 %
μ = 463± 1
σ = 44.7± 0.9

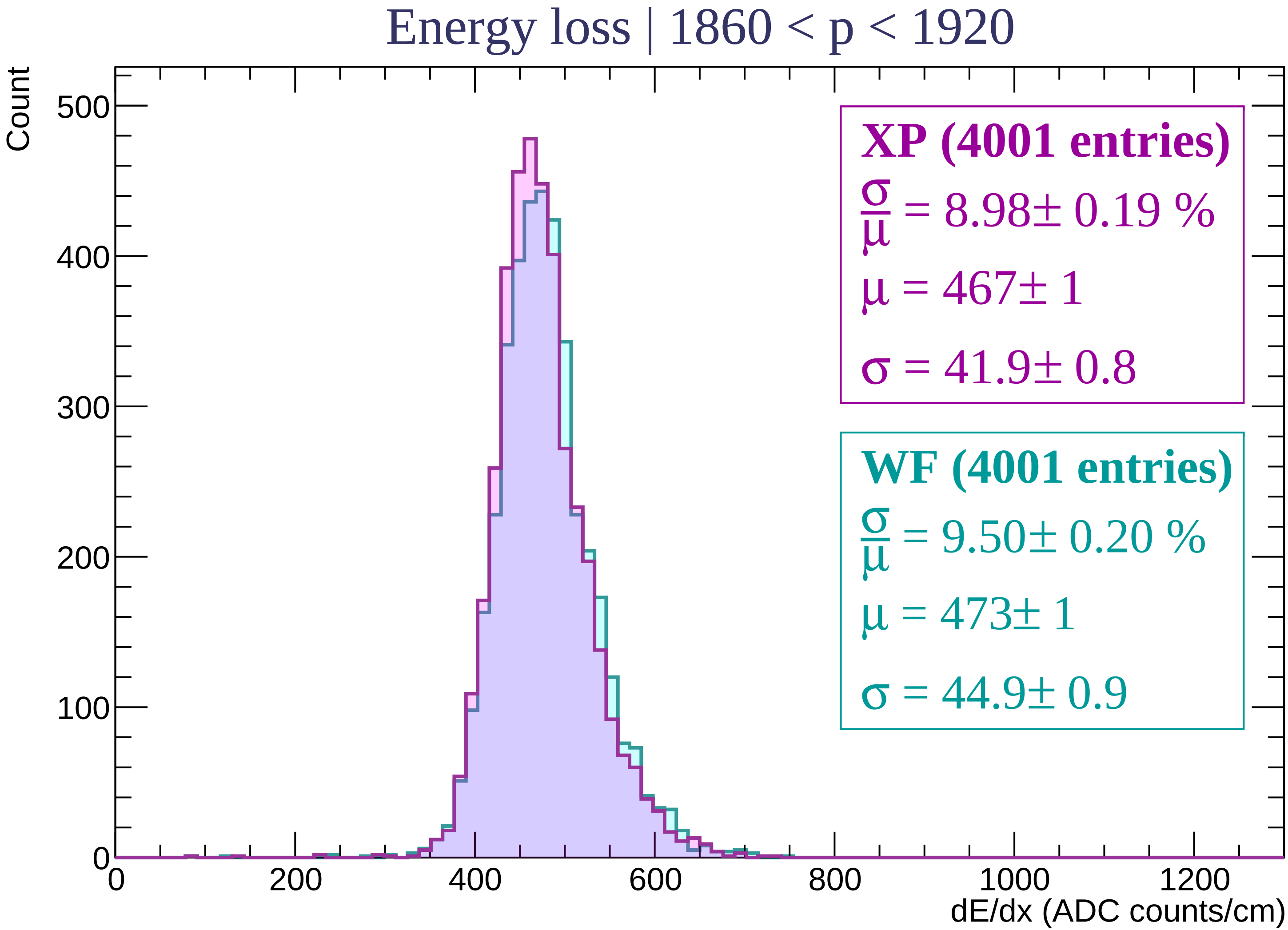

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4539 entries)
σ/μ = 9.03± 0.19 %
μ = 458± 1
σ = 41.3± 0.8
WF (4539 entries)
σ/μ = 9.59± 0.20 %
μ = 464± 1
σ = 44.5± 0.8


Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4367 entries)
$\frac{\sigma}{\mu}$ = 9.24± 0.19 %
μ = 462± 1
σ = 42.6± 0.8
WF (4367 entries)
$\frac{\sigma}{\mu}$ = 9.34± 0.19 %
μ = 466± 1
σ = 43.6± 0.8

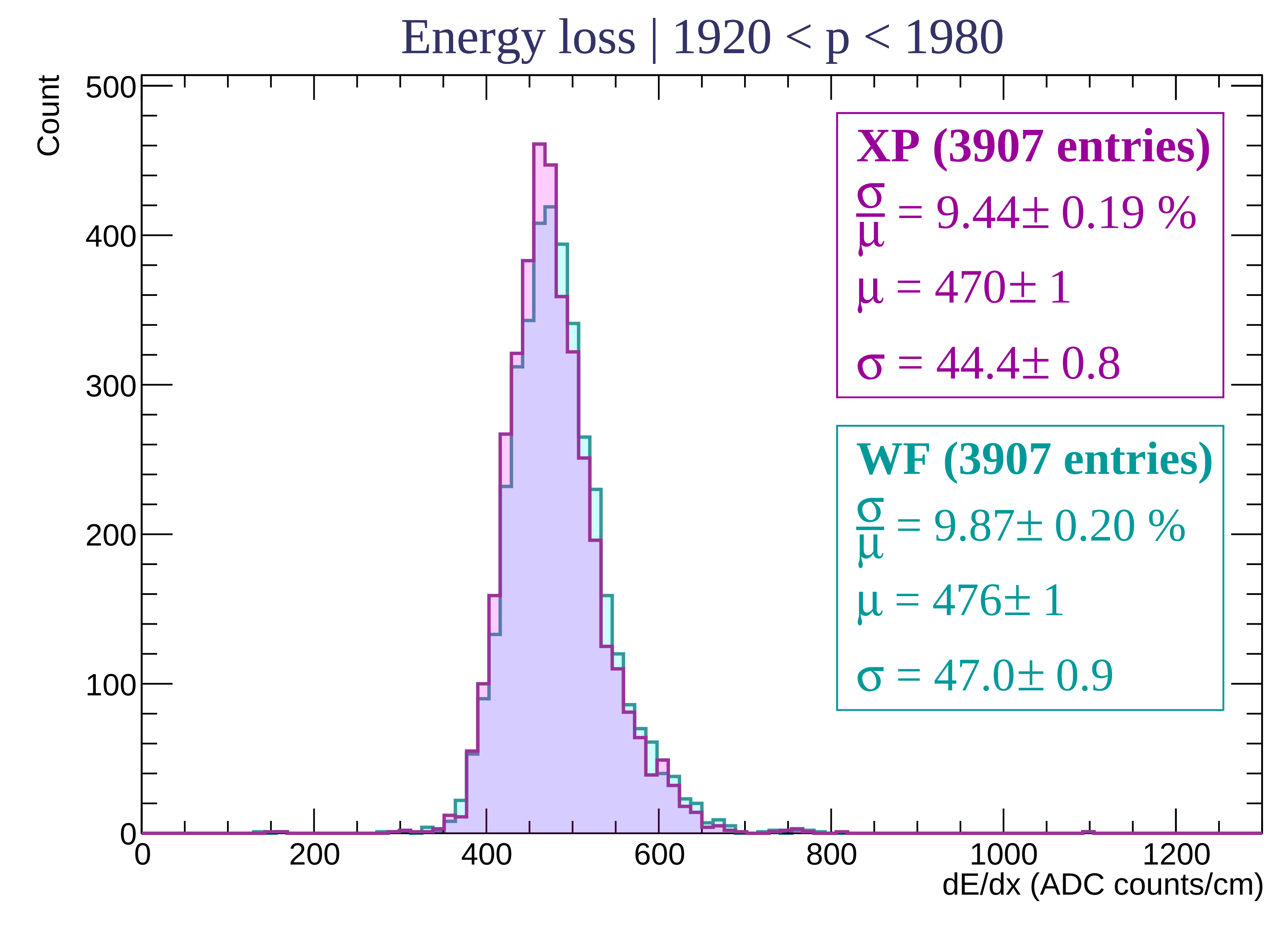

Energy loss | 1740 < p < 1800
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4265 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.18 %
μ = 463± 1
σ = 42.5± 0.8
WF (4265 entries)
$\frac{\sigma}{\mu}$ = 9.69± 0.19 %
μ = 469± 1
σ = 45.4± 0.8



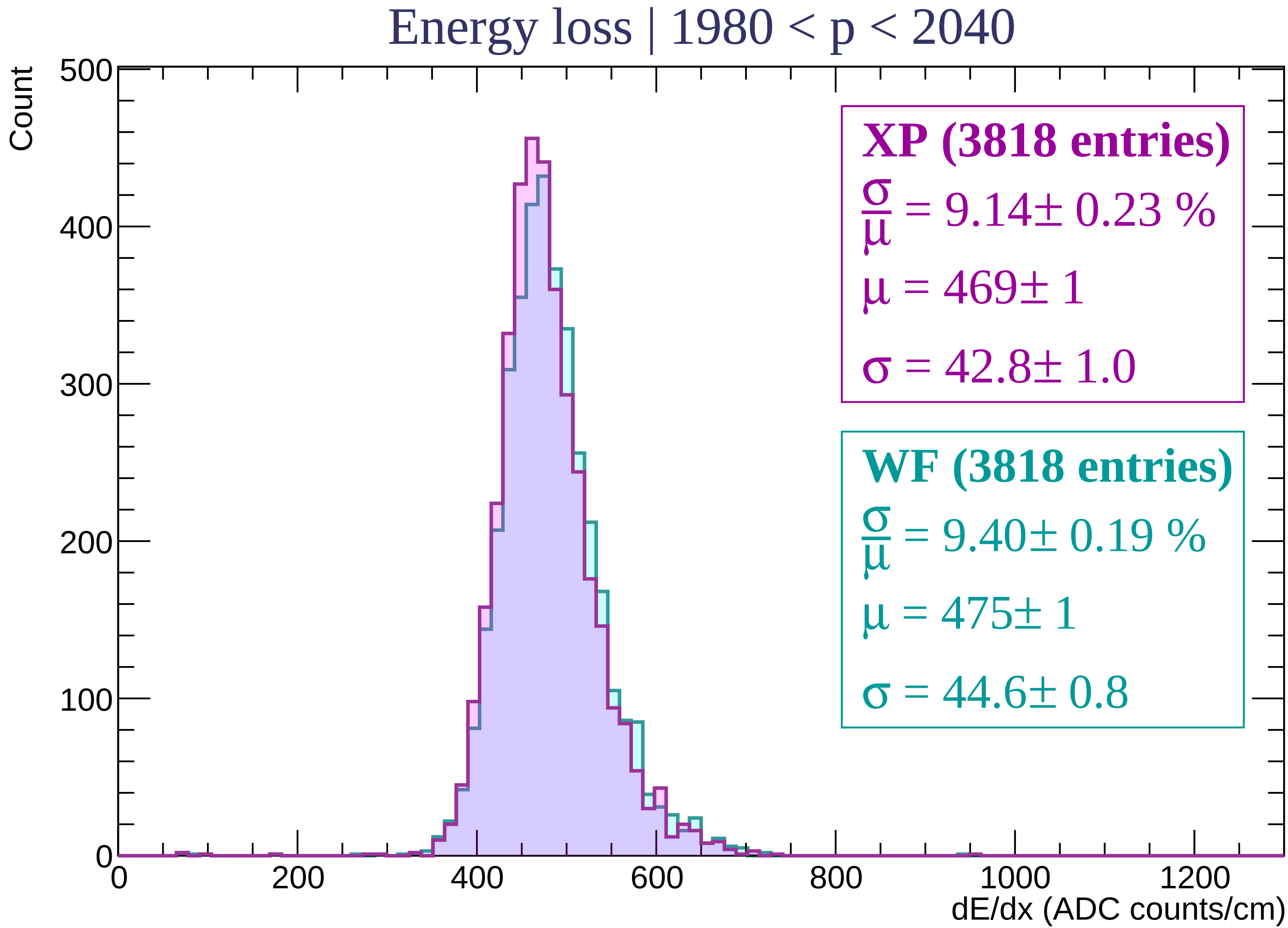
Energy loss | 1800 < p < 1860
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4037 entries)
$\frac{\sigma}{\mu}$ = 9.07± 0.18 %
μ = 466± 1
σ = 42.3± 0.8
WF (4037 entries)
$\frac{\sigma}{\mu}$ = 9.56± 0.19 %
μ = 472± 1
σ = 45.1± 0.8



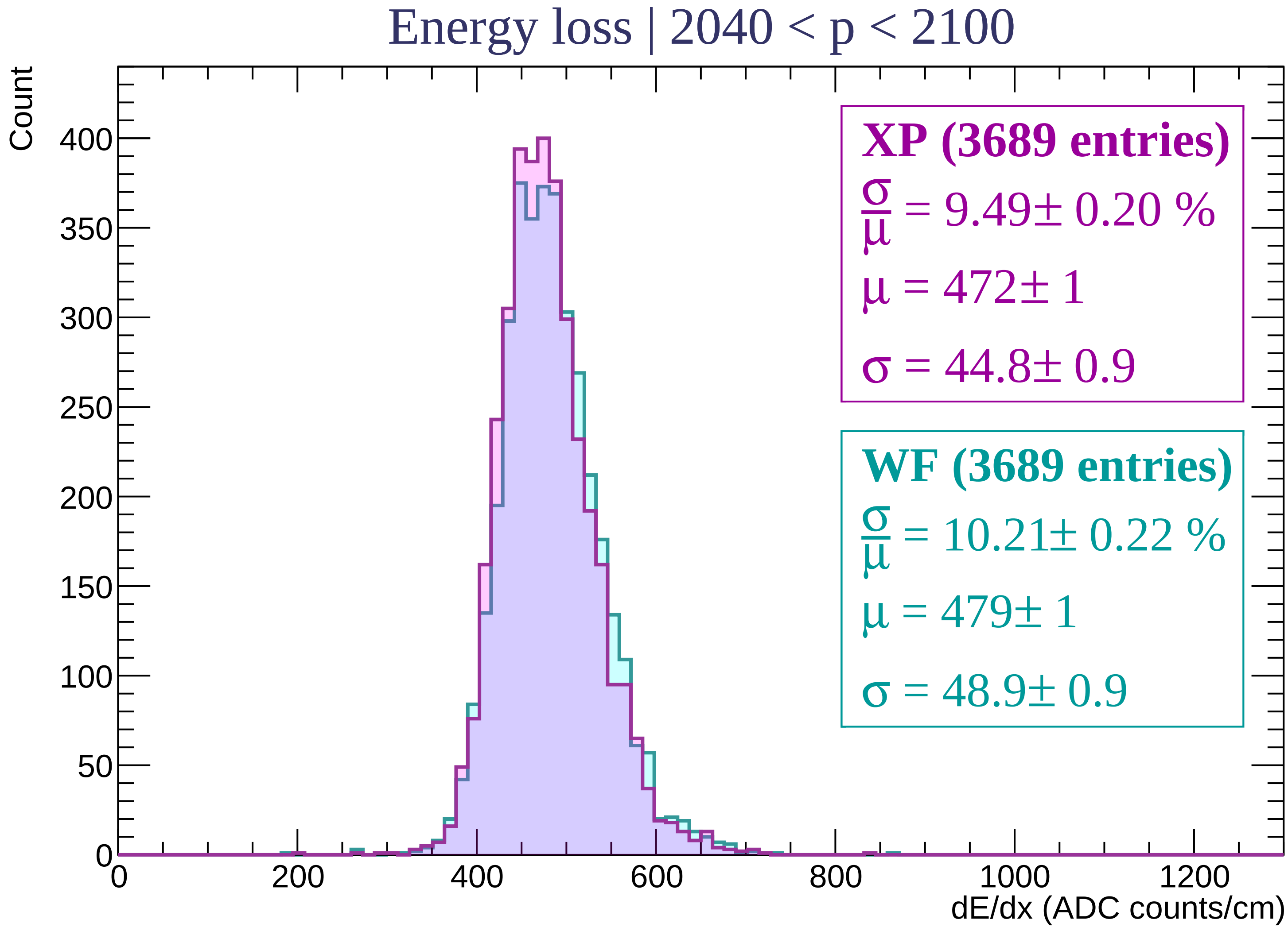
Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4001 entries)
σ/μ = 8.98± 0.19 %
μ = 467± 1
σ = 41.9± 0.8
WF (4001 entries)
σ/μ = 9.50± 0.20 %
μ = 473± 1
σ = 44.9± 0.9



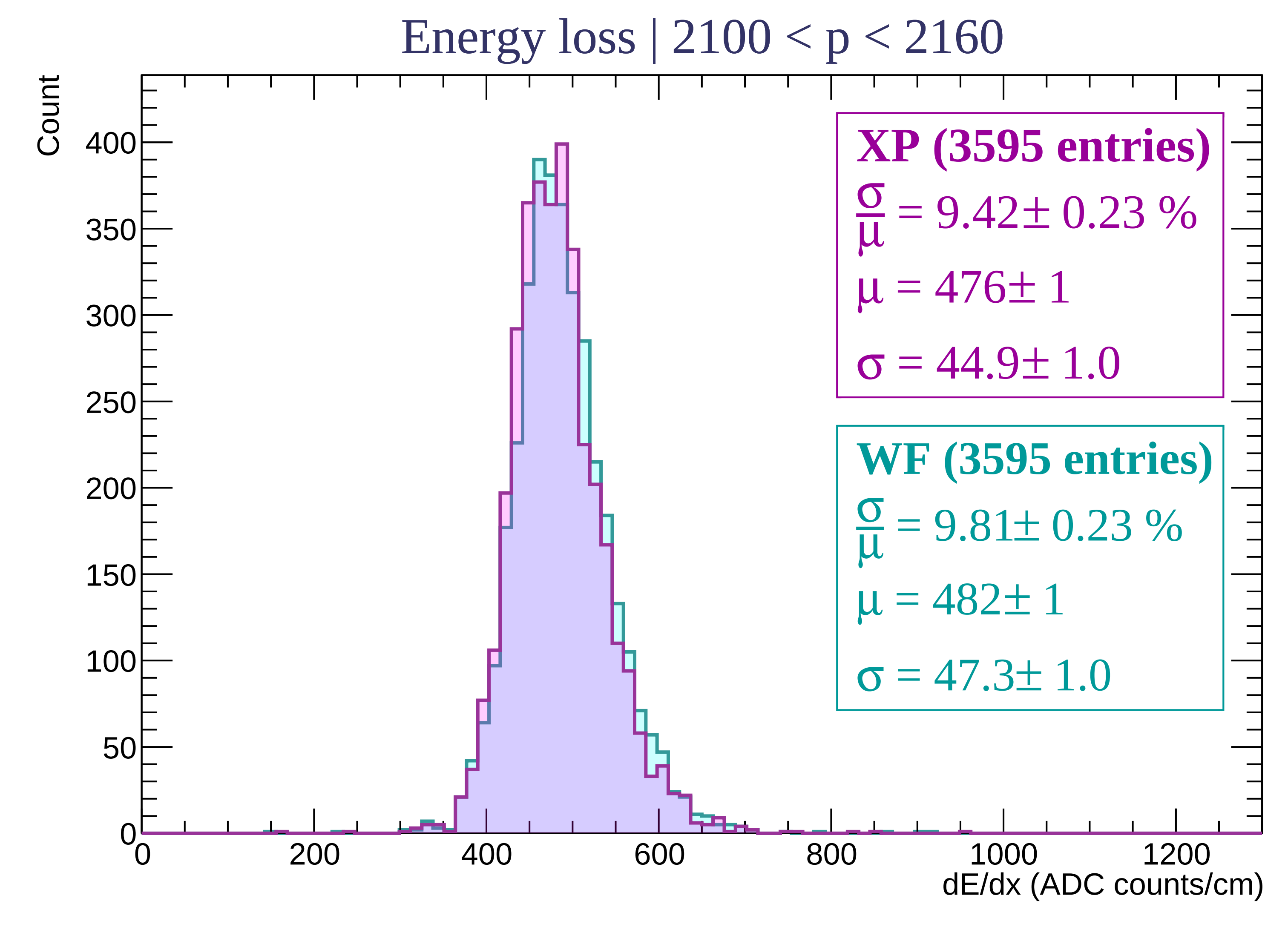
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP (3907 entries)
$\frac{\sigma}{\mu}$ = 9.44± 0.19 %
μ = 470± 1
σ = 44.4± 0.8
WF (3907 entries)
$\frac{\sigma}{\mu}$ = 9.87± 0.20 %
μ = 476± 1
σ = 47.0± 0.9



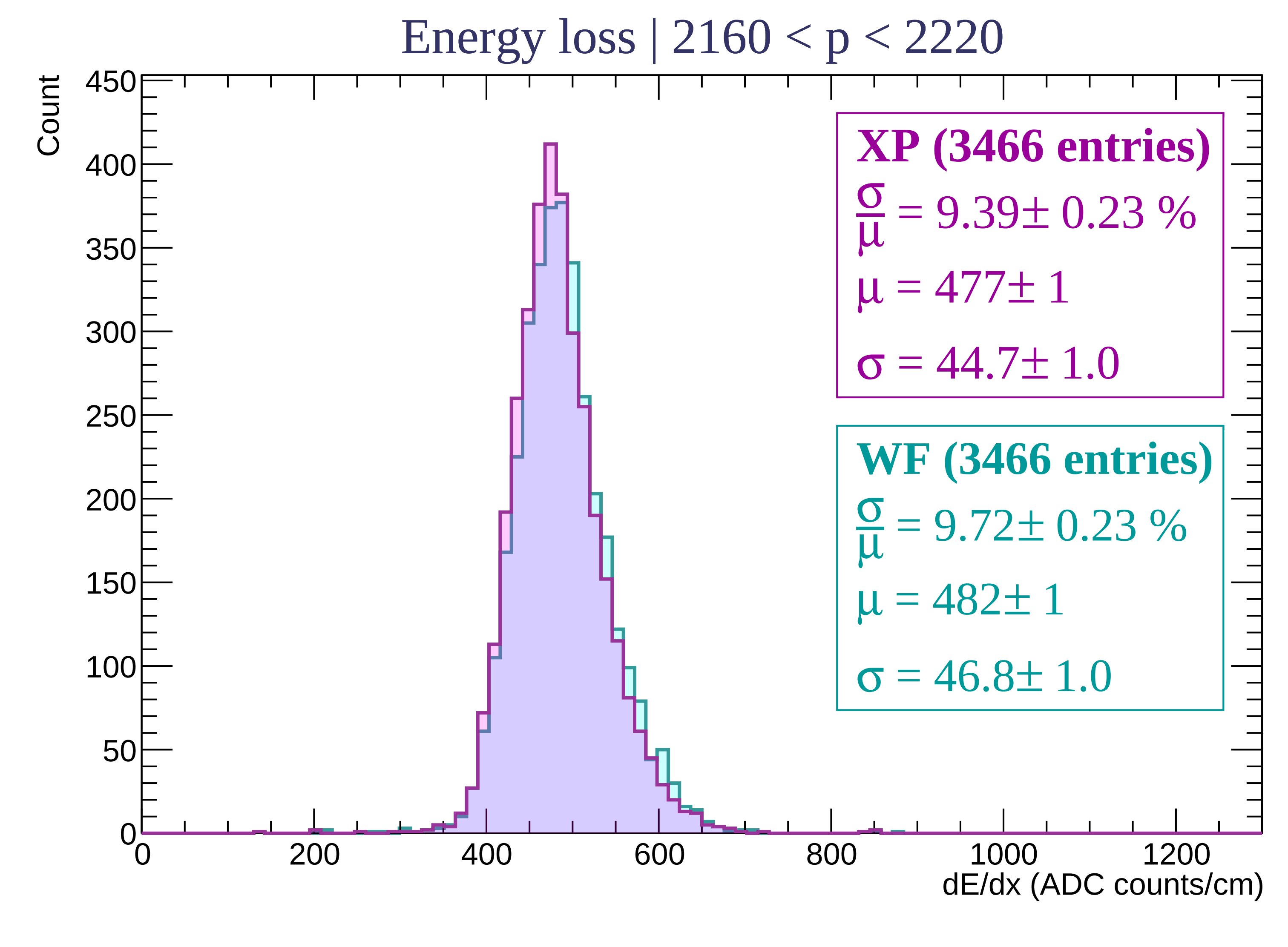
Energy loss | 1980 < p < 2040
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3818 entries)
σ/μ = 9.14± 0.23 %
μ = 469± 1
σ = 42.8± 1.0
WF (3818 entries)
σ/μ = 9.40± 0.19 %
μ = 475± 1
σ = 44.6± 0.8

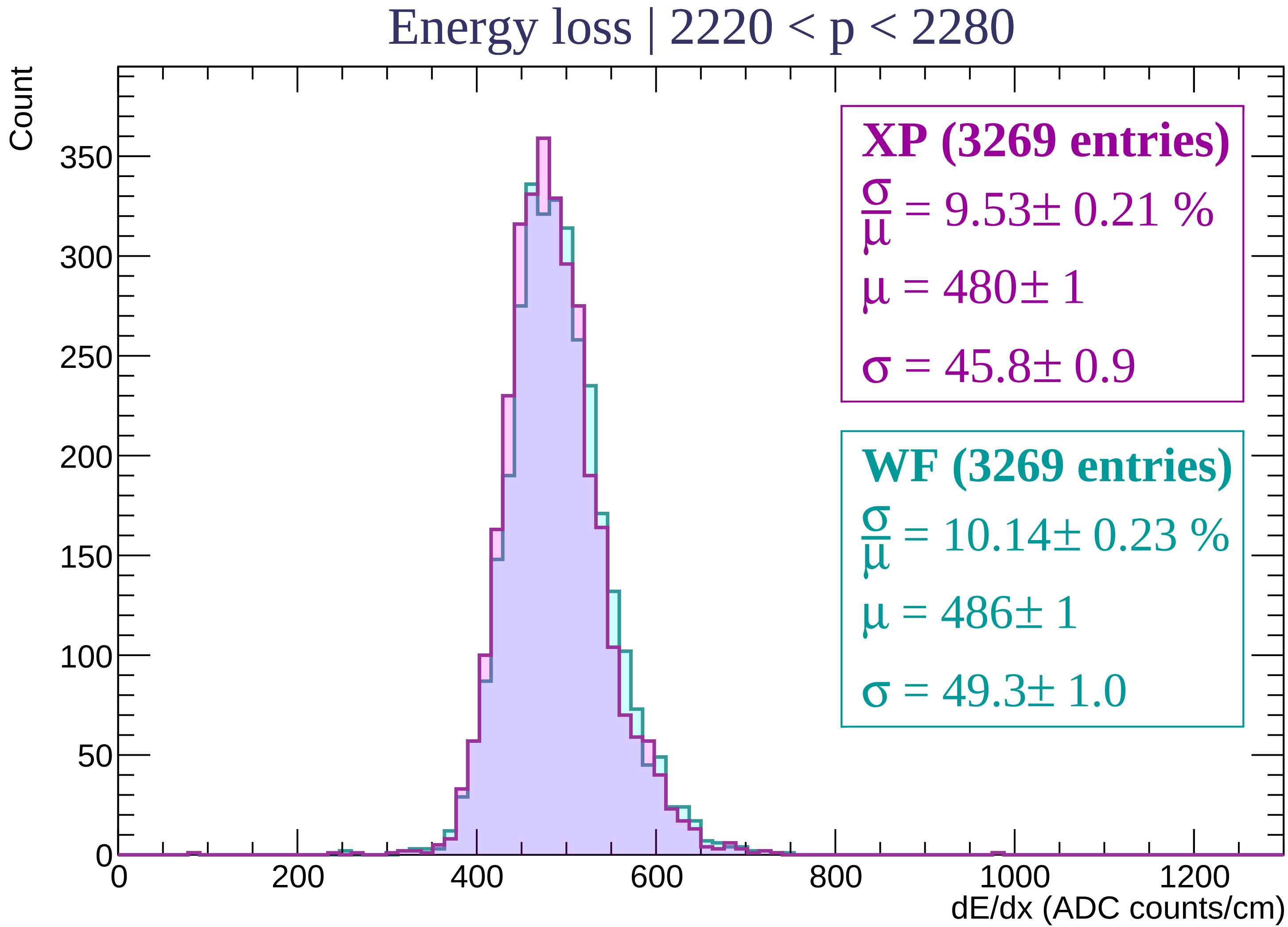


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
XP (3689 entries)
$\frac{\sigma}{\mu}$ = 9.49± 0.20 %
μ = 472± 1
σ = 44.8± 0.9
WF (3689 entries)
$\frac{\sigma}{\mu}$ = 10.21± 0.22 %
μ = 479± 1
σ = 48.9± 0.9

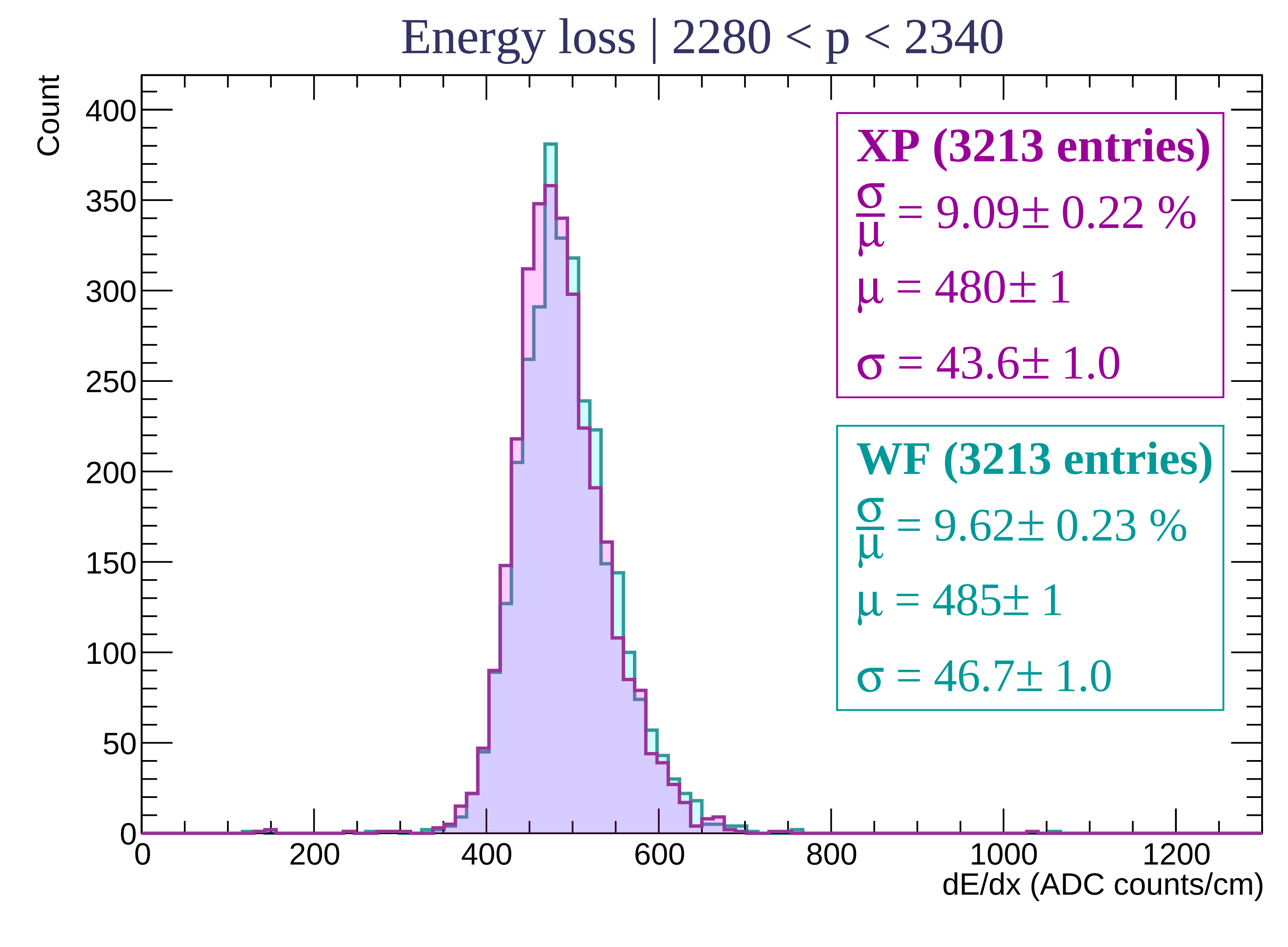


Energy loss | 2100 < p < 2160
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3595 entries)
$\frac{\sigma}{\mu}$ = 9.42± 0.23 %
μ = 476± 1
σ = 44.9± 1.0
WF (3595 entries)
$\frac{\sigma}{\mu}$ = 9.81± 0.23 %
μ = 482± 1
σ = 47.3± 1.0

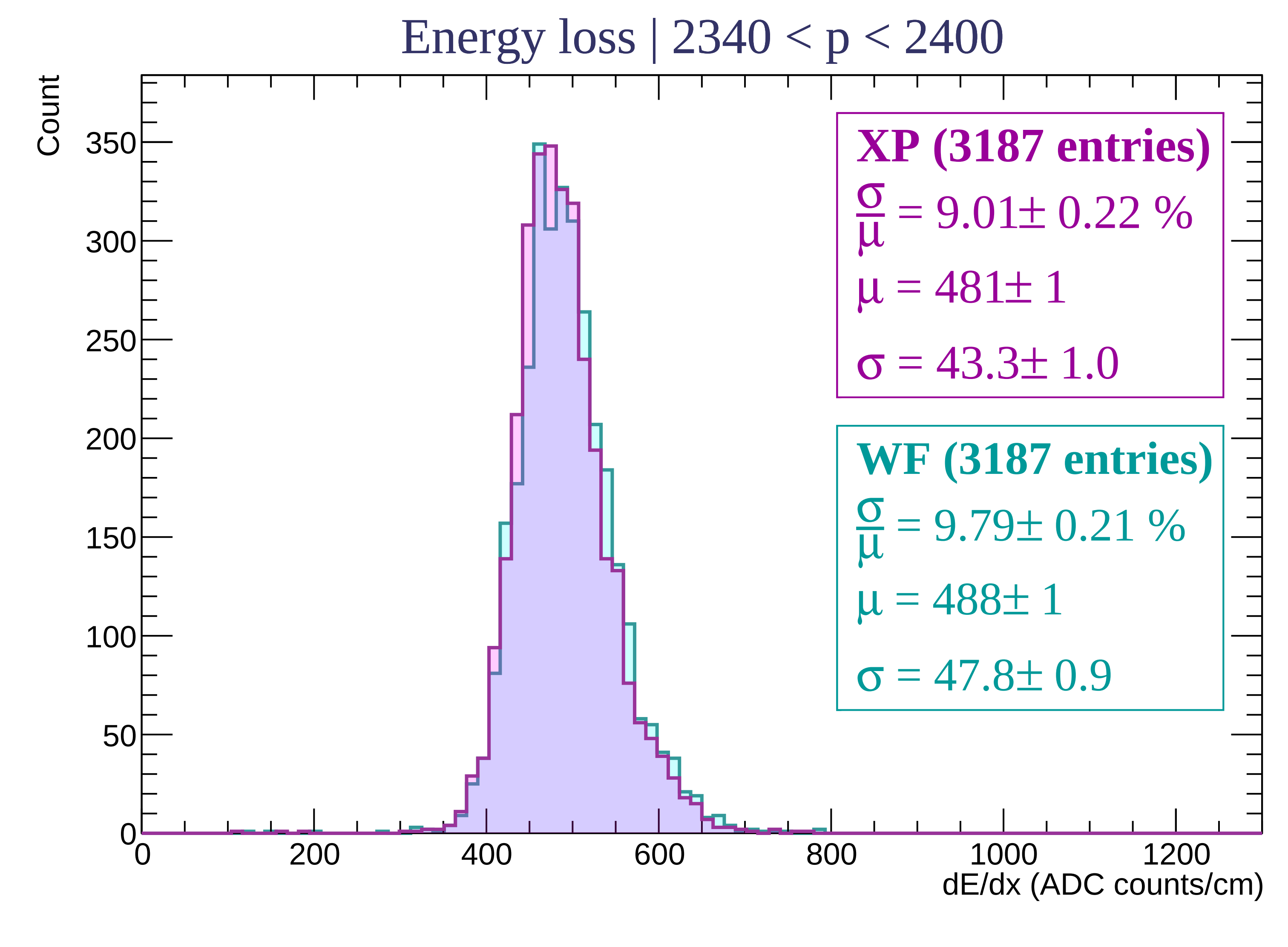

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
XP (3466 entries)
σ/μ = 9.39± 0.23 %
μ = 477± 1
σ = 44.7± 1.0
WF (3466 entries)
σ/μ = 9.72± 0.23 %
μ = 482± 1
σ = 46.8± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



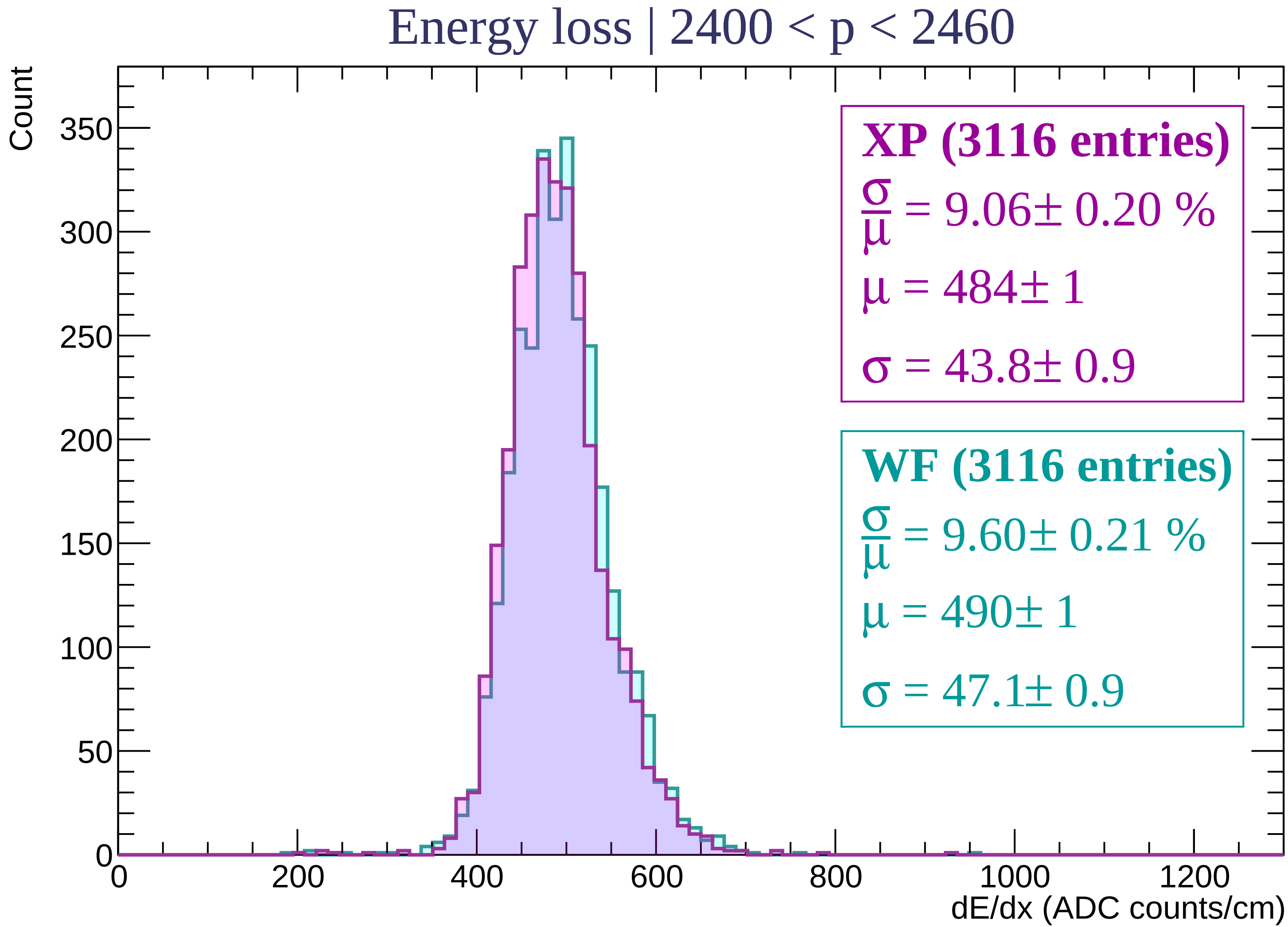
Energy loss | 2220 < p < 2280
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3269 entries)
$\frac{\sigma}{\mu}$ = 9.53± 0.21 %
μ = 480± 1
σ = 45.8± 0.9
WF (3269 entries)
$\frac{\sigma}{\mu}$ = 10.14± 0.23 %
μ = 486± 1
σ = 49.3± 1.0



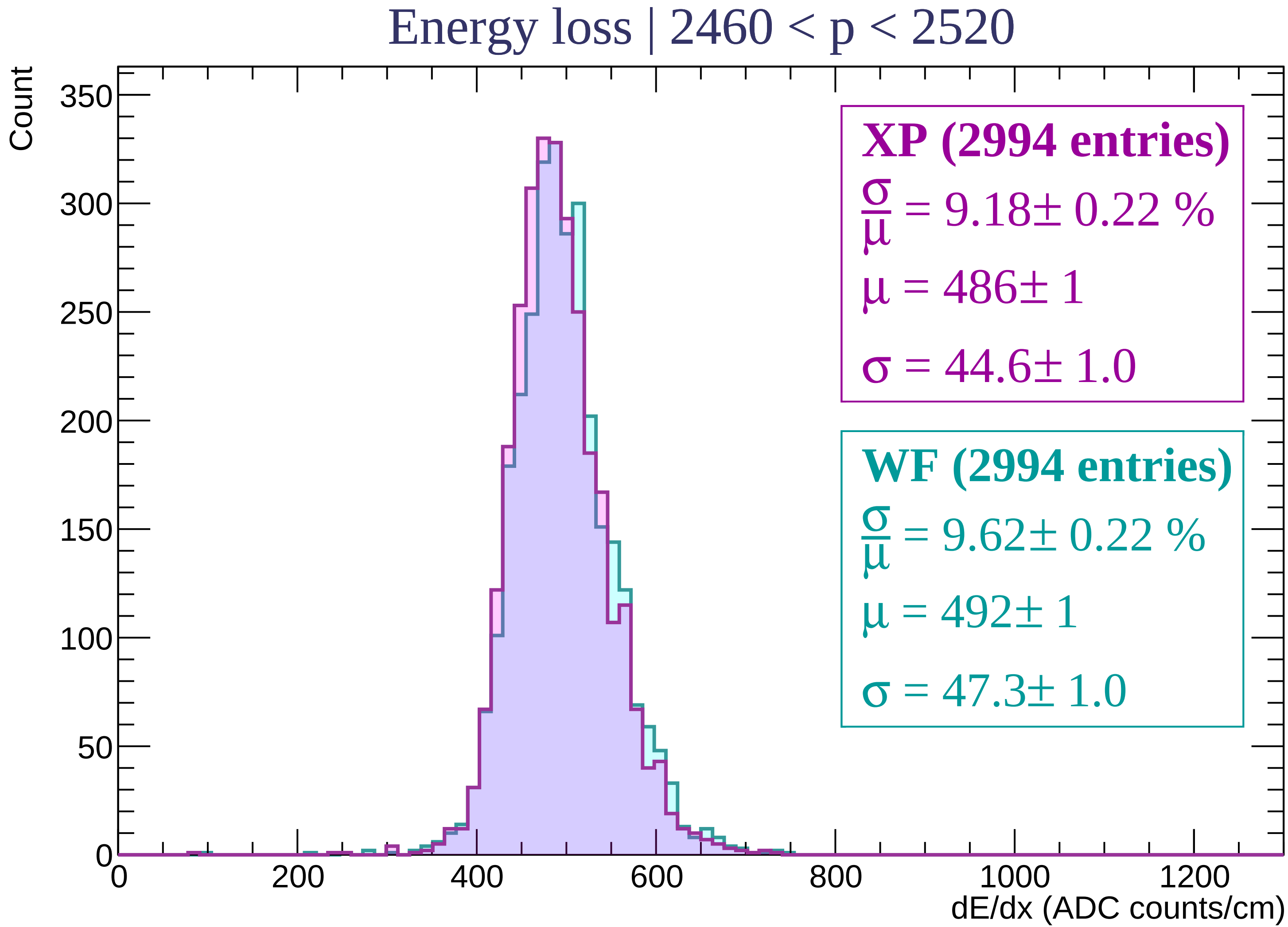
Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
XP (3213 entries)
$\frac{\sigma}{\mu}$ = 9.09± 0.22 %
μ = 480± 1
σ = 43.6± 1.0
WF (3213 entries)
$\frac{\sigma}{\mu}$ = 9.62± 0.23 %
μ = 485± 1
σ = 46.7± 1.0
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400



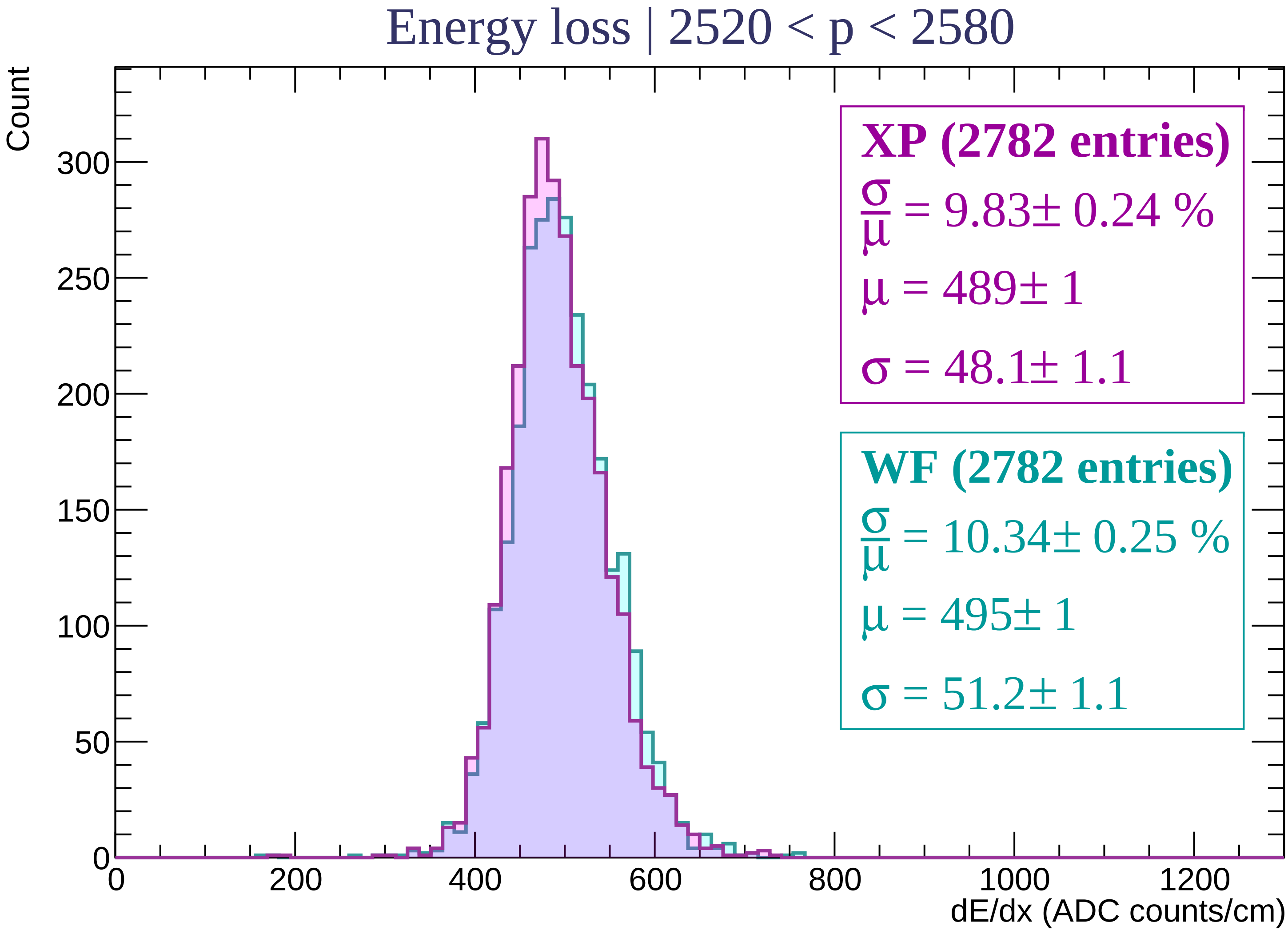
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3187 entries)
σ/μ = 9.01± 0.22 %
μ = 481± 1
σ = 43.3± 1.0
WF (3187 entries)
σ/μ = 9.79± 0.21 %
μ = 488± 1
σ = 47.8± 0.9



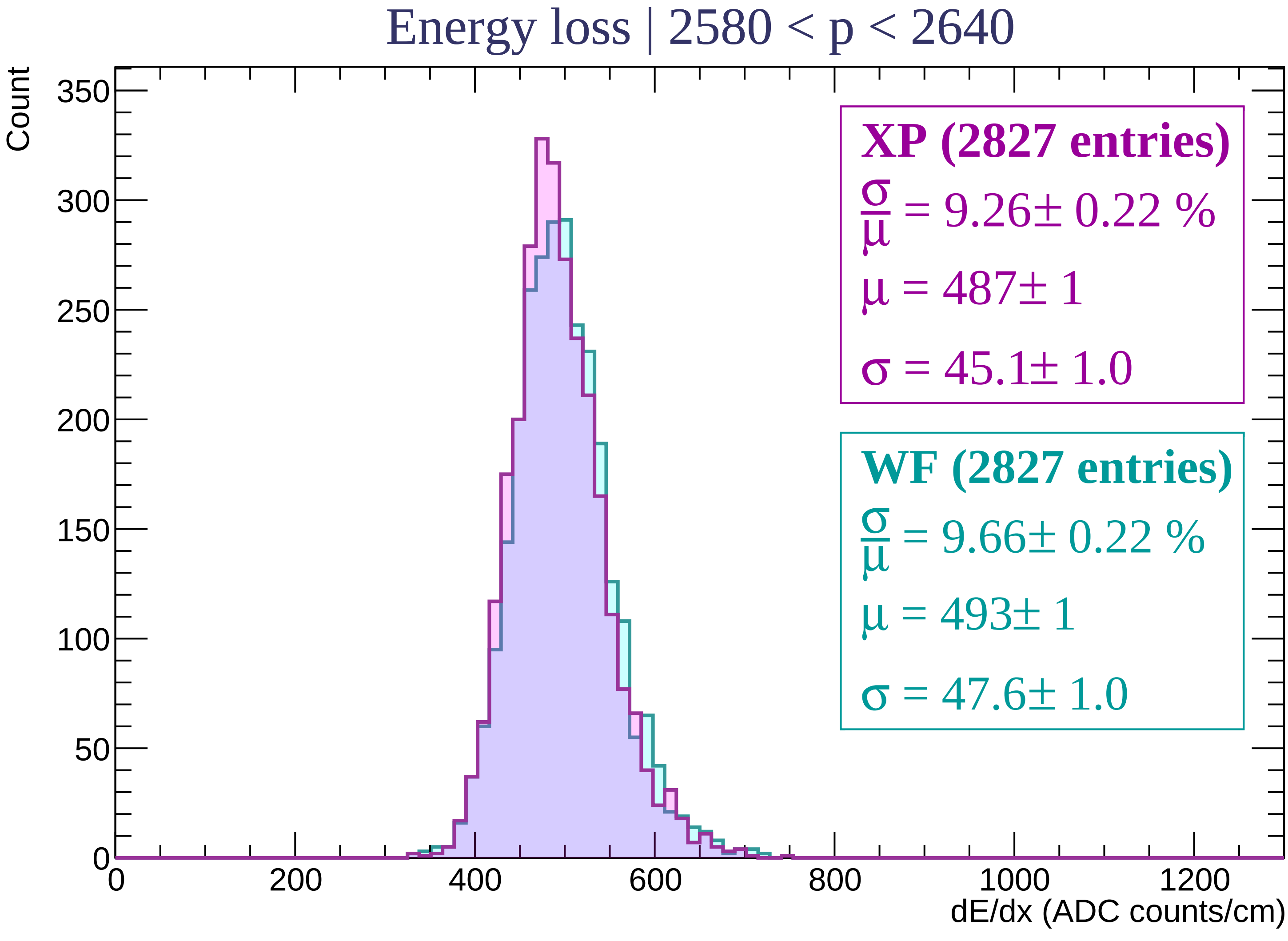
Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3116 entries)
σ/μ = 9.06± 0.20 %
μ = 484± 1
σ = 43.8± 0.9
WF (3116 entries)
σ/μ = 9.60± 0.21 %
μ = 490± 1
σ = 47.1± 0.9



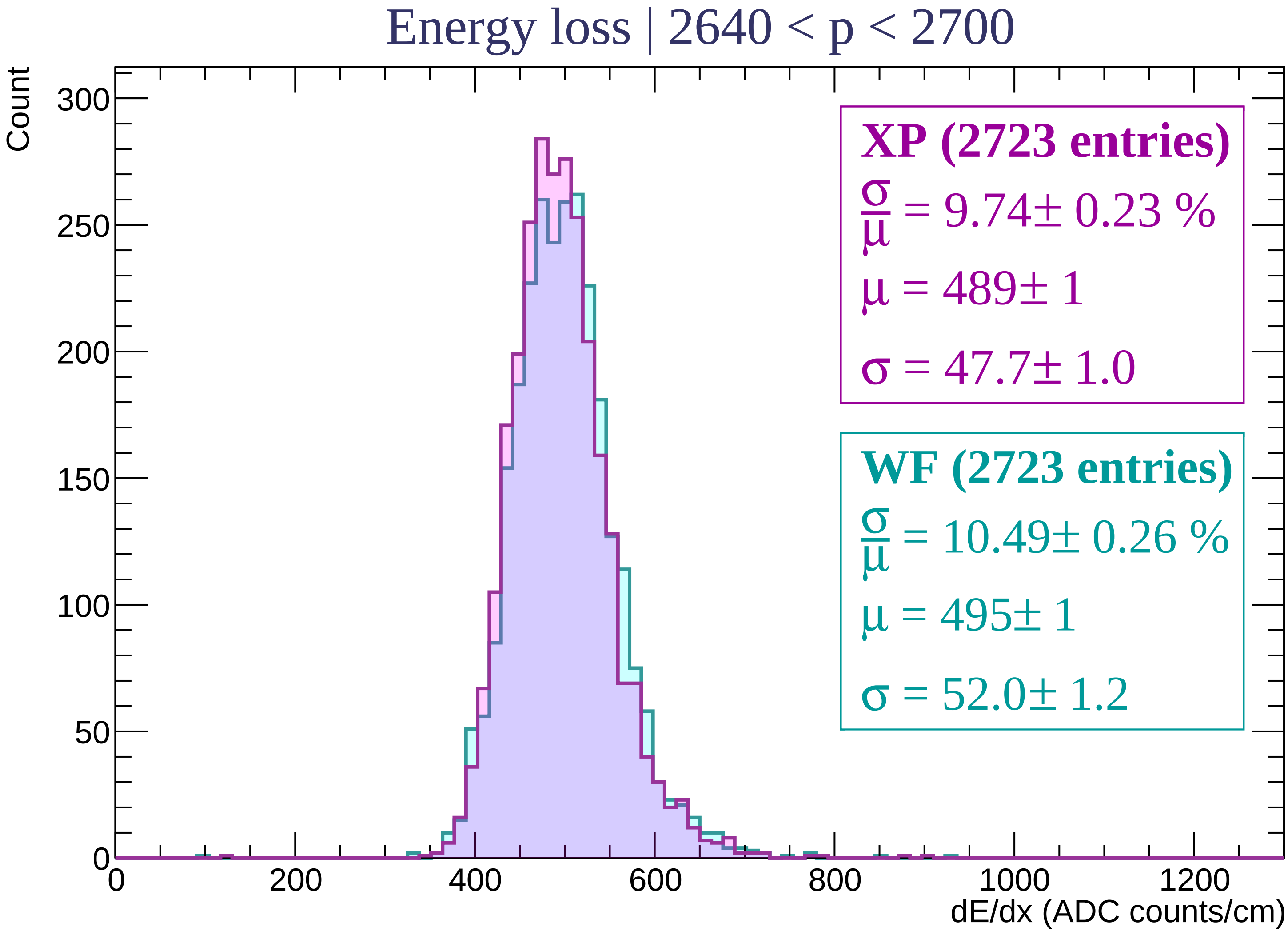
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (2994 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.22 %
μ = 486± 1
σ = 44.6± 1.0
WF (2994 entries)
$\frac{\sigma}{\mu}$ = 9.62± 0.22 %
μ = 492± 1
σ = 47.3± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



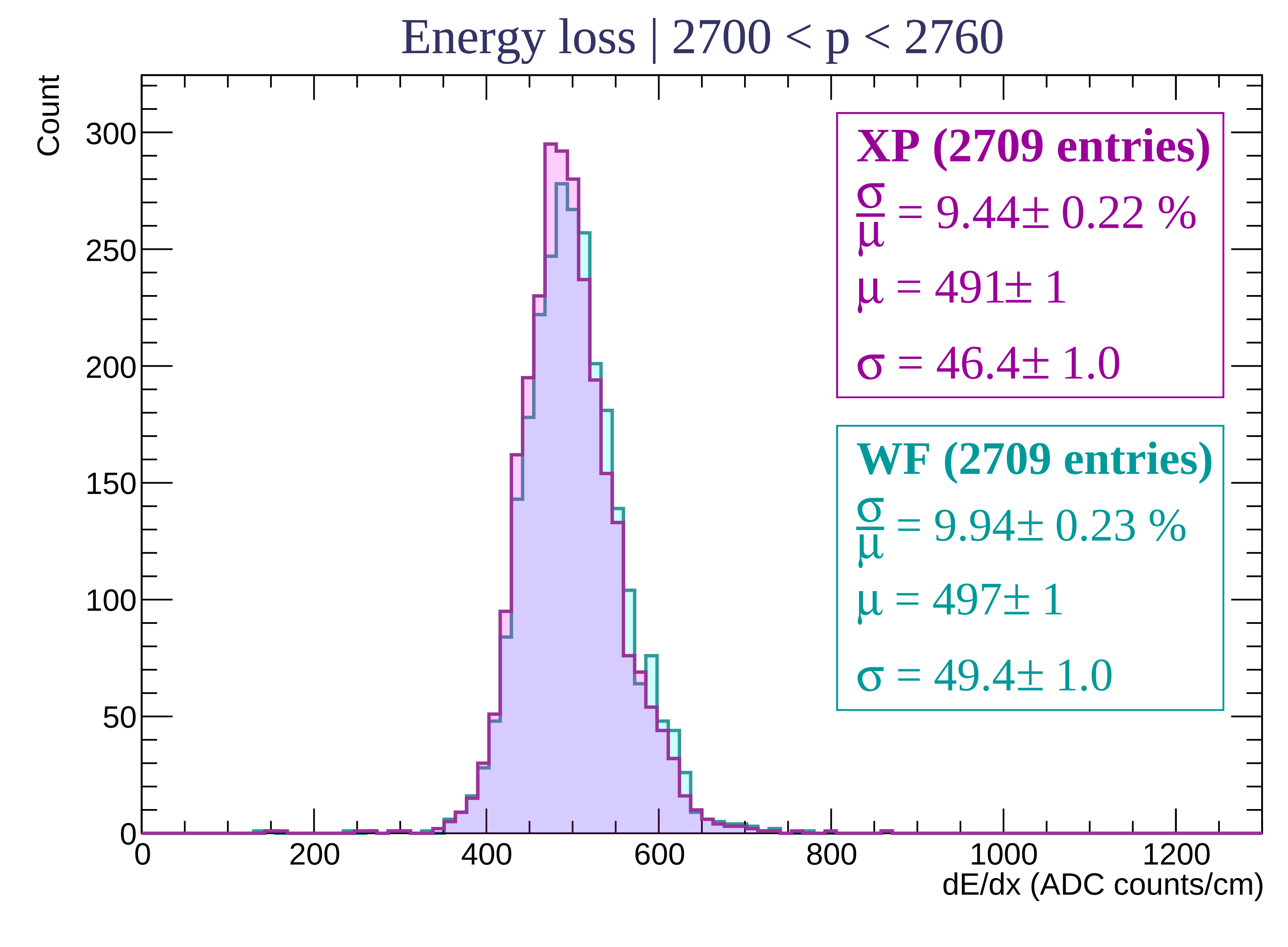
Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2782 entries)
$\frac{\sigma}{\mu}$ = 9.83± 0.24 %
μ = 489± 1
σ = 48.1± 1.1
WF (2782 entries)
$\frac{\sigma}{\mu}$ = 10.34± 0.25 %
μ = 495± 1
σ = 51.2± 1.1



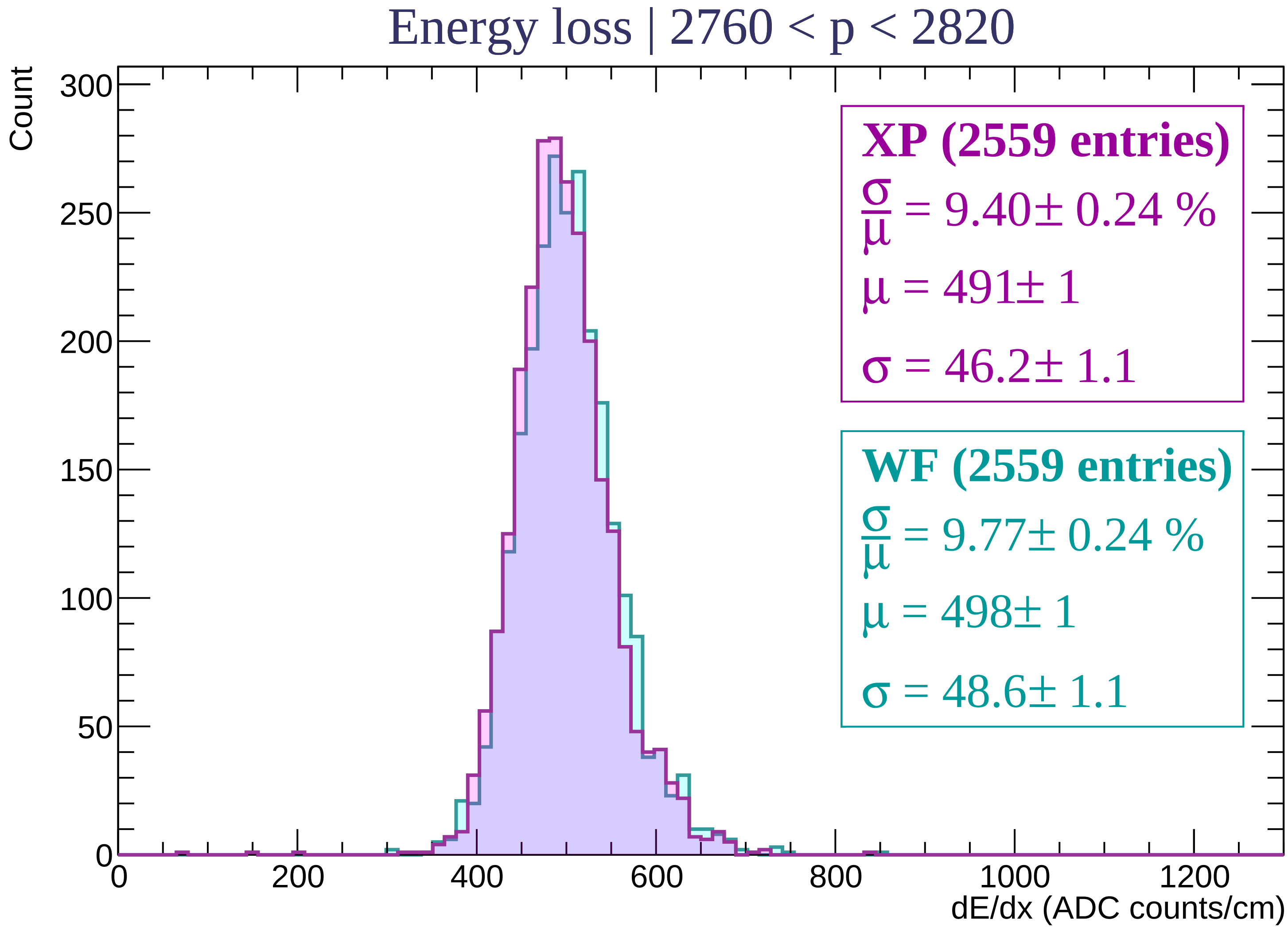
Energy loss | 2580 < p < 2640
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2827 entries)
$\frac{\sigma}{\mu}$ = 9.26± 0.22 %
μ = 487± 1
σ = 45.1± 1.0
WF (2827 entries)
$\frac{\sigma}{\mu}$ = 9.66± 0.22 %
μ = 493± 1
σ = 47.6± 1.0



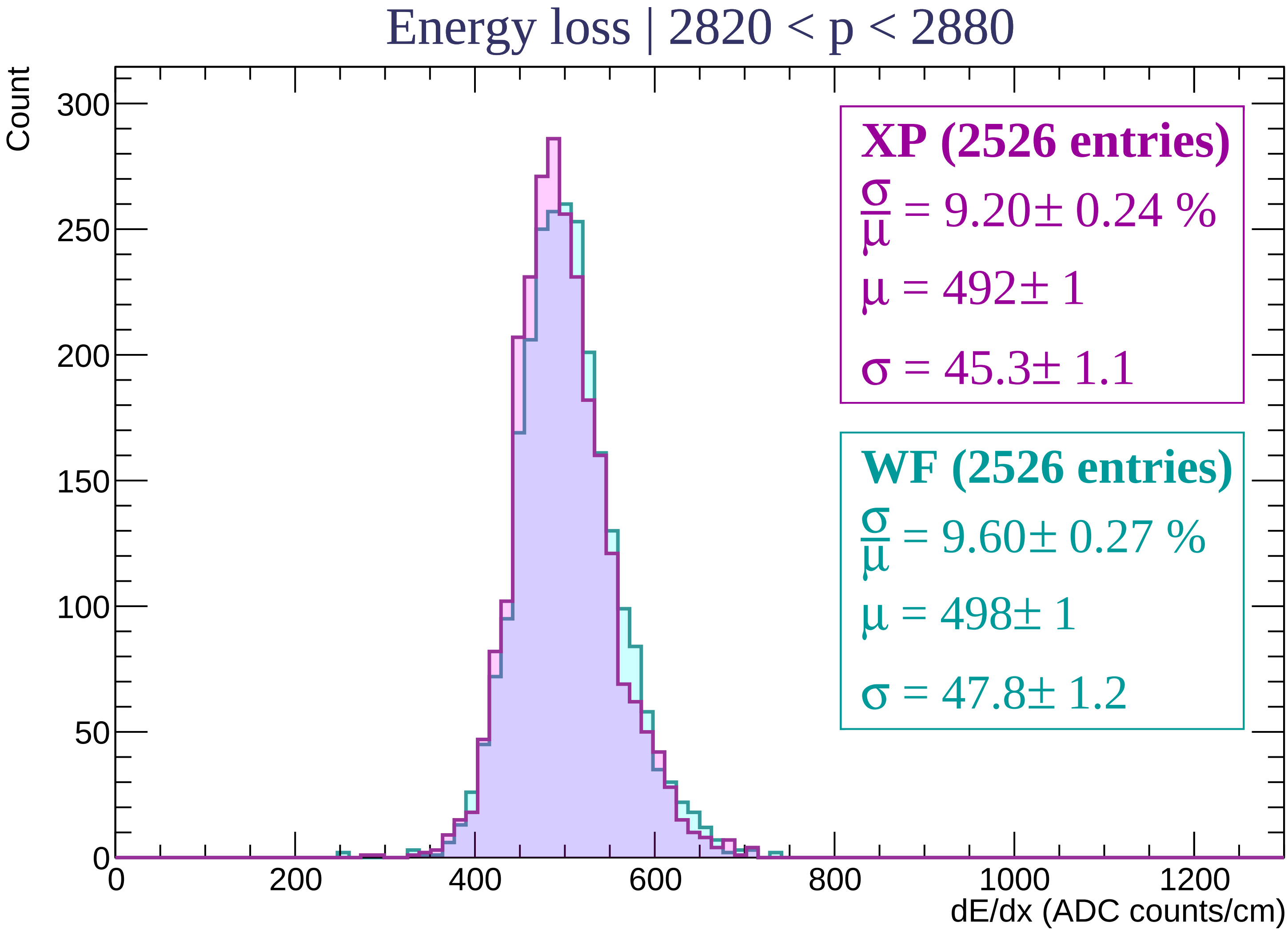
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2723 entries)
$\frac{\sigma}{\mu}$ = 9.74± 0.23 %
μ = 489± 1
σ = 47.7± 1.0
WF (2723 entries)
$\frac{\sigma}{\mu}$ = 10.49± 0.26 %
μ = 495± 1
σ = 52.0± 1.2

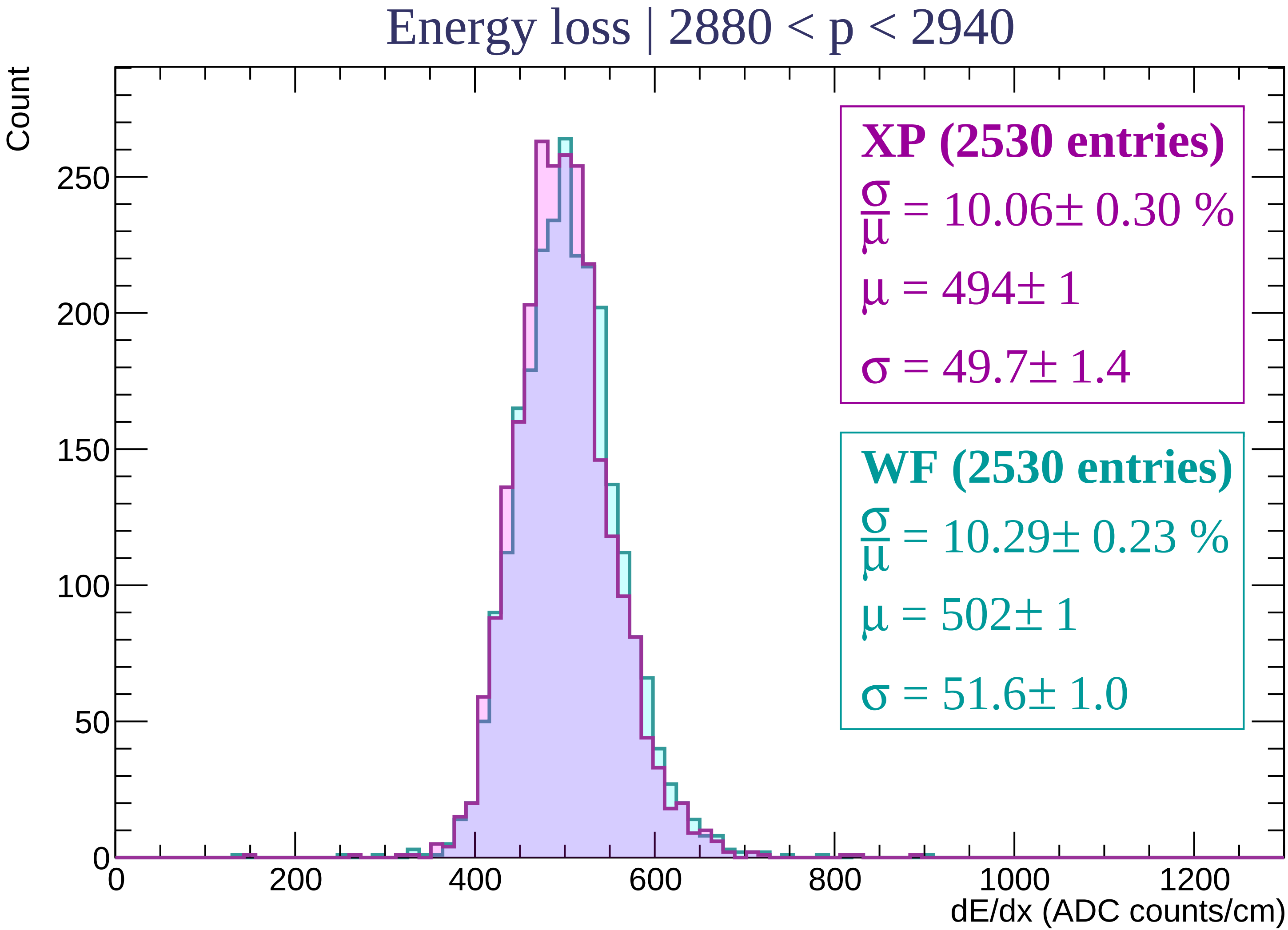


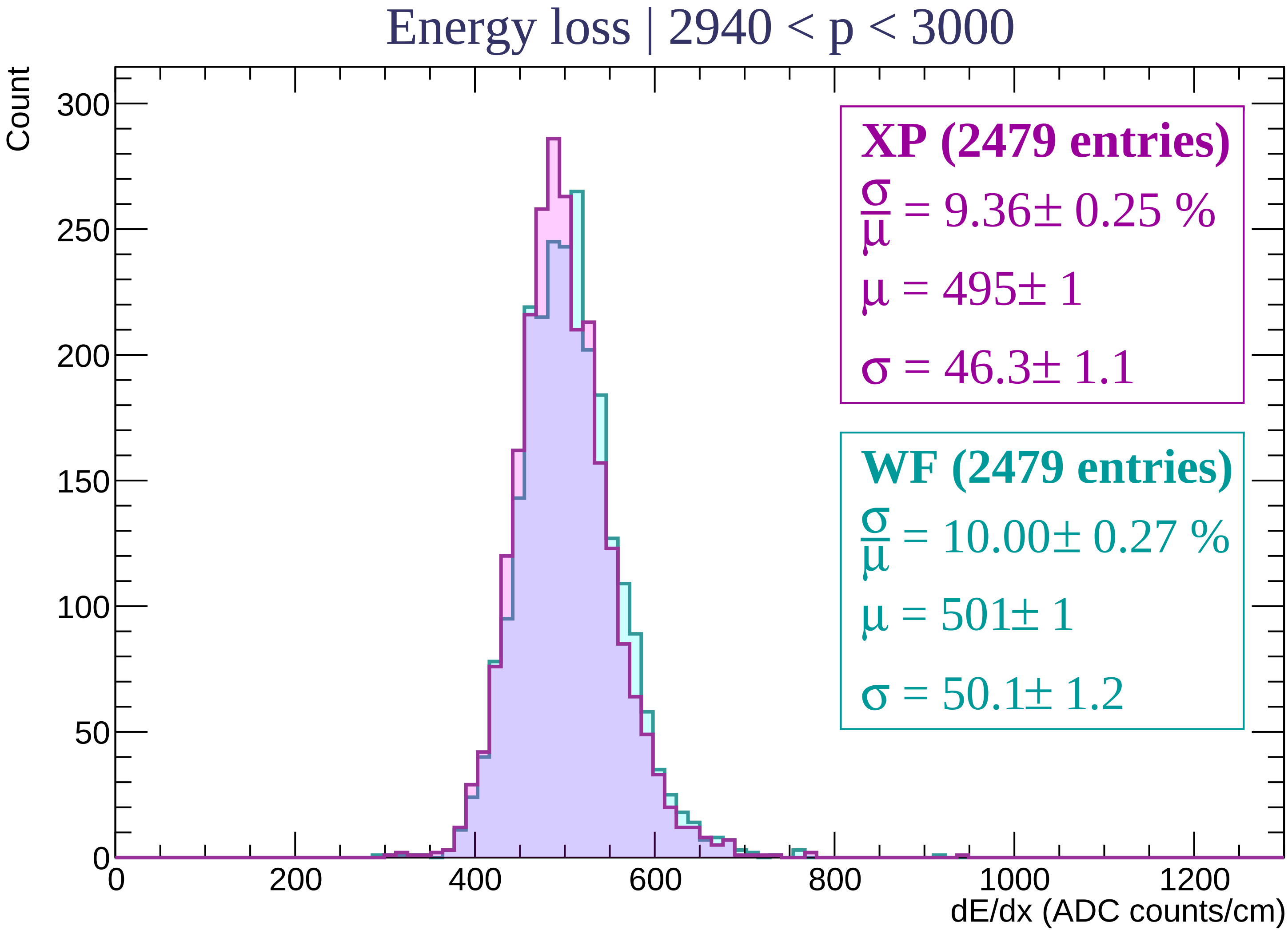
Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
XP (2709 entries)
$\frac{\sigma}{\mu}$ = 9.44± 0.22 %
μ = 491± 1
σ = 46.4± 1.0
WF (2709 entries)
$\frac{\sigma}{\mu}$ = 9.94± 0.23 %
μ = 497± 1
σ = 49.4± 1.0

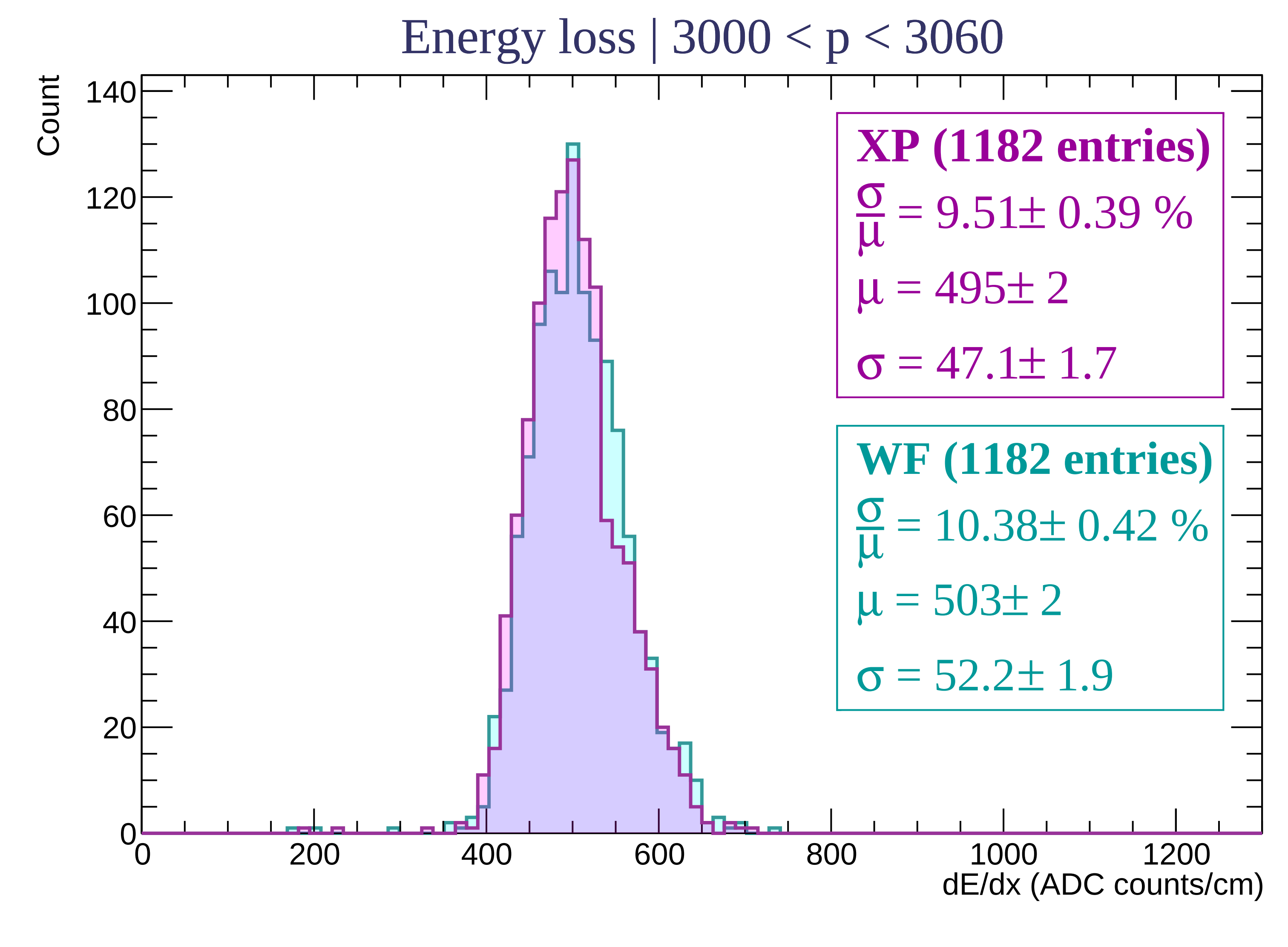


Energy loss | 2760 < p < 2820
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2559 entries)
$\frac{\sigma}{\mu}$ = 9.40± 0.24 %
μ = 491± 1
σ = 46.2± 1.1
WF (2559 entries)
$\frac{\sigma}{\mu}$ = 9.77± 0.24 %
μ = 498± 1
σ = 48.6± 1.1

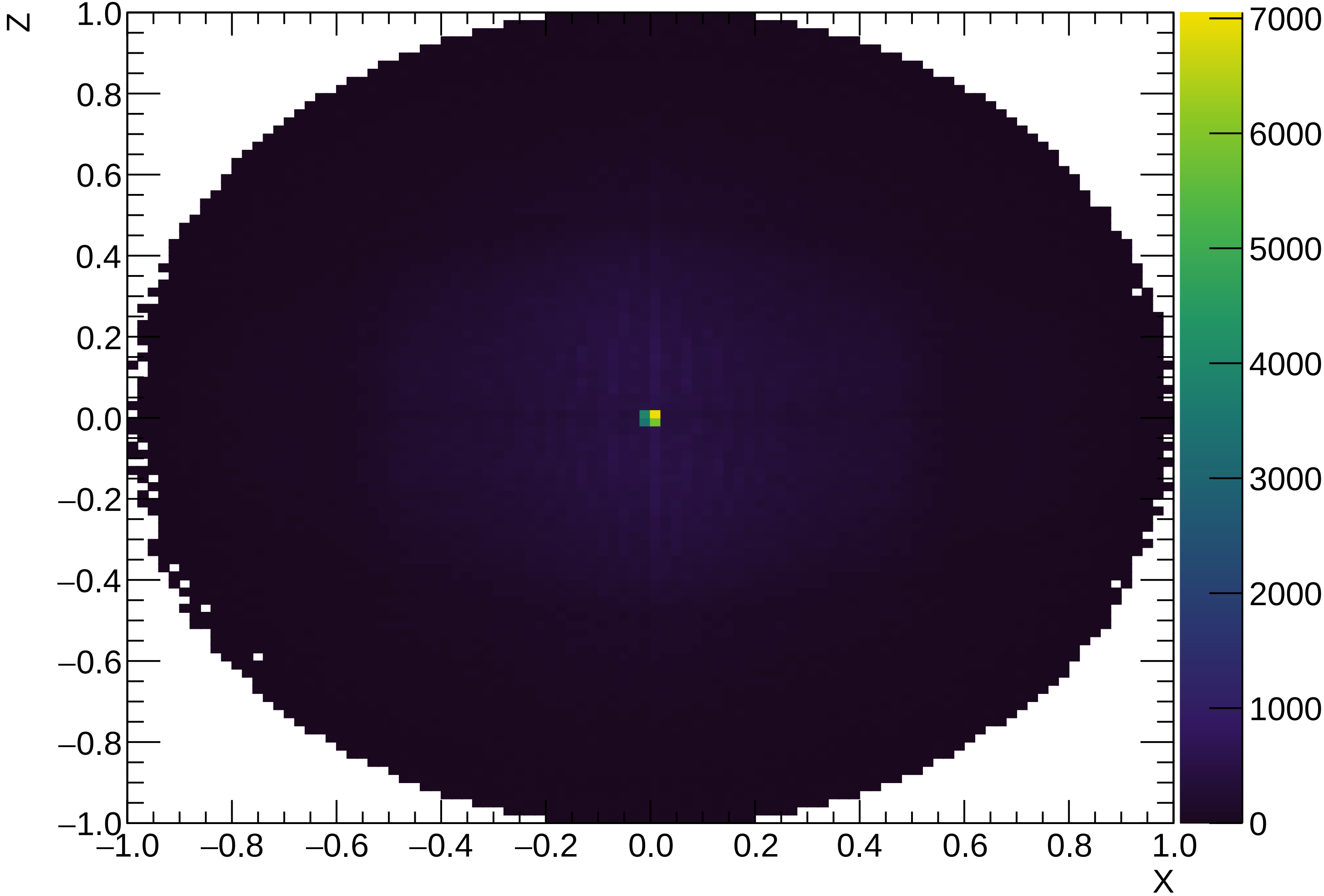

Energy loss | 2820 < p < 2880
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2526 entries)
σ/μ = 9.20± 0.24 %
μ = 492± 1
σ = 45.3± 1.1
WF (2526 entries)
σ/μ = 9.60± 0.27 %
μ = 498± 1
σ = 47.8± 1.2


Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2530 entries)
σ/μ = 10.06± 0.30 %
μ = 494± 1
σ = 49.7± 1.4
WF (2530 entries)
σ/μ = 10.29± 0.23 %
μ = 502± 1
σ = 51.6± 1.0

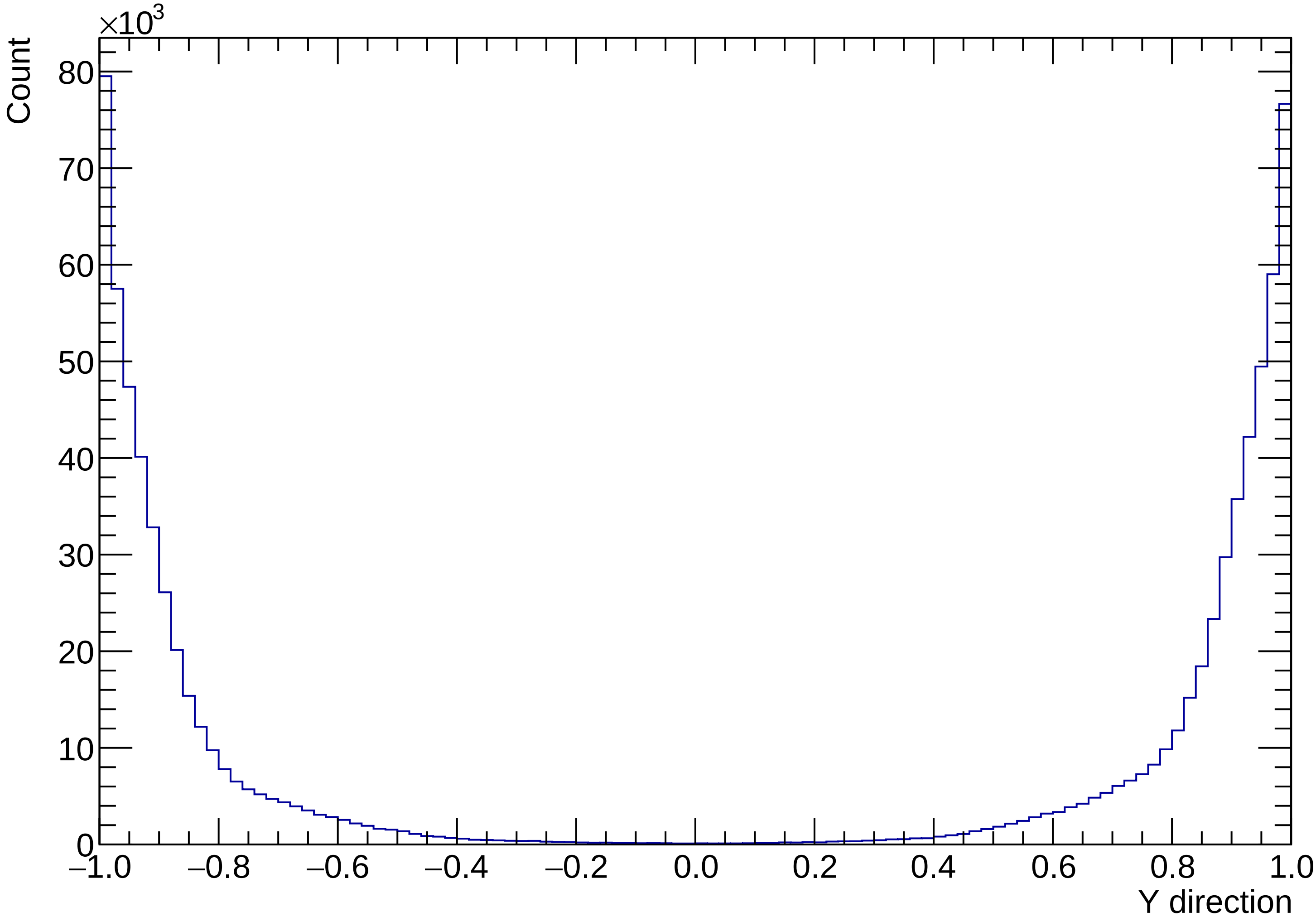

Energy loss | 2940 < p < 3000
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2479 entries)
$\frac{\sigma}{\mu}$ = 9.36± 0.25 %
μ = 495± 1
σ = 46.3± 1.1
WF (2479 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.27 %
μ = 501± 1
σ = 50.1± 1.2

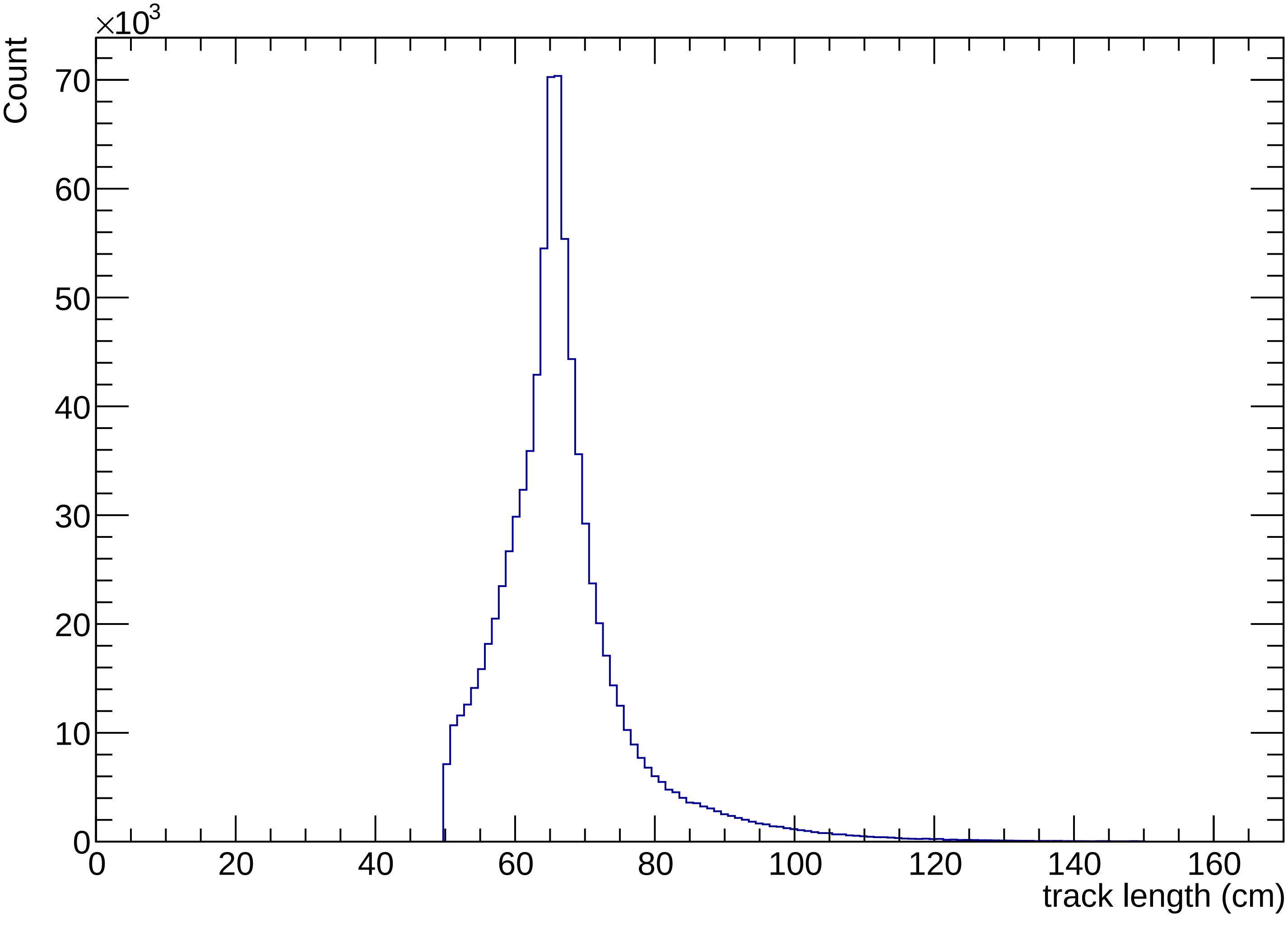


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (1182 entries)
$\frac{\sigma}{\mu}$ = 9.51± 0.39 %
μ = 495± 2
σ = 47.1± 1.7
WF (1182 entries)
$\frac{\sigma}{\mu}$ = 10.38± 0.42 %
μ = 503± 2
σ = 52.2± 1.9

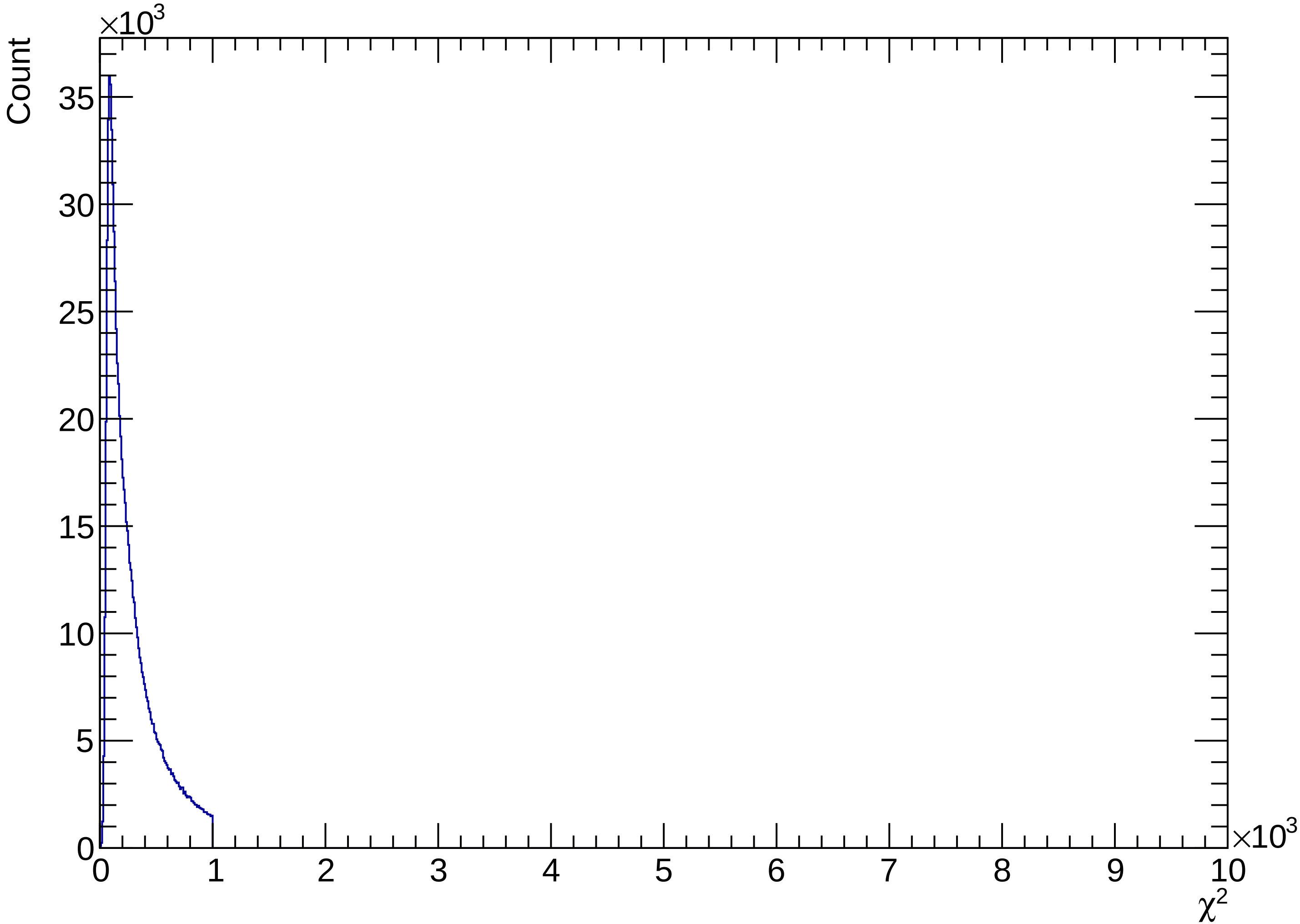


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
7000
6000
5000
4000
3000
2000
1000
0

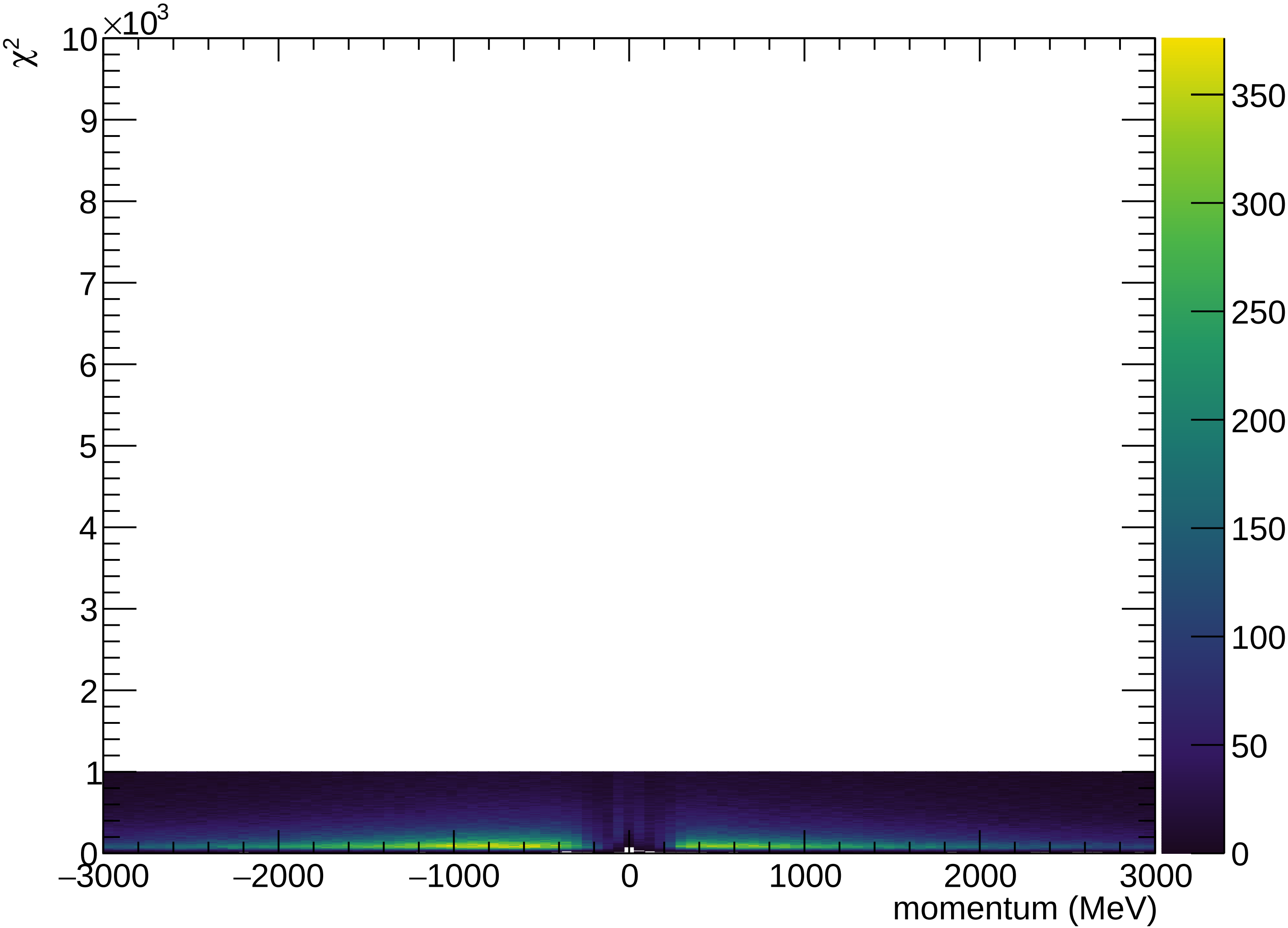

×10^3
Count
80
70
60
50
40
30
20
10
0
–1.0
–0.8
–0.6
–0.4
–0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



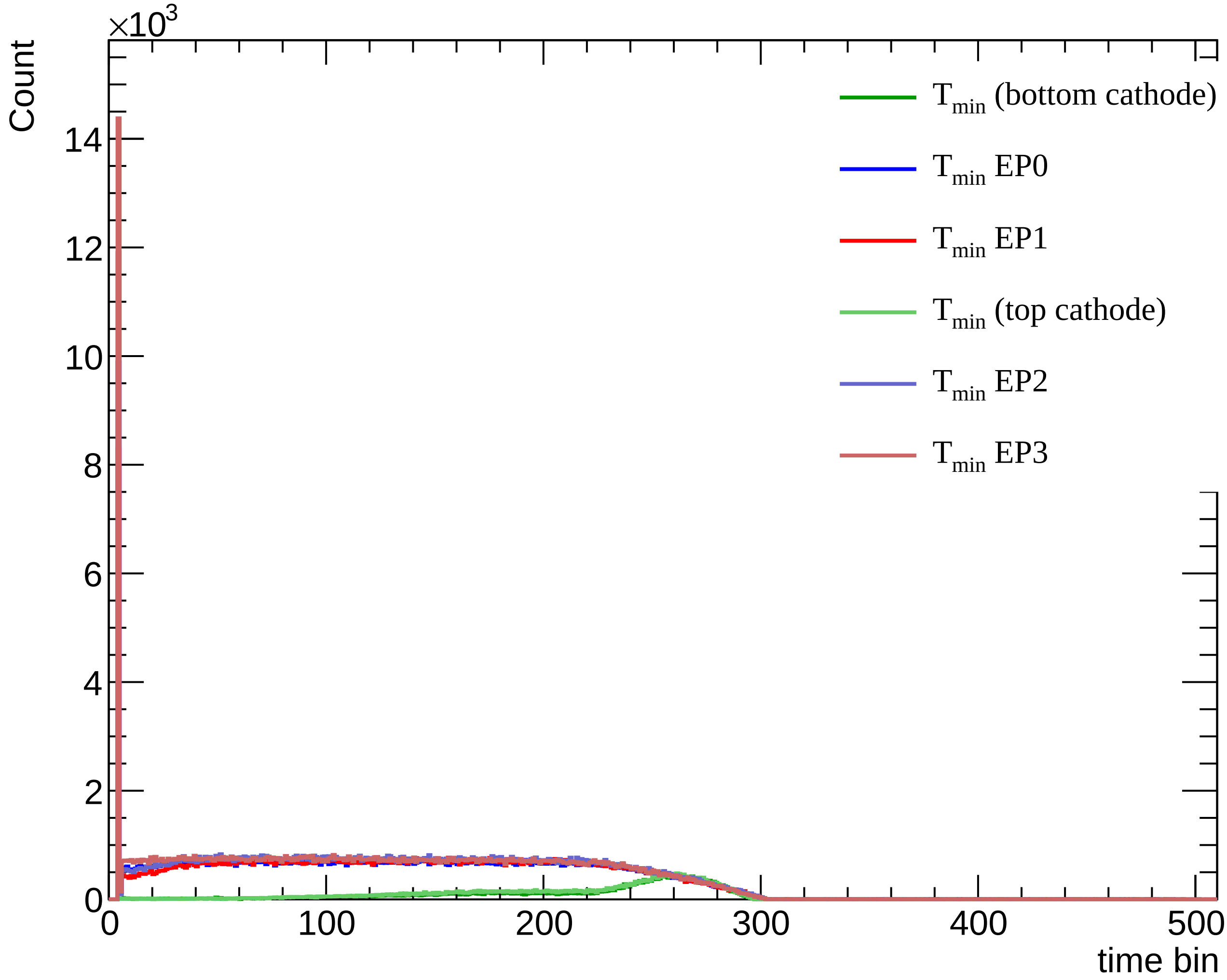
×10^3
Count
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)



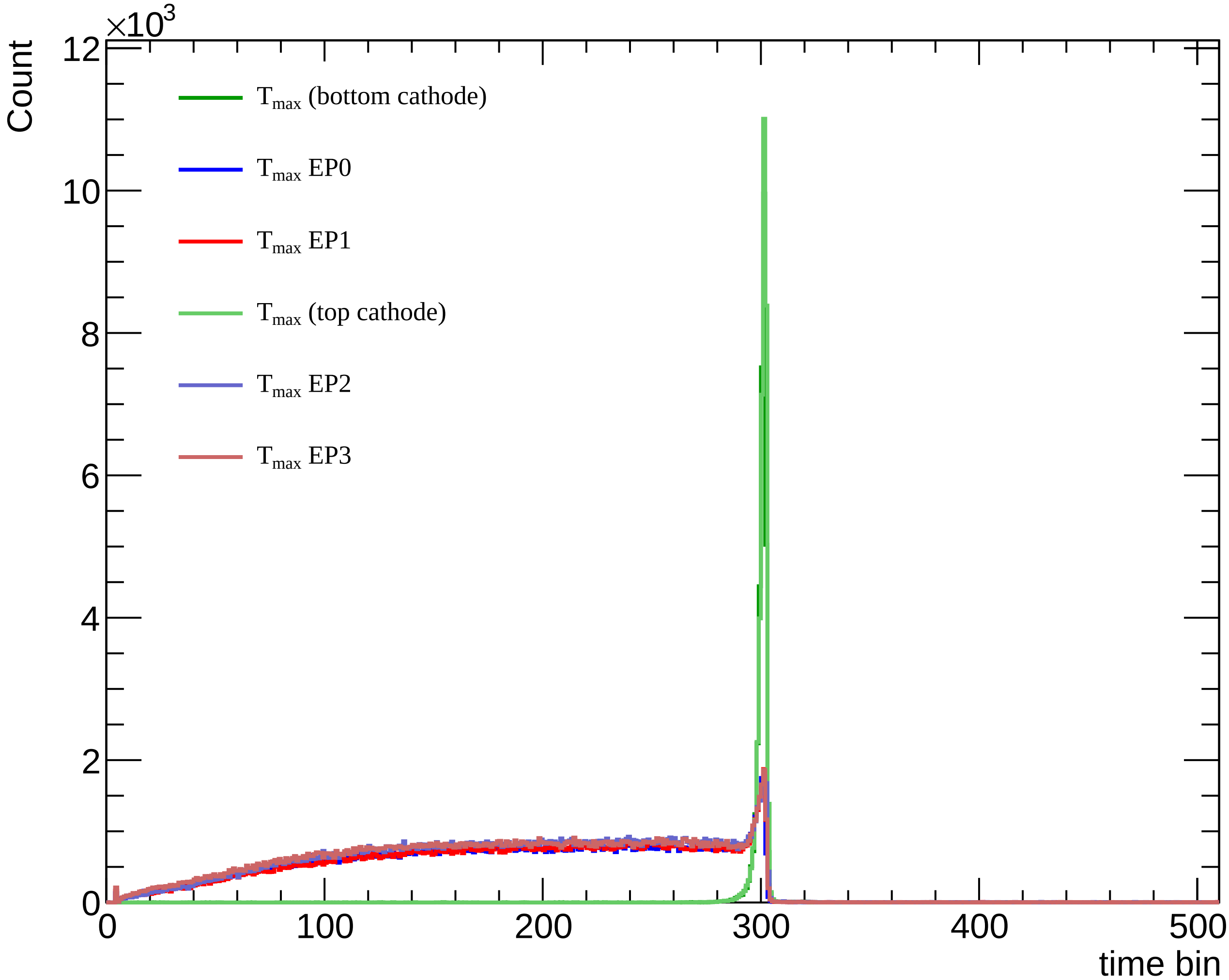
Count
×10^3
35
30
25
20
15
10
5
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
350
300
250
200
150
100
50
0



×10³
Count
time bin
0
2
4
6
8
10
12
14
100
200
300
400
500
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
×10^3
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3